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Department of Physics and Astronomy "A. Righi"

Theoretical Physics

Statistical Field Theory

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Part I

Phase Transitions and Critical Phenomena

Introduction

The study of macroscopic matter in the vicinity of continuous phase transitions requires a fundamental departure from the traditional methods of microscopic statistical mechanics. While standard statistical ensembles excel at describing systems governed by weak interactions or localized correlations, these formalisms become fundamentally insufficient when approaching a critical point. In this regime, thermal fluctuations manifest across all macroscopic length scales, the correlation length diverges asymptotically, and the system exhibits a striking independence from its underlying microscopic Hamiltonian, a profound physical property mathematically formalized as universality.

This first module bridges the conceptual gap between discrete statistical models and the continuous, field-theoretic frameworks necessary to capture critical phenomena. The treatment begins by critically re-examining the foundational postulates of statistical mechanics and the heuristic power of Mean Field Theory. By testing these approximations against the archetypal Ising model, the limitations of neglecting spatial fluctuations become rigorously apparent. The exact analytical solutions of the Ising model in one and two dimensions, achieved through the transfer matrix formalism and Onsager's algebraic framework, serve as a definitive proof of the crucial role that spatial dimensionality plays in the stabilization of macroscopic order and the breaking of symmetries.

To systematically address the breakdown of mean-field predictions, the framework relies on the concept of coarse-graining, smoothly transitioning from discrete lattice spin variables to a continuous order parameter field. This procedure naturally yields the Landau-Ginzburg effective field theory. Within this continuous paradigm, it becomes possible to analyze the saddle-point approximation, the thermodynamics of domain walls, and the spectral nature of local fluctuations. The validity domain of the mean-field regime is then quantitatively delineated by the Ginzburg criterion, which reveals the critical dimensionalities where fluctuation-dominated physics inevitably takes over.

Ultimately, the macroscopic divergence of the correlation length implies that critical systems possess an exact underlying scale invariance. This realization is mathematically formalized through the scaling and hyperscaling hypotheses, which constrain the thermodynamic singularities and interrelate the critical exponents. The theoretical path culminates in the introduction of the Renormalization Group (RG). By constructing transformations in real space (effectively tracking the flow of the system's coupling constants under variations of the observation scale) the RG framework provides the definitive mathematical explanation for universality classes. Through the rigorous analysis of flow diagrams, fixed points, anomalous dimensions, and linear perturbations around the Gaussian model, the following pages establish the modern theoretical machinery required to understand continuous scale symmetry in statistical field theory.

1 Statistical Mechanics Review

In this chapter we will review some basic concepts of statistical mechanics, which will be useful for the rest of the course. We will start with the microcanonical ensemble, which is the simplest ensemble, and then we will move to the canonical ensemble, which is more general and more useful for practical applications. Finally, we will briefly discuss quantum statistical mechanics, which is a generalization of classical statistical mechanics to quantum systems.

1.1 | Equal a Priori Probabilities

We begin recalling the foundamental principles of statistical mechanics, starting from a classical isolated system of N particles, with total energy E and volume V.

The **microstates** of the system are the points in the phase space of the system, which is a $6N$-dimensional space spanned by the coordinates and momenta of the particles. The **macrostates** of the system are defined by the macroscopic variables, such as energy, volume, and number of particles. The **ensemble** is a collection of microstates that are compatible with a given macrostate. The **probability distribution** of the system is a function that assigns a probability to each microstate in the ensemble.

It is useful to recall the Hamiltonian formalism, which is a powerful tool for describing the dynamics of classical systems. The Hamiltonian H is a function of the coordinates and momenta of the particles, and it encodes the energy of the system. The equations of motion for the system can be derived from the Hamiltonian using Hamilton's equations, which describe how the coordinates and momenta evolve in time. For each particle i we have

$$(q_1, q_2, q_3, p_1, p_2, p_3) \in \mathcal{M},$$

where $\mathcal{M}$ is the phase space of the system, which is a $6N$-dimensional manifold. The Hamiltonian H is a function that maps points in the phase space to real numbers, and it encodes the energy of the system

$$H = H(q_1, \ldots, q_{3N}, p_1, \ldots, p_{3N}) : \mathcal{M} \to \mathbb{R}.$$

The equations of motion for the system can be derived from the Hamiltonian using Hamilton's equations, which describe how the coordinates and momenta evolve in time:

$$\begin{cases} \dot{q}_i = \dfrac{\partial H}{\partial p_i}, \\ \dot{p}_i = -\dfrac{\partial H}{\partial q_i}. \end{cases}$$

We can visualize the phase space of the system as a high-dimensional space, where each point represents a microstate of the system. The energy shell defined by the total energy E is a hypersurface in this phase space, and the trajectories of the system are represented by curves that lie on this hypersurface.

TODO: add a picture of the phase space and the energy shell

If we suppose that

1. we know positions and momenta of all particles at a given time t, thus we know the microstate of the system at time t,

2. we have an "infinite-powerful" computer that can solve the equations of motion for the system, thus we can predict the microstate of the system at any time t' given the microstate at time t,

we should be able to predict the future evolution of the system with perfect accuracy. However, in practice, we do not have access to the microstate of the system, and we do not have an infinite-powerful computer that can solve the equations of motion for the system. Specially if we think about a system with $N \sim N_A = 6 \times 10^{23}$ particles, it is impossible to know the microstate of the system, and it is impossible to solve the equations of motion for the system. We need to introduce approximations and assumptions in order to be able to make predictions about a large collection of particles.

A standard way to introduce statistical mechanics is through the so-called **postulate of equal a priori probabilities**:

Postulate 1.1 (Equal A Priori Probabilities)**.** *Macroscopic observables can be computed defining a **probability distribution** on the phase space of the system, which assigns probabilities to all microstates, and taking the expectation value of the observable with respect to this probability distribution. In an isolated system, all microstates are equally probable, thus the probability distribution is uniform on the energy shell defined by the sub-manifold of the phase space where the energy is equal to E:*

$$\mathcal{M}_E = \{(q_1, \ldots, q_{3N}, p_1, \ldots, p_{3N}) \in \mathcal{M} : H(q_1, \ldots, q_{3N}, p_1, \ldots, p_{3N}) = E\}.$$

Let us analyze this postulate in more detail through an exemplification.

Example. Suppose we have a system of N non-interacting particles in a box of volume V, with total energy E. We are interested in computing the average particle velocity $\overline{v}_x$ along some axis x.

We should compute the time dependent function $v_x(t)$ and then take the time average of it:

$$\overline{v}_x(t) = \overline{v}_x(\{q_j(t)\},\, \{p_j(t)\}) = \frac{1}{N}\sum_{j=1}^{N} \frac{v_{x,j}(t)}{m},$$

which is very complicated in general, since we have to solve the equations of motion for the system in order to find the trajectories of the particles in the phase space. However, we can use the postulate of equal a priori probabilities to compute the average particle velocity without solving the equations of motion.

The function $\overline{v}_x(\{q_j\},\, \{p_j\})$ is defined on the phase space of the system, and it assigns a value to each microstate of the system. Given the knowledge on some initial microstate, the system will evolve along a curve in the fixed energy shell defined by E:

$$\mathbf{R}(t) = \{q_j(t),\, p_j(t)\}, \quad \overline{v}_x(t) = \overline{v}_x(\mathbf{R}(t)).$$

We can now assume the system to be at the equilibrium, thus we can take the time average of the function $\overline{v}_x(t) \sim \overline{v}_x$ over a long time interval T as

$$\overline{v}_x = \frac{1}{T}\int_0^T \mathrm{d}t\, \overline{v}_x(\mathbf{R}(t)).$$

Now we can use the postulate of equal a priori probabilities to replace the time average with an average over the energy shell defined by E:

$$\overline{v}_x = \frac{1}{|\mathcal{M}_E|}\int_{\mathcal{M}_E} \mathrm{d}\mathcal{M}\, \overline{v}_x(\{q_j\},\, \{p_j\}),$$

where we are using the previous definition of the energy shell $\mathcal{M}_E$ and $|\mathcal{M}_E|$ is the volume of the energy shell in the phase space

$$|\mathcal{M}_E| = \int_{\mathcal{M}_E} \mathrm{d}\mathcal{M}.$$

Thus we have computed the average particle velocity without solving the equations of motion for the system, but using only the postulate of equal a priori probabilities: it told us that *averaging over the system trajectory is equivalent to averaging over the whole manifold at fixed energy.*

Let us now analyze the postulate of equal a priori probabilities in more detail. We have to stress some important points about it:

- In typical equilibrium situations, macroscopic quantities (such as $\overline{v}_x$ from the example) usually are constant in time only up to small fluctuations
$$\overline{v}_x(t) = \overline{v}_x + \delta v_x(t).$$
We have previously neglected these fluctuations, but they are actually present in the system, and they can be formally removed by studying the time averages. Indeed we expect those fluctuations to be small and to occur at microscopic time scales, much smaller than the time scales of the macroscopic observables, thus they will not affect the macroscopic properties of the system. Thus measuring (i.e. averaging) over a long time interval T will remove these fluctuations:
$$\frac{1}{\tau}\int_0^\tau \mathrm{d}t\,(\overline{v}_x + \delta v_x(t)) = \frac{1}{\tau}\left(\overline{v}_x\tau + \int_0^\tau \mathrm{d}t\,\delta v_x(t)\right) = \overline{v}_x,$$
where we have used the fact that the time average of the fluctuations is zero.

- The traditional definition of the postulate of equal a priori probabilities is simpler than the one we have given, since it is usually stated as *in an isolated system, all microstates are equally probable*, without specifying that the microstates are the points in the energy shell defined by E. We have given a more modern and precise definition of the postulate, which is more useful for our purposes, since it allows us to compute the average of macroscopic observables as an average over the energy shell in the phase space.

- In the study of dynamical systems, it is possible to mathematically formalize the problem of proving the postulate of equal a priori probabilities, and to find conditions under which it holds. This is a very active field of research, called **ergodic theory**, and in some cases one can prove its validity, while in other cases one can find counterexamples where it does not hold. There exist complicated systems for which the postulate of equal a priori probabilities does not hold, but these systems are not relevant for our purposes, thus we will assume the validity of the postulate for the rest of the course, and we will use it as a fundamental principle.

- We have assumed above that the system is at equilibrium, thus we can take the time average of the macroscopic observable over a long time interval T. If the system is not at equilibrium, then the time average of the macroscopic observable will not be constant in time, and it will depend on the initial conditions of the system. Studying how a system reaches the equilibrium (or even defining it rigorously) is non-trivial, and it is a very active field of research, called **non-equilibrium statistical mechanics** or **thermalization**. Typically the approach to equilibrium is obtained via a weak coupling among theh system and the environment, which allows the system to exchange energy and information with the environment, so that the system is not completely isolated.

With the above discussion at hand, it is useful to define the concept of **typical behaviour**. Given a function defined on the phase space (an observable) $f(\{q_j, p_j\})$ associated with some macroscopic quantity, we can define a **typical point** $\mathbf{R} = \{q_j, p_j\}$ as the point in which the function f takes a value that is close to the average value of the function over the energy shell defined by E:

$$f(\mathbf{R}) = \frac{1}{|\mathcal{M}_E|} \int_{\mathcal{M}_E} \mathrm{d}\mathcal{M}\, f(\{q_j\},\, \{p_j\}) + \delta f,$$

where δf is a small fluctuation. The postulate of equal a priori probabilities implies that the typical behaviour of the system is close to the average behaviour of the system, thus we can say that the typical point in the phase space of the system is close to the average value of the function f over the energy shell defined by E.

1.2 | Statistical Ensembles

[...]

1.2.1 | Microcanonical Ensemble

The postulate of equal a priori probabilities defines the microcanonical ensemble (in general an ensemble is a collection of all possible microstates that are consistent with a given set of macroscopic constraints), which is the ensemble of microstates that are compatible with a given macrostate defined by the energy E, volume V, and number of particles N. The probability distribution of the system in the microcanonical ensemble is uniform on the energy shell defined by E, thus we can write it as

$$\rho_{mc}(\{q_j\},\,\{p_j\}) = \begin{cases} \dfrac{1}{|\Gamma_\Delta(E)|}, & \text{if } E \leq H(\{q_j\},\,\{p_j\}) \leq E+\Delta, \\ 0, & \text{otherwise,} \end{cases} \tag{1.2.1}$$

where Δ is a small energy interval that defines the width of the energy shell, and $|\Gamma_\Delta(E)|$ is the volume of the energy shell in the phase space, which can be computed by considering the normalization condition for the probability distribution

$$\int \rho_{mc}\,\mathrm{d}^{3N}q\,\mathrm{d}^{3N}p = 1 \implies \rho_{mc} = \frac{1}{\int_{\Gamma_\Delta(E)} \mathrm{d}^{3N}q\,\mathrm{d}^{3N}p} = \frac{1}{|\Gamma_\Delta(E)|}.$$

Given a function f defined on the phase space of the system, we can compute its **ensamble average**, denoted by $\langle f\rangle_{mc}$ or $\mathbb{E}_{mc}[f]$, as the average of the function over the energy shell defined by E with respect to the probability distribution of the microcanonical ensemble:

$$\langle f\rangle_{mc} = \int f(\{q_j\},\,\{p_j\})\rho_{mc}(\{q_j\},\,\{p_j\})\,\mathrm{d}^{3N}q\,\mathrm{d}^{3N}p. \tag{1.2.2}$$

A *consistency condition* for the microcanonical ensemble (or any other ensemble) is that the statistical fluctuations of macroscopic observables are small in the thermodynamic limit $N \to \infty$. This means that the variance of a macroscopic observable f should vanish in the thermodynamic limit, thus we can write

$$\sigma_f^2 = \frac{\langle f^2\rangle_{mc} - \langle f\rangle_{mc}^2}{\langle f^2\rangle_{mc}} \ll 1 \quad \text{for } N \to \infty.$$

Now we can use these conditions to further analyze this ensamble; in order to do so let us introduce the entropy S of the system, which can be used to derive the thermodynamic properties of the system, and to find the most probable configuration of the system.

Entropy

As seen in previous courses, in this framework we are outlining a microscopic basis for the thermodynamic properties of the system, thus we need to introduce a microscopic definition of the entropy S of the system, which is a measure of the number of microstates available to the system at a given energy E. The **Boltzmann formula** for the entropy is given by

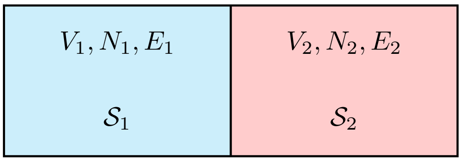
$$S(E) = k_B \log|\Gamma_\Delta(E)|, \tag{1.2.3}$$

where k_B is the Boltzmann constant, and $|\Gamma_\Delta(E)|$ is the volume of the energy shell in the phase space. The Boltzmann formula for the entropy tells us that the entropy of the system is proportional to the logarithm of the number of microstates available to the system at a given energy E. This is a fundamental result in statistical mechanics, and it allows us to derive the thermodynamic properties of the system from its microscopic properties.

We will not derive the Boltzmann formula for the entropy, but we will use it as a fundamental principle to derive the thermodynamic properties of the system. For example, we can derive a definition of temperature T of the system from the entropy S.

Temperature

Let us consider two systems $\mathcal{S}_1$ and $\mathcal{S}_2$ in thermal contact such that they can exchange energy, but they are isolated from the rest of the universe. The total system is isolated, as well as the subsystems, thus we can apply the microcanonical ensemble to them.



$$\begin{cases} N = N_1 + N_2, \\ V = V_1 + V_2, \\ E = E_1 + E_2. \end{cases}$$

Thermal contact means that the two systems can exchange energy, thus the total energy E is fixed, but the energy of each subsystem E_1 and E_2 can fluctuate. The total number of particles N and the total volume V are also fixed, since there is no exchange of matter or volume between the two systems.

The total energy can now be computed from the Hamiltonian of the system, sum of the subsystems' Hamiltonians plus an interaction term

$$H(\{q_j\}, \{p_j\}) = H_1(\{q_{1j}\}, \{p_{1j}\}) + H_2(\{q_{2j}\}, \{p_{2j}\}) + \delta H_{12}(\{q_{1j}\}, \{p_{1j}\}, \{q_{2j}\}, \{p_{2j}\}),$$

where the last term accounts for the interaction between the two systems, which we will neglect in the following. The energy of the total system is fixed, but the energy of each subsystem can fluctuate. The probability of finding the system in a state with energy E_1 is proportional to the number of states available to the second system with energy $E_2 = E - E_1$. If we fix an energy shell of width Δ, we can write

$$E_k^{(1)} = k\Delta, \qquad E_k^{(2)} = E - E_k^{(1)} = E - k\Delta,$$

and with this "discretization" we can count the number of states available to the total system at a fixed energy E as

$$\Gamma(E) = \sum_{k=0}^{E/\Delta} \Gamma_1(E_k^{(1)})\Gamma_2(E - E_k^{(1)}),$$

where $\Gamma_k(E) = \int_{E<H_k<E+\Delta} \mathrm{d}\{q_{kf}\}\mathrm{d}\{p_{kf}\}$ is the number of states available to the k-th subsystem at energy E.

Thus we are saying that *in order to count the number of microstates at energy E we have to multiply the number of microstates of the subsystem S_1 with energy ϵ_1 to the number of microstates of S_2 at energy $\epsilon_2 = E - \epsilon_1$, then sum over all possible values of ϵ_1.*

The **most probable configuration** of the system is the one that maximizes the number of states available to the total system, which is equivalent to **maximizing the entropy** of the total system. Thus we can compute the entropy of the total system as

$$S(E) = k_B \log \left[\sum_{k=0}^{E/\Delta} \Gamma_1(E_k^{(1)}) \Gamma_2(E - E_k^{(1)}) \right].$$

Now, in order to maximize it, we can take the derivative of the entropy with respect to E_1 and set it to zero: let us denote $\bar{E}_1$ the energy of the subsystem S_1 that maximizes the entropy, then $\bar{E}_2 = E - \bar{E}_1$ is the energy of the subsystem S_2 that maximizes the entropy. We have

$$\Gamma(\bar{E}^{(1)})\Gamma(\bar{E}^{(2)}) \le \Gamma(E) \le \frac{E}{\Delta}\Gamma(\bar{E}^{(1)})\Gamma(\bar{E}^{(2)}),$$

where the first inequality comes from the fact that $\Gamma(E)$ is a sum over all the values of E_1, thus it is greater than or equal to the single term with $E_1 = \bar{E}_1$, while the second inequality comes from the fact that there are at most E/Δ terms in the sum, thus we can upper bound it by multiplying the maximum term by the number of terms. Taking the logarithm of these inequalities we arrive at

$$S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2) \le S(E, V) \le S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2) + k_B \log \left(\frac{E}{\Delta} \right),$$

where both inequalities are saturated in the thermodynamic limit $N \to \infty$: entropy and energy are extensive quantities, thus $S(E, V) \sim N$ and $E \sim N$, so the last term in the upper bound is negligible in the thermodynamic limit

$$k_B \log \left(\frac{E}{\Delta} \right) \sim k_B \log(N) \ll N \sim S(E, V).$$

Thus we can write, in the thermodynamic limit, the entropy of the total system as the sum of the entropies of the subsystems, computed at the most probable values of the energy of the subsystems:

$$S(E,\, V) = S(\bar{E}^{(1)}, V_1) + S(\bar{E}^{(2)}, V_2).$$

The first consequence of this result is that the **entropy is an extensive quantity**, since it scales with the number of particles N in the system. The second consequence is that the *most probable configuration of the system is the one that maximizes the entropy*, which is equivalent to maximizing the number of states available to the total system. Thus we can write

$$\Gamma(E) = \Gamma_1(\bar{E}^{(1)})\Gamma_2(\bar{E}^{(2)}).$$

Therefore terms with different values of E_1 are negligible compared to the term with $E_1 = \bar{E}_1$, which is the one that maximizes the number of states available to the total system. This is a consequence of the fact that the number of states available to the total system grows exponentially with the number of particles N, thus even a small difference in energy between different configurations can lead to a huge difference in the number of states available to the total system. $\bar{E}^{(1)}$ and $\bar{E}^{(2)}$ are the most probable values of the energy of the subsystems, which are the ones that maximize the entropy of the total system: the *equilibrium energy values*. Mathematically, we can obtain $\bar{E}^{(1)}$ from the saddle-point condition:

$$\frac{\partial}{\partial \epsilon} \left[\Gamma_1(\epsilon) \Gamma_2(E - \epsilon) \right]_{\epsilon = \bar{E}^{(1)}} = 0,$$

which, expanding the derivative, can be written as

$$\left[\frac{\partial}{\partial \epsilon} \Gamma_1(\epsilon) \right]_{\epsilon = \bar{E}^{(1)}} \Gamma_2(E - \epsilon) + \Gamma_1(\epsilon) \left[-\frac{\partial}{\partial \epsilon} \Gamma_2(E - \epsilon) \right]_{\epsilon = \bar{E}^{(2)}} = 0,$$

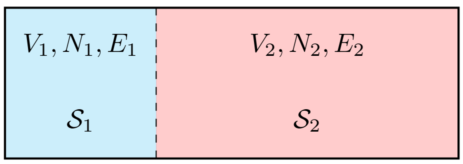
which now, dividing by $\Gamma_1(\bar{E}^{(1)})\Gamma_2(\bar{E}^{(2)})$ (in order to build the logarithmic derivative), and using the definition of entropy we arrive at

$$\frac{\partial}{\partial\epsilon}S_1(\epsilon)\bigg|_{\epsilon=\bar{E}^{(1)}} = \frac{\partial}{\partial\epsilon}S_2(\epsilon)\bigg|_{\epsilon=\bar{E}^{(2)}}.$$

This is a fundamental result, which tells us that at equilibrium the derivative of the entropy with respect to the energy is the same for both subsystems. This allows us to define a thermodynamic quantity which is equal in systems in thermal equilibrium, called **temperature** T (recovering the first principle of thermodynamics). The temperature is defined as the inverse of the derivative of the entropy with respect to the energy, thus we can write

$$\frac{1}{T} = \frac{\partial S}{\partial E}(E, V).$$

1.2.2 | Canonical Ensemble



$$\begin{cases} N = N_1 + N_2, \\ V = V_1 + V_2, \\ E = E_1 + E_2, \end{cases} \quad \text{and} \quad \begin{cases} N_2 \gg N_1, \\ V_2 \gg V_1, \\ E_2 \gg E_1. \end{cases}$$

The second system S_2 is called the *heat bath*, and it is so large that it can exchange energy with the first system S_1 without changing its thermodynamic properties. The total system is isolated, thus we can apply the microcanonical ensemble to it. The probability of finding the system in a state with energy E_1 is proportional to the number of states available to the second system with energy $E_2 = E - E_1$, which is given by $\Omega_2(E - E_1)$.

Technically speaking, we want to find the probability distribution of the subsystem S_1 when it is in contact with a heat bath S_2, knowing the total probability distribution: the total system is isolated, thus we can apply the microcanonical ensemble to it, and we know that it will be evolving with a trajectory in the phase space that is uniformly distributed on the energy shell defined by the total energy E. This means that if we analyze the evolution of the subsystem S_1 in its phase space, we will find that it will not be limited to fixed energy states, but it will be able to explore all the energy states available to it, with a probability that is proportional to the number of states available to the heat bath S_2 at the corresponding energy. we are interested in finding the **induced probability distribution** of the subsystem S_1.

Let us visualize the phase space of the total system as a high-dimensional space, where each point represents a microstate of the total system. The energy shell defined by the total energy E is a hypersurface in this phase space, and the trajectory of the total system is uniformly distributed on this hypersurface. The subsystem S_1 can be thought of as a projection of the total system onto a lower-dimensional subspace, where each point represents a microstate of the subsystem S_1. The probability distribution of the subsystem S_1 is then given by the projection of the uniform distribution on the energy shell onto this lower-dimensional subspace. Differents probabilities correspond to different energy states of the subsystem S_1, and the probability of finding the subsystem S_1 in a state with energy E_1 is proportional to the number of states available to the heat bath S_2 at energy $E - E_1$.

The idea is to fix a point in the phase space of the subsystem S_1 with energy E_1, and then count the number of points in the phase space of the heat bath S_2 that are compatible with this point, i.e. that have energy $E - E_1$. This will give us the number of states available to the heat bath S_2 at energy $E - E_1$, which is proportional to the probability of finding the subsystem S_1 in a state with energy E_1.

So we fix an energy ϵ for the subsystem S_1, and we want to find the number of states available to the heat bath S_2 at energy $E - \epsilon$. *The number of points in the phase space of S that corresponds to P is equal to the number of points in the phase space of S_2 that corresponds to $E - \epsilon$.* Thus we can write

$$\rho(\{q_f\}, \{p_f\}) \propto \frac{\Gamma_2(E-\epsilon)}{\Gamma(E)}.$$

We know that the entropy of the heat bath S_2 is related to the number of states available to it by the Boltzmann formula (we are using $k_B = 1$ for simplicity), such that

$$\Gamma_2(E-\epsilon) = e^{S_2(E-\epsilon)} \sim e^{S_2(E) - \epsilon \frac{\partial S_2}{\partial E}\epsilon},$$

where we have used the fact that the entropy is an extensive quantity, thus we can write it as a function of the energy per particle, and we have expanded it around the total energy E of the heat bath S_2. We can now recall the temperature as a thermodynamic quantity defined as the inverse of the derivative of the entropy with respect to the energy, while reinserting the Boltzmann constant k_B for clarity, we have

$$\rho(\{q_f\}, \{p_f\}) \propto e^{-\beta\epsilon} = e^{-\frac{\epsilon}{k_B T}},$$

where $\beta = \frac{1}{k_B T}$ is the inverse temperature. Thus we have derived the Boltzmann distribution for the subsystem S_1 in contact with a heat bath S_2. The probability of finding the subsystem S_1 in a state with energy ϵ is proportional to the exponential of the negative of the energy divided by the temperature of the heat bath S_2. This is the canonical ensemble, where the subsystem S_1 is in thermal equilibrium with a heat bath S_2 at temperature T.

We can now define the partition function Z as the normalization constant for the probability distribution of the subsystem S_1, such that

$$Z = \int \frac{\mathrm{d}^{3N}q\,\mathrm{d}^{3N}p}{h^{3N}N!}\, e^{-\beta H(\{q\},\{p\})},$$

which can be used as a generating function for the thermodynamic properties of the subsystem S_1. For example, the free energy F can be computed from the partition function as

$$F = -k_B T \log Z = -\frac{1}{\beta}\log Z,$$

and the internal energy U can be computed as

$$U = -\frac{\partial}{\partial\beta}\log Z,$$

while the entropy S can be computed as

$$S = -\frac{\partial F}{\partial T} = k_B \beta^2 \frac{\partial F}{\partial \beta}.$$

Discrete System

A discrete system is a system that can only take a finite number of states, such as a spin system or a lattice gas. In this case, let us imagine a collection of N particles that can only occupy a finite number of states, let us imagine two values of spin, up and down, thus we have 2^N possible configurations of the system. The phase space of the system is now a discrete set of points, configurations (a set of sets) $\{c_j\}_{j=1}^N$, and the probability distribution of the system is defined on this discrete set of points. The partition function can be written as a sum over all possible configurations of the system, such that

$$Z = \sum_{\{c_j\}} e^{-\beta H(\{c_j\})}.$$

1.3 | Quantum Statistical Mechanics

Considering a general quantum mechanical system, we can study it with different tools, such as the hamiltonian operator, the density matrix, the partition function, etc. The Hamiltonian operator H is a self-adjoint operator that acts on the Hilbert space of the system, and it encodes the energy of the system. The density matrix ρ is a positive semi-definite operator that acts on the Hilbert space of the system, and it encodes the state of the system. The partition function Z is a scalar quantity that encodes the thermodynamic properties of the system.

$$
\begin{aligned}
\hat{H}\left|n\right\rangle &= E_n \left|n\right\rangle, \\
\left|\psi\right\rangle &\in \mathcal{H}, \\
\{\left|E_n\right\rangle\}_{n=1}^{D} \implies \left|\psi\right\rangle &= \sum_{n=1}^{D} c_n \left|E_n\right\rangle, \\
\left\langle\psi\right| \hat{A} \left|\psi\right\rangle &= \sum_{n,m} c_n^* c_m \left\langle E_n\right| \hat{A} \left|E_m\right\rangle, \\
\left|\psi(t)\right\rangle = \sum_n c_n(t) \left|E_n\right\rangle, \quad c_n(t) &= c_n e^{-i\frac{E_n}{\hbar}t} \\
\left\langle\psi(t)\right| \hat{A} \left|\psi(t)\right\rangle &= \sum_{n,m} c_n^* c_m e^{i\frac{E_n - E_m}{\hbar}t} \left\langle E_n\right| \hat{A} \left|E_m\right\rangle, \\
\implies = \frac{1}{T}\int_0^T \mathrm{d}t\, \left\langle\psi(t)\right| \hat{A} \left|\psi(t)\right\rangle &= \sum_n |c_n|^2 \left\langle E_n\right| \hat{A} \left|E_n\right\rangle.
\end{aligned}
$$

Let us report some postulates of quantum statistical mechanics:

- Postulate of equal a priori probabilities: in an isolated system, all microstates are equally probable,
$$
|\bar{c_n}|^2 = \begin{cases} \frac{1}{D}, & \text{if } E_n \in [E, E+\Delta], \\ 0, & \text{otherwise.} \end{cases}
$$
- postulate of random phases: the phases of the coefficients c_n are random variables uniformly distributed in the interval $[0, 2\pi]$, thus
$$
\frac{1}{T}\int_0^T \mathrm{d}t\, c_n^*(t) c_m(t) = \frac{1}{T}\int_0^T \mathrm{d}t\, c_n^* c_m\, e^{i\frac{E_n - E_m}{\hbar}t} = \begin{cases} |c_n|^2, & \text{if } n = m, \\ 0, & \text{otherwise.} \end{cases}
$$

The time average of an observable $\hat{A}$ is given by

$$
\overline{\left\langle\psi(t)\right| \hat{A} \left|\psi(t)\right\rangle} = \frac{1}{T}\int_0^T \mathrm{d}t\, \left\langle\psi(t)\right| \hat{A} \left|\psi(t)\right\rangle = \sum_n |c_n|^2 \left\langle E_n\right| \hat{A} \left|E_n\right\rangle.
$$

[...]

We can define the density matrix ρ as

$$
\rho = \frac{1}{Z} \sum_{E \le E_n \le E+\Delta} \left|E_n\right\rangle \left\langle E_n\right|,
$$

which is a positive semi-definite operator that encodes the state of the system. The expectation value of an observable $\hat{A}$ can be then computed as

$$
\langle\hat{A}\rangle = \mathrm{Tr}(\rho\hat{A}).
$$

We have to note that dealing with quantum statistical mechanics is more complicated than dealing with classical statistical mechanics, since we have to deal with operators instead of functions, and we have to deal with the non-commutativity of operators. However, the general principles of statistical mechanics still apply, and we can still define ensembles, partition functions, and thermodynamic quantities in the quantum case. The main difference is that we have to use the language of operators and Hilbert spaces instead of the language of functions and phase space.

Defining an ensemble in quantum statistical mechanics automatically defines a density matrix, which encodes the state of the system, and from which we can compute the expectation values of observables and the thermodynamic properties of the system. We are governing our set of eigenstates with a probability distribution, which is encoded in the density matrix, and we can compute the expectation values of observables and the thermodynamic properties of the system from this density matrix.

For a quantum system at thermal equilibrium, we have the following identification of the thermodynamic free energy F with the partition function Z:

$$F = -k_B T \log Z,$$

which is the same relation that we have in classical statistical mechanics. These postulates are like operational tools: justify them as they are can be rally difficult and it is an actual field of research with a lot of literature, but they are very useful for computing the thermodynamic properties of quantum systems at thermal equilibrium. For us, justifying why a precise postulate workd in the actual setting will be enough.

2 Phase Transitions and Mean Field Theory

Phase Transitions are fundamental phenomena in statistical physics, where a system undergoes a qualitative change in its macroscopic properties as external parameters (like temperature or pressure) are varied. These transitions can be classified into different types based on their characteristics and the nature of the change in the system. Let us start from the definition of a phase of a many-particle system, which is crucial for understanding phase transitions.

Definition 2.1 (Phase of Many-Particle System)**.** A phase of a many-particle system is a region in the parameter space (e.g. temperature, pressure) where the system exhibits distinct physical properties and behavior: it is a "type" of collective state of matter. They are defined in terms of macroscopic properties such as density, magnetization, or symmetry.

A phase transition occurs when a system undergoes an *abrupt change from one phase to another* (states) *as we smoothly vary an external parameter.* Often PTs are accompanied by a discontinuity or singularity in some thermodynamic quantity, which serves as an order parameter to characterize the transition.

A phase transition is always a result of a competition between "order" and "disorder" in the system. For example, in a ferromagnetic material, the ordered phase corresponds to a state where the magnetic moments are aligned, while the disordered phase corresponds to a state where the magnetic moments are randomly oriented. The transition between these two phases can be driven by temperature, where thermal fluctuations disrupt the order at high temperatures.

We are now going to define PTs in a mathematical framework, which will allow us to classify them into different types. To start we have to highlight how PTs are related to **singularities in the partition function** of the system at the thermodynamic limit, or equivalently in the free energy. The **free energy** is a thermodynamic potential that encodes the equilibrium properties of a system, and its behavior near a phase transition can reveal important information about the nature of the transition:

$$F = -\frac{1}{\beta} \ln Z = U - TS,$$

where Z is the partition function of the system and $\beta = 1/(k_B T)$ is the inverse temperature. A phase transition is typically associated with a non-analytic behavior of the free energy as a function of the external parameters, such as temperature or pressure. Thus PTs are defined as singularities in the thermodynamic limit of the partition function Z or equivalently of the free energy F.

Definition 2.2 (Phase Transition)**.** A phase transition is a non-analytic behavior of the free energy F as a function of external parameters (e.g., temperature, pressure) in the thermodynamic limit.

This non-analyticity manifests as a discontinuity or singularity in some thermodynamic quantity, which serves as an order parameter to characterize the transition.

Example (Finite system and phase transitions). If we consider a finite set of spins, the partition function Z is a finite sum of exponentials, which is an analytic function of the parameters:

$$Z = \sum_{\{\mathrm{conf}\}} e^{-\beta E[\mathrm{conf}]}.$$

Therefore, in a finite system, there can be no true phase transition, as the free energy will always be analytic. However, as we take the thermodynamic limit (i.e., let the number of spins go to infinity), the partition function can develop singularities, leading to non-analytic behavior in the free energy and thus allowing for phase transitions to occur.

In physics we never encounter real infinite systems, but we can approximate them by considering large systems and taking the thermodynamic limit. This is a common approach in statistical mechanics to study phase transitions, as it allows us to capture the essential features of the transition while avoiding the complications that arise from finite-size effects.

2.1 | Classification of Phase Transitions

Phase transitions can be classified into different types based on their characteristics and the nature of the change in the system. One of the first classifications was proposed by *Paul Ehrenfest*, who categorized phase transitions based on the order of the derivative of the free energy that exhibits a discontinuity at the transition point. Today, we have a more refined classification, which is based on the concept of order parameters and symmetry breaking.

Phase transitions can be classified depending on the type of singularity in the free energy. We can distinguish between first-order and continuous (second-order) phase transitions:

- **First-order phase transitions** are characterized by a discontinuity in the first derivative of the free energy with respect to some external parameter (e.g., temperature). This means that there is a **latent heat** associated with the transition, and the order parameter changes discontinuously.[1] Examples include the liquid-gas transition and the melting of a solid.

- **Continuous (second-order) phase transitions** are characterized by a continuous first derivative of the free energy, but a discontinuity in the second or higher derivatives. In this case, there is **no latent heat**, and the order parameter changes continuously. Examples include the ferromagnetic transition in the Ising model and the superconducting transition in certain materials. These are the most interesting PTs and will drive most of our discussion in the following chapters.

The latter group of PTs is also called **critical phenomena**, and they are associated with the emergence of long-range correlations and *scale invariance* at the critical point. The study of critical phenomena has led to the development of powerful theoretical tools, such as the *renormalization group*, which allows us to understand the universal behavior of systems near criticality. Another important phenomenon associated with continuous phase transitions is the concept of **spontaneous symmetry breaking**. In many cases, the ordered phase of a system exhibits a lower symmetry than the disordered phase. This spontaneous symmetry breaking is a key feature of many continuous phase transitions and plays a crucial role in determining the properties of the system near criticality.

2.1.1 | Phase Diagram of Water

Looking at the phase diagram of water, we can see that there are three main phases: solid (ice), liquid (water), and gas (vapor). The **phase boundaries** between these phases are represented by lines in the pressure-temperature (p-T) diagram, also called coexistence lines. The point where the liquid-gas phase boundary ends is called the **critical point**, which is a second-order phase transition (only "crossing" the specific point). Over this point, the properties of the liquid and gas phases become indistinguishable, suggesting that the two phases are not fundamentally different, and the system exhibits critical phenomena such as large fluctuations and long-range correlations.

[1] This happens because we can express order parameters as derivatives of the free energy, which is continuous but has discontinuous first derivative.

Moving along the phase boundaries, we can observe first-order phase transitions, where the system undergoes a discontinuous change in density and other properties. For example, when water freezes into ice or boils into vapor, there is a latent heat associated with the transition, and the order parameter (density) changes discontinuously. The first derivative of the free energy with respect to temperature (which is related to the entropy) and pressure (which is related to the volume) will show a discontinuity at these transitions, confirming their first-order nature.

img/pt_water_diagram.png

Figure 2.1: Phase diagram of water in the pressure-temperature (p-T) plane. The lines represent the phase boundaries between solid, liquid, and gas phases, with the critical point marking the end of the liquid-gas phase boundary.

Decreasing the pressure at a fixed temperature can also lead to phase transitions. For example, if we decrease the pressure of liquid water at a constant temperature, we can reach the point where it transitions into vapor. This is another example of a first-order phase transition, as there is a discontinuity in the density and other properties of the system.

Each point in the (p-T) phase diagram corresponds to a specific value of $v = \frac{V}{N}$ specific volume, except for the points on the phase boundaries (I order transition lines), where we have a coexistence of phases and the specific volume can take on different values depending on the relative proportions of the coexisting phases. For example, along the liquid-gas phase boundary, we can have a mixture of liquid and gas phases, and the specific volume will depend on the ratio of liquid to gas in the mixture. This can be seen in the following phase diagram.

The phase diagram of water in the pressure-volume (p-v) plane shows indeed the different phases and the transitions between them, with the critical point marking the end of the liquid-gas phase boundary.

img/pv_water_diagram.png

If we are below the critical temperature, at a fixed T we can observe all three phases by varying the pressure or volume: this isothermal lines form a mesoscopic domain where we can observe the coexistence of phases and the transitions between them. This is a rich area of study in statistical mechanics, as it allows us to explore the behavior of systems near criticality and understand the underlying mechanisms driving phase transitions.

Figure 2.2: Phase diagram of a pure substance in the pressure-volume (p-v) plane. The lines represent the phase boundaries between different phases, with the critical point marking the end of the liquid-gas phase boundary.

The isotherms in the previous diagram are not to be confused with the coexistence lines in the p-T diagram, which represent the phase boundaries. In the p-v diagram, the isotherms represent the behavior of the system at a fixed temperature as we vary the pressure and volume, and we encounter phase I order phase transitions as we cross the coexistence region ("dome" of saturated liquid/gas). This region is characterized by a coexistence of liquid and gas phases, where the specific volume can take on different values depending on the relative proportions of the coexisting phases

$$\rho_g \sim \frac{1}{v_g}, \quad \rho_l \sim \frac{1}{v_l}, \quad \text{at } T < T_c.$$

The critical point marks the end of this coexistence region, where the properties of the liquid and gas phases become indistinguishable.

The full phase diagram of water in the pressure-volume-temperature (p-v-T) space provides a comprehensive view of the different phases and the transitions between them. It shows how the phases of water change as we vary both temperature and pressure, with the critical point marking the end of the liquid-gas phase boundary.

img/pvt-water-diagram.png

img/pvt_water_phase_diagram.png

Figure 2.3: Phase diagram of water in the pressure-volume-temperature (p-v-T) space. On the right, the surfaces represent the different phases of water, with the critical point marking the end of the liquid-gas phase boundary. On the left, different isotherms are shown, illustrating how the phases change as we vary temperature and pressure (only near the critical point).

Transitions may be direction dependent in the phase space, meaning that the nature of the transition can change depending on the path taken in the parameter space. At the critical point, the system exhibits critical phenomena such as large fluctuations and long-range correlations. The properties of the system near the critical point can be described by critical exponents, which characterize how various physical quantities diverge or vanish as the critical point is approached. These critical exponents are universal, meaning that they depend only on the dimensionality of the system and the symmetry of the order parameter, rather than on the specific details of the microscopic interactions. This universality is a key feature of critical phenomena and allows us to classify different systems into universality classes based on their critical behavior.

Phase transitions are well defined when described with a curve in the parameter space, but we can also have transitions that occur at a single point, such as the critical point in the phase diagram of water. In this case, the transition is not associated with a curve but rather with a specific set of conditions (temperature and pressure) where the properties of the system change dramatically.

We are only going to discuss PTs which occurs with a smooth variation of the parameters, but there are also PTs that can occur with a sudden change in the parameters, such as a quench or a rapid cooling. These non-equilibrium phase transitions can lead to interesting phenomena such as the formation of topological defects and the emergence of new phases that are not present in equilibrium conditions. The study of non-equilibrium phase transitions is an active area of research in statistical mechanics and condensed matter physics.

Remark (Critical Opalescence). *The **milky appearence** (or critical opalescence) of a system near the critical point is a direct consequence of the divergence of the correlation length ξ (for example in a liquid-gas mixture). As the correlation length increases, fluctuations in the density or order parameter occur over larger and larger distances, leading to the scattering of light and giving rise to a milky or cloudy appearance. This phenomenon is a visual manifestation of the critical behavior of the system and is often observed in fluids near their critical point, such as in the case of water at its critical temperature and pressure.*

2.1.2 | Magnetic System

In this setting, we are considering a universal magnetic system, which can be described by a Hamiltonian that includes interactions between magnetic moments (spins) and an external magnetic field. The behavior of the system can be analyzed in terms of its phase transitions, which are characterized by changes in the order parameter (magnetization) as we vary the temperature and the external magnetic field.

We have few control parameters over the system: the temperature T and the external magnetic field h. The phase diagram of the magnetic system can be represented in the $T - h$ plane, where we can identify different phases based on the behavior of the magnetization.

img/h-T_phase_diagram.png

Figure 2.4: Phase diagram of a magnetic system in the (h-T) plane, showing the ferromagnetic phase at low temperatures, cut in half by the first-order phase transition line, which ends at the critical point. Over the critical point, the system is in the paramagnetic phase.

- At high temperatures and low magnetic fields, the system is in a disordered phase where the magnetization is zero: **paramagnetic phase**.
- At low temperatures and high magnetic fields, the system is in an ordered phase where the magnetization is non-zero: **ferromagnetic phase**.
- The transition between these two phases can be either first-order or second-order, depending on the specific interactions in the system. If we cross the phase buondary under the critical point, we observe a first-order phase transition, while if we cross the critical point, we observe a second-order phase transition. Over it we have only the paramagnetic phase, which is a disordered phase with zero magnetization.
- At the critical point, the system exhibits critical phenomena such as large fluctuations in magnetization and long-range correlations. The properties of the system near the critical point

can be described by critical exponents, which characterize how various physical quantities diverge or vanish as the critical point is approached.

Introducing the **magnetization** as the order parameter of the system, we can analyze how it changes across the phase transitions.

Definition 2.3 (Order Parameter)**.** An order parameter is a quantity that serves as a measure of the degree of order in a system and can be used to characterize different phases. It typically takes on a non-zero value in the ordered phase and vanishes in the disordered phase. It is expressed generally as

$$m(T) = \frac{1}{V} \lim_{h \to 0} \langle M(h, T) \rangle,$$

where the notation is deliberately describing the case of magnetization, but we can interpret more generally as the order parameter of a system, which can be defined in terms of the expectation value of some observable that characterizes the order in the system.

Thus we interpret in general as

$$\phi(T) = \frac{1}{V} \lim_{h \to 0} \langle O(h, T) \rangle,$$

where $O(h, T)$ is an observable that characterizes the order in the system, and V is the volume of the system, h is an external field that couples to the order parameter, and the limit $h \to 0$ ensures that we are measuring the spontaneous magnetization (or order parameter) in the absence of an external field.

In the paramagnetic phase, the magnetization is zero, while in the ferromagnetic phase, it takes on a non-zero value. The behavior of the magnetization near the critical point can be described by a power law, with a critical exponent that characterizes how the magnetization vanishes as we approach the critical point from the ordered phase.

Figure 2.5: Phase diagram of a magnetic system in the (M-h) plane, showing the behavior of the magnetization as an order parameter as a function of the external magnetic field at different temperatures. The coexistence line at the critical temperature is shown.

img/M-h_phase_diagram.png

This is equivalent to the jump of the volume in the liquid-gas transition (p-v water diagram), which is the order parameter of that system. Indeed, the red line in the previous figure represents the coexistence line, where the system can exist in both phases. The magnetization serves as a measure of the degree of order in the magnetic system, and it is associated to a first derivative of the free energy, which is discontinuous at a first-order phase transition and continuous at a second-order phase transition.

At any temperature below the critical temperature, we can observe a first-order phase transition by varying the external magnetic field. As we increase the magnetic field from zero, the system transitions from the paramagnetic phase to the ferromagnetic phase, with a discontinuous change in magnetization. This is a first-order phase transition, as there is a latent heat associated with the transition and the order parameter (magnetization) changes discontinuously.

The critical temperature separates two phases with different symmetries: the paramagnetic phase has a higher symmetry (rotational symmetry) than the ferromagnetic phase (which has a lower symmetry due to the alignment of spins). This is an example of spontaneous symmetry breaking, where the ordered phase exhibits a lower symmetry than the disordered phase. The study of this magnetic system and its phase transitions provides valuable insights into the behavior of many other systems that exhibit similar phenomena, such as superconductors and liquid crystals.

2.2 | Critical Behaviour

When the order parameter approaches zero continuously at the critical point, we can describe its behavior using a power law:

$$m(T) = \begin{cases} 0 & \text{for } T > T_c, \\ (T_c - T)^\beta & \text{for } T < T_c, \end{cases}$$

where β is the critical exponent that characterizes how the magnetization vanishes as we approach the critical point from the ordered phase. The value of β depends on the universality class of the system, which is determined by factors such as the dimensionality of the system and the symmetry of the order parameter. For example, in the three-dimensional Ising model, which is a prototypical model for ferromagnetic phase transitions, the critical exponent β is approximately 0.326.

The surprising thing is that if we compute the same exponent for a completely different system, such as the liquid-gas transition, using some difference in densities as the order parameter, we find the same value of β:

$$\begin{cases} |\rho_l - \rho_c| \sim (T - T_c)^\beta, \\ |\rho_c - \rho_g| \sim (T - T_c)^\beta, \end{cases} \quad \beta \approx 0.326,$$

where ρ_l and ρ_g are the densities of the liquid and gas phases, respectively, ρ_c at the critical point, and the critical exponent β is exactely the same as in the magnetic system, despite the fact that these systems have different microscopic details and interactions.

This is a manifestation of the **universality of critical phenomena**, which states that systems with different microscopic details can exhibit the same critical behavior near the phase transition, as long as they belong to the same universality class. This universality allows us to classify different systems based on their critical exponents and understand their behavior near criticality without needing to know the specific details of their interactions.

2.2.1 | Critical Exponents and Universality Classes

The critical exponents are a set of numbers that characterize the behavior of physical quantities near the critical point. They describe how various properties of the system diverge or vanish as the critical point is approached. Some of the most common critical exponents include:

- β: describes the behavior of the **order parameter** *as a function of temperature* near the critical point, $m(T) \sim (T_c - T)^\beta$ for $T < T_c$. We have already discussed this exponent in the context of the magnetization and the liquid-gas transition, where it characterizes how the order parameter vanishes as we approach the critical point from the ordered phase.

- α: describes the divergence of the specific heat (heat capacity) at constant volume and zero external field as a function of temperature near the critical point,

  $$C_V \sim |T - T_c|^{-\alpha},$$

  and the heat capacity is defined as the second derivative of the free energy with respect to temperature:

  $$C_V = \frac{\partial U}{\partial T} = \frac{\partial}{\partial T}(F + TS) = T\frac{\partial S}{\partial T} = -T\frac{\partial^2 F}{\partial T^2}.$$

If we denote by $c_{\pm}(T, h = 0)$ the heat capacity at zero external field above/below the critical point ($t = \frac{T-T_c}{T_c}$), we can express the critical behavior of the heat capacity as

$$c_{\pm}(t, h = 0) \sim |t|^{-\alpha_{\pm}},$$

where $\alpha_{\pm}$ are the critical exponents for the heat capacity, in principle different above and below the critical point.

- δ: describes the behavior of the **order parameter** *as a function of the external field* at the critical temperature,

  $$m(h) \sim h^{1/\delta} \quad \text{at } T = T_c,$$

  where h is the external magnetic field. This means that the magnetization, along the critical isotherm, behaves as a power law in the external field, going to zero as the field goes to zero with the previus dependence on δ.

- γ: describes the divergence of the *zero-field* **susceptibility** *as a function of temperature*:

  $$\chi \sim |T - T_c|^{-\gamma},$$

  where the susceptibility χ is defined as the derivative of the order parameter with respect to the external field:

  $$\chi_{\pm} = \chi(t \to 0^{\pm}) = \left(\frac{\partial m}{\partial h}\right)_{h=0}, \quad \chi_{\pm} \sim |t|^{-\gamma_{\pm}},$$

  with $t = (T - T_c)/T_c$ being again the reduced temperature. The susceptibility measures how responsive the system is to changes in the external field, and its divergence at the critical point indicates that even a small change in the magnetic field can lead to a large change in the magnetization. This is a hallmark of criticality and is associated with the emergence of long-range correlations in the system.

2.2.2 | Correlation Length and Universality

Before continuing with the definitions of ν and η, we have to introduce the concept of **correlation length**, which is a measure of how far correlations in the system extend. Near the critical point, the correlation length diverges, indicating that fluctuations occur over increasingly large distances. Let us introduce it through a little model: let us consider a system of spins on a lattice, where each spin s_i can take on values of +1 or -1. The magnetization m is defined as the average spin per site:

$$m = \frac{1}{N}\sum_{i=1}^{N}\langle s_i\rangle,$$

where the angle brackets denote the thermal average:

$$\langle s_i\rangle = \langle s_i\rangle_c = \frac{1}{Z}\sum_{\{s_k\}} s_i e^{-\beta H(\{s_k\},h)},$$

with Z being the partition function and H the Hamiltonian of the system. The correlation function $G(r)$ is defined as the average relative alignment of spins, i.e. the average product of spins at a distance r:

$$G^{(2)}_{ij} = \langle s_i s_j\rangle = \frac{1}{Z}\sum_{\{s_k\}} s_i s_j e^{-\beta H(\{s_k\},h)} = G^{(2)}(\mathbf{r}), \quad \text{where } \mathbf{r} = |\mathbf{r}_i - \mathbf{r}_j|.$$

Since $G_{ij}^{(2)}$ depends only on the distance between spins for translational invariance, it depends only on the distance r. We can then define the **connected correlation function**, which measures the correlations between fluctuations in the system:

$$G_c^{(2)}(\mathbf{r}) = \langle (s_i - \langle s_i \rangle)(s_j - \langle s_j \rangle) \rangle = \langle s_i \rangle \langle s_j \rangle - \langle s_i \rangle \langle s_j \rangle.$$

The correlation length ξ is defined as the characteristic length scale over which correlations decay in the system, away from the critical point:

$$G_c^{(2)}(\mathbf{r}) \sim e^{-r/\xi} \quad \text{for } r \gg \xi.$$

Near the critical point, the correlation function typically decays as a power law, which can be described by the critical exponent η. Let us introduce two more critical exponents describing the behavior of the correlation function and the correlation length near the critical point:

- ν: describes the divergence of the correlation length, $\xi_\pm \sim |T - T_c|^{-\nu_\pm}$, with $\nu_\pm > 0$.
- η: describes the behavior of the correlation function at the critical point, $G(r) \sim r^{-(d-2+\eta)}$ at $T = T_c$, where d is the dimensionality of the system.

Remark (Critical Opalescence and Correlation Length). *Going back to the phenomenon of critical opalescence, we can understand it better after having introduced the correlation length. Since the wavelength of visible light is much larger than the microscopic length scales of the system (i.e. atomic distances), the scattering of light must be related to density fluctuations that occur over large distances. Near the critical point, the correlation length diverges, which means that fluctuations occur over increasingly large distances. This leads to a significant scattering of light, giving rise to the milky or cloudy appearance known as critical opalescence. When the correlation length diverges, the system becomes increasingly sensitive to fluctuations, and exhibits macroscopic collective behavior that allow us to ignore the microscopic details of the system and focus on the universal properties that emerge near criticality.*

Now we are ready to define and discuss the concept of **universality classes**. Systems that share the same critical exponents and scaling behavior near the critical point are said to belong to the same universality class. The universality class of a system is determined by factors such as the dimensionality of the system, the symmetry of the order parameter, and the range of interactions between particles.

Definition 2.4 (Universality Class)**.** A universality class is a category of systems that share the same critical exponents and scaling behavior near the critical point, regardless of their microscopic details. Systems that belong to the same universality class exhibit the same critical behavior and can be described by the same effective field theory near the critical point. The universality class of a system is determined by factors such as the dimensionality of the system, the symmetry of the order parameter, and the range of interactions between particles.

Let us now make some observations about the critical behavior of systems near phase transitions, which will be important for our understanding of critical phenomena and the renormalization group.

Remark. *There is a connection between the divergence of the correlation length the response funcion. The connection relies on the fact that near the critical point, the system exhibits long-range correlations, which means that fluctuations occur over increasingly large distances. This leads to a divergence in the correlation length ξ, which in turn causes the response function (such as susceptibility) to diverge as well.*

TODO: *add reference to Kardar - stat ph of fields, 1.4*

The response function measures indeed how the system responds to external perturbations, and its divergence at the critical point indicates that even a small change in the external field can lead to a large change in the order parameter, which is a hallmark of criticality.

***TODO:** add reference to J. Cardy - scaling and renormalization in statistical physics, 1.2*

Remark. *Second order phase transitions are also called **continuous phase transitions** because the order parameter changes continuously across the transition, and there is no latent heat associated with the transition. As we will see, these PTs are associated with the **spontanoeus symmetry breaking**. In some cases, the symmetry is only **emergent**, meaning that it is not present in the microscopic description of the system but emerges as a result of the collective behavior of the system near the critical point (approximate symmetry which becomes exact only near the critical point). This emergent symmetry can play a crucial role in determining the properties of the system and its critical behavior.*

2.3 | Mean Field Theory

Mean field theory is a powerful approximation method used to study phase transitions and critical phenomena in many-body systems. The basic idea behind mean field theory is to replace the complex interactions between particles in the system with an average or "mean" field that each particle experiences. This simplification allows us to derive self-consistent equations for the order parameter and analyze the behavior of the system near the critical point.

TODO: reference: D. Tong - statistical field theory, 1 | D. Tong - statistical physics, 5

Let us concentrate on a classical model, the Ising model, which is a prototypical model for studying ferromagnetic phase transitions.

2.3.1 | Ising Model

The Ising model consists of a lattice of spins, where each spin s_i can take on values of +1 or -1. The Hamiltonian of the Ising model is given by:

TODO: insert images of spins on a lattice

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i,$$

where J is the coupling constant that determines the strength of the interaction between neighboring spins, h is the external magnetic field, and the sum $\langle i, j \rangle$ runs over pairs of **nearest-neighbor** spins. The first term in the Hamiltonian represents the interaction between neighboring spins, which favors alignment (ferromagnetic order) when $J > 0$. The second term represents the coupling of the spins to the external magnetic field, which tends to align the spins in the direction of the field.

Definition 2.5 (Coordination Number)**.** The coordination number z of a lattice is the number of nearest neighbors that each site has. For example, in a two-dimensional square lattice, each site has 4 nearest neighbors, so the coordination number is $z = 4$. In a three-dimensional cubic lattice, each site has 6 nearest neighbors, so the coordination number is $z = 6$. The coordination number plays an important role in determining the behavior of the system and the nature of the phase transition.

Our aim is to study the equilibrium properties (where observables can effectively be calculated through thermal averaging) of the Ising model and understand the nature of the phase transition it exhibits. To do this, we will apply mean field theory to derive self-consistent equations for the magnetization (order parameter) and analyze the behavior of the system near the critical point.

Each configuration of the system is specified by the set of spin values $\{s_i\}_{i=1}^N$, and it is associated to a probability given by the Boltzmann distribution:

$$P(\{s_i\}) = \frac{1}{Z} e^{-\beta H(\{s_i\})},$$

where Z is the **partition function**, defined as the sum over all possible configurations:

$$Z = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})}.$$

Example (Pair of spins). When we have only two spins, the partition function can be calculated explicitly. The Hamiltonian for two spins s_1 and s_2 is given by:

$$H = -J s_1 s_2 - h(s_1 + s_2),$$

where the sum is removed since we only have two spins. We can now compute the energy for each of the four possible configurations of the spins:

- $s_1 = +1$, $s_2 = +1$: $H = -J - 2h$.
- $s_1 = +1$, $s_2 = -1$: $H = J$.
- $s_1 = -1$, $s_2 = +1$: $H = J$.
- $s_1 = -1$, $s_2 = -1$: $H = -J + 2h$.

The partition function is then given by:

$$Z = e^{\beta J + 2\beta h} + 2e^{-\beta J} + e^{\beta J - 2\beta h}.$$

If we now wanted to compute the probability of finding the system in the configuration $s_1 = +1$, $s_2 = +1$, we would use the Boltzmann distribution:

$$P(s_1 = +1, s_2 = +1) = \frac{e^{\beta J + 2\beta h}}{Z}.$$

For the magnetization, we can compute the average spin per site:

$$M = \left\langle \sum_i s_i \right\rangle_c, \quad m = \frac{M}{N} = \frac{1}{N} \langle s_1 + s_2 \rangle_c ,$$

where the summations run over the configuration space.

$$m = \sum_{\langle s_i \rangle} p(\{s_i\}) \left(\frac{1}{N} \sum_i s_i \right),$$

where the sum runs over all possible configurations of the spins. In this case, we can explicitly compute the magnetization by summing over the four configurations.

If we were now to consider a system with $N = 100$, we would have 2^{100} possible configurations for a chain with $z = 2$, which is an astronomically large number. This makes it impossible to compute the partition function and the magnetization by explicitly summing over all configurations. This is where mean field theory comes into play, as it allows us to approximate the behavior of the system without needing to consider every possible configuration.

2.3.2 | Mean Field Approach - Euristic

Let us now apply the mean field approximation to the Ising model. The key idea is to replace the interaction term in the Hamiltonian with an average or "mean" field that each spin experiences. This allows us to derive a self-consistent equation for the magnetization.

It is a *euristic* approach, which is not exact but provides a good approximation, remaining not systematic, not rigorous and hard to fully justify. We will see how it can be derived from a more rigorous approach, which is the **Landau-Ginzburg effective field theory**.

There are many versions of this approach (different implementations of the approximation), but the idea is almost the same: we want to replace some microscopic degrees of freedom with an average field, which is the mean field. The mean field is defined as the average value of the spin at a given site:

Let us rewrite the Hamiltonian of the Ising model

$$H = -J \sum_{\langle s_i, s_j \rangle} s_i s_j - h \sum_i s_i.$$

If we now consider the case in which the coupling constant J is positive ($J > 0$ which corresponds to a ferromagnetic interaction), and a null external magnetic field ($h = 0$), the Hamiltonian simplifies to:

$$H = -J \sum_{\langle s_i, s_j \rangle} s_i s_j.$$

In this case the system exhibits a phase transition at a critical temperature T_c, below which the spins tend to align and the system enters an ordered ferromagnetic phase, while above T_c the spins are disordered and the system is in a paramagnetic phase.

Being interested in the behavior of the magnetization, we can express it as a thermal average in the canonical ensemble:

$$\left\langle \frac{1}{N} \sum_i s_i \right\rangle_c = \frac{1}{N} \sum_{\{s_i\}} \left(\sum_i s_i \right) \frac{e^{-\beta H(\{s_i\})}}{Z},$$

where the sum runs over all possible configurations of the spins. The mean field approximation consists in building a term which encodes the average effect of the neighboring spins on a given spin: it is natural to think about building this average field from the magnetization, so let us analyze it in more detail. The magnetization can assume the following values:

$$M \in [-N, \, -N+2, \, -N+4, \, \dots, \, +N] \, ,$$

where the values change by 2 because each spin can flip from +1 to -1 or vice versa. The magnetization per site is then given by:

$$m = \frac{M}{N} \in \left[-1, \, -1 + \frac{2}{N}, \, -1 + \frac{4}{N}, \, \dots, \, +1 \right].$$

The average field we are building has to be proportional to the magnetization m, which serves as a measure of the degree of order in the system. Thus we can express the sum over configurations of the spins as a sum over the magnetization M and then a sum over all configurations that correspond to that magnetization:

$$\sum_{\{s_i\}_{i=1}^N} [\dots] = \sum_{M=-N}^{N} \sum_{\{s_i\}_{i=1}^N} \delta \left(\sum_i s_i - M \right) [\dots],$$

where we have introduced a delta function to enforce the constraint that the sum of the spins equals the magnetization M. This allows us to rewrite the partition function as a sum over the magnetization, but let's understand better the meaning of this delta function through an example.

Example ($N = 2$). If we have only two spins, the possible configurations and their corresponding magnetizations are:

- A: $s_1 = +1$, $s_2 = +1$: $M = 2$.
- B: $s_1 = +1$, $s_2 = -1$: $M = 0$.
- C: $s_1 = -1$, $s_2 = +1$: $M = 0$.
- D: $s_1 = -1$, $s_2 = -1$: $M = -2$.

The delta function $\delta\left(\sum_i s_i - M\right)$ ensures that we only sum over configurations that correspond to a specific magnetization M.

Let us analyze the contribution of each configuration to the partition function: starting from a sum over all configurations, we get

$$Z = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})} = e^{-\beta H(A)} + e^{-\beta H(B)} + e^{-\beta H(C)} + e^{-\beta H(D)},$$

where $H(A)$, $H(B)$, $H(C)$, and $H(D)$ are the Hamiltonian evaluated at each configuration: considering the case $h = 0$ for simplicity, we have

$$H(A) = -J, \quad H(B) = J, \quad H(C) = J, \quad H(D) = -J.$$

Thus the partition function can be rewritten as

$$Z = 2(e^{\beta J} + e^{-\beta J}).$$

We now want to get the same result using the sum over magnetization, which is given by

$$Z = \sum_{M=-2}^{2} \sum_{\{s_i\}} \delta\left(\sum_i s_i - M\right) e^{-\beta H(\{s_i\})} = \sum_{M=-2}^{2} e^{-\beta H(M)} \sum_{\{s_i\}} \delta\left(\sum_i s_i - M\right) = 2(e^{\beta J}+e^{-\beta J}),$$

where we have used the fact that the Hamiltonian depends only on the magnetization M and not on the specific configuration of the spins, and that the sum over configurations for each magnetization value is just the number of configurations with that value of magnetization. This example illustrates how the sum over configurations can be rewritten as a sum over magnetization, with the delta function ensuring that we only consider configurations that correspond to a specific magnetization.

Now we can implement the mean field approximation by replacing the interaction term in the Hamiltonian with an average field that each spin experiences. This average field is proportional to the magnetization m, which serves as a measure of the degree of order in the system: in a fixed magnetization sector (m-sector) we can replace the interaction term with an effective field that depends on the magnetization (reinserting the external magnetic field h):

$$H_{mf}(\{s_i\}, m) = -J \sum_{\langle i,j \rangle} s_i\, m - h \sum_i s_i,$$

where we are summing over single sites (only s_i appears in the sum), but we kept the notation of the sum over nearest neighbors to indicate that the effective field is proportional to the number of nearest neighbors (coordination number z) and the magnetization m. This is the point where we have to accept that this method works even if it is hard to justify: there exist some ad-hoc argument, but it would be meaningless to use those since we will anyway present the Landau-Ginzburg effective field theory in the following sections, which then will make us able to justify this approximation in a systematic way.

Remark. *The hamiltonian H_{mf} is independent of the specific configuration of the spins $\{s_i\}$, as long as*

$$\sum_i s_i = M,$$

So that we have efficiently replaced the interaction term with an average field that depends only on the magnetization m. This allows us to derive a self-consistent equation for the magnetization and analyze the behavior of the system near the critical point without needing to consider every possible configuration of the spins.

If we continue the computation of the mean field hamiltonian we get

$$H_{mf}(\{s_i\}, m) = -Jm \sum_{\langle i,j \rangle} s_i - h \sum_i s_i = -(Jzm + h) \sum_i s_i = -(Jzm + h)Nm,$$

where we have used the fact that the sum over the spins is equal to the magnetization $M = Nm$ and the sum over the nearest neighbors gives a factor of the coordination number $\frac{Nz}{2}$.[2] This mean field Hamiltonian describes a system of non-interacting spins in an *effective magnetic field* that depends on the magnetization m and the external field h.

The average magnetization per spin in the canonical ensamble can now be performed easily applying the mean field approximation:

$$\begin{aligned}\left\langle \frac{1}{N} \sum_i s_i \right\rangle_c &= \sum_{m \in [-1,\dots,+1]} m \sum_{\{s_i\}} \delta \left(\sum_i s_i - Nm \right) \frac{e^{-\beta H_{mf}(m)}}{Z} \\ &= \frac{1}{Z} \sum_m m e^{-\beta H_{mf}(m)} \left[\sum_{\{s_i\}} \delta \left(\sum_i s_i - Nm \right) \right],\end{aligned}$$

where the last term identifieas a sum over all the configurations of the spins that correspond to a given magnetization m. Note that a configuration of $N_\uparrow$ of up spins and $N_\downarrow$ of down spins corresponds to a magnetization

$$m = \frac{N_\uparrow - N_\downarrow}{N} = \frac{2N_\uparrow - N}{N} = 2\frac{N_\uparrow}{N} - 1.$$

The number of such configurations is given by the binomial coefficient:

$$\binom{N}{N_\uparrow} = \frac{N!}{N_\uparrow!(N - N_\uparrow)!} = \Omega(N_\uparrow),$$

which we can approximate using Stirling's formula for large N:

$$\log(\Omega(N_\uparrow)) \approx N \log N - N_\uparrow \log N_\uparrow - (N - N_\uparrow) \log(N - N_\uparrow),$$

which after some algebraic manipulations can be rewritten as

$$\log(\Omega(N_\uparrow)) \approx N \left[\log(2) - \frac{(1+m)}{2} \log(1+m) - \frac{(1-m)}{2} \log(1-m) \right].$$

Thus we have found an expression for the number of configurations that correspond to a given magnetization m, which can be used to compute the final form of the magnetization:

$$\left\langle \frac{1}{N} \sum_i s_i \right\rangle_c = \frac{1}{Z} \sum_m m e^{-\beta N f(m)} = \frac{1}{Z} \sum_m m e^{-\beta H_{mf}(m)} \Omega(m),$$

where we have defined the function $f(m)$ as

$$f(m) = -hm - \frac{Jzm^2}{2} - T \left[\log(2) - \frac{(1+m)}{2} \log(1+m) - \frac{(1-m)}{2} \log(1-m) \right],$$

which can be interpreted as a **free energy density** (per site) that depends on the magnetization m. Indeed we have the last sum over the magnetization m getting dominated by the minimum of

[2]When we count the same site for each of its n.n., we are counting each s_i a number of z times, but we have to divide by 2 to avoid double-counting. Thus the total is $N \cdot \frac{z}{2}$.

the function $f(m)$ in the thermodynamic limit $N \to \infty$, while the other terms get suppressed by the factor $e^{-\beta N f(m)}$.[3] Thus the average magnetization can be approximated as

$$\left\langle \frac{1}{N} \sum_i s_i \right\rangle_c \approx \frac{1}{e^{-\beta N f(m^*)}} \left(m^* e^{-\beta N f(m^*)} \right) = m^*, \qquad \left. \frac{\partial f(m)}{\partial m} \right|_{m^*} = 0,$$

TODO: Add a figure of a parabula with the minimum at m-star.

where m^* is the value of m that minimizes the function $f(m)$. This is known as a *saddle point approximation*, which is a common technique used in statistical mechanics to evaluate integrals or sums that are dominated by the contribution of a single point (the minimum of the free energy in this case). We can use the same approximation to compute the partition function Z as well, which is given by

$$Z = \sum_m e^{-\beta N f(m)} \approx e^{-\beta N f(m^*)}.$$

Self-consistency relation. Now computing the derivative of $f(m)$ with respect to m and setting it to zero, we get the self-consistent equation for the magnetization:

$$\beta(h + Jzm) = \frac{1}{2} \log \left(\frac{1+m}{1-m} \right) = \tanh^{-1}(m),$$

where the last step is obtaines after some algebraic manipulations. This equation can be rewritten as

$$m = \tanh\left(\beta h_{eff}\right), \quad h_{eff} = h + Jzm,$$

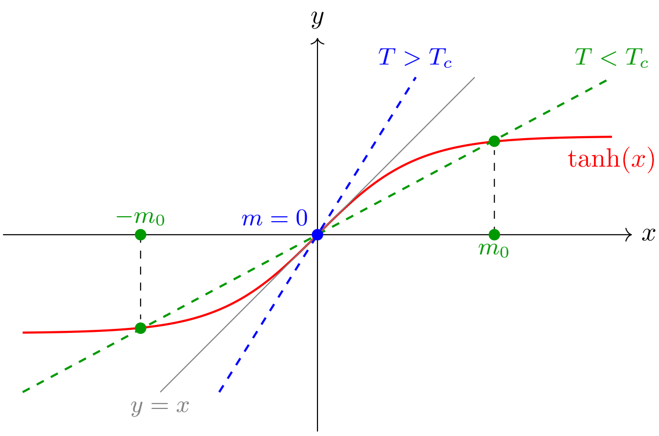
where h_{eff} is the effective magnetic field that each spin experiences due to the mean field approximation. This self-consistent equation can be solved numerically to find the magnetization m as a function of temperature T and external field h, allowing us to analyze the behavior of the system near the critical point and understand the nature of the phase transition. The effective field h_{eff} captures the influence of the neighboring spins on a given spin, and the self-consistent equation reflects the fact that the magnetization m depends on itself through the effective field. This is a hallmark of mean field theory, where the order parameter (magnetization) is determined by a self-consistent equation that takes into account the average effect of the interactions in the system.

[3] Since we always have to consider an enormously large N (the real world is at the thermodynamic limit), even small variations in $f(m)$ will be exponentially suppressed.

2.4 | Self-Consistency and Phase Transitions

In order to solve the self-consistent equation, we can analyze it graphically by plotting the function m on the left-hand side and the function $\tanh(\beta h_{eff})$ on the right-hand side as functions of m. The points where these two functions intersect correspond to the solutions of the self-consistent equation. By varying the temperature T and the external field h, we can observe how the number and nature of the solutions change, which provides insights into the phase transition and critical behavior of the system. Let us start from the case $h = 0$:

TODO: fix the plot



The graphical solution shows that for $T > T_c$ the only intersection is at the origin, yielding $m = 0$: this corresponds to the **paramagnetic phase**. For $T < T_c$, instead, the line intersects the curve at two symmetric non-zero points, corresponding to two stable ordered states with opposite magnetization: this is the **ferromagnetic phase**.

- For $T > T_c$, there is only one solution at $m = 0$, which corresponds to the paramagnetic phase where the spins are disordered and there is no net magnetization.
- For $T < T_c$, there are three solutions: one at $m = 0$ and two symmetric solutions at $m = \pm m^*$, which correspond to the ferromagnetic phase where the spins are aligned and there is a non-zero magnetization. The solution at $m = 0$ becomes unstable (not a minimum) in this regime, while the two symmetric solutions represent the stable ordered phases.
- At $T = T_c$, the solution at $m = 0$ becomes marginally stable, and the system undergoes a second-order phase transition from the paramagnetic phase to the ferromagnetic phase. The magnetization m changes continuously across the transition, and there is no latent heat associated with the transition.

The critical temperature T_c is determined by the condition that the slope of the tanh function at $m = 0$ equals 1, which gives

$$\beta_c J z = 1 \quad \Rightarrow \quad T_c = \frac{Jz}{k_B},$$

where k_B is the Boltzmann constant. Thus the condition on the critical temperature can be expressed as $\beta_c J z = 1$.

We can now study the beaviour of the magnetization as a function of temperature. As we decreas the temperature T below T_c, the magnetization m increases continuously from zero, indicating the onset of ferromagnetic order. We got two symmetric solutions for the magnetization, which correspond to the two possible orientations of the spins in the ferromagnetic phase. The magnetization m serves as an order parameter that characterizes the phase transition: it is zero in the paramagnetic phase and becomes non-zero in the ferromagnetic phase, indicating the spontaneous breaking of symmetry.

TODO: add plot of m vs T

We can obtain the plot of the free energy as a function of the magnetization m for different temperatures to visualize the phase transition. For $T > T_c$, the free energy has a single minimum at $m = 0$, corresponding to the stable paramagnetic phase. As we decrease the temperature below T_c, the free energy develops two symmetric minima at $m = \pm m^*$, corresponding to the stable ferromagnetic phases, while the minimum at $m = 0$ becomes a local maximum, indicating that it is no longer a stable solution. This can be seen studying the expression for the free energy density $f(m)$ that we derived earlier as $h = 0$:

$$f(m) = \dots$$

TODO: add plot of f(m) vs m for different T

and we get then two different expressions for $T > T_c$ and $T < T_c$:

- For $T > T_c$, the free energy has a single minimum at $m = 0$, which corresponds to the stable paramagnetic phase: we have a parabula with a minimum at the origin, which is the only stable solution in this regime.
- For $T < T_c$, the free energy develops two symmetric minima at $m = \pm m^*$, which correspond to the stable ferromagnetic phases, while the minimum at $m = 0$ becomes a local maximum, indicating that it is no longer a stable solution: we have a double-well potential with two minima at $m = \pm m^*$ and a local maximum at $m = 0$. As the temperature decreases further below T_c, the minima at $m = \pm m^*$ become deeper, indicating that the ferromagnetic phases become more stable.

If we were to plot the free energy as a function of the magnetization m for different temperatures, we would see how the shape of the free energy changes across the phase transition, with the emergence of two minima below T_c and the disappearance of the minimum at $m = 0$. This visualization helps to understand the nature of the phase transition and the role of the magnetization as an order parameter.

If we now imagine to "turn on" the external magnetic field h, we would see that the symmetry between the two minima at $m = \pm m^*$ is broken, and one of the minima becomes deeper than the other, depending on the sign of h. The easier way is to consider an external field which is positive but very small in comparison to the coupling constant J, so that we can treat it as a perturbation.

TODO: add plot of f(m) vs m for different h

- For $h > 0$, the minimum at $m = +m^*$ becomes deeper than the minimum at $m = -m^*$, indicating that the ferromagnetic phase with positive magnetization is more stable than the one with negative magnetization. The system will preferentially align its spins in the direction of the external field, leading to a net positive magnetization.
- For $h < 0$, the minimum at $m = -m^*$ becomes deeper than the minimum at $m = +m^*$, indicating that the ferromagnetic phase with negative magnetization is more stable than the one with positive magnetization. The system will preferentially align its spins in the opposite direction of the external field, leading to a net negative magnetization.

This means that the external magnetic field h breaks the $\mathbb{Z}_2$ symmetry between the two ferromagnetic phases and selects one of them as the stable phase, depending on the sign of h. This is a manifestation of the fact that the external field acts as a symmetry-breaking field, which can influence the behavior of the system near the critical point and affect the nature of the phase transition, especially in the presence of a small external field where the system can exhibit critical behavior and scaling properties.

Remark (On Symmetries and Spontaneous Symmetry Breaking). *A symmetry of a system is a transformation that leaves the Hamiltonian (and so the physical properties and evolution of the system) invariant.*

In the case of the Ising model with $h = 0$, there is a $\mathbb{Z}_2$ symmetry corresponding to the transformation $s_i \to -s_i$ for all spins, which leaves the Hamiltonian unchanged:

$$H(\{s_1, s_2, \ldots, s_n\}) = H(\{-s_1, -s_2, \ldots, -s_n\}).$$

When we flip all spins, is like changing the sign of the magnetization m, but the Hamiltonian remains the same; in the graph of the free energy, this corresponds to the symmetry of the double-well potential around $m = 0$. This symmetry is a fundamental property of the system and plays a crucial role in determining the nature of the phase transition and the behavior of the system near the critical point.

This symmetry is ***spontaneously broken*** *as $T > T_c$, since ...*

Once the system is in one of the two ferromagnetic phases (with $m = +m^*$ or $m = -m^*$), we could expect the system to be able to transition between these two phases by flipping all the spins, since the states are completely symmetric: same energy, entropy etc. However, in the thermodynamic limit $N \to \infty$, the system becomes trapped in one of the two minima of the free energy, and it cannot transition to the other minimum due to the presence of an infinite energy barrier between them: the transition probability between the two minima approches zero as a power law, and the system remains in one of the two ferromagnetic phases.

2.4.1 | Critical Exponents

The mean field theory allows us to compute the critical exponents that characterize the behavior of the system near the critical point. By analyzing the self-consistent equation for the magnetization and the free energy, we can derive expressions for the critical exponents α, β, γ, and δ in terms of the parameters of the model. The mean field theory predicts specific values for these critical exponents, which can be compared to experimental results and more sophisticated theoretical approaches to understand the limitations and applicability of mean field theory in describing critical phenomena.

We start from the analysis of the **free energy** $f(m)$ and the magnetization m as a function of temperature T near the critical point T_c. We can expand the the free energy in a Taylor series around $m = 0$ to analyze the behavior of the system near the critical point. The expansion takes the form:

TODO: correct the expansion

$$f(m) \sim T\left[\frac{m^2}{2}\left(1 - \frac{T}{T_c}\right) + \frac{m^4}{12}T_c^3 + O(m^6)\right],$$

where the interesting feature is that the coefficient of the m^2 term changes sign at $T = T_c$, which indicates the onset of the phase transition. We can now find an analytic expression for the derivative of the free energy with respect to m and set it to zero to find the equilibrium magnetization m as a function of temperature T:

$$\frac{\partial f}{\partial m} \propto m(T - T_c) + Bm^3 + O(m^5) = 0, \quad B \sim O(1) > 0,$$

which can be solved to find the behavior of the magnetization near the critical point. For $T > T_c$, the only solution is $m = 0$, while for $T < T_c$, we have two symmetric solutions given by

$$m = \pm\sqrt{\frac{T_c - T}{B}} \sim (T_c - T)^{1/2},$$

which indicates that the magnetization m behaves as a power law near the critical point with a critical exponent

$$m \propto (T_c - T)^{\beta}, \quad \beta = \frac{1}{2}.$$

This critical exponent β characterizes the behavior of the magnetization as the system approaches the critical temperature from below, and it is a key quantity in understanding the nature of the phase transition.

We can now analyze the behavior of the **magnetic susceptibility** χ near the critical point. The magnetic susceptibility is defined as the derivative of the magnetization with respect to the external magnetic field h at $h = 0$:

$$\chi = N \frac{\partial m}{\partial h}\bigg|_{h=0}.$$

Recovering the expression for the magnetization

$$m = \tanh\left(\beta(h + Jzm)\right),$$

we know that near the critical point (approached from above)T_c, the magnetization m is small, so we can expand the tanh function to first order in m:

$$m \approx \beta(h + Jzm).$$

Thus the magnetic susceptibility can be computed as

$$\chi = \frac{N\beta}{1 - \beta Jz},$$

which diverges as T approaches T_c from above, since $\beta Jz = 1$ at $T = T_c$. The critical exponent γ characterizes the divergence of the magnetic susceptibility near the critical point, and it can be defined as

$$\chi \propto |T - T_c|^{-\gamma}, \quad \gamma = 1.$$

These critical exponents helps us to understand the behavior of the system near the critical point and to classify the universality class of the phase transition. Usually when quantities diverge at the critical point, we can characterize their behavior by a power law with a critical exponent, which provides insights into the nature of the fluctuations and correlations in the system as it undergoes a phase transition. Indeed in an experiments, we can measure also **finite size effect**: all of our computations are true in the thermodynamic limit $N \to \infty$, but in a real experiment we have a finite number of particles, so we can observe how the critical behavior is affected by the finite size of the system, which can lead to rounding and shifting of the critical point, as well as modifications to the scaling behavior of physical quantities near the transition. All of these is heavily connected to the concept of **correlation length**, which is a measure of how far correlations between particles extend in the system, and it diverges at the critical point, leading to long-range correlations and critical fluctuations that dominate the behavior of the system near the transition.

How can we concile the mean field predictions for the critical exponents with experimental results? The mean field theory provides is just an approximation, and it is known to give incorrect predictions for the critical exponents in low-dimensional systems (e.g., 2D and 3D), where fluctuations play a significant role (speaking about the Ising model). In higher dimensions (e.g., 4D and above), the mean field theory becomes more accurate, and the critical exponents predicted by mean field theory are closer to experimental results. For Ising

- In 1D, the Ising model does not exhibit a phase transition at finite temperature, and the critical exponents are not defined in the usual sense. The mean field theory incorrectly predicts a phase transition in 1D, which is a clear indication of its limitations in low-dimensionality.
- In 2D, the exact solution of the Ising model gives $\beta = 1/8$ and $\gamma = 7/4$, which are different from the mean field predictions.
- In 3D, numerical simulations and experiments suggest that $\beta \approx 0.326$ and $\gamma \approx 1.237$, which again differ from the mean field values.
- In 4D and above, the critical exponents approach the mean field values of $\beta = 1/2$ and $\gamma = 1$.

Here we can see that the mean field theory provides a qualitative understanding of the phase transition and critical behavior, but it fails to capture the correct critical exponents in low-dimensional systems due to the neglect of fluctuations, which are crucial in determining the behavior near the critical point. More sophisticated approaches, such as renormalization group theory, are needed to accurately describe critical phenomena and compute critical exponents in low-dimensional systems.

We will solve the Ising model in 1D in the next section, and we will give some arguments about the behavior of the model in 2D, which will allow us to understand better the limitations of mean field theory and the importance of fluctuations in low-dimensional systems.

3 Exact Solution of the Ising Model

The Ising model represents one of the fundamental paradigms of statistical mechanics and lattice field theory. Originally formulated to describe the behavior of ferromagnetic materials, over the decades it has assumed a much broader role, becoming the essential testing ground for understanding phase transitions and critical phenomena. In this chapter, we will undertake a rigorous study of the model, moving from topologically simpler cases to some of the most sophisticated achievements of twentieth-century theoretical physics.

The discussion will begin with the one-dimensional Ising model. Although van Hove's theorem precludes the existence of a finite-temperature phase transition in 1D, the analysis of this system remains crucial for pedagogical and methodological reasons. By introducing the transfer matrix formalism, we will be able to exactly compute the partition function, the magnetization, and the behavior of the two-point correlation function, highlighting the role of the correlation length and its exponential decay.

In the second part of the chapter, we will shift our focus to the two-dimensional Ising model, a physically much richer system. Unlike the one-dimensional case, the 2D lattice admits the existence of an order-disorder phase transition at a finite temperature. After a heuristic analysis based on the competition between energy and entropy in the formation of *domain walls*, we will provide a rigorous proof using the celebrated Peierls argument.

Finally, we will present two of the most mathematically elegant results in all of statistical mechanics. First, we will discuss the Kramers-Wannier duality which, by exploiting the deep topological symmetries between high- and low-temperature expansions on the dual lattice, allows for the exact determination of the system's critical point. To culminate this journey, we will explore the exact analytical solution of the 2D model by Lars Onsager: a monumental achievement that, by explicitly demonstrating a logarithmic singularity in the specific heat, forever revolutionized our theoretical understanding of phase transitions.

3.1 | Ising Model in One Dimension

The hamiltonian of the 1D Ising model is given by

$$H = -J\sum_{i=1}^{N} s_i s_{i+1} - h\sum_{i=1}^{N} s_i,$$

where we have periodic boundary conditions $s_{N+1} = s_1$. The partition function depends upon the temperature T, the external magnetic field h and the number of spins N, and it can be written as

$$Z_N(\beta, h) = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})}.$$

It is important since we can derive an expression for the ensemble average of the magnetization per site as a function of the partition function:

$$\left\langle \frac{1}{N}\sum_{i=1}^{N} s_i \right\rangle_c = -\frac{1}{N\beta}\frac{\partial \log Z_N}{\partial h}.$$

If we manipulate the expression of the hamiltonian, to express two symmetric sums, we get

$$H = -J\sum_{i=1}^{N} s_i s_{i+1} - \frac{h}{2}\sum_{i=1}^{N} s_i s_{i+1}$$

$$Z = \sum_{\{s_i\}}\prod_{i=1}^{N}(e^{\beta J s_i s_{i+1} + \frac{\beta h}{2} s_i s_{i+1}})\dots$$

Now we can introduce the **transfer matrix** T, which is a 2×2 matrix that encodes the interactions between neighboring spins and the effect of the external magnetic field. The transfer matrix is defined as

$$T_{s_i, s_{i+1}} = e^{\beta J s_i s_{i+1} + \frac{\beta h}{2}(s_i + s_{i+1})},$$

where s_i and s_{i+1} can take values of ± 1. The partition function can then be expressed as a product of transfer matrices:

$$Z = \sum_{\{s_i\}} T_{s_1,s_2} T_{s_2,s_3} \cdots T_{s_N,s_1} = \mathrm{Tr}(T^N),$$

where the trace is taken over the possible spin states. The transfer matrix method allows us to compute the partition function and other thermodynamic quantities by diagonalizing the transfer matrix and finding its eigenvalues. The largest eigenvalue of the transfer matrix dominates the behavior of the partition function in the thermodynamic limit, and it can be used to derive expressions for the free energy, magnetization, and other physical quantities of interest. This method is particularly powerful for solving one-dimensional models like the 1D Ising model, where it provides an exact solution for the partition function and allows us to analyze the behavior of the system at different temperatures and external fields.

Transfer Matrix

Expressing the partition function in terms of the transfer matrix, we can notice how the computational complexity of the problem is reduced from summing over 2^N configurations to diagonalizing a 2×2 matrix multiplied N times with itself, which is a much more tractable problem. The eigenvalues of the transfer matrix can be computed explicitly, and they determine the thermodynamic properties of the system. We can write the transfer matrix explicitly as

$$T = \begin{pmatrix} e^{\beta J+\beta h} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J-\beta h} \end{pmatrix},$$

with the idea of diagonalizing this matrix to find its eigenvalues λ_1 and λ_2. Let us note that $T = T^\dagger$, thus we know there exist an orthogonal matrix $O = O^T$ such that $T = ODO^T$, where D is a diagonal matrix containing the eigenvalues of T. The diagonal maatrix D can be expressed as

$$D = \begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix},$$

and we can compute the the Nth power of the transfer matrix as

$$T^N = (ODO^T)^N = (ODO^T)(ODO^T)\cdots(ODO^T) = OD^NO^T,$$

where we have used the fact that $O^TO = I$ to simplify the expression. The partition function can then be expressed as

$$Z = \mathrm{Tr}(T^N) = \mathrm{Tr}(OD^NO^T) = \mathrm{Tr}(D^N) = \lambda_+^N + \lambda_-^N,$$

where λ_+ and λ_- are the eigenvalues of the transfer matrix T and we have used the cyclic property of the trace.

Now we can also compute the determinant of the transfer matrix, which is given by

$$\det(T) = \lambda_+\lambda_- = e^{2\beta J} - e^{-2\beta J} = 2\sinh(2\beta J),$$

while for the trace we had

$$\mathrm{Tr}(T) = \lambda_+ + \lambda_- = e^{\beta J+\beta h} + e^{\beta J-\beta h} + 2e^{-\beta J} = 2e^{\beta J}\cosh(\beta h) + 2e^{-\beta J}.$$

The eigenvalues of the transfer matrix can be computed using the characteristic polynomial, which is given by

$$x^2 - \mathrm{Tr}(T)x + \det(T) = 0,$$

which, inserting the expressions for the trace and determinant, can be rewritten as

$$\ldots$$

and in the end we get the eigenvalues as

$$\lambda_\pm = e^{\beta J}\left(\cosh(\beta h) \pm \sqrt{\sinh^2(\beta h) + e^{-4\beta J}}\right),$$

which gives

$$|\frac{\lambda_-}{\lambda_+}| < 1, \quad |\frac{\lambda_-}{\lambda_+}|^N \ll 1.$$

thus we can write the partition function in the thermodynamic limit as

$$Z = \lambda_+^N + \lambda_-^N = \lambda_+^N\left(1 + \left(\frac{\lambda_-}{\lambda_+}\right)^N\right) \approx \lambda_+^N,$$

which allows us to compute the free energy per site as

$$f = -\frac{1}{\beta} \lim_{N\to\infty} \frac{1}{N} \log Z,$$

where $f = \lim_{N\to\infty} \frac{F}{N}$; thus we have

$$f = -\frac{1}{\beta} \log \lambda_+ = -J - \frac{1}{\beta} \log \left(\cosh(\beta h) + \sqrt{\sinh^2(\beta h) + e^{-4\beta J}} \right).$$

from this we can understand that there are no singularities in the free energy as a function of temperature T and external field h, which indicates that there is no phase transition in the 1D Ising model at finite temperature. This is consistent with the known result that the 1D Ising model does not exhibit a phase transition at finite temperature, and it highlights the importance of dimensionality and fluctuations in determining the behavior of the system near critical points.

3.1.1 | Magnetization

Let us now compute the magnetization per site as a function of temperature T and external field h. We can use the expression for the partition function to compute the magnetization as

$$\langle s_i \rangle_c = m = -\frac{1}{N} \frac{\partial}{\partial h} \log Z = -\frac{1}{\beta} \frac{\partial}{\partial h} (\frac{F}{N}).$$

Inserting the expression for the free energy per site, we get

$$m = \frac{\sinh(\beta h)}{\sqrt{\sinh^2(\beta h) + e^{-4\beta J}}}.$$

This expression shows that the magnetization m is a smooth function of temperature T and external field h, and it does not exhibit any singularities or discontinuities, which is consistent with the absence of a phase transition in the 1D Ising model at finite temperature.

`img/ising_1d_m-h_plot.png`

Figure 3.1: Magnetization per site m as a function of the external field h at different values of temperature T.

As T increases, the curve becomes more steep, but it never exihibit a discontinuity or a singularity, which confirms that there is no phase transition in the 1D Ising model at finite temperature. Only in the limit $T \to 0$ we have a discontinuity in the magnetization at $h = 0$, where the system transitions from a fully magnetized state with $m = +1$ for $h > 0$ to a fully magnetized state with $m = -1$ for $h < 0$:

$$\lim_{T\to 0} m(T, h) = \operatorname{sgn}(\sinh(\beta h)) = \operatorname{sgn}(h).$$

img/ising_1d_chi-T_plot.png

Figure 3.2: Magnetic susceptibility χ as a function of temperature T at zero external field $h = 0$.

We could repeat the computation for χ the **magnetic susceptibility**, which is given by

$$\chi = \frac{\partial}{\partial h} m(T, h) = \dots$$

and it can be shown that χ does not diverge at any finite temperature, which is consistent with the absence of a phase transition in the 1D Ising model at finite temperature. Only in the limit $T \to 0$ we have a divergence in the magnetic susceptibility at $h = 0$, which indicates a critical point at zero temperature.

The euristic derivation of mean field theory has to have some assumptions which do not hold at low dimensions, and this is the reason why it fails to capture the correct critical behavior in 1D and 2D systems. Only in higher dimensions, where fluctuations are less important, the mean field theory provides a more accurate description of the critical behavior and the critical exponents.

The exact solution of the 1D Ising model using the transfer matrix method allows us to understand the limitations of mean field theory and the importance of fluctuations in determining the behavior of the system near critical points.

3.1.2 | Two Point Correlation Function

It is now time to compute the two-point correlation function for the 1D Ising model: the correlation function is defined as

$$\langle s_i s_{i+r} \rangle = \frac{1}{Z} \sum_{\{s_k\}} s_i s_{i+r} e^{-\beta H(\{s_k\})},$$

which can become a connected correlation function if we subtract the product of the magnetizations:

$$\langle s_i s_{i+r} \rangle - \langle s_i \rangle \langle s_{i+r} \rangle = \frac{1}{Z} \sum_{\{s_k\}} s_i s_{i+r} e^{-\beta H(\{s_k\})} - m^2.$$

To compute the correlation function, we can use the transfer matrix method again. We can express the correlation function in terms of the transfer matrix as follows:

$$\langle s_i s_{i+r} \rangle = \frac{1}{Z} \sum_{\alpha_1=1}^{2} \sum_{\alpha_2=1}^{2} \cdots \sum_{\alpha_N=1}^{2} \sigma_{\alpha_j} \sigma_{\alpha_{j+r}} T_{\alpha_1,\alpha_2} T_{\alpha_2,\alpha_3} \cdots T_{\alpha_N,\alpha_1},$$

where σ_{α_j} is the spin value corresponding to the state α_j (i.e., $\sigma_1 = +1$ and $\sigma_2 = -1$). We can rewrite this expression in a more compact form using matrix notation. Let us define a diagonal matrix D such that

$$D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \text{ and } D_{\alpha,\beta} = \delta_{\alpha,\beta} \cdot \sigma_\alpha,$$

meaning that D is a diagonal matrix with entries corresponding to the spin values. Then we can express the correlation function as

$$\langle s_j s_{j+r} \rangle = \frac{1}{Z} \sum_{\alpha_1=1}^{2} \sum_{\alpha_2=1}^{2} \cdots \sum_{\alpha_N=1}^{2} T_{\alpha_1,\alpha_2} T_{\alpha_2,\alpha_3} \cdots T_{\alpha_{j-1},\alpha_j} D_{\alpha_j,\beta_1} T_{\beta_1,\alpha_{j+1}} \cdots T_{\alpha_{j+r-1},\alpha_{j+r}} D_{\alpha_{j+r},\beta_2} T_{\beta_2,\alpha_{j+r+1}} \cdots T_{\alpha_N,\alpha_1},$$

where we have inserted the diagonal matrices D at the positions corresponding to the spins s_j and s_{j+r}. This expression can be further simplified by recognizing that the sums over the indices α_k correspond to matrix multiplications, and we can express

$$\langle s_j s_{j+r} \rangle = \frac{1}{Z} \mathrm{Tr}(T^{j-1} D T^r D T^{N-r+1-j}),$$

where we have used the cyclic property of the trace to rearrange the order of the matrices. This expression allows us to compute the two-point correlation function in terms of the transfer matrix and the diagonal matrix D, which encodes the spin values. By diagonalizing the transfer matrix with a unitary matrix U:

$$T = U \begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix} U^\dagger,$$

where U can be computed explicitly, we can express the correlation function in terms of the eigenvalues and eigenvectors of the transfer matrix, which allows us to analyze its behavior as a function of the distance r and the temperature T. Let us focus on the case without external field: the unitary transformation becomes

$$\dots$$

and so we can write the transfer matrix in the diagonal basis as

$$\dots$$

Now putting everything together, we can express the correlation function as

$$\dots$$

In the end we find a close analytical form for the correlation function, which can be expressed as

$$\langle s_j s_{j+r} \rangle_{h=0} = \frac{\lambda_+^{N-r} \lambda_-^r + \lambda_-^{N-r} \lambda_+^r}{\lambda_+^N + \lambda_-^N},$$

and again if we use tha fact that $|\frac{\lambda_-}{\lambda_+}| < 1$, we can simplify this expression in the thermodynamic limit as

$$\langle s_j s_{j+r} \rangle_{h=0} = \dots \sim e^{-\frac{r}{\xi}},$$

effectively defining the correlation length ξ as

$$\xi^{-1} = \log\left(\frac{\lambda_+}{\lambda_-}\right) > 0,$$

which is a measure of how far correlations between spins extend in the system. Now considering the absence of an external field, we can express the eigenvalues of the transfer matrix as ...

and in the end we get the correlation length in the $T \to 0$ limit without external field as

$$\xi \sim e^{2\beta J},$$

so that it diveres exponentially as $T \to 0$, which indicates that the system exhibits long-range correlations at low temperatures, and it confirms the presence of a critical point at zero temperature in the 1D Ising model. This behavior of the correlation length is consistent with the absence of a phase transition at finite temperature, as the correlation length remains finite for all $T > 0$ and only diverges as $T \to 0$.

The correlation lenght correspond to the distance after which we can consider the spins as independent, since the correlation function decays exponentially with distance r as $e^{-r/\xi}$. As the temperature approaches zero, the correlation length diverges, indicating that the spins become correlated over longer distances, which is a signature of critical behavior at zero temperature. This analysis of the two-point correlation function and the correlation length provides insights into the nature of correlations in the 1D Ising model and helps to understand the absence of a phase transition at finite temperature. But since this divergence occur at $T = 0$ then everything is in accord to the fact that there is no phase transition at finite temperature, and the critical point is located at zero temperature.

3.2 | Ising Model in Two Dimensions

We will not solve exactly the 2D Ising model, but we will give some arguments about its behavior and the nature of the phase transition it exhibits, and then in the end we will present the proof of the existence of a phase transition in the 2D Ising model at finite temperature, which is a non-trivial result that highlights the importance of dimensionality and fluctuations in determining the behavior of the system near critical points.

3.2.1 | Phase Transition - Euristic

Consider the 2D Ising model on a square lattice, where each spin interacts with its four nearest neighbors. Take the limit for a very low temperature $T \to 0$, and suppose that there is a region in the first 1D raw where the system is in a fully magnetized state, with all l spins aligned (e.g., $m = +1$). We focus for a moment only on this 1D raw, and we ask what is the energy cost of creating a droplet of flipped spins in this region, which corresponds to a local change in the magnetization. We can consider:

- Case A: $\uparrow\uparrow\ \dots\ \uparrow$, where we have no domain walls, since all the l spins are aligned. The energy cost is zero, since there are no domain walls and the system is in a fully magnetized state.
- Case B: $\uparrow\ \dots\ \uparrow\downarrow\ \dots\ \downarrow$, where we have one domain wall separating k up spins and $l-k$ down spins. The energy cost of creating this domain wall is proportional to the length of the domain wall, which is given by the only two nearest neighbor pairs that are affected by the change in configuration, thus we have

  $$\Delta E = 2J,$$

  where J is the coupling constant of the Ising model: the hamiltonian of the model depends on the interaction between nearest neighbor spins, and the energy cost of creating a domain wall is proportional to the number of nearest neighbor pairs that are affected by the change in configuration, which in this case is two (the pair of spins at the boundary of the droplet).
- Case C: $\uparrow\ \dots\ \uparrow\downarrow\ \dots\ \downarrow\uparrow\ \dots\ \uparrow$ where we have two domain walls separating three regions of spins. Since the energy cost of creating a domain wall is proportional to the number of nearest neighbor pairs that are affected by the change in configuration, which in this case is four (the pairs of spins at the boundaries of the droplet), we have

  $$\Delta E = 4J,$$

  which is twice the energy cost of creating a single domain wall, since we have two domain walls in this case.

Thus the configurations with the least energy cost are those with a single domain wall (with respect to Case A in which there are no domain walls), which corresponds to the case.

Remark (Entropy of domains). *Since the entropy is related to the number of configurations, we can estimate the entropy gain associated with the formation of the domain wall by counting the number of ways to arrange such wall. In a string with l spins, we can choose the position* *of the domain wall in $l-1$ ways (since we cannot place the domain wall at the very beginning or at the very end of the string), thus we can estimate the entropy gain as*

$$\Delta S = k_B \log(l-1).$$

In more dimensions, the number of ways to arrange a domain wall of length l can be approximated as the path we can take to create the domain wall, which is given by three choices of direction at each step (since we cannot go back), but we will see this in a moment.

If we now compute the free energy cost of creating such a droplet, we need to take into account both the energy cost and the entropy gain associated with the formation of the droplet. The free energy cost can be then expressed as

$$\Delta F = \Delta E - T\Delta S = 2J - k_B T \log(l-1),$$

since the energy cost of creating the droplet in case B is $2J$ and the entropy gain is proportional to the logarithm of the number of ways to arrange the droplet, which is approximately $\log(l-1)$. As l increases, the energy cost dominates over the entropy gain, and the free energy cost becomes positive, which means that it is energetically unfavorable to create such a droplet

$$\lim_{l\to\infty} \Delta F = -\infty.$$

It seems that breaking a domain wall is energetically unfavorable, but there is a maximum value of l for which the entropy does not make unfavorable the creation of the droplet, and this maximum value of l can be estimated by setting the free energy cost to zero:

$$\Delta F = 0 \implies 2J - k_B T \log(l-1) = 0 \implies l \sim e^{\frac{2J}{k_B T}} = \xi,$$

which indicates that there is a *characteristic length scale l* above which the domain wall are not favorable anymore, and this length scale can be identified with the **correlation length** ξ of the system (the expression is indeed the same we found in the previous section, when we solved it exactely). As the temperature approaches the critical temperature T_c from below, the correlation length diverges, indicating that the system exhibits long-range correlations and critical behavior at the phase transition.

Thus this simple argument suggests that in one dimension, the system does not exhibit a phase transition at finite temperature: the average 1D Ising system will be a sequence of domains walls of size $\sim \xi$, which ends up being a disordered phase with zero magnetization in the thermodynamic limit. This is consistent with the exact solution of the 1D Ising model, which shows that there is no ferromagnetic phase transition at finite temperature.

Euristic arguments in statistical physics can provide valuable insights into the behavior of complex systems and help to understand the underlying mechanisms driving phase transitions and critical phenomena. Sometimes, when a problem is mathematically intractable, we can use physical intuition and qualitative reasoning to make predictions about the behavior of the system, which can then be used to address the problem in new ways and in the end find the analytical solution.

3.2.2 | Domain Walls - Euristic

Let us compare the free energy of the two configurations:

img/ising_2d_plus_grid.png

Figure 3.3: Configuration A: no domain walls, fully magnetized.

img/ising_2d_minus_domain.png

Figure 3.4: Configuration B: one domain wall, with a droplet of flipped spins.

We ask what is the difference in energy of the two 2D configurations described. Each set of nearest neighbours of the subdomain considered in the configuration contributes to the change in energy from one configuration to the other. In particular

$$\Delta U = U_B - U_A = 2Jl,$$

where now in 2D, l represents the lenght of the perimeter of the flipped subdomain, which is proportional to the number of nearest neighbor pairs that are affected by the change in configuration. The energy cost of creating a droplet of flipped spins is proportional to the length of the domain wall, which is $2Jl$ in this case. This energy cost is positive, which means that it is energetically unfavorable to create such a droplet, and it suggests that the system will tend to remain in a magnetized state with non-zero magnetization, indicating the presence of a phase transition at finite temperature.

But we have always to take into account the entropy gain associated with the formation of the droplet, which can be estimated after counting the number of ways to arrange the droplet of flipped spins.

img/ising_2d_different_domain_walls.png

Figure 3.5: Three different configurations of domain walls of the same length $l = 4$.

The number of ways to arrange a droplet of perimeter l can be approximated as the path we can take to create the domain wall, which is given by three choices of direction at each step (since we cannot go back),[1] thus we can estimate the number of possible domain walls (i.e. the number of configurations) as

$$N_D \sim 3^l,$$

thus if the number of domains wall of length l is given by N_D, then the entropy gain associated with the formation of the droplet can be estimated as

$$\Delta S = k_B \log(N_D) = k_B l \log(3),$$

[1] Extimating like this is exagerated, since we are counting also "snakes" that do not close on a region, but let us do this anyway and then reason on it.

which is proportional to the length of the domain wall.

Thus in the end, the free energy cost of creating such a droplet can then be expressed as

$$\Delta F = \Delta U - T\Delta S = 2Jl - Tk_B l \log(3) = (2J - Tk_B \log(3))l,$$

which indicates that there is a critical temperature T_c at finite temperature, at which we observe a phase transition.

> **Remark.** *Contrary to the one dimensional case, energy and entropy scale both with the same power of l (i.e., linearly), thus there is a critical temperature at which the two contributions balance each other, and we observe a phase transition. In the one dimensional case, energy scales with l while entropy scales with $\log(l)$, thus energy dominates over entropy for large l, and there is no phase transition at finite temperature.*

In 2D, for a sufficiently low temperature, the energy cost of creating a droplet of flipped spins dominates over the entropy gain (it is not convenient to "break" large domains of magnetization), and the system tends to remain in a spontaneous magnetized state with non-zero magnetization, indicating the presence of a phase transition at finite temperature. As the temperature increases, the entropy gain becomes more significant, and at the critical temperature T_c, the two contributions balance each other, leading to a phase transition from a more ordered phase to a disordered phase.

3.2.3 | Peierls's argument

TODO: Add reference to 14.3 Huang's book stat mech.

Let us consider a system with **fixed open boundary conditions**, where we have an Ising model defined on a square lattice of size $N = A \times B$ (we are not interested in the specific values of A and B), and we fix the spins on the boundary to be all up (i.e., $s_i = +1$ for all spins on the boundary). We want to show that there is a phase transition at finite temperature, which can be done by comparing the free energy of two configurations: one with all spins up (the ordered phase) and one with a droplet of flipped spins (the disordered phase).

img/ising_2d_oriented_domain_walls.png

Figure 3.6: Example of configuration with fixed open boundary conditions, where the spins on the boundary are all up (red), and three droplets of flipped spins inside the (blue) oriented domain walls.

Let us consider configurations with all the boundary spins fixed to be up; then it will be uniquely determined by the set of "oriented domain walls" that separate the up spins from the down spins. An **oriented domain wall** is a closed loop of nearest neighbor pairs of spins that are different (i.e., one spin is up and the other is down), where the orientation of the loop is defined by the direction of the spins: we define it as the one which keeps the up spins on the left and the down spins on the right.

The hamiltonian of the system is unchanged, and it is given by

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i,$$

where the first sum is taken over nearest neighbor pairs of spins, and the second sum is taken over all spins in the lattice.

We can fix the temperature and compute the partition function and magnetization for the configuration. Taking the thermodynamic limit $N \to \infty$ and studying the number of flipped spins N_-, we can distinguish two cases:

- $\lim_{N\to\infty} \frac{N_-}{N} = \frac{1}{2}$: in this case the system is in a disordered phase with zero magnetization, since the number of up spins and down spins are equal in the thermodynamic limit.
- $\lim_{N\to\infty} \frac{N_-}{N} < \frac{1}{2}$: in this case the system is in an ordered phase with non-zero magnetization, since the number of up spins is greater than the number of down spins in the thermodynamic limit.

We used a preferred direction setting the boundary conditions, but we could have done the same analysis with all spins down at the boundary, and we would have found the same result. Exact numbers will fluctuate, but these ratios will converge to the same value, and in the TD limit we will have a well defined phase transition at finite temperature. In this case, fixing the boundary to be up spins is enough to align the majority of the spins in the system along "+" in the ordered phase; with down spin boundaries the same results would hold, but the majority of the spins would be aligned along "-" in the ordered phase.

We know that in 1D the second case never happens, since the system is always in a disordered phase with zero magnetization, so our goal is to show that in 2D the second case can happen, which means that there can be a phase transition at finite temperatures, $\forall T < T_c$.

Classification of Domain Walls

For any boundary condition, the domain walls are closed loops of nearest neighbor pairs of spins that are different. They can be classified according to their length l (i.e., the number of nearest neighbor pairs that are different), and within the class of lenght-l domain walls, we can make use of an additional index i to distinguish between different configurations of domain walls. Thus we can label the domain walls with a pair of indices (l, i).

Then let us define:

- The **area** $A(l, i)$ of a domain wall (l, i) as the number of flipped spins that are enclosed by the domain wall. The area is a measure of the size of the droplet of flipped spins, and it is easy to show that at it has to be surely smaller than the area of a square of side l, thus we have
$$A(l, i) < l^2.$$
- The **number** $M(l)$ of domain walls with the same length l as the number of different configurations of domain walls with length l. The number of domain walls with length l can be estimated as the number of paths we can take to create a closed loop of length l, which is given by three choices of direction at each step, except for the first step where we have

four choices of direction; furthermore, the number of starting point has to be less than the number of sites N. In the end we can estimate

$$M(l) < 4N3^{l-1}.$$

- The **caracteristic function** for the lattice configurations. Let us consider a set of configurations of the system $\{C\}$, then for each of those we can define

$$\chi_c(l,i) = \begin{cases} 1 & \text{if } C \text{ contains the DW } (l,i), \\ 0 & \text{otherwise.} \end{cases}$$

Statistics of Domain Walls

We can now use the classification of domain walls to compute the average number of flipped spins in the system. With our choice of boundary conditions, we know that every flipped spin is enclosed by at least one domain wall, thus we can write the number of flipped spins N_- for a configuration C as

$$N_-(C) \leq \sum_{(l,i)} A(l,i)\chi_c(l,i),$$

where the sum is taken over all possible domain walls (l,i), and we are maximizing the number of flipped spins with the area of a domain wall multiplied by the function checking if such domain wall is present in the configuration. To see this more clearly, we can write

$$N_-(C) = \sum_{j:s_j=\downarrow} 1 \leq \sum_{j:s_j=\downarrow} \sum_{(l,i)}{}' \chi_c(l,i),$$

where the second sum is taken over all the domain walls (l,i) in the configuration C that enclose the flipped spin s_j (i.e., the domain walls for which $\chi_c(l,i) = 1$). Now we can exchange the order of the sums as follows

$$\sum_{j:s_j=\downarrow} \sum_{(l,i)}{}' \chi_c(l,i) = \sum_{(l,i)}{}' \chi_c(l,i) \sum_{j\in(l,i)} = \sum_{(l,i)} A(l,i)\chi_c(l,i),$$

where in the first step the change of order is allowed by the fact that we are counting on both sides the number of flipped spins in C by the number of domain walls that enclose them; we then used the fact that the number of flipped spins enclosed by a domain wall (l,i) is exactely the definition its area $A(l,i)$.

Thus using the previously shown inequalities, we can maximise the number of flipped spins as

$$N_-(C) \leq \sum_{(l,i)} A(l,i)\chi_c(l,i) < \sum_{(l,i)} l^2\chi_c(l,i) = \sum_{l} l^2 \sum_{i=1}^{M(l)} \chi_c(l,i),$$

where we used the fact that the area of a domain wall is smaller than l^2 to get the first inequality, and then we grouped the sum over domain walls by their length l to get the second equality.

Now we can take the **thermal average** over all configurations of the system, on both sides, in order to obtain an upper bound for the average number of flipped spins and thus getting our proof of the existence of a phase transition at finite temperature. The first term we are considering is the thermal average of the caracteristic function, which can be computed as

$$\langle \chi_c(l,i) \rangle = \frac{\sum_C' e^{-\beta E_c}}{\sum_C e^{-\beta E_c}},$$

where the sum in the numerator is taken over all configurations C that contain the domain wall (l,i), while the sum in the denominator is taken over all configurations of the system. In order to understand how to compute the energy of such configurations, we can compare the energy of a configuration C that contains the domain wall (l,i) with the energy of a configuration $\tilde{C}$ that is identical to C except for the fact that it does not contain the domain wall (l,i) (i.e., the spins enclosed by the domain wall (l,i) are flipped in $\tilde{C}$ with respect to C).

img/ising_2d_energy_conf1.png

Figure 3.7: Configuration C: a set of domain walls, including the domain wall (l,i) which we are considering.

img/ising_2d_energy_conf2.png

Figure 3.8: Configuration $\tilde{C}$: the same configuration without the domain wall (l,i): the spins inside of it are flipped.

The energy of the two configurations are related by

$$E_c = E_{\tilde{c}} + 2Jl,$$

so that now we can rewrite the sum over the configurations that contain the domain wall (l,i) as a sum over the configurations $\tilde{C}$ that do not contain the domain wall (l,i) (but which are built from a certain C containing it) as follows

$$\sum_C{}' e^{-\beta E_c} = \sum_{\tilde{C}} e^{-\beta(E_{\tilde{c}}+2Jl)} = e^{-2\beta Jl} \sum_{\tilde{C}} e^{-\beta E_{\tilde{c}}}.$$

Now we can be sure of the fact that the sum over all configurations C in the denominator is greater than the sum over the configurations $\tilde{C}$ (since the latter is a subset of the former), thus we can write

$$\langle \chi_c(l,i) \rangle = \frac{\sum_C' e^{-\beta E_c}}{\sum_C e^{-\beta E_c}} \leq \frac{e^{-2\beta Jl} \sum_{\tilde{C}} e^{-\beta E_{\tilde{c}}}}{\sum_{\tilde{C}} e^{-\beta E_{\tilde{c}}}} = e^{-2\beta Jl}.$$

Thus we have shown that the thermal average of the caracteristic function is upper bounded by an exponentially decaying function of the length of the domain wall, which means that the probability of finding a domain wall of length l in the system decreases exponentially with l.

Now we can use this result to compute the thermal average of the number of flipped spins as follows

$$\langle N_-(C) \rangle \leq \sum_l l^2 \sum_{i=1}^{M(l)} e^{-2\beta Jl} = \sum_l l^2 M(l) e^{-2\beta Jl} \leq 4N \sum_l l^2 3^{l-1} e^{-2\beta Jl},$$

such that the fraction of flipped spins can be estimated as

$$\frac{\langle N_-(C) \rangle}{N} \leq \frac{4}{3} \sum_l l^2 3^l e^{-2\beta Jl} = \frac{4}{3} \sum_l l^2 e^{-(2\beta J - \log(3))l}.$$

We can study the convergence of the series on the right hand side, which is a power series in l with an exponential decay factor. The series converges if the exponential decay factor dominates over the polynomial growth of l^2, which happens if the exponent is negative, i.e., if

$$2\beta J - \log(3) > 0 \implies T < \frac{2J}{k_B \log(3)} = T_c.$$

Thus we have shown that for temperatures below the critical temperature T_c, the average fraction of flipped spins is less than $\frac{1}{2}$, which means that the system is in an ordered phase with non-zero magnetization, and thus there is a phase transition at finite temperature.

We can also compute the sum explicitly to further convice ourselves: the allowed values of l are even numbers, since we need to have an even number of steps to create a closed loop in the lattice, thus we can write

$$l = 2, 4, 8 \cdots = 2k + 2, \quad k = 0, 1, 2 \ldots, +\infty,$$

and by taking the thermodinamic limit $N \to \infty$ after having inserted this expression for l in the sum, we can obtain

$$\lim_{N\to\infty} \frac{\langle N_-(C)\rangle}{N} \leq \frac{4}{3}\sum_{k=1}^{\infty}(2k+2)^2 e^{-(2\beta J - \log(3))(2k+2)} = \frac{64}{3}\frac{x^2}{(1-x)^3}\left(1 - \frac{3}{4}x + \frac{1}{4}x^2\right),$$

where we have defined $x = 9e^{-4\beta J}$. Looking at the expression on the right hand side, it is easy to convince ourselves that there exists a critical temperature T_c such that

$$\lim_{N\to\infty} \frac{\langle N_-(C)\rangle}{N} \leq \frac{1}{2} \quad \text{for } T < T_c,$$

which means that the system is in an ordered phase with non-zero magnetization for temperatures below T_c, and thus there is a phase transition at finite temperature. This completes our proof of the existence of a phase transition in the 2D Ising model at finite temperature. The Peierls's argument is a powerful tool in statistical physics that allows us to establish the existence of phase transitions in systems with discrete symmetries, such as the Ising model, by analyzing the energetics and statistics of domain walls. It provides a rigorous way to show that the system can exhibit long-range order and spontaneous symmetry breaking at finite temperatures, which is a fundamental aspect of phase transitions in statistical mechanics.

3.2.4 | Kramers-Wannier Duality

It is not difficult to prove, even if we will only comment the result, that the partition function of the 2D Ising model displays a special symmetry, called **Kramers-Wannier duality**, which relates the partition function of the model at a given temperature T to the partition function of a "dual model" at a different temperature T^*.

The fundamental idea behind the **dual lattice approach** in the two-dimensional Ising model lies in the geometric equivalence between its low-temperature and high-temperature series expansions. At low temperatures, the partition function is formulated as a sum over closed contours, or domain walls, separating regions of opposite spins. These contours naturally lie on the edges of the dual lattice. Conversely, the high-temperature expansion expresses the partition function as a sum over closed loop graphs formed by the bonds of the original lattice. Because the dual of a square lattice is simply another square lattice shifted by half a lattice spacing, the topological structures of the two expansions are formally identical.

By equating the statistical weight of a domain wall segment, $e^{2\beta^* J}$, to the weight of a high-temperature bond, $\tanh(\beta J)$, one establishes a direct mathematical mapping between the low-temperature regime β^* and the high-temperature regime β. This symmetry ultimately leads to the Kramers-Wannier duality condition:

$$\frac{1}{N}\log(Z_N(\beta)) = \frac{1}{N}\log(Z_N(\beta^*)) - \log(\sinh(2\beta^* J)),$$

where $\beta^* = \beta^*(\beta)$ is a monotonically decreasing function of β defined by the relation

$$\sinh(2\beta J)\sinh(2\beta^* J) = 1.$$

The Kramers-Wannier duality relates the high-temperature phase of the model (where β is small) to the low-temperature phase of the dual model (where β^* is large), and vice versa. In particular if $\log(Z_N)$ is analytic for some value of β, then it must also be analytic for the corresponding value of β^*. Similarly, if there is a non-analyticity in $\log(Z_N)$ at some value of β, then there must also be a non-analyticity at the corresponding value of β^* (thus a phase transition).

This implies that, assuming there to be only one single phase transition, it must coincide with the *self-dual point* $\beta_c = \beta_c^*$, then the critical temperature T_c of the phase transition must satisfy the condition

$$\sinh^2(2\beta_c J) = 1 \implies \sinh(2\beta_c J) = 1.$$

If we know define a variable $x = e^{2\beta J}$, we can rewrite the condition for the critical temperature as

$$\frac{1}{2}(x - \frac{1}{x}) = 1 \implies x^2 - 2x - 1 = 0 \implies x = 1 + \sqrt{2},$$

and by substituting back the definition of x, we can obtain the exact expression for the critical temperature T_c as

$$T_c = \frac{2}{\log\left(1+\sqrt{2}\right)}\frac{J}{k_B}.$$

This is a remarkable result, since it provides an exact expression for the critical temperature of the phase transition in the 2D Ising model, which is a non-trivial result that highlights the power of duality transformations in statistical physics. The Kramers-Wannier duality is a fundamental symmetry of the 2D Ising model that allows us to relate the behavior of the system at different temperatures and to derive exact results about its critical properties.

3.2.5 | Onsager's Exact Solution

Finally let us mention that the 2D Ising model was solved exactly by **Lars Onsager** in 1944, who derived an exact expression for the free energy of the system and showed that it exhibits a phase transition at a finite temperature T_c. The exact solution of the 2D Ising model is a landmark result in statistical physics, as it provides a complete understanding of the critical behavior of the system and serves as a benchmark for testing approximate methods and numerical simulations. The solution involves advanced mathematical techniques, such as transfer matrix methods and complex analysis, and it has been extensively studied and generalized to other models and higher dimensions.

The **Onsager's partition function** for the 2D Ising model can be expressed as

$$Z_N = \left[\sqrt{2}\cosh(2\beta J)\right]^N e^{\frac{N}{\pi}\int_0^{\pi/2} \mathrm{d}\varphi \log\left(1+\sqrt{1-\kappa^2\sin^2\varphi}\right)},$$

where

$$\kappa = \frac{2\sinh(2\beta J)}{\cosh^2(2\beta J)}.$$

From this expression, we can derive the value of the **critical exponents** of the phase transition in the 2D Ising model, which are given by

$$\beta = \frac{1}{8}, \qquad \nu = 1, \gamma = \frac{7}{4},$$
$$\alpha = 0, \qquad \eta = \frac{1}{4}, \delta = 15.$$

These exponents have a simple rational form, which is a consequence of the integrability of the model, but they are substantially different from the mean-field exponents, which highlights the importance of fluctuations and correlations in determining the critical behavior of the system.

The exact solution of the 2D Ising model has been a source of inspiration for many developments in statistical physics, including the study of conformal field theories and the renormalization group, and it continues to be an active area of research in theoretical physics.

4 Landau-Ginzburg Effective Field Theory

We are now ready to introduce the Landau-Ginzburg effective field theory, along with the field theory description of phase transitions. We will frame the discussion in the language of a magnetic system, with the simple Ising model as a paradigmatic example, but the discussion can be easily generalized to other systems and other order parameters.

The Landau-Ginzburg effective field theory is a **mesoscopic description** of phase transitions, which allows us to describe the behavior of the system near the critical point. Consider a ferromagnet such as iron, which can exhibit a phase transition from a paramagnetic to a ferromagnetic phase below a critical temperature T_c (known as the *Curie temperature*). Close to the critical point, we may expect that it is not necessary to know all the details about the microscopic costituent of the system and their interactions: indeed, as the correlation length diverges, the system becomes scale invariant, and the relevant degrees of freedom will become the collective excitations of the system, which are described by a mesoscopic parameters.

Therefore it makes sense to change paradigm from the microscopic description of the system, which is given by the Hamiltonian of the system, to a mesoscopic description, which will be given by a **free energy functional**. The mesoscopic degrees of freedom are much larger than the lattice spacing, but much smaller than the system size, so that they can be considered as continuous variables. We can thus imagine to define a field $m(\mathbf{x})$ which gives the local average of the magnetization in a small region around the point $\mathbf{x}$. The free energy functional will then be a functional of the field $m(\mathbf{x})$, and it will encode the behavior of the system near the critical point.

4.1 | Mesoscopic Description of Phase Transitions

Let us exemplify the concept of mesoscopic description of phase transitions by considering a trivial one dimensional chain of classical spins. Suppose the spins to be non interacting, so that the Hamiltonian of the system is simply given by the interaction with an external magnetic field h:

$$H[\{s_i\}] = -h \sum_i s_i,$$

and the partition function is given by

$$Z = \sum_{\{s_i\}} e^{-\beta H[\{s_i\}]} = \sum_{\{s_i\}} e^{\beta h \sum_i s_i} = 2\cosh(\beta h)^N.$$

Let us now show how we can write the partition function in terms of a mesoscopic degrees of freedom. We can consider the groupings of M spins, with $1 \ll M \ll N$, into N/M blocks of M spins each:

$$\cdots \underbrace{\uparrow\uparrow\downarrow\uparrow}_{\sigma_1=1/2}\ \underbrace{\downarrow\downarrow\uparrow\downarrow}_{\sigma_2=-1/2}\ \underbrace{\uparrow\downarrow\downarrow\uparrow}_{\sigma_3=0}\ \underbrace{\uparrow\uparrow\uparrow\uparrow}_{\sigma_4=1}\cdots$$

We associate to each block a mesoscopic degree of freedom σ_k defined as the average of the spins in the block:

$$\sigma_k = \frac{1}{M} \sum_{j=kM}^{M(k+1)} s_j,$$

which is not anymore a microscopic degree of freedom, since it is defined as an average over a block of spins, but instead a mesoscopic degree of freedom, since it is defined on a larger scale than the lattice spacing, but still very small compared to the system size. The variable σ_k can take values in the set $\{-1, -1+\frac{2}{M}, -1+\frac{4}{M} \ldots, 1-\frac{4}{M}, 1-\frac{2}{M}, 1\}$, which are the possible values of the average magnetization in a block of M spins.

Then we can rewrite the partition function as

$$\begin{aligned}
Z &= \sum_{\{s_i\}} e^{-\beta H[\{s_i\}]} \\
&= \sum_{\{\sigma_k\}} \sum_{\{s_i\}} \left[\prod_k \delta\left(\sigma_k - \frac{1}{M} \sum_{j=kM}^{M(k+1)} s_j \right) \right] e^{-\beta H[\{s_i\}]} \\
&=: \sum_{\{\sigma_k\}} e^{-\beta F[\{\sigma_k\}]},
\end{aligned}$$

where the second line is just a mathematical identity (since we inserted an integrated delta function), and we have defined the free energy functional $F[\{\sigma_k\}]$ in the third line. Let us highlight the nature of the first summation: it goes over all the sequences

$$\{\sigma_k\}_{k=1}^{N/M}, \text{ with } \sigma_k \in \{-1, -1+\tfrac{2}{M}, -1+\tfrac{4}{M} \ldots, 1-\tfrac{4}{M}, 1-\tfrac{2}{M}, 1\}.$$

Thus, computing the functional $F[\{\sigma_k\}]$ explicitly, we find

$$
\begin{aligned}
Z &= \sum_{\{\sigma_k\}} \sum_{\{s_i\}} \left[\prod_k \delta\left(\sigma_k - \frac{1}{M} \sum_{j=kM}^{M(k+1)} s_j \right) \right] e^{\beta h \sum_i s_i} \\
&= \sum_{\{\sigma_k\}} \sum_{\{s_i\}} \left[\prod_k \delta\left(\sigma_k - \frac{1}{M} \sum_{j=kM}^{M(k+1)} s_j \right) \right] e^{\beta M h \sum_k \sigma_k} \\
&= \sum_{\{\sigma_k\}} e^{\beta M h \sum_k \sigma_k} \prod_k \left[\sum_{\{s_j\}_{j=kM+1}^{M(k+1)}} \delta\left(\sigma_k - \frac{1}{M} \sum_{j=kM}^{M(k+1)} s_j \right) \right] \\
&=: \sum_{\{\sigma_k\}} e^{\beta M h \sum_k \sigma_k} \prod_k e^{S(\sigma_k)} \\
&= \sum_{\{\sigma_k\}} e^{\beta M h \sum_k \sigma_k + \sum_k S(\sigma_k)} =: \sum_{\{\sigma_k\}} e^{-\beta H[\{\sigma_k\}] + S[\{\sigma_k\}]},
\end{aligned}
$$

where we have defined an entropic factor $S(\sigma_k)$ as the logarithm of the number of configurations of the spins in the block k that give rise to a given value of σ_k:

$$
S(\sigma_k) = \log \left[\sum_{\{s_j\}_{j=kM+1}^{M(k+1)}} \delta\left(\sigma_k - \frac{1}{M} \sum_{j=kM}^{M(k+1)} s_j \right) \right],
$$

and in the end we named

$$
H[\{\sigma_k\}] = -\beta M h \sum_k \sigma_k, \quad S[\{\sigma_k\}] = \sum_k S(\sigma_k).
$$

Now we can compute the entropic factor $S(\sigma_k)$ explicitly. For $\sigma = 1 - \frac{2p}{M}$, with the correct values of p to make σ an allowed value, we have that the exponential of the entropy is given by the number of ways to choose p spins down in a block of M spins, which is given by the binomial coefficient $\binom{M}{p}$. Thus we have

$$
e^{S(\sigma)} = \binom{M}{p} = \binom{M}{\frac{M}{2}(\sigma + 1)} \implies S(\sigma) = \log \binom{M}{\frac{M}{2}(\sigma + 1)}.
$$

Putting everything together, we have that the partition function can be written in terms of the mesoscopic degrees of freedom σ_k as

$$
Z = \sum_{\{\sigma_k\}} e^{-\beta H[\{\sigma_k\}] + S[\{\sigma_k\}]} = \sum_{\{\sigma_k\}} e^{-\beta F[\{\sigma_k\}]}.
$$

4.1.1 | Principles and Free Energy Functional

Thus we have found a way to write the partition function in terms of a mesoscopic degrees of freedom, and we have defined the free energy functional $F[\{\sigma_k\}]$ as

$$
F[\{\sigma_k\}] = H[\{\sigma_k\}] - \frac{1}{\beta} S[\{\sigma_k\}] = -Mh \sum_k \sigma_k - \frac{1}{\beta} \sum_k \log \binom{M}{\frac{M}{2}(\sigma_{k+1})}.
$$

In this simple case, the Hamiltonian $H[\{\sigma_k\}]$ in terms of the collective variables σ_k has conserved the original functional form, with just a rescaling of the magnetic field $h \to Mh$. However, the patition function has acquired a non trivial entropic contribution of the form $S[\{\sigma_k\}] = \sum_k S(\sigma_k)$, which is a non trivial function of the mesoscopic degrees of freedom σ_k. In general, it may not be as simple to find the explicit form of the free energy functional given any microscopic Hamiltonian, but in principle it is possible to do so by following the same procedure we have just outlined.

Close to criticality, we expect the density σ_k to vary smoothly across the system, allowing us to approximate the sequence $\{\sigma_k\}$ with a continuous magnetization field $m(\mathbf{x})$. We can then write the free energy functional as a functional of the continuous field $m(\mathbf{x})$, where $\mathbf{x}$ is the spatial coordinate in the system. This is a mathematically delicate prcedure, which starts by identifying

$$m(\mathbf{x}) \simeq \frac{M\sigma_k}{\Delta x},$$

which in a one dimensional system can be pictured as follows:

`img/LG_average_magnetization_field.png`

Figure 4.1: Schematic representation of the *coarse-graining procedure* in a one-dimensional system. The discrete mesoscopic degrees of freedom $\{\sigma_k\}$, defined on blocks of finite size Δx and centered at x_k, are approximated by a continuous magnetization field $m(x)$ under the assumption that the magnetization varies slowly over spatial scales comparable to Δx.

The idea is to "replace sum by integrals" in the partition function, and to write the free energy functional as a functional of the continuous field $m(\mathbf{x})$:

$$\sum_k \sigma_k = \sum_k \Delta x \frac{m(x_k)}{M} \simeq \frac{1}{M} \int \mathrm{d}^d\mathbf{x}\, m(\mathbf{x}),$$

therefore letting us write the partition function as

$$Z = \int \mathcal{D}m(\mathbf{x}) e^{-\beta F[\{m(\mathbf{x})\}]}.$$

In general, finding the explicit form of the free energy functional $F[\{m(\mathbf{x})\}]$ is a very difficult task, and it may not be possible to do so in a closed form. However, the LG theory gives us a prescription on how to describe it close to criticality, in terms of a few fenomenological parameters, by imposing some general principles on the form of the free energy functional.

The Landau-Ginzburg theory is very general. So far we have examined an uniaxial ferromagnet, where the order parameter is a scalar field $m(\mathbf{x})$ that describes the local magnetization. In general, we can consider systems with phase transitions described by a vectorial order parameter field

$$\mathbf{m}(\mathbf{x}) \in \mathbb{R}^n, \text{ i.e. } \begin{cases} \mathbf{x} = (x_1, \ldots, x_d) \in \mathbb{R}^d, \\ \mathbf{m}(\mathbf{x}) = (m_1(\mathbf{x}), \ldots, m_n(\mathbf{x})) \in \mathbb{R}^n, \end{cases}$$

where d is the spatial dimension and n is the number of components of the order parameter.

Example. Taking in consideration some examples of other systems which can be described by the Landau-Ginzburg theory may help to clarify the generality of the theory:

- ($n = 1$): uniaxial ferromagnet, liquid-gas transition, binary mixtures, etc.
- ($n = 2$): superfluidity, superconductivity, planar magnets, etc.
- ($n = 3$): Heisenberg ferromagnet, classical magnets, etc.

Tipically, we consider systems in 3 spatial dimensions, so that $d = 3$, but it may happen to consider phase transitions appearing on surfaces, which are effectively two dimensional, or on lines, which are effectively one dimensional. In these cases, we will have $d = 2$ or $d = 1$ respectively.

Landau-Ginzburg Prescription

The Landau-Ginzburg theory gives us a prescription on how to describe the free energy functional $F[\{m(\mathbf{x})\}]$ close to criticality, in terms of a few fenomenological parameters, by imposing some general principles on the form of the free energy functional. These principles are:

- **Locality and uniformity**: different positions in the system are equivalent. Therefore, we must be abe to write the free energy functional as an integral in space of a free energy density $\Phi(m(\mathbf{x}), \nabla m(\mathbf{x}), \dots)$ that depends on the field $m(\mathbf{x})$ and its derivatives at the same point $\mathbf{x}$:
$$\beta F[\{m(\mathbf{x})\}] = \int \mathrm{d}^d\mathbf{x}\, \Phi(m(\mathbf{x}), \nabla m(\mathbf{x}), \nabla^2 m(\mathbf{x}), \dots),$$
where integrating we are effectively removing the spatial dependence of the free energy functional.
- **Analiticity**: the function Φ should be analytic in its arguments, since we do not expect any singularity in the free energy functional at mesoscopic scales (we are not at the critical point yet, where the correlation length diverges, but we are close to it). Close to criticality, the order parameter $m(\mathbf{x})$ is small, so we can expand the free energy density Φ in powers of $m(\mathbf{x})$ and its derivatives, and keep only the leading terms in the expansion (we will later state this more precisely).
- **Symmetry**: the functional form of Φ should reflect the symmetries of the model. In particular, a given term in the Taylor expansion of Φ is allowed only if it does not break any of the symmetries of the model. For example, in the case of a uniaxial ferromagnet, the free energy functional should be invariant under the transformation $m(\mathbf{x}) \to -m(\mathbf{x})$, since the system is symmetric under the inversion of the magnetization. This means that only even powers of $m(\mathbf{x})$ are allowed in the expansion of Φ.

We now have all the ingredients to put the LG principle into pactice, and to write down the free energy functional close to criticality. We will do this in the next section, where we will apply the LG theory to the d-dimensional Ising model.

4.1.2 | Application on Ising Model

Consider the d-dimensional Ising model, which is a generalization of the one-dimensional chain of classical spins we have considered before. Since the order parameter is a scalar mgnetization

$m \in \mathbb{R}$, we can follow the LG prescription and consider the scalar field $m(\mathbf{x})$ ($\mathbf{x} \in \mathbb{R}^d$), the local (mesoscopic) magnetization. Thus we can build the mesoscopic partition function as a function of the field $m(\mathbf{x})$ as

$$Z = \int \mathcal{D}m(\mathbf{x}) e^{-\beta F[\{m(\mathbf{x})\}]},$$

where we know

$$\beta F[\{m(\mathbf{x})\}] = \int \mathrm{d}^d\mathbf{x}\, \Phi(m(\mathbf{x}), \nabla m(\mathbf{x}), \nabla^2 m(\mathbf{x}), \dots).$$

Close to the critical point, the configurations contribuiting the most to the partition function are those with small magnetization, so we can expand the free energy density Φ in powers of $m(\mathbf{x})$ and its derivatives, and keep only the leading terms in the expansion:

$$\Phi(m(\mathbf{x}), \nabla m(\mathbf{x}), \nabla^2 m(\mathbf{x}), \dots) = a_1 + a_2 m^2(\mathbf{x}) + \dots + \sum_k b_k \partial_{x_k} m(\mathbf{x}) \dots$$

The next prescription of the LG theory is to impose the symmetries of the model on the free energy density Φ. In the case of the Ising model, in the absence of the external field ($h = 0$), the system is symmetric under a $\mathbb{Z}_2$ transformation, mapped by $m(\mathbf{x}) \to -m(\mathbf{x})$; thus only even powers of $m(\mathbf{x})$ are allowed in the expansion of Φ, canceling out a_1 as well as the other coefficients in front of odd powers. Thus we can write

$$\Phi[m(\mathbf{x}), \nabla m(\mathbf{x}), \nabla^2 m(\mathbf{x}), \dots]\big|_{h=0} = a_2 m^2(\mathbf{x}) + a_4 m^4(\mathbf{x}) + \dots + b_2(\nabla m(\mathbf{x}))^2 + \dots$$

Let us highlight that if we are in presence of the external field h, the $\mathbb{Z}_2$ symmetry is broken, and odd powers of $m(\mathbf{x})$ are allowed in the expansion of Φ, but they must vanish with h. Furthermore, close to criticality, even h has to be small, so we can keep terms linear in h in the expansion of Φ, and thus we can write

$$\begin{aligned}\Phi[m(\mathbf{x}), \nabla m(\mathbf{x}), \nabla^2 m(\mathbf{x}), \dots] &= a_2 m^2(\mathbf{x}) + a_4 m^4(\mathbf{x}) + \dots + b_1(\nabla m(\mathbf{x}))^2 + \dots - h(m(\mathbf{x}) + \dots) \\ &\simeq \frac{t}{2} m^2(\mathbf{x}) + u m^4(\mathbf{x}) + \frac{\kappa}{2}(\nabla m(\mathbf{x}))^2 - h m(\mathbf{x}),\end{aligned}$$

where we kept only a few of the first orders in the expansion, and we renamed the coefficients as $a_2 = t/2$, $a_4 = u$, $b_1 = \kappa/2$, using conventional notation.

At this point, the parameters t, u and κ are just phenomenological parameters, which depend on the microscopic details of the model, and in principle can also depend on the temperature T. These are not universal parameters, and they should be fixed by experimental observations. One could also make assumptions based on the behavior of the system and the meaning of the powers associated to the parameters, but in the end these assumptions should be verified by experiments. For example, we can assume that $u > 0$ in order to have a stable free energy functional, and we can also assume that t is a linear function of the temperature T, such that $t = 0$ at the critical temperature T_c, so that $t < 0$ for $T < T_c$ and $t > 0$ for $T > T_c$. However, these assumptions may not hold in general.

Putting everything together, we have significantly simplified the problem of finding a form for the partition function close to criticality, by reducing it to the problem of finding a few phenomenological parameters t, u and κ that describe the free energy functional:

$$Z = \int \mathcal{D}m(\mathbf{x}) e^{-\int \mathrm{d}^d\mathbf{x}\left[\frac{t}{2}m^2(\mathbf{x}) + u m^4(\mathbf{x}) + \frac{\kappa}{2}(\nabla m(\mathbf{x}))^2 - h m(\mathbf{x})\right]}.$$

The price we have paid for this simplification is that we have lost the connection with the microscopic details of the model, and we have introduced some phenomenological parameters that need to be fixed by experiments: they have non trivial functional dependence on the temperature T, the external field h and on the microscopic details of the model, and they are not universal parameters. However, the LG theory is very powerful, since it allows us to describe the behavior of the system close to criticality in terms of a few parameters.

4.2 | Saddle Point Approximation

Continuing our study of the Landau-Ginzburg theory, let us report the form of the partition function we have found for the d-dimensional Ising model near criticality:

$$Z = \int \mathcal{D}m(\mathbf{x}) e^{-\int \mathrm{d}^d\mathbf{x}\left[\frac{t}{2}m^2(\mathbf{x})+um^4(\mathbf{x})+\frac{\kappa}{2}(\nabla m(\mathbf{x}))^2-hm(\mathbf{x})\right]}.$$

Apart from the comment we already made about the phenomenological parameters t, u and κ, we can also notice that the partition function has a very appealing form: it is a **path integral**, which is one of the most studied objects in theoretical physics, and it is the starting point for many powerful techniques to study the behavior of the system close to criticality. Keep in mind that the exact evaluation is still far out of reach, thus making non-trivial to extract quantitative predictions from the partition function.

However, we can still extract some qualitative predictions from the partition function by making some approximations. The simplest approximation we can make is the **saddle point approximation**, which substantially consists in approximating the partition function by the contribution of the configuration $m(\mathbf{x})$ that minimizes the free energy functional $F[\{m(\mathbf{x})\}]$. This is a good approximation when the fluctuations around the minimum are small, which is typically the case far from criticality, where the correlation length is finite and the system is not scale invariant.

Let us first employ this approximation in a simple context, in order to get a better intuition on how it works, and then we will see how to generalize it to the case of the partition function we have found for the d-dimensional Ising model near criticality.

Example (Saddle Point Approximation: a simple integral). Let us consider the following simple integral:

$$I = \int_a^b \mathrm{d}x\, e^{-R\,g(x)}, \quad R \in \mathbb{R},$$

where we can suppose the parameer R to be large ($R \gg 1$) and $g(x)$ to have a unique minimum at some point $\overline{x} \in [a,b]$. In this way we can expand $g(x)$ around the minimum $\overline{x}$ as

$$g(x) \sim g(\overline{x}) + \frac{1}{2}g''(\overline{x})(x-\overline{x})^2 + \mathcal{O}((x-\overline{x})^3),$$

where in the minimum we have $g'(\overline{x}) = 0$ and $g''(\overline{x}) > 0$. Thus we can write the integral as

$$I \sim e^{-R\,g(\overline{x})} \int_a^b \mathrm{d}x\, e^{-\frac{R}{2}g''(\overline{x})(x-\overline{x})^2} \sim e^{-R\,g(\overline{x})} \int_{-\infty}^{+\infty} \mathrm{d}x\, e^{-\frac{R}{2}g''(\overline{x})(x-\overline{x})^2},$$

where in the last step we have extended the limits of integration to infinity, since the main contribution to the integral comes from the region around the minimum $\overline{x}$. The remaining integral is a Gaussian integral, which can be evaluated explicitly after the change of variable $y = (x-\overline{x})$:

$$I \sim e^{-R\,g(\overline{x})} \int_{-\infty}^{+\infty} \mathrm{d}y\, e^{-\frac{R}{2}g''(\overline{x})y^2} = e^{-R\,g(\overline{x})}\sqrt{\frac{2\pi}{R\,g''(\overline{x})}}.$$

If we now take the logarithm of the integral, we find

$$\log I \sim -R\,g(\overline{x}) - \frac{1}{2}\log(R\,g''(\overline{x})) + \frac{1}{2}\log(2\pi) \sim -R\,g(\overline{x}) + \mathcal{O}(\log R),$$

which means that up to subleading corrections in R, the integral is dominated by the contribution of the minimum of the function $g(x)$: we can replace the integral with the maximal value of the integrand, which is given by $e^{-R\,g(\overline{x})}$

$$I = \int_a^b e^{-R\,g(x)} \sim e^{-R\,g(\overline{x})}.$$

This is the essence of the saddle point approximation: the integral is dominated by the contribution of the minimum of the function $g(x)$, and we can approximate the integral by the value of the integrand at the minimum, up to subleading corrections in R. Of course, untill R remains finite, the saddle point approximation is not exact, but is a very powerful tool to extract qualitative predictions from the integral, and can be applied in many different contexts, such as the evaluation of path integrals, which is the case we are interested in.

4.2.1 | Application to the LG Ising Partition Function

Thus we can apply the saddle point approximation to the partition function we have found for the d-dimensional Ising model near criticality, which is again given by

$$Z = \int \mathcal{D}m(\mathbf{x}) e^{-\int \mathrm{d}^d\mathbf{x}\,\left[\frac{t}{2}m^2(\mathbf{x})+um^4(\mathbf{x})+\frac{\kappa}{2}(\nabla m(\mathbf{x}))^2-hm(\mathbf{x})\right]}.$$

Firstly, let us underline that the phenomenological parameter κ has to be positive: if it were negative, the free energy functional would not be stable, since it would be unbounded from below:

$$\int \mathrm{d}^d\mathbf{x}\,\left[\frac{t}{2}m^2(\mathbf{x}) + um^4(\mathbf{x}) + \frac{\kappa}{2}(\nabla m(\mathbf{x}))^2 - hm(\mathbf{x})\right],$$

we could choose an oscillating configuration of the field $m(\mathbf{x})$ with a very large gradient, such that the contribution of the gradient term becomes very negative, and thus the free energy functional becomes unbounded from below, resulting in a divergence of the path-integral.

Thus we have to impose $\kappa > 0$ in order to have a stable free energy functional, and the field configurations that maximize the previous integral is therefore constant in space

$$m(\mathbf{x}) = m, \quad \forall \mathbf{x} \in \mathbb{R}^d.$$

Considering a system of finite size L, the saddle point approximation gives us the following expression of the partition function:

$$Z \simeq e^{-L^d\left[\frac{t}{2}m^{*2}+um^{*4}-hm^*\right]},$$

where $L^d = V$ is the volume of the system and we have defined m^* as the constant value of the field $m(\mathbf{x})$ that minimizes the integral

$$\Phi[m(\mathbf{x})] = \int \mathrm{d}^d\mathbf{x}\,\left[\frac{t}{2}m^2(\mathbf{x}) + um^4(\mathbf{x}) + \frac{\kappa}{2}(\nabla m(\mathbf{x}))^2 - hm(\mathbf{x})\right].$$

Indeed, for a constant configuration of the field $m(\mathbf{x})$, the contribution of the gradient term vanishes, and the integral reduces to

$$\Phi[m(\mathbf{x}) = m] = V\left[\frac{t}{2}m^2 + um^4 - hm\right],$$

Recognizing that this expression has the same functional form of the free energy density that we computed in the mean field approximation, which is non surprising since, as we will see, the saddle point approximation is a generalization of the mean field approximation.

In particular, the form of the free energy density depends on the sign of the parameter t:

img/spa_free_functional_t-g-0.jpg

Figure 4.2: Schematic representation of the free energy density $\Phi(m)$ as a function of the magnetization m for $t > 0$. In this case, the free energy density has a single minimum at $m = 0$, which corresponds to the disordered phase of the system.

img/spa_free_functional_t-l-0.jpg

Figure 4.3: Schematic representation of the free energy density $\Phi(m)$ as a function of the magnetization m for $t < 0$. In this case, the free energy density has two symmetric minima at $m = \pm m^*$, which correspond to the ordered phase of the system. The minimum at $m = 0$ becomes a local maximum, indicating that the disordered phase is unstable.

Thus, it is natural to postulate the following functional dependence of the parameter t on the temperature T:

$$t(T) = a(T - T_c) + \mathcal{O}((T - T_c)^2),$$

since we want t to vanish at the critical temperature T_c, and to be positive for $T > T_c$ and negative for $T < T_c$. With this assumption, we can see that the system undergoes a phase transition at the critical temperature T_c, where the free energy density changes its shape, going from a single minimum at $m = 0$ for $T > T_c$ to two symmetric minima at $m = \pm m^*$ for $T < T_c$.

With this identification, we have fully recovered the mean field approximation for the Ising model, and its predictions. Thus it is now clear that the mean field approximation is a particular case of the saddle point approximation, where we restrict the field configurations to be constant in space, and we look for the value of the constant field that minimizes the free energy functional. The saddle point approximation is a generalization of the mean field approximation, since it allows us to consider non-constant configurations of the field $m(\mathbf{x})$, and thus to take into account spatial fluctuations of the order parameter.

Now in principle, we could go back to the path integral and estimate the corrections to the saddle point approximation by considering **field fluctuations** around the minimum of the free energy functional.

4.2.2 | Lower and Upper Critical Dimension

Let us underline the substantial difference between the first simple example of the saddle point approximation we have considered, which is a simple integral, and the second example, which is a path integral.

- In the first case, the saddle point approximation is a good approximation when the parameter R is large, since in this case the fluctuations around the minimum of the function $g(x)$ are suppressed by the factor R in the exponent.

- In the second case, we did not use any "large parameter" to justify the saddle point approximation, and it is still not clear at the moment under which conditions the saddle point approximation is a good approximation near criticality. In the next section, we will see how this is exactely done estimating the fluctuations around the minimum of the free energy functional, and we will find that the validity of the saddle point approximation depends on the spatial dimension d of the system.

In particular, we will find that there exists a **lower critical dimension** D_l and an **upper critical dimension** D_c such that:

- For $d < D_l$, the SPA gives wrong predictions.
- For $D_l < d < D_c$, there may be some divergences in the corrections to the SPA, but the qualitative predictions of the SPA are still correct; the same cannot be said for quantitative results, which are modified by the fluctuations around the minimum of the free energy functional.
- For $d > D_c$, the fluctuations around the minimum of the free energy functional are weak enough that they do not modify the critical behavior of the system, resulting in a correctness of the SPA both qualitatively and quantitatively: the critical exponents predicted by the SPA are exact for $d > D_c$.

In the case of the Ising model, we will find that the lower critical dimension is $D_l = 1$ and the upper critical dimension is $D_c = 4$. This means that the SPA gives wrong predictions for the one-dimensional Ising model (for which we have an analytical solution), it gives correct qualitative predictions but wrong quantitative predictions for the two- and three-dimensional Ising model,[1] and it gives correct qualitative and quantitative predictions, giving us the exact critical exponents.

Before proceeding, let us summarize the emergent picture of a generic continuous phase transition that we have obtained so far:

- The system undergoes a continuous phase transition at the critical temperature T_c, where the free energy density changes its shape, describing a transition from a disordered phase to an ordered one.
- Above the critical temperature T_c, the system is in a disordered phase, and it is invariant under the action of a certain symmetry group G (for example, in the case of the Ising model, the system is invariant under a $\mathbb{Z}_2$ transformation).
- Under the critical temperature T_c, the system is in an ordered phase, and **the symmetry** G **is spontaneously broken**, since the system chooses one of the minima of the free energy density, which are not invariant under the action of the symmetry group G: an **order parameter** m emerges, which is zero in the disordered phase and non-zero in the ordered phase, and which transforms non-trivially under the action of the symmetry group G.
- Based on physical principles, one can describe the behaviour of this order parameter at mesoscopic scales by a **free energy functional** $F[\{m(\mathbf{x})\}]$, with the *path integral formulation of the LG theory.*

[1] The qualitative structure of the phase diagram is correct, but the quantitative values of the critical exponents are modified by fluctuations by the mean field approximation (which, as we have already seen, is just a particular case of the saddle point approximation).

- At the zeroth order, we can apply the **saddle point approximation** to the partition function, which can gives us the mean field approximation for the system, and allows us to extract some qualitative predictions (exact if $d > D_c$) about the behavior of the system close to criticality.

In the following, we will complete our study of LG theory by analyzing the fluctuations. This will allow us to determine the upper critical dimension for the Ising model (note that we have previously established $D_l = 1$, based on its exact solution). Before embarking on this, let us show that, even when working at the level of the saddle-point approximation, the LG free-energy contains more information than the mean-field approach, as the LG theory takes into account the spatial dependence: we will be able to describe non-uniform configurations of the order parameter, which are not captured by the mean-field approximation, and which are relevant for the description of phenomena such as domain walls, interfaces, and critical fluctuations.

4.2.3 | Domain Walls

Taking again into consideration the Ising model in d dimensions in the ordered phase ($t < 0$) and **fix the boundary conditions** $m(x_1 \to \pm\infty) = \pm\overline{m}$, where x_1 is the first coordinate of the position vector $\mathbf{x} = (x_1, \dots, x_d)$. Here, $\pm\overline{m}$ are the two equilibrium values at temperature t. Let us use LG theory along with the saddle-point approximation to predict the spatial dependence of $m(\mathbf{x})$ in this setting: we want to find the configuration of the field $m(\mathbf{x})$ that minimizes the free energy functional, given the boundary conditions we have imposed. This configuration will describe a **domain wall**, which is a non-uniform configuration of the order parameter that interpolates between the two equilibrium values $-\overline{m}$ and $+\overline{m}$ across space.

The starting point is the LG partition function we previously derived for the d-dimensional Ising model near criticality, which is given by

$$\mathcal{Z} = \int \mathcal{D}m(\mathbf{x}) e^{-\int \mathrm{d}^d\mathbf{x}\left[\frac{t}{2}m^2(\mathbf{x}) + um^4(\mathbf{x}) + \frac{\kappa}{2}(\nabla m)^2 - hm(\mathbf{x})\right]},$$

where now the path integral is only over functions $m(\mathbf{x})$ such that

$$\lim_{x_1 \to \pm\infty} m(x_1, x_2 \dots x_d) = \pm\overline{m}.$$

We now apply the saddle-point approximation. However, the minimum of the free energy is not given by $m(\mathbf{x})$ constant, because this does not satisfy the boundary condition (it has to give two opposite values at the boundaries, therefore it cannot be constant). To find the correct minimum, we need to take a <u>functional derivative</u> of the free energy. This is a tool in functional analysis which generalizes the derivative, and it is used, for instance, to derive the Euler-Lagrange equations of statistical mechanics from the least-action principle. To see how it works, take a field configuration $m(\mathbf{x})$ and perturb it by a small amount

$$m(\mathbf{x}) \mapsto m(\mathbf{x}) + \delta m(\mathbf{x}), \text{ with } \delta m(\mathbf{x}) = 0 \text{ at the boundaries.}$$

The change in the LG free energy (up to $O(\delta m(\mathbf{x}))$) can be computed by inserting the variation of the field configuration into the free energy functional and expanding to first order in $\delta m(\mathbf{x})$: we are computing the functional derivative of the free energy functional with respect to the field $m(\mathbf{x})$

$$\delta\left[\int \mathrm{d}^d\mathbf{x}\left[\frac{t}{2}m^2(\mathbf{x}) + um^4(\mathbf{x}) + \frac{\kappa}{2}(\nabla m)^2\right]\right]$$
$$= \int \mathrm{d}^d\mathbf{x}\left[tm(\mathbf{x})\delta m(\mathbf{x}) + 4um^3(\mathbf{x})\delta m(\mathbf{x}) + \kappa(\nabla m)\cdot\nabla(\delta m)\right],$$

where now the last term can be integrated by parts to give

$$\int \mathrm{d}^d\mathbf{x}(\nabla m)\cdot\nabla(\delta m) = -\int \mathrm{d}^d\mathbf{x}(\nabla^2 m)\delta m + \underbrace{\int \mathrm{d}^d\mathbf{x}\nabla\cdot(\nabla m\delta m)}_{(*)}.$$

We have $(*) = 0$, since we choose $\delta m(\mathbf{x}) = 0$ at the boundary of the integration domain. Therefore, substituting back into the previous expression, we find

$$\begin{aligned}&\delta\left[\int \mathrm{d}^d\mathbf{x}\left[\frac{t}{2}m^2(\mathbf{x}) + um^4(\mathbf{x}) + \frac{\kappa}{2}(\nabla m)^2\right]\right]\\ &= \int \mathrm{d}^d\mathbf{x}\left[tm(\mathbf{x}) + 4um(\mathbf{x})^3 - \kappa\nabla^2 m(\mathbf{x})\right]\delta m(\mathbf{x}).\end{aligned}$$

The saddle-point condition is $\delta F/\delta m(\mathbf{x}) = 0$, which finally gives

$$\kappa\nabla^2 m(\mathbf{x}) = tm(\mathbf{x}) + 4um(\mathbf{x})^3.$$

These are the **Euler-Lagrange equations** for our field theory.

Coming back to our problem, we can reduce the Euler-Lagrange equations to a 1D equation (since we are only considering spatial variation along x_1) giving

$$\kappa\frac{\mathrm{d}^2}{\mathrm{d}x^2}m(x) = tm(x) + 4um(x)^3$$

where we renamed $x_1 = x$. Using

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\tanh(ax) = a^2\tanh(ax)[1-\tanh^2(ax)]$$

we find the solution from the previous equation, which is given by

$$m(x) = \overline{m}\tanh\left(\frac{x-x_0}{w}\right),$$

where

$$w = \sqrt{\frac{2\kappa}{-t}}, \quad \overline{m} = \sqrt{\frac{-t}{4u}},$$

are the **width of the domain wall** and the equilibrium value of the magnetization, respectively.

img/spa_domain_wall.png

Figure 4.4: Schematic representation of the spatial dependence of the magnetization field $m(x)$ in the presence of a domain wall, as predicted by the Landau-Ginzburg theory. The magnetization field $m(x)$ interpolates smoothly between the two equilibrium values $-\overline{m}$ and $\overline{m}$ across a region of finite width w, which characterizes the domain wall. The position of the domain wall is given by the free parameter x_0.

Here x_0 is a free parameter, which represents the position of the domain wall, that emerges between $-\overline{m}$ at $x = -\infty$ and $\overline{m}$ at $x = +\infty$. The width of the domain wall is given by w, which diverges as we approach the critical temperature T_c (where $t \to 0^-$, we will see this more clearly later). This is a manifestation of the critical fluctuations that occur near the phase transition, and it is a feature that cannot be captured by the mean-field approximation, which predicts a sharp domain wall with zero width.

Thus we were able to find a non-uniform configuration of the order parameter $m(\mathbf{x})$ that minimizes the free energy functional, and which describes a domain wall. This is an example of how the Landau-Ginzburg theory allows us to describe phenomena that are not captured by the mean-field approximation, by taking into account the spatial dependence of the order parameter and its fluctuations. The mean-field approximation, on the other hand, would predict a sharp domain wall with zero width, which is not physical and does not capture the critical fluctuations that occur near the phase transition.

4.3 | Fluctuations

We now want to study the path integral going beyond the saddle-point approximation and understand the impact of fluctuations. We start from the LG partition function

TODO: Add reference to Kardar book (Ch. 3) and the notes statistical field theory by D. Tong.

$$Z = \int \mathcal{D}m(\mathbf{x}) e^{-\int \mathrm{d}^d\mathbf{x}\left[\frac{t}{2}m^2(\mathbf{x})+um^4(\mathbf{x})+\frac{\kappa}{3}(\nabla m)^2\right]}.$$

We have already saw that the saddle-point approximation to this partition function gives a good description of the phase transition, and allows us to compute the critical exponents. However, it is not clear whether the saddle-point approximation is reliable, and whether the critical exponents obtained from it are correct. To answer these questions, we need to study the fluctuations around the saddle-point solution.

For $u > 0$ and without imposing fixed boundary conditions, we saw that the saddle-point solution corresponds to

$$m(\mathbf{x}) \equiv \overline{m} = \begin{cases} 0 & t > 0, \\ \sqrt{\dfrac{-t}{4u}} & t < 0. \end{cases}$$

Since we want to examine the fluctuations around the saddle-point solution, we write the field $m(\mathbf{x})$ as the sum of the saddle-point solution $\overline{m}$ and a fluctuation field $\phi(\mathbf{x})$:

$$m(\mathbf{x}) = \overline{m} + \phi(\mathbf{x}).$$

Substituting this expression into the LG free energy, we can expand to the 2nd order in $\phi(\mathbf{x})$ the powers of $m(\mathbf{x})$ and its derivatives, to obtain

$$\begin{cases} (\nabla m)^2 = \nabla\phi(\mathbf{x})^2, \\ m^2 = \overline{m}^2 + 2\overline{m}\phi(\mathbf{x}) + \phi(\mathbf{x})^2, \\ m^4 = \overline{m}^4 + 4\overline{m}^3\phi(\mathbf{x}) + 6\overline{m}^2\phi^2(\mathbf{x}) + \mathcal{O}(\phi(\mathbf{x})^3). \end{cases}$$

Therefore, substituting these expressions back into the free energy, we have

$$\begin{aligned} F_{LG} &\simeq V\left(\frac{t}{2}\overline{m}^2 + u\overline{m}^4\right) + \int \mathrm{d}^d\mathbf{x}\left[\frac{\kappa}{2}(\nabla\phi)^2 + \frac{t+12u\overline{m}^2}{2}\phi^2\right] \\ &\simeq F[\overline{m}] + \int \mathrm{d}^d\mathbf{x}\left[c_1\phi^2(\mathbf{x}) + c_2(\nabla\phi)^2\right], \end{aligned}$$

where $V = \int \mathrm{d}^d\mathbf{x}$ is just the volume of the system, and we have defined $c_1(t)$ and c_2 as follows:

$$c_1(t) = \frac{t+12u\overline{m}^2}{2} > 0, \quad c_2 = \frac{\kappa}{2} > 0.$$

Note that there are no linear terms in ϕ, which follows from the fact that we are expanding around the minimum of the free energy (where $\frac{\partial F}{\partial m} = 0$).

Since we have defined $m(\mathbf{x}) = \overline{m} + \phi(\mathbf{x})$, we can perform explicitly the functional integral over $\phi(\mathbf{x})$

$$\mathcal{D}m(\mathbf{x}) \mapsto \mathcal{D}\phi(\mathbf{x}),$$

to compute the partition function, and thus the free energy. This will allow us to understand how fluctuations modify the critical behavior of the system with respect to the saddle-point prediction.

In order to rewrite the free energy in a more convenient form, let us remember that in general

$$Z = \int \mathcal{D}m(\mathbf{x}) e^{-F[m(\mathbf{x})]},$$

which is a functional integral over all possible field configurations $m(\mathbf{x})$. We have moved onto an integration over the fluctuations $\phi(\mathbf{x})$, and now it is useful to consider the Fourier transform of the field $\phi(\mathbf{x})$ with the idea of diagonalizing the free energy in momentum space:

$$\phi_{\mathbf{q}} = \int \mathrm{d}^d\mathbf{x}\, e^{-i\mathbf{q}\cdot\mathbf{x}}\phi(\mathbf{x}) \implies \phi(\mathbf{x}) = \int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\, e^{i\mathbf{q}\cdot\mathbf{x}}\phi_{\mathbf{q}}.$$

where $\phi(\mathbf{x}) \in \mathbb{R}$ has to remain a real field, so that we need to impose $\phi^*_{\mathbf{q}} = \phi_{-\mathbf{q}}$, and $\mathbf{q}$ are the momenta of the field. Remember that $\phi(\mathbf{x})$ was obtained by **coarse-graining**[2] a microscopic spin configuration. Therefore, we can assume that $\phi(\mathbf{x})$ does not vary over length scales that are smaller than the spin-spin distance a (the lattice spacing). In turn, it means we can introduce a cutoff Λ in momentum space, and require

$$\phi_{\mathbf{q}} = 0 \text{ if } |\mathbf{q}| > \Lambda = 1/a.$$

Indeed, if the field $\phi(\mathbf{x})$ does not vary over length scales smaller than a, then its Fourier transform $\phi_{\mathbf{q}}$ must vanish for momenta larger than $1/a$, since such momenta must correspond to fluctuations that vary on length scales smaller than a. Therefore, if $\phi(\mathbf{x})$ is smooth and does not have any sharp features on length scales smaller than a, then its Fourier transform $\phi_{\mathbf{q}}$ must be concentrated around low momenta, and must vanish for momenta larger than $1/a$.

Next, if we we plug the Fourier transform into the free energy, we can effectively diagonalize it in momentum space. In fact, we have

$$F_{LG} - F[\overline{m}] = \int \frac{\mathrm{d}^d\mathbf{q}_1}{(2\pi)^d} \int \frac{\mathrm{d}^d\mathbf{q}_2}{(2\pi)^d} \int \mathrm{d}^d\mathbf{x}\, (-c_2\mathbf{q}_1\cdot\mathbf{q}_2 + c_1)\, \phi_{\mathbf{q}_1}\phi_{\mathbf{q}_2} e^{i(\mathbf{q}_1+\mathbf{q}_2)\cdot\mathbf{x}},$$

where now we have two integrations in momentum space (since the field $\phi(\mathbf{x})$ appeared always squared), and one integration in real space. Using the definition of the Dirac delta function

$$\int \mathrm{d}^d\mathbf{x}\, e^{i(\mathbf{q}_1+\mathbf{q}_2)\cdot\mathbf{x}} = (2\pi)^d\delta^d(\mathbf{q}_1+\mathbf{q}_2),$$

we can perform the integration in real space to enforce the constraint $\mathbf{q} = \mathbf{q}_1 = -\mathbf{q}_2$, and we finally arrive at

$$F_{LG} - F[\overline{m}] = \int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}(c_2q^2 + c_1)\phi_{\mathbf{q}}\phi_{-\mathbf{q}} = \int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}(c_2q^2 + c_1)|\phi_{\mathbf{q}}|^2,$$

since $\phi_{\mathbf{q}}\phi_{-\mathbf{q}} = \phi_{\mathbf{k}}\phi^*_{\mathbf{k}} = |\phi_{\mathbf{q}}|^2$ is the definition of the squared modulus of a complex number. We have thus succeeded in diagonalizing the free energy in momentum space and we can now perform explicitly the functional integral to compute the partition function.

To compute explicitly the partition function, we need to change variables in the original functional path-integral in order to express it in terms of the Fourier modes $\phi_{\mathbf{q}}$. We have

$$\int \mathcal{D}\phi(\mathbf{x}) \propto \int \mathcal{D}\phi_{\mathbf{q}} = \prod_{\mathbf{q}} \int \mathrm{d}\phi_{\mathbf{q}}, \qquad (*)$$

[2] With *coarse-graining* we mean the process of averaging over short-distance fluctuations, which is exactely what we have been doing in the Landau-Ginzburg theory to obtain a mesoscopic description of the system.

where we need to impose the additional reality constraint $\phi^*_{\mathbf{q}} = \phi_{-\mathbf{q}}$. Finally, we get

$$Z = \left(\prod_{\mathbf{q}} \int \mathrm{d}\phi_{\mathbf{q}}\right) e^{-\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}(c_2 q^2 + c_1)|\phi_{\mathbf{q}}|^2},$$

where we neglected possible overall normalization constants, which may appear as the Jacobian of the change of variable $(*)$. At this point, it is useful to adopt a further simplification and imagine that we are working in a finite volume $V = L^d$, such that the integral over $\mathbf{q}$ is then replaced by a sum over the quantized momenta $\mathbf{q} = \frac{2\pi}{L}\mathbf{n}$, with $\mathbf{n} \in \mathbb{Z}^d$:

$$\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}(\cdots) \longrightarrow \frac{1}{V}\sum_{\mathbf{q}}(\cdots),$$

which in some way is equivalent to a sort of spatial cutoff. Making this substitution we arrive at

$$Z = \prod_{\mathbf{q}}{}' \left[\int \mathrm{d}\phi_{\mathbf{q}}\, e^{-\frac{1}{V}(c_2 q^2 + c_1)|\phi_{\mathbf{q}}|^2}\right]$$

Here, $\prod'_{\mathbf{q}}$ means that we are taking the product over $\mathbf{q} = (q_1, \ldots, q_d)$ such that $q_1 > 0$. Since the original real-space field is real-valued, its Fourier components are constrained by $\phi_{-\mathbf{q}} = \phi^*_{\mathbf{q}}$, which implies that $\phi_{\mathbf{q}}$ and $\phi_{-\mathbf{q}}$ are not independent degrees of freedom. To avoid double-counting when computing the partition function, the product must be restricted to half of the momentum space, which is denoted by the primed product $\prod'_{\mathbf{q}}$ (e.g., considering only $q_1 > 0$, choice which allows us to consider only positive frequencies). Furthermore, because the integrand depends only on q^2 and $|\phi_{\mathbf{q}}|^2$, it is entirely symmetric under the exchange $\mathbf{q} \to -\mathbf{q}$. Therefore, evaluating the product over the restricted half-space is mathematically equivalent to taking the square root of the product over the entire momentum space:

$$\prod_{\mathbf{q}}{}' [\ldots] = \sqrt{\prod_{\mathbf{q}} [\ldots]}.$$

So finally we can use these results to rewrite the partition function as

$$Z = \sqrt{\prod_{\mathbf{q}} \left[\int \mathrm{d}\phi_{\mathbf{q}}\, e^{-\frac{1}{V}(c_2 q^2 + c_1)|\phi_{\mathbf{q}}|^2}\right]},$$

which is now an easy product of decoupled gaussian integrals over complex variables. Using

$$\int_{\mathbb{C}} \mathrm{d}z\, e^{-|z|^2/a} = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \mathrm{d}x\mathrm{d}y\, e^{-(x^2+y^2)/a} = \pi a,$$

we arrive at the final expression for the partition function

$$Z = \prod_{\mathbf{q}} \sqrt{\frac{\pi V}{c_2 q^2 + c_1}}.$$

Therefore, neglecting irrelevant overall constants and using

$$\frac{1}{V}\log\left(\prod_{\mathbf{q}}(\ldots)\right) = \frac{1}{V}\sum_{\mathbf{q}}\log(\ldots) \simeq \int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\log(\ldots),$$

we can rewrite

$$\boxed{\frac{F_{LG} - F[\overline{m}]}{V} = -\frac{1}{2}\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\log\left(\frac{1}{c_2 q^2 + c_1}\right)},$$

which is the contribution to the free energy coming from the fluctuations around the saddle-point solution. Equivalently, we can write the total free energy as

$$F_{LG} = V\left(\frac{t}{2}\overline{m}^2 + u\overline{m}^4\right) + \frac{V}{2}\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\log\bigl(c_2 q^2 + c_1\bigr).$$

The contribution we found is non-trivial, and it modifies the critical behavior of the system with respect to the mean-field (or, more generally, the SPA) prediction. In particular, it can lead to a divergence of the heat capacity at the critical temperature T_c, which is a signature of a second-order phase transition, but we will see this more explicitly in the next section.

4.3.1 | Heat Capacity

We will now use the previous form of the free energy to compute the heat capacity close to the critical temperature T_c. Comparing the saddle-point prediction with the fluctuation contribution will allow us to identify the upper critical dimension for the Ising model: when $d > D_c$ is above the upper critical dimension, the saddle-point approximation is reliable and the critical exponents are given by the mean-field predictions. Instead, when $d < D_c$ is below the upper critical dimension, fluctuations become important and modify the critical exponents with respect to their mean-field values.

Remember that we defined the **heat capacity** by the second derivative of the free energy with respect to the temperature T:

$$C = -T\frac{\partial^2}{\partial T^2}F.$$

Recall we identified the parameter t in the LG free energy by

$$t = a(T - T_c) + \mathcal{O}((T - T_c)^2),$$

since we needed t to vanish at the critical temperature T_c and change sign across the transition. Therefore, we can express the heat capacity in terms of derivatives with respect to t, which at the leading order, relates to derivatives with respect to T as

$$\frac{\partial}{\partial T} = a\frac{\partial}{\partial t}.$$

We can thus rewrite the heat capacity as

$$C \simeq Ta^2\frac{\partial^2}{\partial t^2}\left[\frac{1}{\beta}\log Z\right] \simeq T_c a^2\frac{\partial^2}{\partial t^2}\left[T_c \log Z(t)\right].$$

The idea is now to compute the heat capacity using the partition function obtained by including the fluctuations, and compare it with the saddle-point prediction. This will allow us to understand whether the saddle-point approximation is reliable, and whether the critical exponents obtained from it are correct.

- First, let us recall the **saddle-point approximation** for Z_{LG} at $h = 0$, given by

$$\overline{m} = \begin{cases} 0 & t > 0, \\ \sqrt{\dfrac{-t}{4u}} & t < 0, \end{cases}$$

which was derived in the previous section. Substituting this solution into the LG free energy,

$$\log Z_{LG} \simeq -V\left(\frac{t}{2}\overline{m}^2 + u\overline{m}^4\right) = \begin{cases} 0 & \text{for } t > 0, \\ +V\dfrac{t^2}{16u} & \text{for } t < 0, \end{cases}$$

$$\Rightarrow \quad C = \begin{cases} C_0 & \text{for } t > 0, \\ C_0 + Va^2T_c^2\dfrac{1}{8u} & \text{for } t < 0. \end{cases}$$

Here C_0 is a constant which comes from the regular part of the partition function (namely, $\log Z = \log Z_0 + \log Z_{LG}$, where $\log Z_0$ contains all the m-independent terms that we neglected in the LG free energy, since they do not contribute to the critical behavior).

Therefore, the saddle-point (or mean-field) prediction is

`img/fluctuations_C.png`

Figure 4.5: Plot of the heat capacity in absence of external field $C(h=0)$ versus reduced temperature t, showing a step discontinuity at $t=0$ according to the mean-field approximation.

Recall that the α-exponent is defined by

$$C_\pm(T, h=0) \sim A_\pm |t|^{-\alpha},$$

since it describes the divergence of the heat capacity at the critical temperature T_c. The saddle-point approximation predicts $\alpha = 0$: we do not have an expression depending on t in the heat capacity, but just a step discontinuity at $t=0$.

- Next, let us compute how the prediction is modified using the partition function obtained by including the **fluctuations** up to second order, where again we now include the regular part Z_0 of the partition function, the saddle-point contribution, and the fluctuation contribution:

$$-\log Z = -\log Z_0 + V\left(\frac{t}{2}\overline{m}^2 + u\overline{m}^4\right) + \frac{V}{2}\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\log\bigl(c_2 q^2 + c_1\bigr),$$

where we had

$$c_2 = \frac{\kappa}{2} \quad \text{and} \quad c_1 = \frac{t + 12u\overline{m}^2}{2} = \begin{cases} \dfrac{t}{2} & \text{for } t > 0, \\ -t & \text{for } t < 0, \end{cases}$$

recalling the SPA solution for $\overline{m}(t)$. Therefore, the new contribution to the heat capacity is given by the second derivative with respect to t of the fluctuation contribution to the free energy:

$$\frac{V}{2}\frac{\partial^2}{\partial t^2}\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\log\bigl(c_2 q^2 + c_1\bigr) = \begin{cases} \dfrac{V}{2}\displaystyle\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\frac{1}{(\kappa q^2 + t)^2} & \text{for } t > 0, \\ 2V\displaystyle\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\frac{1}{(\kappa q^2 - 2t)^2} & \text{for } t < 0. \end{cases}$$

If we now (after some algebra and dimensional analysis) introduce the following length scale

$$\xi^{-2} = \begin{cases} \dfrac{t}{\kappa} & \text{for } t > 0, \\ -\dfrac{2t}{\kappa} & \text{for } t < 0, \end{cases}$$

we can rewrite the previous expression as

$$\frac{V}{2}\frac{\partial^2}{\partial t^2}\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\log\bigl(c_2 q^2 + c_1\bigr) \sim \frac{1}{\kappa^2}\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\frac{1}{(q^2+\xi^{-2})^2}$$

where we did not keep track of the numerical prefactors. We now have to explicit some considerations from dimensional analysis:

- The integral now has dimension $\sim L^{4-d}$. In fact, the integration measure $\mathrm{d}^d\mathbf{q}$ has dimension $\sim L^{-d}$, while the integrand has dimension $\sim L^4$ (since q^2 and ξ^{-2} have dimension $\sim L^{-2}$).

- For $d \geq 4$ the integral diverges at large q, and is dominated by the momentum cutoff Λ: we would have an integrand behaving as q^{-4} at large q, and thus the integral would diverge as Λ^{d-4}.

- For $d < 4$ the integral is convergent. It can be made dimensionless by rescaling $q \mapsto q' = q\xi^{-1}$, since we saw that in the momentum space $[q_i] \sim L^{-1}$ (keeping in mind the example of spherical coordinates, where $\mathrm{d}\mathbf{q} = q^{d-1}\,\mathrm{d}q\,\mathrm{d}\Omega$). After the rescaling, the integral becomes

  $$\xi^{4-d}\int \frac{\mathrm{d}^d\mathbf{q}'}{(2\pi)^d}\frac{1}{(q'^2+1)^2},$$

  which is now dimensionless and convergent. Therefore, the original integral behaves asymptotically as ξ^{4-d} for $d < 4$.

Therefore we have:

$$\frac{\partial^2}{\partial t^2}\frac{V}{2}\int \frac{\mathrm{d}^d\mathbf{q}}{(2\pi)^d}\log\bigl(c_2 q^2 + c_1\bigr) \sim \frac{1}{\kappa^2}\begin{cases} \Lambda^{d-4} & \text{for } d \geq 4, \\ \xi^{4-d} & \text{for } d < 4, \end{cases}$$

which is the asymptotic behavior of the fluctuation contribution to the heat capacity (when $d = 4$, the divergence in Λ is logarithmic).

We obtained that in $d \geq 4$ fluctuation corrections to the heat capacity add a constant term to the heat capacity for both $t > 0$ and $t < 0$. Instead, for $d < 4$, the divergence $\xi \sim t^{-1/2}$ at the transition leads to a divergence of the heat capacity with a critical exponent

$$\alpha = \frac{4-d}{2}$$

img/fluctuations_C_comparison.jpg

Figure 4.6: Two side-by-side plots showing the heat capacity $C(h=0)$ versus reduced temperature t. The left plot illustrates the case $d \geq 4$ with a constant fluctuation correction, while the right plot illustrates the case $d < 4$ where the heat capacity diverges at $t = 0$.

The exponent $\alpha = \frac{4-d}{2}$ is not exact because it only takes into account quadratic fluctuations. However, the fact that C diverges for $d < 4$ means that the saddle-point conclusions are not reliable for $d < 4$, identifying $d = 4 = D_c$ as the upper critical dimension.

4.3.2 | Correlation Functions

The same machinery can be used to compute correlation functions and in particular the connected correlator

$$G^c(\mathbf{x}, \mathbf{y}) = \langle (m(\mathbf{x}) - \overline{m})(m(\mathbf{y}) - \overline{m}) \rangle = \langle \phi(\mathbf{x})\phi(\mathbf{y}) \rangle,$$

where $\langle \dots \rangle$ denotes the ensemble average. The computation is not hard but we will not see it explicitly, giving only the final result, which reads

$$G^c(\mathbf{x}, \mathbf{y}) = -\frac{1}{\kappa} I_d(\mathbf{x} - \mathbf{y}),$$

where

$$I_d(\mathbf{x}) = -\int \frac{\mathrm{d}^d \mathbf{q}}{(2\pi)^d} \frac{e^{i\mathbf{q}\cdot\mathbf{x}}}{q^2 + \xi^{-2}},$$

and ξ is the length scale introduce above

$$\xi^{-2} = \begin{cases} \dfrac{t}{\kappa} & \text{for } t > 0, \\ -\dfrac{2t}{\kappa} & \text{for } t < 0. \end{cases}$$

One can study this function and infer the following behavior

$$I_d(\mathbf{x}) \sim \begin{cases} \dfrac{1}{|\mathbf{x}|^{d-2}} & |\mathbf{x}| \ll \xi, \\ \dfrac{e^{-|\mathbf{x}|/\xi}}{|\mathbf{x}|} & |\mathbf{x}| \gg \xi. \end{cases}$$

For $t \neq 0$, ξ is naturally interpreted as a spatial correlation length. When $t \to 0$, we have consequently $\xi \to \infty$, such that the connected correlators decay as a power law.

4.3.3 | The Ginzburg Criterion

Before moving on, we complete our discussion of LG theory with the Ginzburg criterion. Recall our predictions for the heat capacity using the quadratic corrections to the saddle point for $d < 4$

img/fluctuations_ginzburg_crit_C.png

Figure 4.7: Plot of the heat capacity $C(h = 0)$ versus reduced temperature t, showing the mean-field step ΔC_{SP} and the fluctuation correction C_{corr} diverging at $t = 0$.

Denote by C_{corr} the difference between the calculated value and the SP predictions. Further, denote by ΔC_{SP} the value of "the jump" of the heat capacity predicted by the saddle-point approximation. We can estimate the importance of fluctuations by comparing C_{corr} with ΔC_{SP}, with the idea to find a regime where fluctuations are negligible and the saddle-point approximation is reliable.

Fluctuations are important if their contribution to the heat capacity is larger than the jump predicted by the saddle-point approximation, which alternatively reads $C_{corr} \gg \Delta C_{SP}$; we thus expect to obtain correct results from the saddle-point approximation when

$$C_{corr} \ll \Delta C_{SP}.$$

Recalling that, for $d \leq 4$, we have found a contribution to the heat capacity from fluctuations which behaves as

$$C_{corr} \sim \frac{1}{\kappa}\xi^{4-d} \sim \frac{1}{\kappa}|t|^{-\frac{4-d}{2}},$$

where we used the definition of the length scale ξ introduced above (without considering numerical factors depending on the sign of t). On the other hand, the jump predicted by the saddle-point approximation, as we computed, is proportional to $\frac{1}{u}$.

Therefore, the condition $C_{corr} \ll \Delta C_{SP}$ can be rewritten as

$$\implies \qquad |t|^{\frac{d-4}{2}} \ll \frac{\kappa}{u} \qquad \Rightarrow \qquad |t| \gg t_G \simeq \left(\frac{u}{\kappa}\right)^{\frac{2}{4-d}}.$$

This is Ginzburg's criterion. We see that, as we expected, for $|t|$ sufficiently low the fluctuations are important. However, it gives a temperature t_G above which fluctuations become less important, thus the saddle-point approximation becomes reliable in that regime. In some experiments we may not get below t_G, so we may see a critical behaviour which is similar to the mean-field result. This is just an artifact due to the fact that we are not sufficiently close to T_c, and thus we are not able to see the true critical behavior of the system.

4.4 | The Scaling Hypothesis

Discussing the realations between critical exponents is a foundamental step in the study and comprehension of critical phenomena. The scaling hypothesis is a powerful tool that allows us to derive these relations and understand the behavior of physical quantities near the critical point.

We can start by considering the free energy per unit volume $f(t,h) = F(t,h)/V$ as computed via the saddle point approximation in the Landau-Ginzburg theory (), which is given by

$$f(t,h) = \min_m \left[\frac{t}{2}m^2 + um^4 - hm\right] = \begin{cases} -\dfrac{1}{16}\dfrac{t^2}{u} & (h=0,\, t<0), \\ -\dfrac{3}{4^{4/3}}\dfrac{h^{4/3}}{u^{1/3}} & (h \neq 0,\, t=0), \end{cases}$$

where these singularities of the free energy density f[3] at the critical point are consistent with the assumption that the free energy density f is a **generalized homogeneous function** of the reduced temperature $|t|$ and the external field h when we study it close to the critical point T_c (as we will soon show). This is a non-trivial assumption, which is not satisfied by the mean field theory, but it is satisfied by the true free energy density f of the system, which can be very different from the one predicted by the mean field theory. The scaling hypothesis is based on this homogeneity assumption, which allows us to derive scaling relations between critical exponents.

Let us start by understanding the mathematical definition of a generalized homogeneous function.

Definition 4.1 (Homogeneous function)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is called a homogeneous function of degree k if it satisfies the following scaling property:

$$f(\lambda x_1, \lambda x_2, \ldots, \lambda x_n) = \lambda^k f(x_1, x_2, \ldots, x_n), \quad \forall \lambda > 0.$$

This means that if we scale all the input variables by a factor of λ, the output of the function scales by a factor of λ^k. The degree k can be any real number, and it characterizes how the function responds to scaling transformations.

We can easily verify that the free energy density f computed via mean field theory is indeed a homogeneous function: for $h = 0$ we have

$$f(\lambda t, 0) = -\frac{1}{16}\frac{(\lambda t)^2}{u} = \lambda^2 f(t, 0),$$

and a similar calculation can be performed for $t = 0$ and $h \neq 0$.

Definition 4.2 (Generalized homogeneous function)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is called a generalized homogeneous function if there exist real numbers $d_1, d_2, \ldots, d_n \in \mathbb{R}$ such that it satisfies the following scaling property:

$$f(\lambda^{d_1} x_1, \lambda^{d_2} x_2, \ldots, \lambda^{d_n} x_n) = \lambda f(x_1, x_2, \ldots, x_n), \quad \forall \lambda > 0.$$

In this case, the function scales linearly with respect to the scaling of its arguments, but each argument can scale with a different exponent d_i. The exponents d_i characterize how each variable contributes to the overall scaling behavior of the function.

[3]There is a jump discontinuity (at $t > 0$ we have that $f = 0$), and also higher order derivatives of f with respect to h (such as the magnetic susceptibility) or t are discontinuous at $t = 0$.

Now taking a generalized homogeneous function of two variables $f(x, y)$ such that:

$$f(\lambda^{d_1} x, \lambda^{d_2} y) = \lambda f(x, y),$$

if we set

$$\lambda = \left(\frac{1}{x}\right)^{\frac{1}{d_1}},$$

we can notice how the function f can be rewritten as

$$f(1, \frac{1}{x^{d_2/d_1}} y) = \left(\frac{1}{x}\right)^{\frac{1}{d_1}} f(x, y).$$

Now if we define $g(z) = f(1, z)$, we can invert the relation above to obtain

$$f(x, y) = x^{d_1} g\left(\frac{y}{x^{d_2/d_1}}\right),$$

which therefore shows how the generalized homogeneous function f can be expressed in terms of a single variable, which is the ratio $\frac{y}{x^\Delta}$ (where $\Delta = \frac{d_2}{d_1}$) of the two original variables, up to an overall prefactor x^{d_1}. This is the essence of the scaling hypothesis, which states that near the critical point, physical quantities can be expressed as homogeneous functions of the relevant variables, leading to scaling relations between critical exponents; we will see this more explicitly in the following.

If we assume that the true free energy density f is a generalized homogeneous function of the reduced temperature $|t|$ and the external field h when we study it close to the critical point T_c, then it must satisfy the following scaling property:

$$f(|t|, h) = |t|^{d_1} g\left(\frac{h}{|t|^\Delta}\right),$$

for some d_1, Δ and g (notice how we are now treating d_1 and Δ as independent parameters). In general, the true free energy density f of the system can be very different from the one predicted by the mean field theory.

> **Remark.** *It has to be clarified that we are considering only the singular part of the free energy density f in the homogeneity assumption, since the regular part of f does not contribute to the singularities of physical quantities at the critical.*

Knowing the singularities of the free energy density $f(t, h)$, we can fix the values of d_1 and Δ and therefore derive the scaling relations between critical exponents. Thus we can start from the singularities of the free energy density f as predicted by the saddle point approximation in the Landau-Ginzburg theory, and see how they are consistent with the homogeneity assumption that we made on the free energy density f.

- For $(h = 0,\, t \neq 0)$, we must have

  $$f(t, 0) \sim \frac{|t|^2}{u}, \qquad \Longrightarrow \qquad d_1 = 2,$$

  where we used the limit

  $$\lim_{x \to 0} g(x) \sim \frac{1}{u}.$$

- Instead, for $(h \neq 0,\, t \to 0)$, we must have

  $$\lim_{t \to 0} f(t, h) \sim \frac{h^{4/3}}{u^{1/3}},$$

which can be verified by setting

$$\lim_{x\to\infty} g(x) \sim \frac{x^{4/3}}{u^{1/3}}.$$

Now, using the last relation, we can fix the value of Δ as follows:

$$\lim_{t\to 0} f(t,h) \sim |t|^2 \frac{h^{4/3}}{u^{1/3}\, t^{4\Delta/3}} \qquad \Longrightarrow \qquad \Delta = \frac{3}{2}.$$

We can conclude that the homogeneity assumption in the singular part of the free energy density f is consistent with the singularities predicted by the saddle point approximation, with

$$d_1 = 2, \quad \Delta = \frac{3}{2}.$$

We can now state the **scaling hypothesis** as follows: *going beyond the saddle point approximation, the singular part of the free energy density f remains a generalized homogeneous function of the reduced temperature $|t|$ and the external field h when we study it close to the critical point T_c.* This is a non-trivial assumption, which is not satisfied by the mean field theory, but it is satisfied by the true free energy density f of the system, which can be very different from the one predicted by the mean field theory.

4.4.1 | Scaling Relations

Assuming thus that the singular part of the free energy density f is a generalized homogeneous function of the reduced temperature $|t|$ and the external field h when we study it close to the critical point T_c, we can again write

$$f(|t|,h) = |t|^d g\left(\frac{h}{|t|^\Delta}\right),$$

for some d, Δ and g. This does not imply the free energy of the mean field theory, which is a particular approximation of the true free energy (which can even have this form in general, but with different exponents), but it is a more general assumption that we make on the true free energy itself. Let us now find a way to relate the exponents d and Δ to the critical exponents.

Critical Exponents: α

We start by relating d with the critical exponent α that describes the singularity of the heat capacity C at the critical point. We know that

$$C = -T\frac{\partial^2 f}{\partial T^2},$$

so that we have to compute the second derivative of the free energy density f with respect to the temperature T. Computing the first derivative of f with respect to T:

$$\begin{aligned}
\frac{\partial f}{\partial T} &\sim d|t|^{d-1} g\left(\frac{h}{|t|^\Delta}\right) - \Delta |t|^{d-\Delta-1} h g'\left(\frac{h}{|t|^\Delta}\right) \\
&= |t|^{d-1}\left[d\, g\left(\frac{h}{|t|^\Delta}\right) - \Delta \frac{h}{|t|^\Delta} g'\left(\frac{h}{|t|^\Delta}\right)\right] \\
&=: |t|^{1-\alpha}\tilde{g}\left(\frac{h}{|t|^\Delta}\right),
\end{aligned}$$

since in the previous computation h appeared always divided by $|t|^\Delta$, in order to let us define the function $\tilde{g}$.

Remark. *More generally, partial derivative of an generalized homogeneous function will again be an generalized homogeneous function, with different prefactors and a different functional form $\tilde{g}$ (which is related to the original function g by differentiation).*

Repeating the same procedure to compute the second derivative of f with respect to T, we obtain

$$C \propto -\frac{\partial^2}{\partial t^2} f \sim |t|^{-\alpha} \tilde{\tilde{g}} \left(\frac{h}{|t|^{\Delta}} \right),$$

where $\tilde{\tilde{g}}$ is another appropritely defined function.

Now, since we need the critical exponent α to describe the singularity of the heat capacity C at the critical point in absence of external field h, we need to have

$$C(t, h = 0) \xrightarrow{t \to 0} |t|^{-\alpha} \tilde{\tilde{g}}(0) = \text{const} \cdot |t|^{-\alpha}.$$

Therefore, we can conclude that the critical exponent α is related to the exponent d by the following relation:

$$\lim_{x \to 0} \tilde{\tilde{g}}(x) = \text{const} \implies \alpha = 2 - d,$$

and thus we found the relation between the critical exponent α and the exponent d that appears in the homogeneity assumption of the free energy density f: $d = 2 - \alpha$. The logic behind this is that we need to now how this function behaves at the critical point, where it could go to 0, diverge to infinity or go to a constant; since

$$\lim_{t \to 0} C(t, h = 0) \sim |t|^{-\alpha},$$

then we need to have $\lim_{x \to 0} \tilde{\tilde{g}}(x) = \text{const}$ in order to have the correct singularity of the heat capacity C at the critical point.

Critical Exponents: β

Now the only unknown parameter is Δ, which we call the **gap exponent**: it is related to the critical exponent β that describes the singularity of the order parameter m at the critical point. We are saying that the theory in which we compute the critical exponent β is not important (mean field, Landau-Ginzburg, etc), but it is important that the true free energy density f has the homogeneity property that we assumed above, with some value of Δ. Thus let us relate Δ to β by computing the order parameter m as a function of the reduced temperature t at zero external field $h = 0$: remember the definition of the critical exponent β is

$$m(t, h = 0) \sim |t|^{\beta},$$

then we can compute m by taking the derivative of the free energy density f with respect to the external field h:

$$m = -\frac{\partial f}{\partial h} \sim |t|^{d-\Delta} g' \left(\frac{h}{|t|^{\Delta}} \right),$$

and therefore we can conclude that the critical exponent β is related to the gap exponent Δ by the following relation:

$$\lim_{x \to 0} g'(x) = \text{const} \implies \beta = d - \Delta,$$

which is the second scaling relation that we were looking for

$$\beta = d - \Delta = 2 - \alpha - \Delta.$$

Critical Exponents: δ

Now we can consider the critical exponent δ, which describes the singularity of the order parameter m at the critical point as a function of the external field h:

$$m(t=0,h) \sim h^{1/\delta},$$

then we can assume that the function g behaves as

$$\lim_{x\to\infty} g(x) \sim x^{p+1}, \quad \Longrightarrow \quad \lim_{x\to\infty} g'(x) \sim x^{p},$$

which is the only reasonable behavior that we can assume for the function g in order to have a power law singularity of the order parameter m at the critical point. Therefore, we can compute the order parameter m at the critical point $t=0$ as a function of the external field h:

$$\lim_{t\to 0} m(t,h) \sim |t|^{2-\alpha-\Delta} \left(\frac{h}{|t|^{\Delta}}\right)^{p},$$

which is well-defined only if $p\Delta = 2-\alpha-\Delta$ leading us to

$$\lim_{t\to 0} m(t,h) \sim h^{\frac{2-\alpha-\Delta}{\Delta}},$$

which recalling the definition of the critical exponent δ leads us to the following scaling relation:

$$\delta = \frac{\Delta}{2-\alpha-\Delta} = \frac{\Delta}{\beta}.$$

Critical Exponents: γ

For the susceptibility χ, we can use the same arguments and procedures, computing it as the derivative of the magnetization m with respect to the external field h:

$$\chi(t,h) \propto \frac{\partial m}{\partial h} = -\frac{\partial^2 f}{\partial h^2} \sim |t|^{2-\alpha-2\Delta} g''\left(\frac{h}{|t|^{\Delta}}\right),$$

which, after making the usual assumptions on the behavior of the function g as before, leads us to

$$\chi(t,h=0) \sim |t|^{2-\alpha-2\Delta}, \quad \chi(t=0,h) \sim |t|^{-\gamma},$$

obtaining, from the previous definition of the critical exponent γ, the following scaling relation:

$$\gamma = 2\Delta + \alpha - 2.$$

Scaling Identities

After these computations, we have found the following scaling relations between critical exponents:

$$\begin{cases} \alpha = 2-d, \\ \beta = d-\Delta = 2-\alpha-\Delta, \\ \delta = \dfrac{\Delta}{\beta} = \dfrac{\Delta}{2-\alpha-\Delta}, \\ \gamma = 2\Delta+\alpha-2, \end{cases}$$

showing how the gap exponent Δ appear in all the expressions for the critical exponents: it is a fundamental exponent that characterizes the scaling behavior of the free energy density f and it is related to all the critical exponents.

Since we have only two unknown parameters d and Δ, we can express all the critical exponents in terms of these two parameters, and therefore we can derive scaling identities between critical exponents by eliminating d and Δ from the previous relations.

For example, from the scaling relations of β, δ and γ, we can derive the following scaling identities:

$$\begin{cases} \alpha + 2\beta + \gamma = 2, & \textbf{Rushbrooke's identity,} \\ \delta - 1 = \dfrac{\gamma}{\beta}, & \textbf{Widom's identity.} \end{cases}$$

Now, the first identity is simply obtained by summing the expressions of α, 2β and γ; the second one, by direct computation we have:

$$\delta - 1 = \frac{\Delta}{\beta} - 1 = \frac{\Delta - 2 + \alpha + \Delta}{\beta},$$

and by recalling the expression of γ we can rewrite the previous expression as

$$\delta - 1 = \frac{2\Delta + \alpha - 2}{\beta} = \frac{\gamma}{\beta}.$$

These relations can be tested experimentally and numerically, and they are indeed satisfied with good precision, despite the fact that the individual exponents are different than the one predicted by the mean field theory.

α	β	γ	δ	ν	η
0.11	0.32	1.24	4.9	0.63	0.04

Table 4.1: Critical exponents for the three-dimensional Ising model, which is a paradigmatic example of an uniaxial ferromagnet. These exponents are different from the ones predicted by the mean field theory, but they satisfy the scaling identities with precision.

So it seems that even in systems with very different exponents, the scaling relations are still satisfied, which is a strong evidence in favor of the scaling hypothesis and the homogeneity assumption that we made on the true free energy density f. Historically, these identities and relations among critical exponents were derived experimentally and numerically before the scaling hypothesis was formulated, which in the end explains their universal validity.

4.4.2 | Hyperscaling Hypothesis

The homogeneity assumption do not apply only to the free energy density f, but also to other physical quantities derived from it. But it says nothing about the correlation functions. In order to obtain identities related to exponents of the correlation functions, we need to make two additional assumptions which will seem to be more foundamental and more general than the homogeneity assumption on the free energy density f: the **hyperscaling hypothesis**.

- The correlation length ξ is a generalized homogeneous function of the reduced temperature $|t|$ and the external field h when we study it close to the critical point T_c:

$$\xi(t, h) = |t|^{-\nu} g\left(\frac{h}{|t|^{\Delta}}\right).$$

- Close to criticality, ξ is the most important lenght scale in the system and is solely responsible for the singular contributions to thermodynamic quantities.

The idea is now to show euristically how to use these assumptions to derive the scaling form of the free energy density f and therefore the scaling relations between critical exponents.

`img/hyperscaling.png`

Figure 4.8: The system can be divided into blocks of linear size ξ and volume ξ^d, so that we have $N_b = V/\xi^d$ uncorrelated subsystems in total.

We can start by considering a system of linear size L and volume $V = L^d$, where d is the spatial dimension of the system. We can divide the system into blocks of linear size ξ and volume ξ^d, so that we have $N_b = V/\xi^d$ blocks in total.
Now, since we are assuming that ξ is the most important length scale in the system, we can assume that each block contributes independently to the free energy density f, so that points from different blocks can be considered uncorrelated.

Since we are interested in the computation of the free energy density f, we can start from the **partition function** Z of the system: up to a first approximation (if we are interested only in the diverging behavior of Z), we can write the partition function Z as a sum of local partition functions

$$\log Z \sim \left(\frac{L}{\xi}\right)^d g_s,$$

where g_s is the single region contribution to the free energy, while the prefactor $(L/\xi)^d$ counts the number of uncorrelated blocks in the system. Furthermore, since Z is dimensionless, we can assume that g_s is a dimensionless function.

Finally, since by the second assumption of the hyperscaling hypothesis we have that ξ is the only relevant length scale in the system, then it follows that any singular behavior of $\log Z$ must be due to the divergence of ξ at the critical point, so that g_s must be regular at the critical point, and therefore we can write

$$f(t,h) \sim \frac{1}{L^d}\log Z \sim \xi^{-d} \sim |t|^{\nu d} g\left(\frac{h}{|t|^{\Delta}}\right),$$

where the second equality is obtained close to criticality, while the last equality is obtained by using the homogeneity assumption on the correlation length ξ from the hyperscaling hypothesis.

Therefore, we have shown how to derive the scaling form of the free energy density f from the hyperscaling hypothesis, which is a more general and more fundamental assumption than the homogeneity assumption on the free energy density f itself.

Hyperscaling Identities

In addition, we get a new scaling relation between the critical exponent α and the correlation length exponent ν: by equating the previous expression of the free energy density f with the one obtained from the homogeneity assumption, we get

$$2 - \alpha = d\nu, \quad \textbf{Josephson's identity.}$$

Identities like the one above are called **hyperscaling identities**, since they involve the spatial dimension d of the system, and they are a direct consequence of the hyperscaling hypothesis. These identities are satisfied in low dimensions, but they are violated in high dimensions, where the mean field theory becomes exact and the critical exponents take their mean field values. Indeed, the previous relation among α and ν is satisfied by the real exponents in two or three dimensions (as one can see from the Table 4.1), however it does not agree with the saddle point (mean field) solution for $d > 4$: in this case, the mean field theory becomes exact and the critical exponents take their mean field values, which do not satisfy the previous relation (indeed we would have $\alpha = 0$ and $\nu = 1/2$).

This is a strong evidence that the hyperscaling hypothesis is not valid in high dimensions, where the mean field theory becomes exact, while it is valid in low dimensions, where the critical exponents take non-mean field values. Any theory that aims to explain the critical behavior of systems must therefore satisfy the hyperscaling hypothesis in low dimensions, while taking into account its violation in high dimensions. This is one of the main challenges in the study of critical phenomena, and it is one of the reasons why the renormalization group theory is so important, since it provides a framework to understand the validity of the hyperscaling hypothesis and its violation in high dimensions.

Before embarking in the study of the renormalization group theory, we will first discuss the last critical exponent η that describes the singularity of the correlation function at the critical point.

Correlation Functions and η

The critical exponent η describes the singularity of the correlation function $G(\mathbf{x})$ at the critical point; we already derived the connected correlation function $G^c(\mathbf{x})$ keeping the second order corrections to the saddle point approximation in the Landau-Ginzburg theory. We found that at the critical point $t = 0$ the correlation lenght diverges $\xi \to \infty$, obtaining

$$G^c(\mathbf{x}) = \langle \phi(\mathbf{x})\phi(\mathbf{0})\rangle \sim \frac{1}{|\mathbf{x}|^{d-2}}.$$

We know this behavior to be correct in high dimensions ($d \geq 4$ for ising, $d \geq D_c$ in general), where the mean field theory becomes exact, but it is not correct in low dimensions, where the critical exponents take non-mean field values.

However, we know that the correlation function $G(\mathbf{x})$ must still vanish at large distances $|\mathbf{x}| \to \infty$, since the correlation leght diverges as we said; this vanishing must be a power law, involving some finite lenght scale ζ that is different from the correlation length ξ (which diverges at the critical point), so that we can write

$$\lim_{|\mathbf{x}|\to\infty} G(\mathbf{x}) \sim e^{-\frac{|\mathbf{x}|}{\zeta}},$$

but this lenght scale is finite at the critical point, so that it is different from ξ; but we can account for this vanishing by introducing a new critical exponent η that describes the singularity of the correlation function at the critical point as follows:

$$G(\mathbf{x}) \sim \frac{1}{|\mathbf{x}|^{d-2+\eta}}.$$

This is the definition of the critical exponent η, which describes how the correlation function vanishes at large distances at the critical point, and it is called the **anomalous dimension**.[4]

[4] The critical exponent η is related to the anomalous dimension of the field ϕ in the renormalization group theory, and it is a measure of how much the correlation function deviates from the mean field behavior at the critical point.

Now, even if we did not see this explicitly, it is easy to relate χ the susceptibility to the connected correlation functions $G^c(\mathbf{k})$ as follows:

$$\chi \sim \int \mathrm{d}^d\mathbf{x}\, G^c(\mathbf{x}),$$

so that in the end we find that η can be related to the exponent appearing in the singularity of the susceptibility χ at the critical point, γ, since

$$\chi \sim |t|^{-\gamma}, \quad \text{as } t \to 0.$$

Notice how the expression connecting the susceptibility to the correlation functions is valid only up to distances $\sim \xi$, since for distances larger than ξ the correlation function $G^c(\mathbf{x})$ vanishes exponentially.

We can now plug the previous expression of the correlation function $G^c(\mathbf{x})$ at the critical point into the expression of the susceptibility χ to obtain[5]

$$\chi \sim \int_0^\xi \mathrm{d}^d\mathbf{x}\, \frac{1}{|\mathbf{x}|^{d-2+\eta}} \sim \xi^{2-\eta} \sim |t|^{-\nu(2-\eta)},$$

where we have used the critical behavior of the correlation length ξ at the critical point, which is given by

$$\xi \sim |t|^{-\nu}.$$

Now, recalling the critical behavior of the susceptibility, we are lead to the following hyperscaling relation between the critical exponents γ, ν and η:

$$\gamma = (2-\eta)\nu, \quad \textbf{Fisher's identity.}$$

This is another hyperscaling identity, since it involves the critical exponent ν that describes the divergence of the correlation length ξ at the critical point, and it is a direct consequence of the hyperscaling hypothesis. Again, this identity is satisfied in low dimensions, but it is violated in high dimensions, where the mean field theory becomes exact and the critical exponents take their mean field values.

Remark. *The scaling hypothesis is a general assumption that can be applied to a wide range of systems and phenomena, and gives us predictions (such as the scaling relations between critical exponents) that can be tested experimentally and numerically to be true for any value of the spatial dimension d.*

The hyperscaling hypothesis is a more specific assumption that applies to systems near criticality, and it allows us to derive additional scaling relations which we used to derive the hyperscaling identities for the exponents ν and η. But this assumption gives correct predictions only in low dimensions, while it is violated in high dimensions, where the mean field theory becomes exact and the critical exponents take their mean field values.

Scale Invariance

Until now, we have made more of a phenomenological analysis of the critical behavior of systems, by making assumptions on the form of the free energy density f and the correlation length ξ

[5] Here, we have assumed that the connected correlation function, for values of $|\mathbf{x}| \gg \xi$, is exponentially suppressed, so that we can neglect its contribution in that regime.

close to the critical point, and by deriving scaling relations between critical exponents from these assumptions. However, in the next chapter, we are going to introduce the renormalization group theory, which will provide us with a more fundamental and more microscopic understanding of the critical behavior of systems, and it will allow us to derive the scaling hypothesis and the hyperscaling hypothesis from first principles. So let us make some observations about the scale invariance of systems at criticality, which will be a key concept in the renormalization group theory.

A consequence of the previous scaling assumptions is that, at criticality, the system acquire an additional symmetry: the **dilatation symmetry**. Namely, under a change of scale, the critical correlation functions transform as

$$G(\lambda \mathbf{x}) = \lambda^p G(\mathbf{x}),$$

where p is some exponent that characterizes the scaling behavior of the correlation function under dilatations. This means that at criticality, the system is invariant under scale transformations, thus we call it **scale invariant** or *self-similar*. This is a very important property of systems at criticality, and it is one of the key ingredients in the renormalization group theory, which will allow us to understand the critical behavior of systems from a more fundamental and more microscopic perspective.

This comment becomes really important when we are studying two dimensional systems, where the scale invariance at criticality is sufficient to directly enforce **conformal invariance** as we will see in the second part of the course, which is a much stronger symmetry that allows us to derive many exact results for two dimensional systems at criticality, such as the exact values of critical exponents and the exact form of correlation functions.

Instead, we will conclude this first module by introducing the renormalization group theory, which will provide us with a more fundamental understanding of the critical behavior of systems, and a unified point of view on the many facets of universal behavior of systems at criticality, providing at the and a justification of the scaling hypothesis and the hyperscaling hypothesis that we have introduced in this chapter.

5 | Renormalization Group

The renormalization group (RG) is a powerful theoretical framework that allows us to understand the behavior of physical systems at different scales, and in particular to analyze criticality and phase transitions. It provides a unified perspective on the many facets of critical phenomena.

The name "renormalization group" is rather misleading: it is a framework in which one applies transformations on a system, but such transformations do not form a group in the mathematical sense. However, the name has been kept for historical reasons, since it was first introduced in the context of quantum field theory, where it was used to understand the behavior of quantum fields at different energy scales; now it is used to refer to a more general framework that can be applied to a wide range of systems, and it is associated to a vast set of ideas and techniques that are used to understand critical phenomena in statistical mechanics.

TODO: Add reference to the John Cardy book, chapter 3.

The renormalization group is based upon the following observation: *critical phenomena are dominated by fluctuations at a scale ξ which diverges at the critical point; the system thus becomes effectively scale-invariant at criticality.* The idea is to iteratively modify our description of the system, eliminating details about the short range correlations at each step, and thus to understand how the system behaves at different scales. In this way, one is eventually left with a simplified description of the system that captures the essential features of the critical behavior through correlations at large scales.

This iterative procedure is called a **coarse-graining transformation**, and it allows us to understand how the parameters of the system change as we look at it at different scales. It is indeed equivalent to a flow in the space of parameters (space of all possible free energies or actions), which is called the **action space**, and the **fixed points** of this flow correspond to critical points of phase transitions. The RG flow allows us to understand the behavior of the system near criticality, and in particular to identify *relevant and irrelevant directions* in the parameter space, associated with the critical exponents that describe the behavior of physical quantities at criticality.

The conceptual framework derived from the RG analysis is very general, and can be implemented in different forms and with different techniques; traditionally, the main distinction is made between **real space RG** and **momentum space RG**: the first one is very useful to introduce the framework, as it involves a more intuitive approach; the latter, instead, provides more powerful tools for quantitative computations (for istance, encoding a systematic approach to derive critical exponents below the upper critical dimension).

However, as we said, real space RG is more intuitive and thus we will start by using this approach to revisit the 1D Ising model, which we have already solved, in order to understand how the RG works in a simple case, and then we will introduce the more general framework that can be applied to more complicated systems.

5.1 | Coarse-Graining in Real Space

Consider again the 1D Ising model, with Hamiltonian

$$H = -J \sum_i s_i s_{i+1} - \tilde{h} \sum_i s_i - \tilde{g} N,$$

were J is the coupling constant, $\tilde{h}$ is the external field and we added a constant term $-\tilde{g}N$ to the Hamiltonian: it does not change the physics of the system, but it will be useful for the RG analysis.

In order to write the partition function, we can introduce the following coefficients:

$$\begin{cases} k = \beta J, \\ h = \beta \tilde{h}, \\ g = \beta \tilde{g}, \end{cases}$$

so that we can write the partition function as

$$\mathcal{Z} = \sum_{\{s_i\}} e^{k \sum_i s_i s_{i+1}} \, e^{h \sum_i s_i} \, e^{gN}.$$

Now we have everything described in terms of the spin configurations (microscopic degrees of freedom), but we expect them to carry a lot of unnecessary details, since we are interested in the behavior of the system at larger scales than the microscopic one. The idea is to change our description of the system in terms of **coarser-grained degrees of freedom**: it naively consists in "thinning" the degrees of freedom by summing over some of the spins.

It is easy to understand that this idea can be implemented in different ways, and thus there is not a unique way to perform the coarse-graining transformation, but in general the procedure involves the separation of the system in different blocks and then the definition of a rule to relate the original degrees of freedom to the new ones. We will employ, for the 1D Ising model, the procedure that makes the subsequent computations easier: the so called **decimation** procedure.

img/RG_decimation.jpg

Figure 5.1: Decimation procedure for the 1D Ising model: the new coarse-grained degrees of freedom s'_i are defined as the majority of the original ones (s_i, s_{i+1}) in each of the two-spins blocks in which we have divided the system.

We divide the system into blocks of two spins, and we define the new coarse-grained degrees of freedom s'_i as the majority of the original ones (s_i, s_{i+1}) in each block: if the spins are equal, then the majority is well defined, while if they are different, we choose to take the first value of the two, since they are equivalent.[1] In this way, we have defined a new set of degrees of freedom that are coarser than the original ones, and we can move on with the RG analysis by writing the partition function in terms of the new degrees of freedom.

[1] In other words, we are summing over the odd spins, since we are basically always taking the value of the first spin of the couple.

In order to do this, let us define the following function of two spins, which will soon come in handy:

$$B(s_1, s_2) = k\, s_1 s_2 + \frac{h}{2}(s_1 + s_2) + g.$$

It can be directly substituted in the expression of the partition function, since the exponent $-\beta H$ can be written as:

$$-\beta H = k \sum_i s_i s_{i+1} + h \sum_i s_i + gN = \sum_i B(s_i, s_{i+1}).$$

Therefore, the partition function can be written as

$$\mathcal{Z} = \sum_{\{s_i\}_{i=1}^N} e^{\sum_i B(s_i, s_{i+1})} = \sum_{\{s'_i\}_{i=0}^{n'}} \sum_{\{s_i\}_{i=1}^N} e^{\sum_i B(s_i, s_{i+1})} \prod_{i=0}^{n'} \delta(s'_i, s_{2i+1}),$$

where the delta function enforces the definition of the new degrees of freedom s'_i as the majority of the original ones in each block. Let us highlight that the final product is over the new degrees of freedom, and it spans the new configurations, which are less than the original ones: indeed we defined $n' = \frac{N}{2} - 1$ since we are considering only the odd-indexed spins in the majority definition we gave for the coarse-graining procedure.

This last identity is not trivial, thus we are going to show it explicitly.

Proof. Let us start from a simple case: we have a system with $N = 4$ spins with periodic boundary conditions, and we want to define the new degrees of freedom as $s'_1 = s_1$ and $s'_2 = s_3$. The total number of configurations is $2^4 = 16$

$$\{s_i\}_{i=1}^4 = \{(1,1,1,1),\, (1,1,1,-1),\, \dots,\, (1,1,-1,-1),\, \dots,\, (-1,-1,-1,-1)\}.$$

We can define $\mathcal{Z}_{lhs}$ as the microscopical partition function, and $\mathcal{Z}_{rhs}$ as the coarse-grained one, thus we want to show that $\mathcal{Z}_{lhs} = \mathcal{Z}_{rhs}$. We can write

$$\begin{aligned} \mathcal{Z}_{lhs} = {} & e^{B(1,1)+B(1,1)+B(1,1)+B(1,1)} + e^{B(1,1)+B(1,1)+B(1,-1)+B(-1,1)} \\ & + \dots + e^{B(-1,-1)+B(-1,-1)+B(-1,-1)+B(-1,-1)}, \end{aligned}$$

while for the new partition function we have to first understand how the coarse-grained degrees of freedom are defined: we have configurations of the form

$$\{s'_i\}_{i=0}^1 = \{(1,1),\, (1,-1),\, (-1,1),\, (-1,-1)\},$$

but we have only to consider the configurations of the original spins that are compatible with the definition of the new ones through the delta function. For instance, considering the first configuration $(s'_0, s'_1) = (1,1)$ we have to consider only the configurations of the original spins such that

$$\delta(s'_0, s_1)\delta(s'_1, s_3) = \delta(1, s_1)\delta(1, s_3),$$

which means that the original configurations compatible with this definition are only four:

$$\{(1,1,1,1),\, (1,1,1,-1),\, (1,-1,1,1),\, (1,-1,1,-1)\},$$

and thus the contribution to the partition function from this configuration of the new degrees of freedom is

$$\begin{aligned} & e^{B(1,1)+B(1,1)+B(1,1)+B(1,1)} + e^{B(1,1)+B(1,1)+B(1,-1)+B(-1,1)} \\ + & e^{B(1,-1)+B(-1,1)+B(1,-1)+B(-1,1)} + e^{B(1,-1)+B(-1,1)+B(1,-1)+B(-1,-1)}, \end{aligned}$$

which is exactly the same contribution we had in the original partition function for the first four configurations of the original spins. We can repeat this procedure for all the other configurations of the new degrees of freedom, and we will find that the contributions to the partition function are exactly the same as in the original one, thus we have $\mathcal{Z}_{lhs} = \mathcal{Z}_{rhs}$.

But we can do better than this: we can prove explicity the identity for a generic system with N spins. We want to prove that

$$\sum_{\{s_i\}_{i=1}^N} e^{\sum_i B(s_i,s_{i+1})} = \sum_{\{s_i'\}_{i=0}^{n'}} \sum_{\{s_i\}_{i=1}^N} e^{\sum_i B(s_i,s_{i+1})} \prod_{i=0}^{n'} \delta(s_i', s_{2i+1}).$$

We can switch the order of the sums on the right hand side, since they are finite sums of indipendent variables, thus we can write

$$r.h.s. = \sum_{\{s_i\}_{i=1}^N} \sum_{\{s_i'\}_{i=0}^{n'}} e^{\sum_i B(s_i,s_{i+1})} \prod_{i=0}^{n'} \delta(s_i', s_{2i+1}),$$

then we can take the exponential out of the sum on the new degrees of freedom, since it does not depend on them, and we can write

$$r.h.s. = \sum_{\{s_i\}_{i=1}^N} e^{\sum_i B(s_i,s_{i+1})} \sum_{\{s_i'\}_{i=0}^{n'}} \prod_{i=0}^{n'} \delta(s_i', s_{2i+1}).$$

Now the last sum of products of delta functions is equal to one, since for each configuration of the original spins there is only one configuration of the new degrees of freedom that is compatible with it, thus we have

$$r.h.s. = \sum_{\{s_i\}_{i=1}^N} e^{\sum_i B(s_i,s_{i+1})} = l.h.s.,$$

which proves the identity. □

After this statistical digression, we can now proceed with the computation of the partition function in terms of the new degrees of freedom. We can write

$$\begin{aligned}
\mathcal{Z} &= \sum_{\{s_i'\}_{i=0}^{n'}} \sum_{\{s_i\}_{i=1}^N} e^{\sum_i B(s_i,s_{i+1})} \prod_{i=0}^{n'} \delta(s_i', s_{2i+1}) \\
&= \sum_{\{s_i'\}_{i=0}^{n'}} \prod_{i=0}^{n'} e^{\sum_{i=1}^{N/2} (B(s_{2i+1},s_{2i+2}) + B(s_{2i+2},s_{2i+3}))} \prod_{i=0}^{n'} \delta(s_i', s_{2i+1}) \\
&= \sum_{\{s_i'\}_{i=0}^{n'}} \sum_{\{s_{2i+2}\}_{i=1}^{N/2}} e^{\sum_i^{N/2} B(s_i', s_{2i+2}) + B(s_{2i+2}, s_{i+1}')} = \sum_{\{s_i'\}_{i=0}^{n'}} e^{-\beta H'[\{s_i'\}]},
\end{aligned}$$

where we have defined H' as a new Hamiltonian that depends on the new degrees of freedom; now we can further simplify the expression as follows:

$$\begin{aligned}
e^{-\beta H'[\{s_i'\}]} &= \sum_{\{s_{2i+2}\}_{i=1}^{N/2}} e^{\sum_i^{N/2} B(s_i', s_{2i+2}) + B(s_{2i+2}, s_{i+1}')} \\
&= \prod_{i=1}^{N/2} \left[\sum_{s_{2i+2}=\pm 1} e^{B(s_i', s_{2i+2}) + B(s_{2i+2}, s_{i+1}')} \right] \\
&= e^{\sum_{i=1}^{N/2} B'(s_i', s_{i+1}')},
\end{aligned}$$

where we have defined the new B' as the function satisfying the following identity:

$$e^{B'(s_1',s_2')} = \sum_{s=\pm 1} e^{B(s_1',s)+B(s,s_2')}.$$

Since this function must be symmetric under the exchange of s_1' and s_2', it can be proven rigorously that its most general form is

$$B'(s_1', s_2') = k' s_1' s_2' + \frac{h'}{2}(s_1' + s_2') + g',$$

where k', h' and g' are new parameters that depend on the original ones, and they are implicitly defined by the above identity.

This is a crucial step in the RG analysis: we have shown that after the coarse-graining transformation, the partition function can be written as

$$\mathcal{Z} = \sum_{\{s_i'\}_{i=0}^{n'}} e^{-\beta H'[\{s_i'\}]},$$

where the new Hamiltonian has the same form as the original one, but with different parameters

$$-\beta H'[\{s_i'\}] = \sum_{i=1}^{N/2} B'(s_i', s_{i+1}') = k' \sum_{i=1}^{N/2} s_i' s_{i+1}' + h' \sum_{i=1}^{N/2} s_i' + g' \frac{N}{2},$$

which means we have *coarse-grained* the description of the system

$$\begin{aligned} \{s_i\}_{i=1}^{N} \quad &\rightarrow \quad \{s_i'\}_{i=0}^{n'}, \\ (k, h, g) \quad &\rightarrow \quad (k', h', g'). \end{aligned}$$

In the thermodinamic limit $N \to \infty$ this transformation corresponds to a flow in the space of parameters, which is a way of thinking about the RG transformation that we will extensively discuss in the following.

To sum up, we have shown that the partition function of the original system can be written as the partition function of a new system with the same form of the Hamiltonian, but with different parameters: we started from the original hamiltonian

$$-\beta H = k \sum_i s_i s_{i+1} + h \sum_i s_i + gN = \sum_i B(s_i, s_{i+1}),$$

inside the exponential of the partition function, and through the coarse-graining transformation we have derived a new hamiltonian

$$-\beta H' = k' \sum_i s_i' s_{i+1}' + h' \sum_i s_i' + g' \frac{N}{2} = \sum_i B'(s_i', s_{i+1}'),$$

and in this new description in terms of the "renormalized" Hamiltonian, depending upon the new parameters k', h' and g'. In this case, the functional form of the Hamiltonian (and thus the partition function) is preserved, but the parameters are changed, and they are implicitly defined by the identity in equation (?), which explicitly reads **TODO:** add reference

$$e^{k' s_1' s_2' + \frac{h'}{2}(s_1'+s_2')+g'} = \sum_{s=\pm 1} e^{ks(s_1'+s_2') + \frac{h}{2}(s_1'+s_2') + hs + 2g}.$$

It is now convenient to employ the following change of variables:

$$\begin{cases} x = e^k, \\ y = e^h, \\ z = e^g, \end{cases} \qquad \begin{cases} x' = e^{k'}, \\ y' = e^{h'}, \\ z' = e^{g'}. \end{cases}$$

We have then four possible choices for $(s_1, s_2) = (\pm 1, \pm 1)$, all of which need to satisfy the above identity, thus we have the following system of equations:

$$\begin{aligned}
(+1, +1) \quad &\implies \quad x'y'z' = z^2 y(x^2 y + x^2 y^{-1}), \\
(+1, -1) \quad &\implies \quad (x')^{-1} z' = z^2 (y + y^{-1}), \\
(-1, +1) \quad &\implies \quad (x')^{-1} z' = z^2 (y + y^{-1}), \\
(-1, -1) \quad &\implies \quad x'(y')^{-1} z' = z^2 y^{-1} (x^{-2} y + x^2 y^{-1}).
\end{aligned}$$

The system is determined, since we have three unknowns x', y' and z' and three independent equations (the second and the third one are the identical). With simple algebra we can find the solution

$$\begin{cases}
(z')^4 = z^8 (x^2 y + x^{-2} y^{-1})(x^{-2} y + x^2 y^{-1})(y + y^{-1})^2, \\
(y')^2 = y^2 \dfrac{x^2 y + x^{-2} y^{-1}}{x^{-2} y + x^2 y^{-1}}, \\
(x')^4 = \dfrac{(x^2 y + x^{-2} y^{-1})(x^{-2} y + x^2 y^{-1})}{(y + y^{-1})^2},
\end{cases}$$

which provides, after taking the logarithm, the **RG flow equations** for the parameters k and h:

$$\begin{cases}
g' = 2g + \delta g(k, h), \\
h' = h + \delta h(k, h), \\
k' = k'(k, h)
\end{cases}$$

where δg, δh and $k'(k, h)$ are implicitly defined by the above equations. This flow equations describe how the parameters of the Hamiltonian change in the **action space** (or *free energy space*) as we perform a **coarse-graining transformation**. The fixed points of this flow correspond to scale-invariant theories, which are often associated with critical points of phase transitions.

Remark. *When we write the Hamiltonian, if we assume directly that $N \to \infty$, the system has an infinite number of degrees of freedom, and the RG transformation is from an infinite-dimensional space to itself, which is not very useful. Instead, we can consider a finite system with N sites, and then take the limit $N \to \infty$ at the end of the calculation.*

***TODO:** check if this is correct*

We are speaking about a space of parameters, not a space of configurations, so the RG transformation is from a finite-dimensional space to itself, which is more manageable.

Notice how during the RG transformation, lenghts in the system are actually rescaled: in the original microscopic description, the distance between two neighboring spins is one, while in the new description, the distance between two neighboring can be two (as an example, s_1 and s_3 are not neighbors, but the new spins s'_0 and s'_2, defined by the previous ones, are neighbors). Thus, spins which where separated by a distance R in the original lattice are now effevtively separated by a distance $R/2$ in the new lattice, and thus the RG transformation is also a **length rescaling** transformation.

5.1.1 | Recursion Relations and Flow Diagrams

The previous set of equations is also called the **RG recursion relations** for the parameters of the Hamiltonian, and they can be used to understand how the system behaves at different scales. In particular, they allow us to identify the fixed points of the RG flow, which correspond to scale-invariant theories, and thus to critical points of phase transitions.

Although the Ising model is a very simple model, the RG analysis is not trivial, and it can be quite involved for more complicated models. However, the presented approach to compute the partition function already reveals the essential and recurring themes of the RG analysis.

In general one proceeds as follows:

1. **Coarse-graining** or "thinning" of the degrees of freedom: we define new degrees of freedom that are coarser than the original ones, and we write the partition function in terms of these new degrees of freedom.
2. **Lenght rescaling**: after the coarse-graining transformation, the new degrees of freedom are defined on a lattice with a different spacing, thus we need to rescale the lengths in order to compare the new system with the original one.
3. Derivation of new **renormalized Hamiltonian**: after the coarse-graining and the length rescaling, we can write the partition function in terms of a new Hamiltonian that has the same form as the original one, but with different parameters.

We will soon come back to these steps to clarify and state them in a more general and rigorous way, but for now we can already see how they work in the case of the 1D Ising model. However, this is the essential procedure to obtain the RG flow equations for the parameters of the Hamiltonian, which are the main tool to derive predictions about the critical behavior of the system (existence of phase transitions, critical exponents, etc.).

Ising RG Flow: No External Field

Now let us set $h = 0$, which implies $y = 1$, and see how the RG flow looks like in the (k, h) plane. The flow equations give us

$$y' = \frac{x^2 + x^{-2}}{x^2 + x^{-2}} = 1 \implies h' = 0,$$

and we can write the recursion relation for k (from $(x')^4$) as

$$e^{4k'} = \left(\frac{e^{2k} + e^{-2k'}}{2}\right)^2 \qquad \implies \qquad k' = \frac{1}{2}\ln\left(\cosh(2k)\right).$$

This recursion relation has two fixed points, which can be found by setting $k' = k$:

$$k = \frac{1}{2}\ln\left(\cosh(2k)\right) \qquad \implies \qquad k = 0 \quad \text{or} \quad k = +\infty.$$

This is because the function $f(k) = \frac{1}{2}\ln\left(\cosh(2k)\right)$ is a monotonically increasing function of k (the hyperbolic cosine is almost a parabola and its logarithm is also monotonically increasing for $k > 0$) that starts from $f(0) = 0$ and diverges as $k \to +\infty$. Thus, the only points where $f(k) = k$ are at $k = 0$ and $k = +\infty$.

The fixed points are the only candidates for critical points (where the system can undergo a phase transition), since they are the only points where the system is scale-invariant.

The first one is unstable, while the second one is stable:

- At $k = 0$, we have $\beta = 0$, and thus an infinite temperature $T \to \infty$: it is a **stable** fixed point. If we perturb it slightly, for a small k, we have

$$k' = \frac{1}{2}\ln\left(\cosh(2k)\right) \approx \frac{1}{2}\ln\Big(1 + \left(\tfrac{2k}{2}\right)^2\Big) \approx k^2 < k,$$

which means that the flow is pushing towards $k = 0$: the system is driven towards the infinite temperature limit, where it is disordered. This is also called **sink** or **attractor**.

- At $k = \infty$, we have $\beta \to \infty$, and thus a zero temperature $T \to 0$: it is an **unstable** fixed point. If we perturb it slightly, for a large k, we have

$$k' = \frac{1}{2}\ln\left(\cosh(2k)\right) \approx \frac{1}{2}\ln\left(\frac{2^{2k}}{2}\right) = k - \ln 2 < k,$$

which means that the flow is pushing towards $k = 0$ and an infinite temperature: indeed, a perturbation to the large coupling constant k will make it smaller, effectively driving away the system from the zero temperature limit, where it is ordered.

In this case, the only critical point is at $k = 0$, which corresponds to the infinite temperature limit, where the system is disordered. The other fixed point at $k = \infty$ corresponds to the zero temperature limit, where the system is ordered, but it is unstable, since any perturbation will drive the system away from it. Thus, there is no phase transition at finite temperature in the 1D Ising model, and the system is always disordered for any finite temperature.

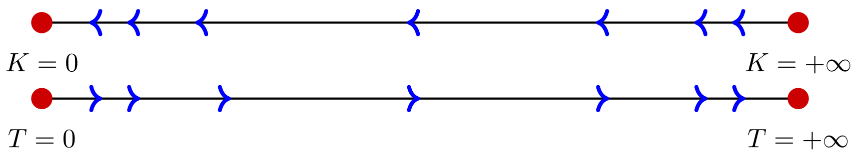


Figure 5.2: One-dimensional renormalization group (RG) flow diagrams. The upper figure depicts the flow of the coupling constant K towards the trivial fixed point at $K = 0$. The lower figure shows the equivalent flow parameterized by temperature T, where the system is driven away from the unstable critical point at $T = 0$ towards the stable high-temperature limit at $T = +\infty$.

Thus the initial problem $\mathcal{Z}(\beta, h = 0)$ is recursively mapped onto new ones $\mathcal{Z}'(\beta', h = 0)$, $\mathcal{Z}''(\beta'', h = 0)$, etc., which by construction have to be all numerically equal to the original one, but with different parameters. The flow of the parameters under the RG transformation allows us to understand the behavior of the system at different scales, and in particular to identify the critical points and the nature of the phase transitions. So under repeated RG transformations, any initial "hard" Hamiltonian (depending upon a generic β) is mapped onto a new simpler one arbitrarily closer to the stable fixed point $\beta \sim 0$. Furthermore we got to identify some fixed points of the flow, which are the only candidates for critical points.

Ising RG Flow: External Field

Going beyond and introducing an external field $h \neq 0$ we can also derive a **flow diagram** for generic values of k and h, which is a graphical representation of the RG flow in the parameter space. The flow diagram allows us to visualize how the parameters evolve under the RG transformation, and to identify the fixed points and their stability properties.

img/RG_flow_temporary.png

Figure 5.3: Flow diagram for the 1D Ising model in the (k, h) plane. The flow lines indicate how the parameters k and h evolve under the RG transformation.

In the precence of an external field, the flow diagram becomes more complex, but we can still identify the fixed points and their stability properties. All points in the parameter space flow towards the stable fixed points at $k = 0$ and arbitrary h, which corresponds to the disordered phase with independent and uncorrelated spins ($\xi = 0$ in this case).

5.1.2 | Linearization and Scaling Form

It is now time to investigate how the analysis of the RG flow allows us to obtain predictions about the behavior of physical quantities near the fixed points, the only candidates for critical points. In particular, the 1D Ising model does not have a phase transition at finite temperature, making this study rather trivial, but the method we are going to use is general and can be applied to more complicated models, where the RG flow can be more complex and the fixed points can have different stability properties.

Thus we can start by linearizing the flow equations around the fixed points since we are interested in the behavior near these. Let us consider in particular $T \sim 0$ and $h \sim 0$, corresponding to $y \sim 1$ and $x \to \infty$. We can write

$$\begin{cases} (x')^4 \simeq \dfrac{x^4}{4}, \\ (y')^2 \simeq y^2 \dfrac{x^2 y}{x^2 y^{-1}} = y^4, \end{cases} \qquad \Longrightarrow \qquad \begin{cases} e^{-k'} = \sqrt{2} e^{-k}, \\ h' = 2h. \end{cases}$$

After each RG iteration, the correlation length gets divided by two, and therefore, we can write ξ as a function of e^{-k} and h:

$$\xi = \xi(k, h) \to \frac{\xi(k, h)}{2} = \xi(k', h'),$$

where the equality is due to the fact that the partition function is invariant under the RG transformation, thus the correlation length must be invariant as well. This is a crucial point: the RG transformation is a change of variables in the partition function, which does not change its value, thus any physical quantity derived from it must be invariant under the RG transformation.

Thus we have the following identity for the correlation length:

$$\frac{1}{2}\xi(e^{-k}, h) = \xi(\sqrt{2}e^{-k}, 2h),$$

which after ℓ RG iterations becomes

$$\xi(e^{-k}, h) = 2^{\ell} \xi(2^{\ell/2} e^{-k}, 2^{\ell} h).$$

Now if we chose ℓ such that

$$2^{\ell/2} e^{-k} = 1 \qquad \Longrightarrow \qquad \ell = 2k,$$

since we are interested in the behavior near $k \to \infty$ (or $T \to 0$) and we want to keep the first argument of the correlation length fixed, we obtain the **scaling form** of the correlation length:

$$\xi(e^{-k}, h) = e^{2k}\xi(1, e^{2k}h) = e^{2k}g_\xi(e^{2k}h),$$

where g_ξ is a scaling function that depends on the combination $e^{2k}h$. This is the most non trivial result of the RG analysis: we are rewriting a new hamiltonian with new parameters, but the partition function is the same, and thus the correlation length is the same. This is what allowed us to derive the scaling form of the correlation length.

Now setting $h \to 0$ to study the behavior of the correlation length at zero magnetic field, we can treat $g_\xi(0)$ as a constant, and we obtain

$$\xi(e^{-k}, 0) \sim e^{2k} \sim e^{2\beta J} \sim e^{\frac{2J}{k_B T}},$$

which is the well-known result for the correlation length of the 1D Ising model at zero magnetic field. This shows how the RG analysis allows us to derive the scaling behavior of physical quantities near critical points, and in this case, it confirms that there is no phase transition at finite temperature, since the correlation length diverges only as $T \to 0$. We had to postulate the scaling form of the correlation length, but now we have an hint the RG ideas are adequate to derive the scaling form of physical quantities near critical points, and in particular to understand the behavior of the correlation length as a function of the parameters of the system.

5.2 | Conceptual Framework

TODO: add ref to chap 4.4-4.5 kardar

Since we have already seen the procedure of the RG transformation in the context of the 1D Ising model, we can now try to understand the general framework of the RG transformation, which is a powerful tool to understand critical phenomena in statistical mechanics. We will start from a more conceptual overview of the real space renormalization group, and then we will move to a more formal framework, which is the end goal of this chapter.

To be as general as possible, from now on we will frame the discussion in terms of the Landau-Ginzburg path-integral: let us imagine to have a partition function of the form

$$\mathcal{Z} = \int \mathcal{D}m(\mathbf{x}) e^{-F[m(\mathbf{x})]},$$

where $m(\mathbf{x})$ is the order parameter field, and $F[m(\mathbf{x})]$ is the free energy functional, which is a functional of the order parameter field. The RG approach is based on defining iterative transformations preserving the functional form of the free energy (and thus $\mathcal{Z}$), but changing the parameters of the theory in a way that eliminates irrelevant informations encoded in microscopic degrees of freedom.

Each RG transformation is composed the following three steps:

1. **Coarse-Grain**: we "decrease the resolution" of each configuration, changing the description of the system defining a new order parameter field $m'(\mathbf{x}')$ as the average of the original order parameter field $m(\mathbf{x})$ over mesoscopic regions of some lenght scale b:

   $$m'(\mathbf{x}) = \frac{1}{b^d} \int_{\mathcal{C}_b(\mathbf{x})} m(\mathbf{x}') \mathrm{d}\mathbf{x}',$$

   where the integral is performed over a cell $\mathcal{C}$ of size b centered around $\mathbf{x}$.

2. **Rescale**: this step consists in rescaling all the lenghts by a factor b in order to preserve the original domain of the system, which is equivalent to a variable change

   $$\mathbf{x} \to \mathbf{x}^* = \frac{\mathbf{x}}{b}.$$

   This step is crucial because we want to compare the new free energy with the original one, and thus we need to have the same domain for both of them.

3. **Renormalize**: Following the coarse-graining and rescaling steps, we might have altered the original "contrast", namely the range over which $m(\mathbf{x}^*)$ varies; thus we have to introduce a further scale factor ζ in order to preserve the original domain for $m'(\mathbf{x}^*)$:

   $$m' \to m^* = \frac{m'}{\zeta}.$$

   Note that in the one dimensional Ising model we had $\zeta = 1$, but in general it can be different from one.

Over all, these three steps are equivalent to the following variable change:

$$m(\mathbf{x}) \xrightarrow{\text{RG step}} m^*(\mathbf{x}^*) = \frac{1}{\zeta b^d} \int_{\mathcal{C}_b(b\mathbf{x}^*)} \mathrm{d}^d\mathbf{x}' m(\mathbf{x}'),$$

where $\mathcal{C}$ is the cell, now centered around $b\mathbf{x}^*$ (which is the original position of $\mathbf{x}$ before the rescaling). This can be read as a mapping from one set of rendom variables ($m(\mathbf{x})$) to another set of random variables ($m^*(\mathbf{x}^*)$), which is a change of variables in the partition function, and thus it does not change its value. Indeed, we can re-express the partition function as a weighted sum over the new field variables $m^*(\mathbf{x}^*)$:

$$\mathcal{Z} = \int \mathcal{D}m^* e^{-F^*[m^*(\mathbf{x}^*)]},$$

where the functional form of the free energy is preserved, but the parameters are changed. The RG transformation is thus a mapping in the space of parameters, which allows us to understand how the system behaves at different scales.

At this point, the crucial observation is the following: since at criticality the system is scale-invariant, then it should be left invariant under an appropriate RG transformation. This implies that the weights (or probabilities) of the renormalized field $m^*(\mathbf{x}^*)$ should be almost the same as the weights of the original field $m(\mathbf{x})$, which means that the functional form of the free energy should be preserved under the RG transformation, but the parameters of the theory will change. This is a key point: the RG transformation is a change of variables in the partition function, which does not change its value, thus any physical quantity derived from it must be invariant under the RG transformation (at least at criticality).

Let us assume than that close to the critical temperature T_c we can neglect all the parameters in $F[m(\mathbf{x})]$ except for the most relevant ones, which are t (the rescaled temperature) and h (the external field), in order to write it as the following functional:

$$F[m(\mathbf{x})] = F[m(\mathbf{x}); t, h].$$

Saying that the configurations of the renormalized field $m^*(\mathbf{x}^*)$ have the same weights as the original field $m(\mathbf{x})$ means that the free energy functional form $F[m(\mathbf{x})]$ has to be preserved under the RG transformation $F^*[m^*(\mathbf{x}^*)]$, but the parameters t and h will change to new values t^* and h^*.

As we coarse-grain and rescale the system, the RG step moves us a bit further away from criticality, since the correlation length gets divided by b, meaning that consequently the parameters t and h have increased, and thus we are looking at a less critical system.

Thus, close to criticality, we expect that the RG transformation will preserve the functional form of the free energy, but it will change the parameters t and h to new values t^* and h^*:

$$F^*[m^*(\mathbf{x}^*); t^*, h^*] = F[m^*(\mathbf{x}^*); t^*, h^*].$$

where the parameters have changed as functions of the original ones:

$$\begin{cases} t^* = t_b(t, h), \\ h^* = h_b(t, h), \end{cases}$$

which will give us the flow equations for t and h. This analysis is really too simplicistic: even if before the RG step we could neglect all of the parameters except for t and h, after the RG step we could have generated new ones, which we cannot neglect anymore; for now we will ignore this conceptual problem, but we will see how to solve it in the next section.

We can infer the functional form of $t_b(t, h)$ and $h_b(t, h)$ by making some symmetry arguments. First, since RG steps involve changes at short lenght scales, then it cannot generate singularities; therefore, close to criticality $[(t, h) = (0, 0)]$, we expect to be able to expand the flow equations in an analytic Taylor series:

$$\begin{cases} t_b(t, h) \sim A(b)t + B(b)h + \mathcal{O}(t^2, h^2), \\ h_b(t, h) \sim C(b)t + D(b)h + \mathcal{O}(t^2, h^2), \end{cases}$$

from which we will have to deduce symmetry relations for the coefficients of the expansion.

Next, we know the free energy to be invariant under the following combined transformations

$$m(\mathbf{x}) \to -m(\mathbf{x}), \quad h \to -h, \quad t \to t,$$

then the hamiltonian will also be invariant under the same transformations; thus the RG step, which is a change of variables in the partition function, will also be invariant under the same transformations, and thus we have to require that $B(b) = 0$ and $C(b) = 0$. Thus we have simplified the flow equations to the following form:

$$\begin{cases} t_b(t, h) = A(b)t, \\ h_b(t, h) = D(b)h. \end{cases}$$

Let us make a couple final considerations. First, we know that for $b = 1$ the RG step is just the identity, thus we have to require that $A(1) = D(1) = 1$. Furthermore, since the RG transformation is a mapping from the parameter space to itself, we can devise the steps in such a way that the composition of two RG steps with parameters b_1 and b_2 is equivalent to a single RG step with parameter $b_1 b_2$. Accordingly, the coefficients $A(b)$ and $D(b)$ have to satisfy:

$$\begin{cases} A(b_1 b_2) = A(b_1)A(b_2), \\ D(b_1 b_2) = D(b_1)D(b_2), \end{cases}$$

which suggest the existence of a **group structure** for the RG transformations.

Taking into account the condition $A(1) = D(1) = 1$ and the group structure, the solution for $A(b)$ and $D(b)$ reads

$$\begin{cases} A(b) = b^{y_t}, \\ D(b) = b^{y_h}, \end{cases}$$

for certain values of the exponents y_t, $y_h \in \mathbb{R}$. Now we want to understand the sign of these exponents y_t and y_h, which will give us the stability of the fixed point.

Since after an RG step the correlation length gets divided by b, it seems that we are looking at a less critical system: it moves us away from criticality, which should correspond to

$$y_t,\, y_h > 0.$$

Thus the parameters t and h transforms under the flow equations near criticality as

$$\begin{cases} t' = t^* = b^{y_t} t, \\ h' = h^* = b^{y_h} h, \end{cases}$$

which means that the flow is pushing the system away from the critical point, which is thus an unstable fixed point.

Now the idea is to demonstrate the scaling and hyperscaling hypotheses, which we encountered in the previous chapters, using the RG framework. We will focus on the free energy and the correlation length in order to do this, and we will be able to explain the emergence of the critical exponents and the scaling relations between them.

5.2.1 | Scaling Relation for Free Energy

In the path integral formulation, the partition function is given by

$$\mathcal{Z} = \int \mathcal{D}m e^{-F[m;\, t, h]} = \int \mathcal{D}m^* e^{-F[m^*; t^*, h^*]},$$

where we have used the fact that near criticality the functional form of the free energy is preserved under the RG transformation, but the parameters t and h are changed to new values t^* and h^*. Since the integration variable is a dummy variable, we can write

$$\mathcal{Z} = \int \mathcal{D}m e^{-F[m;\, t^*, h^*]},$$

and since the free energy is the logarithm of the partition function, we can write its density as

$$f(t,h) = -\frac{1}{V} \ln \mathcal{Z}.$$

We have to take into account that the volume of the system is also changed under the RG transformation, since we are rescaling the lenghts by a factor b. Thus we have to write

$$Vf(t,h) = V^* f(t^*, h^*),$$

and using the fact that $V^*/V = b^{-d}$ (since each dimension is rescaled by b), we can write

$$f(t,h) = b^{-d} f(b^{y_t} t, b^{y_h} h),$$

which is the scaling relation for the free energy, where d is the spatial dimension of the system. We have thus derived the same result that we obtained in the previous chapters using the scaling hypothesis, but now we have a more solid theoretical framework to understand it. We can also write the scaling relation for the free energy in a more convenient form by choosing $b = t^{-1/y_t}$, which gives us

$$f(t,h) = t^{\frac{d}{y_t}} f(1, t^{-\frac{y_h}{y_t}} h) = t^{\frac{d}{y_t}} g\left(\frac{h}{t^{\frac{y_h}{y_t}}}\right),$$

which is the scaling relation for the free energy, where g is a scaling function that depends on the ratio $h/t^{y_h/y_t}$. This scaling relation allows us to understand how the free energy behaves near the critical point, and in particular how it depends on the reduced temperature t and the magnetic field h.

Taking the previous definitions for the critical exponents (obtained from the scaling hypothesis), we can identify the critical exponents α and Δ as

$$\begin{cases} 2 - \alpha = \dfrac{d}{y_t}, \\ \Delta = \dfrac{y_h}{y_t}, \end{cases}$$

completing the identification of the critical exponents in terms of the "anomalous dimensions" y_t and y_h that we obtained from the RG flow equations.

5.2.2 | Scaling Relation for Correlation Length

Given the weights $e^{-F[m;\, t,h]}$, the correlation length is just a function of the parameters t and h:

$$\xi = \Gamma(t,h).$$

By definition, it is the length scale over which the correlations between the spins decay exponentially. Near the critical point, the correlation length diverges, and its behavior can be described by a scaling relation. Under an RG transformation, again the idea is to write the correlation length in terms of the new parameters:

$$\begin{cases} \xi^* = \Gamma(t^*, h^*), \\ \xi^* = \dfrac{\xi}{b} = \dfrac{\Gamma(t,h)}{b}, \end{cases}$$

which gives us the following scaling relation for the correlation length:

$$\Gamma(t,h) = b\Gamma(b^{y_t}t, b^{y_h}h).$$

If we choose $b = t^{-\frac{1}{y_t}}$, this again gives the same result obtained from the scaling hypothesis:

$$\Gamma(t,h) = t^{-\frac{1}{y_t}}\Gamma(1, t^{-\frac{y_h}{y_t}}h).$$

Now we can identify the critical exponent ν that describes the divergence of the correlation length as

$$\nu = \frac{1}{y_t}, \text{ since } \xi \sim t^{-\nu} \text{ for } h = 0.$$

since at $h = 0$ we can treat $\Gamma(1,0)$ as a constant, and thus the correlation length behaves as $\xi \sim t^{-1/y_t} = t^{-\nu}$ near the critical point. This is the scaling relation for the correlation length, which we now derived from the RG framework, and it is consistent with the scaling relation obtained from the scaling hypothesis.

The fact that the functional form of the correlation length is preserved under the RG transformation does not mean that this procedure is valid only for simple cases, but we assumed it for simplicity in the derivation of the scaling relations; we will see the general implications.

In summary, even taking a simplicistic approach, the RG framework is able to explain the scaling hypothesis and the emergence of the critical exponents, which are related to the uniquely determined y_t and y_h that we obtained from the RG flow equations: these coefficients are of paramount importance, since they determine how t and h scale under RG near the critical point and all the critical exponents can be expressed in terms of them.

5.3 | Formal Framework

Our goal is now to rephrase the previous discussion in a more formal way and establish a rigorous framework for the RG transformation, which will allow us to understand how to deal with the problem of the generation of new parameters under the RG transformation, and how to identify the relevant and irrelevant operators in the theory.

In order to make the discussion as general as possible, we have to take into account all the parameters that can appear in the LG free energy; thus we consider the full expansion of the LG free energy (which we derived) as a functional of the order parameter $m(\mathbf{x})$ and the set of all its parameters $\mathbf{s} = (t, u, v_1, v_2, \dots, k, L, h, \dots)$:

$$F[m] = \int \mathrm{d}^d\mathbf{x} \left[\frac{t}{2}m^2 + um^4 + v_1 m^6 + v_2 m^8 + \cdots + \frac{k}{2}(\nabla m)^2 + \frac{L}{2}(\nabla m)^2 + \cdots + hm + hm^{5/7} + \cdots\right],$$

where this functional form is the most general one that we can write, and the RG transformation will give us the flow equations for all the parameters t, u, v_1, v_2, …, k, L and h.

A given system is fully characterized by a point $\mathbf{s}$ in the infinite-dimensional parameter space, which we call the **action space** (or “theory space”), since it is the space of all the possible actions (or free energy functionals, hamiltonians etc.) that we can write for the system. Each point in this space corresponds to a specific theory, and the RG transformation will give us a flow in this space, which allows us to understand how the system behaves at different scales.

When we apply an RG iteration through coarse-graining, we are basically performing a change of variables of the form

$$m^*(\mathbf{x}^*) = \frac{1}{\zeta b^d}\int_{\mathcal{C}} \mathrm{d}^d m(\mathbf{x}'),$$

which does not change the value of the free energy. Indeed this new field will be iself desciribed by a free energy functional of the same form as the original one, but with different parameters:

$$F[m] = \int \mathrm{d}^d\mathbf{x} \left[\frac{t^*}{2}m^2 + u^*m^4 + v_1^* m^6 + v_2^* m^8 + \cdots + \frac{k^*}{2}(\nabla m)^2 + \frac{L^*}{2}(\nabla m)^2 + \cdots + h^*m + h^*m^{5/7} + \cdots\right],$$

which is certainly true, since we also chose the most general functional form for the free energy, including all the possible terms that we can write, and thus it is closed under the RG transformation. Thus the functional form do not change after a parameter change

$$F[m, (t, u, v_1, v_2, \dots, k, L, h, \dots)] \xrightarrow{\text{RG step}} F[m, (t^*, u^*, v_1^*, v_2^*, \dots, k^*, \dots, h^*, \dots)],$$

$$F[m, \mathbf{s}] \xrightarrow{\text{RG step}} F^*[m, \mathbf{s}^*] = F[m^*, \mathbf{s}^*],$$

because in the action space, with the most general functional form, the RG transformation is just a re-mapping from the parameter space to itself:

$$\mathbf{s}_1 \mapsto \mathbf{s}_2 \mapsto \mathbf{s}_3 \mapsto \cdots \in \text{Action Space}, \quad \mathbf{s}' = \mathcal{R}_b(\mathbf{s}),$$

which allows us to understand how the system behaves at different scales, and in particular to identify the critical points and the nature of the phase transitions. The map $\mathcal{R}_b$ is defined to describe the effect of the RG transformation on the parameters of the theory.

We are visualizing the RG transformation as a flow in the action space, and the fixed points of this flow correspond to scale-invariant theories, thus they have to be associated with critical points of phase transitions (at zero or infinite temperature).

5.3.1 | Scaling Directions and Anomalous Dimensions

After an RG step, the correlation length gets divided by b, which means that at the fixed point, the system can have only two options: either the correlation length goes to zero, which corresponds to an infinite temperature limit, or the correlation length goes to infinity, which corresponds to a critical point.

However, we have to take into account that the RG transformation can also generate new parameters if we start neglecting some of the existing ones, and thus we have to consider the full expansion of the free energy: we are studying the full infinite-dimensional action space.

If we imagine to have a system in the vicinity of the fixed point $\bar{\mathbf{s}}$ in the action space, which is left invariant under the RG transformation, we can write

$$\mathbf{s} = \bar{\mathbf{s}} + \delta\mathbf{s}.$$

We can then linearize by Taylor expanding its image under the RG transformation as

$$\mathbf{s}^* = \mathcal{R}_b[\mathbf{s}] = \mathcal{R}_b[\bar{\mathbf{s}}] + \mathcal{R}_b^L \delta\mathbf{s} + \mathcal{O}(|\delta\mathbf{s}|^2),$$

where we have denoted the Jacobian matrix of the map $\mathcal{R}_b$ at the fixed point as $\mathcal{R}_b^L = \frac{\partial \mathcal{R}_b}{\partial \mathbf{s}}\big|_{\bar{\mathbf{s}}}$.

If we now ignore the mathematical subtleties of working with an infinite-dimensional matrix as $\mathcal{R}_b^L$, we can treat it as a linear operator, and thus try to diagonalize it. With this in mind, we start by noting that by our choice of RG procedure, the composition of two RG steps with parameters b and b' is equivalent to a single RG step with parameter bb':

$$\mathcal{R}_b\left[\mathcal{R}_{b'}[\cdot]\right] = \mathcal{R}_{bb'}[\cdot] = \mathcal{R}_{b'}\left[\mathcal{R}_b[\cdot]\right],$$

which is the property justifying the name "group" in RG (even if we already underlined that it is not a proper mathematical group). Therefore, the linearized RG transformation $\mathcal{R}_b^L$ also has to satisfy the same property, and from these we can deduce that $\mathcal{R}_b$ and $\mathcal{R}_{b'}$ (as well as their linearizations) commute with each other:

$$[\mathcal{R}_b, \mathcal{R}_{b'}] = 0 \implies \left[\mathcal{R}_b^L, \mathcal{R}_{b'}^L\right].$$

Following this, we can find a common eigenbasis $\{O_i\}$ for the linearized RG transformation, which diagonalizes it, such that for each eigenvector O_i we have

$$\mathcal{R}_b^L O_i = \lambda_i(b) O_i,$$

and furthermore, since the linearized RG transformation also satisfies the group property, we can apply two RG steps with parameters b and b' to the eigenvector O_i and we have

$$\mathcal{R}_b\mathcal{R}_{b'}O_i = \lambda_i(b)\lambda_i(b') = \mathcal{R}_{bb'}O_i = \lambda_i(bb')O_i,$$

and we obtain the following constraints on the eigenvalues $\lambda_i(b)$:

$$\lambda(b)\lambda(b') = \lambda(bb').$$

Now, recalling that $\lambda(1) = 1$ (since for $b = 1$ the RG transformation has to be just the identity), we can write

$$\lambda(b) = b^{y_i},$$

which let us understand:

$$\mathcal{R}_b^L O_i = \lambda_i(b) O_i \qquad \begin{cases} O_i \text{ are called } \textbf{scaling directions}, \\ y_i \text{ are called } \textbf{anomalous dimensions}. \end{cases}$$

Since the eigenvectors O_i form a complete basis, we can write any Hamiltonian in the vicinity of $\bar{\mathbf{s}}$ as

$$\mathbf{s} = \bar{\mathbf{s}} + \sum_i g_i O_i,$$

which after one RG step is mapped to

$$\mathbf{s}^* = \mathcal{R}_b[\mathbf{s}] = \bar{\mathbf{s}} + \sum_i g_i b^{y_i} O_i.$$

In this expression, the coefficients g_i are basically the coordinates of the point $\mathbf{s}$ in the action space with respect to the basis of scaling directions O_i (the parameters in the new eigenbasis for the linearized RG transformation), and the anomalous dimensions y_i determine how these coordinates change under the RG flow.

We have all we need to understand the stability of the fixed point $\bar{\mathbf{s}}$ under the RG flow: the sign of the anomalous dimensions y_i determines whether the flow is towards or away from the fixed point along the corresponding scaling direction O_i. We can thus classify the scaling directions as follows:

- **Relevant operators** if $y_i > 0$: the flow is pushing the system away from the fixed point, since g_i becomes larger and larger under scaling and thus they are associated with unstable directions.
- **Irrelevant operators** if $y_i < 0$: the flow is directed towards the fixed point, since g_i becomes smaller and smaller under scaling and thus the system is attracted to the fixed point along these directions.
- **Marginal operators** if $y_i = 0$: the flow is neither towards nor away from the fixed point, since g_i does not change under scaling and thus the system is static along these directions.

We call them "operators" even if they are just directions in the parameter space, because they are associated with physical operators in the theory and historically they were called operators in the context of quantum field theory, where the RG was first developed. There is a natural way to associate to each scaling direction a physical operator, since the parameters of the Hamiltonian are associated with physical operators. We accept it as a name for now, but in the second module of the course we will see how to make this association more rigorous.

The saddle point calculation give us exact result when concerned about criticality, because neglecting irrelevant operators not only we make a small error, but we are actually neglecting terms that are going to zero under the RG flow, after looking at an infinite scale of the system. This is the key point of the RG analysis: we are not neglecting terms that are small, but we are neglecting terms that are going to zero.

The subspace spanned by the irrelevant operators is called **basin of attraction** of the fixed points $\bar{\mathbf{s}}$, since any point in this subspace will be attracted to the fixed point under the RG flow. Since the correlation lenght decreases under the RG flow, and it is infinite at the fixed points ($\xi(\bar{\mathbf{s}}) \to \infty$), then it will be infinite for any point in the basin of attraction. For a generic point in the vicinity of $\bar{\mathbf{s}}$, we can repeat the same argument made previously (assuming that near criticality we can consider only h and t) and obtain the following scaling relation for the correlation length:

$$g_i \to b^{y_i} g_i \implies \xi(g_1, g_2, \dots) = b\xi(b^{y_1} g_1, b^{y_2} g_2, \dots).$$

For large b the irrelevant operators will become smaller and smaller, eventually flowing to zero, leaving only the relevant operators to determine the behavior of the system at large scales. We can thus order the operators by decreasing anomalous dimensions $y_1 > y_2 > \cdots$ (such that we get the relevant operators first) and choosing $b = g_1^{-\frac{1}{y_1}}$, leads us to the following expression for the correlation length:

$$\xi(g_1, g_2, \dots) = g_1^{-\frac{1}{y_1}} \xi(1, g_1^{-\frac{y_2}{y_1}} g_2, \dots),$$

where again we can identify the critical exponent ν as $\nu = 1/y_1$, since at $g_2 = g_3 = \cdots = 0$ we can treat $\xi(1, 0, 0, \dots)$ as a constant, and thus the correlation length behaves as $\xi \sim g_1^{-1/y_1} = g_1^{-\nu}$ near the critical point. Namely, we have derived the scaling form of the correlation length, which is a function of the relevant operators, and the irrelevant operators only appear in the scaling function as corrections to scaling.

To reach the repulsive point (infinite correlation) we need to fine-tune the parameters, and thus we need to set $t = 0$ and $h = 0$, which means that we need to set the temperature to the critical temperature and the magnetic field to zero, which is a very special point in the parameter space, and thus it is not surprising that we have a phase transition only at this point.

Historically, the RG was developed in the context of quantum field theory, where it was used to understand the behavior of quantum fields at different energy scales. The idea was coming from Kardanoff, but only after the work of Wilson, who implemented the RG with a more systematic perturbative approach (called the epsilon expansion, which we will not have time to cover), it was understood that the RG is a powerful tool to understand critical phenomena in statistical mechanics, and it has been applied to a wide range of systems, from classical to quantum systems, and from equilibrium to non-equilibrium systems.

5.4 | The Gaussian Model

The Gaussian model is a simple yet powerful statistical model that assumes that the data follows a normal distribution. It is widely used in various fields, including physics, engineering, and machine learning, due to its mathematical tractability and the central limit theorem.

It is a very limited case of application of RG methods, but it is a good starting point to understand the basic concepts and techniques of RG. In this section, we will explore the Gaussian model in detail, including its formulation, properties, and applications.

The gaussian model is defined by keeping only the quadratic term in the LG partition function:

$$\mathcal{Z} = \int \mathcal{D}m(\mathbf{x}) \, \exp\left\{ - \int \mathrm{d}^D\mathbf{x} \, [\cdots] \right\},$$

and it is a bit unphysical since it is only defined for $t \geq 0$, but it is a good starting point to understand the basic concepts and techniques of RG. We would not even need RG to solve this model, but it is a good example to illustrate the RG procedure.

We will provide a **momentum space RG solution** for this model.

[...]

5.4.1 | Exact Solution

The Gaussian model is exactly solvable, and we can find the exact solution by performing a Fourier transform of the field $m(\mathbf{x})$. The Fourier transform allows us to express the field in terms of its momentum components, which simplifies the analysis of the partition function. The Fourier transform of the field $m(\mathbf{x})$ is defined as:

$$\tilde{m}(q) = \int \mathrm{d}^D\mathbf{x} \, m(\mathbf{x}) e^{i\mathbf{q}\cdot\mathbf{x}}, \iff m(\mathbf{x}) = \sum_{\mathbf{q}} e^{-i\mathbf{q}\cdot\mathbf{x}} \tilde{m}(\mathbf{q}) \sim \int \mathrm{d}^D\mathbf{q}(2\pi) \, e^{-i\mathbf{q}\cdot\mathbf{x}} \tilde{m}(\mathbf{q}),$$

where the sum is over the discrete momenta allowed by the finite size of the system, and the integral is over the continuous momenta in the thermodynamic limit. The partition function can be rewritten plugging the Fourier transform of $m(\mathbf{x})$ into the original expression:

$$\mathcal{Z} = e^{\frac{V h^2}{2t}} \ldots$$

If we take the logarithm, the product becomes a sum which we can express as an integral over the Brillouin zone: since the singularities of the integrand are at $q = 0$, we can replace the Brillouin zone with a sphere of radius Λ (the UV cutoff): ...

And in the limit for $t \to 0$ ...

Therefore we can identify

$$\begin{cases} \alpha_+ = 2 - \dfrac{D}{2}, \\ \Delta = \dfrac{1}{2} + \dfrac{D}{4}. \end{cases}$$

[...]

1. **Coarse Grain**

   [...]

2. **Rescale**

   [...]

3. **Renormalize**

   [...]

Relevant and Irrelevant Operators

[...]

Remark (Dangerous Irrelevant Operators).

Infinitesimal Scale Transformations

[...]

Part II

Conformal Invariance and Minimal Models

Introduction

1 Foundamentals and Conformal Symmetry

A deep connection exists between quantum field theory (QFT) and statistical mechanics. This relation becomes particularly powerful in the study of critical phenomena, as one can see from the **Landau-Ginzburg formulation** and by the observation that systems at phase transitions are described by field theories with scale invariance. At criticality the theory is invariant under dilatations of coordinates

$$x_\mu \to \lambda x_\mu.$$

Order parameters (fields) transform as

$$\Phi_i \to \lambda^{\Delta_i} \Phi_i,$$

where the Δ_i determine critical exponents, so their calculation is crucial. This implies power-law behavior for correlation functions at criticality:

$$\langle \Phi_i(x) \Phi_i(0) \rangle \sim \frac{1}{|x|^{2\Delta_i}},$$

so that the Δ_i can be extracted if one is able to compute the correlation functions exactly.

This suggested Polyakov to formulate a strategy to solve scale invariant QFT known as the **Polyakov programme**:

1. show that scale invariance implies, under very general assumptions, a larger **conformal invariance**;

2. conformal invariance and operator product expansions of local fields lead to **conformal bootstrap**, which is a set of conditions to be satisfied by the 4-point correlation functions;

3. conformal bootstrap constraints the correlation functions and in some cases is able to determine them exactly: we can extract the critical exponents from the obtained conformal blocks;

4. correlation functions allow to compute β-functions, renormalization group eigenvalues, scale dimensions and ultimately critical exponents.

The classification of fixed points of Renormalization Group, in practice of the critical points and therefore the possible universality classes is thus equivalent to classify all the conformal field theories (CFT).

A fundamental theorem states that, unless a theory is particularly pathologically defined, the combination of Poincaré and scale invariance automatically implies a larger symmetry: **conformal symmetry**. This symmetry enhancement is the cornerstone of the whole approach.

Once conformal symmetry is established, it can be used to set up the **conformal bootstrap**. This powerful framework leverages the symmetries to impose rigid mathematical conditions that force the exact form of the correlation functions. In particular, conformal invariance heavily constrains their spatial dependence:

- **2-point functions** are fixed exactly, up to a overall normalization;
- **3-point functions** are determined completely, up to a single constant known as the structure constant;
- **4-point functions** (which initially depend on four coordinate variables x_1, x_2, x_3, x_4) are reduced to a fixed kinematic prefactor multiplied by an unknown function of a single invariant variable, called the **anharmonic ratio** (or cross-ratio).

This reduction is particularly intuitive in two dimensions: using a general conformal transformation, it is always possible to map three of the four points to 0, 1, and ∞. The position of the single remaining point corresponds exactly to this anharmonic ratio.

Conformal invariance further dictates that this remaining function must satisfy a specific set of consistency identities. Solving these *bootstrap identities* allows one to decompose the correlation functions into their fundamental building components, the so-called **conformal blocks**.

However, it is not guaranteed that this exact analytical procedure can be carried out completely for every theory. For most theories in generic dimensions $D > 2$, an exact analytical solution is out of reach. Nevertheless, the conformal symmetry still allows us to strongly constrain the properties of these conformal blocks. In higher dimensions, the bootstrap identities yield very complicated functional equations that can be approached *numerically*. By truncating and solving them recursively, it is possible to build highly accurate estimates of the theory's observables. This numerical bootstrap approach has been phenomenally successful for critical theories up to four dimensions, with the 3D Ising model being one of its most celebrated applications.

1.1 | Path Integral Formulation

The path integral formulation of quantum field theory (QFT) shows deep links with statistical mechanics. It is indeed possible, thanks to a Wick rotation on the time axis, to write the generating functional of Green functions in a way that makes evident its equivalence to a partition function of a statistical mechanical system with one more spatial dimension. If the Hamiltonian density of the QFT is quadratic in the conjugated momenta, this statistical system will be purely configurational. In this section we quickly recall the basics of the path integral formulation of a Wick rotated QFT.

1.1.1 | Quantum Mechanics

A system with degrees of freedom $q(t)$ and Lagrangian $L(q, \dot{q})$ has quantum transition **amplitude between** an initial state $|q_i\rangle$ and a final one $|q_f\rangle$ given by:

$$Z(q_f, t_f; q_i, t_i) = \langle q_f| \, e^{-iH(t_f - t_i)} \, |q_i\rangle \,,$$

which represents the probability amplitude for the system to evolve from the state $|q_i\rangle$ at time t_i to the state $|q_f\rangle$ at time t_f. By discretizing the time interval in n small intervals of size ϵ:

$$t_i = t_0 < t_1 < ... < t_n = t_f, \quad t_k - t_{k-1} = \epsilon,$$

and inserting complete sets of states (resolution of the identity) at each time slice:

$$\mathbb{I} = \int \mathrm{d}q \, |q\rangle \, \langle q| \,,$$

one gets:

$$Z = \int \prod_{j=1}^{n-1} \mathrm{d}q_j \prod_{j=0}^{n-1} \langle q_{j+1}| \, e^{-i\epsilon H} \, |q_j\rangle \,.$$

Expanding for small ϵ and inserting momentum eigenstates, one obtains:

$$Z = \int \mathcal{D}q \, e^{iS[q]}, \tag{1.1.1}$$

where

$$S[q] = \int_{t_i}^{t_f} \mathrm{d}t \, L(q, \dot{q})$$

is the action.

In this way, the quantum transition amplitude is expressed as a sum over all possible paths $q(t)$ connecting the initial and final states, weighted by the exponential of the action evaluated on those paths. This formulation is particularly useful for systems where the Hamiltonian is not easily diagonalizable, and it provides a natural framework for perturbation theory.

1.1.2 | Quantum Field Theory

The same construction illustrated above can be done also in QFT. From the Hamiltonian point of view, actually, a field theory is a system with an infinite number of degrees of freedom. To the dynamical variables $q_i(t)$ with $i = 1, ..., N$ one substitutes a field $\phi(x, t)$, where the continuum index x plays the role of the discrete index i; in other words, the field $\phi(x, t)$ represents a dynamical variable for any point x of space. Here, we do not better specify the field, which can have a scalar, vector, tensor or spinor nature, with indices in Minkowski spacetime with metric $g = \mathrm{diag}(1, -1, ..., -1)$ and/or other possible indices for internal degrees of freedom.

The same formulae obtained in quantum mechanics for the time evolution amplitude can be applied to the case of a system with Lagrangian density $\mathcal{L}(\phi, \partial_\mu \phi)$ and Hamiltonian density

$$\mathcal{H} = \pi \cdot \dot{\phi} - \mathcal{L},$$

where π represents the conjugate momentum density

$$\pi = \frac{\delta \mathcal{L}}{\delta \dot{\phi}}.$$

It is important to notice that the Lagrangian density contain all of the information about the dynamics of the field, giving us potentially everything we need to know about the theory, including the spectrum of excitations, the interactions and the symmetries. It literally builds the quantum field theory, and it is the starting point for any calculation in the path integral formalism.

For example, the Lagrangian density of a scalar field of mass m is given by

$$\mathcal{L}(\phi, \partial_\mu \phi) = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi),$$

where the first term is called *kinetic term*, while the term describing the **self interaction** of the field

$$V(\phi) = \sum_{k=1}^{\infty} g_k \phi^k,$$

is called *potential*.

The term quadratic in ϕ usually has, in QFT, the meaning of a mass term and $g_2 = \frac{m^2}{2}$ but, for example, in the LG formulation it has a meaning as (dimensionless) temperature $t = (T - T_c)/T_c$. However, it is important to stress that m here represents the *bare* mass of the field. Whether this corresponds to the actual physical mass depends on the presence of other terms in the potential: if the field is interacting (or even self-interacting), quantum fluctuations will shift this initial value. Consequently, the actual physical mass is obtained only after a renormalization procedure, which systematically redefines these bare parameters to absorb the non-physical infinities that arise in the calculations.

The **conjugated momentum density** $\pi = \dot{\phi}$ (through the Legendre transformation) allows to write the Hamiltonian density as

$$\mathcal{H}(\pi, \phi) = \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} (\nabla \phi)^2 + V(\phi).$$

Notice that the kinetic and spatial gradient terms are strictly positive, being sums of squares. Provided that the potential $V(\phi)$ is also bounded from below (which is guaranteed if we only include even higher powers of the field, such as the ϕ^4 term in the Ising model, making it the simplest physically viable non-trivial theory), the total Hamiltonian—the spatial integral of this density—is bounded from below. This implies that the operator whose eigenvalues represent the total energy of the system has a well-defined lower bound, meaning the theory is mathematically and physically sensible.

With analogy of what is done in the quantum mechanics case, one obtains

$$\langle \phi_f(x) | \, e^{-i\hat{H}(t_f - t_0)} \, | \phi_0(x) \rangle = \int_{\phi(x,t_0)=\phi_0(x)}^{\phi(x,t_f)=\phi_f(x)} \mathcal{D}\phi \mathcal{D}\pi \, e^{i\hbar \int_{t_0}^{t_f} \mathrm{d}t \int \mathrm{d}^{D-1}x [\pi \cdot \dot{\phi} - \mathcal{H}]}.$$

Physically, this object represents the probability amplitude that an initial configuration of fields at time t_0 will evolve into a final configuration of fields at time t_f.

In QFT one often considers the limit $t_0 \to -\infty$ and $t_f \to +\infty$, therefore considering an interaction running on the whole spacetime. For Hamiltonians quadratic in $\pi(x)$, it is possible to apply the same considerations of the quantum mechanical case and Gauss-integrate on momenta, getting the functional Z:

$$Z = \int \mathcal{D}\phi \, e^{\frac{i}{\hbar} S[\phi]}, \tag{1.1.2}$$

where the action is now the integral over the whole spacetime of the Lagrangian density

$$S[\phi] = \int \mathrm{d}t \mathrm{d}^{D-1}x \, \mathcal{L}(\phi, \partial_\mu \phi).$$

We can identify Z as the generating functional of Green's functions, but it is mathematically hard to give a rigorous meaning to this object due to the strongly oscillating behavior of the complex exponential integrand. To resolve this normalization problem, one typically has to regularize the integral by introducing a small parameter $\epsilon \to 0^+$. A more powerful way to deal with this is to perform a **Wick rotation**:[1] by rotating to imaginary time, the oscillating phase becomes a real damping exponential, making the integral strictly analogous to a statistical mechanics partition function.

Once this setup is generalized, we can express any n-point correlation function using this path integral formulation:

$$\langle \phi(x_1) \cdots \phi(x_n) \rangle = \frac{\int \mathcal{D}\phi \, \phi(x_1) \cdots \phi(x_n) e^{iS[\phi]}}{\int \mathcal{D}\phi \, e^{iS[\phi]}}. \tag{1.1.3}$$

1.1.3 | Wick Rotation

The oscillatory nature of the path integral makes it difficult to define rigorously. For this reason, it is usual to perform a Wick rotation:

$$t \to -i\tau, \implies \begin{cases} \mathrm{d}t \to -i\mathrm{d}\tau, \\ \partial_t \to i\partial_\tau. \end{cases} \tag{1.1.4}$$

In the example of a scalar field considered above, the Lagranian density becomes, after the Wick rotation

$$\mathcal{L} = -\frac{1}{2}\partial_\mu \phi \partial^\mu \phi - V(\phi) = -\mathcal{H},$$

where now the metric is Euclidean $g_{\mu\nu} = \delta_{\mu\nu}$ with the index μ running from 1 to D and setting $\tau = x_D$. Thus, at this point, the action is conveniently substituted by the so called **Euclidean action**

$$S = iS_E, \quad S_E[\phi] = \int \mathrm{d}^D x \left[\frac{1}{2}(\partial_\mu \phi)^2 + V(\phi)\right],$$

or, more generally

$$S_E[\phi] = \int \mathrm{d}^D x \, \mathcal{H}(\phi, \nabla\phi). \tag{1.1.5}$$

The possibility of doing this analytic continuation to imaginary times can easily be established in perturbative QFT. Indeed in the perturbative regime the most general Green function

$$G(x_1, \ldots, x_n) = \int \prod_{i=1}^{n} \frac{\mathrm{d}^D p_i}{(2\pi)^D} e^{i \sum_{i=1}^{n} p_i \cdot x_i} \tilde{G}(p_1, \ldots, p_n),$$

[1] Named after Gian Carlo Wick, an Italian physicist of the historical Via Panisperna group.

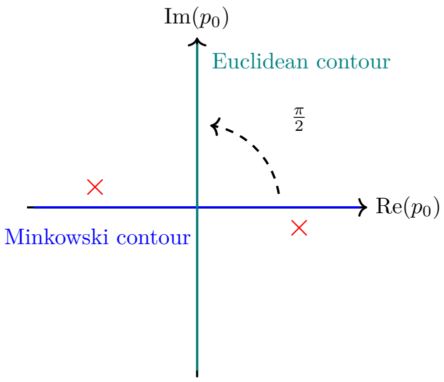
is formed by sums of Feynman diagrams, built with vertices and propagators of the free theory. This latter can be obtained by putting $V(\phi) = 0$. In Minkowski spacetime, giving meaning to the functional in (1.1.2), we add a term of the form

$$\frac{i\epsilon}{2}\phi^2$$

to the action (with ϵ arbitrarily small) and we obtain for the free propagator in momentum space (using units with $\hbar = 1$)

$$\frac{i}{p_0^2 - |\mathbf{p}|^2 - m^2 + i\epsilon}.$$

In the complex plane of $p_0 = \frac{E}{c}$, such propagator has a pole just above the negative part and another just below the positive part of the real axis.



Because of this specific pole structure, we can simultaneously rotate the integration contours of all propagators within a Feynman diagram by an angle of $\pi/2$ counter-clockwise. This procedure safely avoids the singularities, allowing for a rigorous analytic continuation of the Green's functions from the real (Minkowski) axis to the imaginary (Euclidean) axis.

Figure 1.1: Representation of the Wick rotation in the complex p_0 plane.

Thus the analytic continuation of the Green function to imaginary times builds corresponding Euclidean Green function:

$$G(x_1, \dots, x_n) \to G(\xi t_1, \mathbf{x}_1, \dots, \xi t_n, \mathbf{x}_n), \quad \xi = \rho e^{i\theta},$$

given by the usual Green's function of the Euclidean theory, with $\rho > 0$ and $\theta \in [0, \pi]$ and all the times t_i replaced by ρt_i. This function is analytical in the half plane $\mathrm{Im}(\rho) < 0$ and the original Green functions are recovered for $\rho \to 1$. Introducing the new integration variables $p_i^0 \to \frac{p_i^0}{\rho}$ in the particular case of $\rho = -1$ we get

$$G(-it_1, \mathbf{x}_1, \dots, -it_n, \mathbf{x}_n) \propto \int \prod_{i=1}^{n} \frac{\mathrm{d}^D p_i}{(2\pi)^D} e^{-\sum_{i=1}^{n} p_i \cdot x_i} \tilde{G}(p_1, \dots, p_n),$$

Therefore, in the perturbative expansion the results of the euclidean theory are analytically connected to those in the Minkowski spacetime. Practically, in many cases the results obtained in the euclidean space can be directly used to extract physical information and it is pointless to explicitly continue analytically to the Minkowski spacetime.

There are rigorous theorems as the Osterwalder-Schrader positivity condition, on the validity of such analytic continuation even in non-perturbative regime for almost all QFT of physical interest (with the remarkable exception of Einstein gravity). In the following we shall always assume that the theories we deal with admit the analytic continuation of Wick to an euclidean metric.

From now on, we will always refer to the Euclidean version of QFT, which is much more mathematically well defined and easier to handle, especially in the non-perturbative regime. We will drop the subscript E for simplicity, but it is important to keep in mind that we are working in Euclidean space.

1.2 | Connection with Statistical Mechanics

The great advantage of Wick rotation and of the redefinition of QFT on euclidean space consists in the link we can establish between field theory and statistical mechanics. The Euclidean path integral has the same structure as a partition function:

$$Z = \sum_{\text{conf.}} e^{-\beta H}.$$

This single powerful observation leads to the following correspondence:

QFT	Statistical Mechanics
$\phi(x)$	LG Field or microscopical conf.
$S_E[\phi]$	Hamiltonian H
Z	Partition function
$\langle \phi(x_1) \cdots \phi(x_n) \rangle$	Correlation functions

1.2.1 | Generating Functionals

The generating fuctional of Green functions is defined from the partition function Z by introducing a source $J(x)$, similar to coupling a statistical model to an external magnetic field:

$$Z[J] = \int \mathcal{D}\phi \, e^{-S_E[\phi] + \int \mathrm{d}^D x J(x)\phi(x)}. \tag{1.2.1}$$

It is called generating functional since all of the n-point correlation functions are generated by its successive functional derivatives

$$\langle \phi(x_1) \cdots \phi(x_n) \rangle = \left. \frac{\delta^n Z[J]}{\delta J(x_1) \cdots \delta J(x_n)} \right|_{J=0}.$$

This generation includes also disconnected functions. What is physically relevant is the set of **connected n-point functions**, generated by the functional:

$$W[J] = \log Z[J]. \tag{1.2.2}$$

Being the logarithm of a partition function, it looks like the free energy in Statistical Mechanics: it is not a surprise that it generates connected correlation functions, since the free energy is the generating function of connected correlation functions in Statistical Mechanics.

1.2.2 | Effective Action

In particular the one point function, or expectation value, of a field, is called classical field in QFT. The name is inspired by **Ehrenfest theorem**: *the average of an operator evoleves in time with the classical equations of motion*, i.e.

$$\phi_{cl}(x) = \frac{\delta W}{\delta J(x)} = \langle \phi(x) \rangle.$$

This definition of classical field has, as analog is Statistical Mechanics: the **order parameter**. The effective action is the Legendre transform of the generating functional of connected correlation functions, defined as

$$\Gamma[\phi_c] = W[J] - \int \mathrm{d}^D x J(x) \phi_c(x).$$

It plays a role analogous to the **Gibbs free energy** in statistical mechanics. In QFT $\Gamma[\phi_c]$ is the generating function of the so-called *1-particle irreducible graphs*, which are very important in the Feynman diagrams perturbative expansion and in its partial resummation through the Dyson equation. The conclusion is that the equivalence of Statistical Mechanics and (Wick rotated) QFT is on a solid ground and shows the wide generality of QFT as a mathematical tool for the systems with an infinite numebr of degrees of freedom, being able to describe a large range of models in many areas of physics.

1.2.3 | Classical Dimensions and Relevance of Operators

Consider a scalar field theory, whose Euclidean action is given by the integral of the Lagrangian density

$$S = \int \mathrm{d}^D x \left[\frac{1}{2}(\partial_\mu \phi)^2 + \sum_n g_n \phi^n \right]. \tag{1.2.3}$$

If we perform dimensional analysis, knowing that with our units the action has dimension $[S] = 0$, we can infer the dimension of the field ϕ from the kinetic term:

$$[\phi] = \frac{D-2}{2},$$

while the dimension of the coupling can be inferred from the potential term:

$$[g_n] = D - n\frac{D-2}{2}.$$

From the **renormalization group analysis**, we know that the fields in the potential are:

- $[g_n] > 0$ *relevant*: they drive the theory out of the fixed point of RG (which corresponds to a critical point). They do not converge in general so we need to renormalize them.
- $[g_n] = 0$ *marginal*: they can be relevant or irrelevant depending on the sign of their β-function.
- $[g_n] < 0$ *irrelevant*: they are suppressed by the renormalization group flow. They originate non-renormalizable theories, which are not predictive and therefore not physically viable.

Thinking of the potential as perturbation of a Gaussian model (free massless boson), we can see that the irrelevant perturbations leave us on the same critical surface and the fluctuations become, in most cases, uncontrollably wild. The perturbation series by these sort of operators turns out to be non-renormalizable, i.e. plugged by infinities that cannot be cured by some regularization procedure.

When an operator is classified as *marginal* ($[g_n] = 0$), it does not induce a Renormalization Group (RG) flow at first order in perturbation theory. Consequently, the system appears to remain precisely at the critical fixed point. One might then ask: does adding such a marginal perturbation change the physical properties of the theory at all? While it might induce finite shifts in parameters such as the mass (up to a suitable renormalization procedure), understanding its true dynamical effect requires going beyond the linear approximation.

To do so, one must examine higher-order quantum corrections by computing the β-function. If the second-order (or higher) contributions cause the coupling to grow, the operator dynamically breaks scale invariance and is classified as *marginally relevant*.

However, in some special cases, the β-function continues to vanish exactly at all orders in perturbation theory. Such fields are called *truly* (or *exactly*) *marginal* operators. When a theory possesses a truly marginal operator, varying its continuous coupling does not destroy conformal invariance but rather smoothly deforms the theory into a new, physically distinct CFT. In this scenario, instead of an isolated critical point, one finds a continuous manifold (a line or surface) of fixed points, continuously parameterized by the marginal coupling.

The "good" theories are the renormalizable ones, where the dominating field in the potential is relevant, driving the theory out of the fixed point of RG (which corresponds to a critical point) towards a RG flow to some *"infrared" destiny* (critical or trivial). So what field theorists call renormalizability of a theory corresponds, in Statistical Mechanics, adopting the RG language, to relevant perturbations.

If $V(\phi)$ is a polynomial whose term with maximal power is ϕ^n, there exists a space-time dimension

$$D_{crit} = \frac{2n}{n-2}$$

called **upper critical dimension** such that, for $D > D_{crit}$, $[g_n]$ is irrelevant or marginal and the theory becomes non-renormalizable (indeed we derived the critical dimension by imposing $[g_n] = 0$, i.e. it is marginal). This puts a bound on which theories one can be renormalizable at a given spacetime dimension. For a single scalar theory like (1.2.3) we have:

Dimension	Max. field in the potential
$D = 4$	$n = 4 \mapsto \phi^4$
$D = 3$	$n = 6 \mapsto \phi^6$
$D = 2$	any power ($n = \infty$)

In $D = 2$ a scalar field admits any sort of potential, meaning we could add infinitely many relevant perturbations to the free massless boson theory, which is the simplest CFT in two dimensions.

As an example we cite the celebrated *Sine-Gordon theory*

$$S = \int \mathrm{d}^2 x \left(\frac{1}{2} (\partial_\mu \phi)^2 - \frac{m^2}{\beta^2} \cos \beta\phi \right)$$

a very important theory in many aspects, ranging from soliton theory to being the prototypical example of integrable QFT. It is called Sine-Gordon because of the cosine potential: the equations of motion are very similar to Klein-Gordon equation, but with a sine non-linearity

$$\propto \sin\phi.$$

The Sine-Gordon theory is renormalizable in two dimensions, but it becomes non-renormalizable in higher dimensions due to the presence of the cosine potential, which can be expanded into an infinite series of powers of ϕ.

1.3 | Conformal Symmetry in Field Theories

At a critical point, the correlation length diverges:

$$\xi \to \infty.$$

In the language of Quantum Field Theory (QFT), the inverse of the correlation length corresponds to the mass of the lightest excitation in the spectrum ($m \sim \xi^{-1}$), which defines the **energy gap** above the vacuum. When $\xi \to \infty$, this mass gap vanishes, resulting in a *gapless* or *massless* theory (analogous to Quantum Electrodynamics, where the fundamental photon is massless). Since a non-zero mass introduces a fundamental length or energy scale, a massless theory cannot distinguish between different observation scales. Therefore, a diverging correlation length inherently implies that the theory is *scale invariant*.

As a consequence, the system becomes invariant under global scale transformations of the spacetime coordinates (including time):

$$x^\mu \to \lambda x^\mu.$$

Under this rescaling, the fundamental objects of the theory (the local fields, regardless of their scalar, vector, or tensor nature) transform according to their **scaling dimension** Δ:

$$\phi(x) \to \lambda^{-\Delta}\phi(\lambda x).$$

In general, the space of all possible fields can be thought of as a vector space. It is always possible to select a preferred basis of fields such that every element has a definite scaling dimension. Any arbitrary field can then be expressed as a linear combination of these basis fields, a procedure conceptually identical to finding the eigenoperators of the Renormalization Group (RG) flow.

Away from criticality, correlation functions typically exhibit an exponential decay governed by the correlation length, $\sim e^{-|x|/\xi}$. However, at a critical point ($\xi \to \infty$), this exponential suppression factor evaluates to 1. Consequently, the scale invariance rigidly dictates that correlation functions must exhibit a pure power-law behavior:

$$\langle \phi(x)\phi(0)\rangle \sim \frac{1}{|x|^{2\Delta}}.$$

It is therefore very important to deepen our knowledge of scale transformations. As we shall see, a crucial result (the classical Polyakov theorem, a powerful extension of Noether's theorem) states that, under quite general assumptions, scale invariance in field theory is almost always enhanced to a much larger symmetry: *conformal invariance*. Thus, the first necessary step is to formally understand what conformal invariance is and how it governs the theory.

1.3.1 | Conformal Transformations

General coordinate transformations $x \to x'$ change the metric tensor as

$$g'_{\mu\nu}(x') = \frac{\partial x'^\lambda}{\partial x^\mu}\frac{\partial x'^\sigma}{\partial x^\nu} g_{\lambda\sigma}(x).$$

Conformal transformations are coordinate transformations of type

$$\frac{\partial x'^\lambda}{\partial x^\mu} = \Omega(x) g^{\lambda\mu},$$

preserving the metric up to a local rescaling:

$$g'_{\mu\nu}(x') = \Omega^2(x) g_{\mu\nu}(x), \quad \Omega^2(x) = e^{f(x)} > 0.$$

Physically, this represents a strictly *local* rescaling of the coordinates. Unlike a global scale transformation that stretches the entire space uniformly, a conformal transformation allows the scale factor $\Omega(x)$ to vary from point to point. Geometrically, if one imagines a coordinate grid, this operation allows you to fix a point and locally "inflate" or "deflate" the grid around it. The coordinates change scale in different ways at different positions in space, but without arbitrarily mixing the coordinate axes in a way that would alter the angles between intersecting lines.

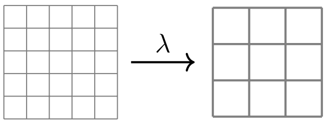


Figure 1.2: Global scale transformation: the grid is scaled uniformly at every point by a constant factor λ.

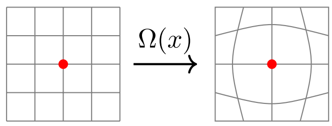


Figure 1.3: Local conformal transformation: the grid undergoes a local "inflation" determined by the position-dependent factor $\Omega(x)$.

This mechanism becomes particularly interesting when applied to a spacetime characterized by a **flat metric**, where the original $g_{\mu\nu}$ is a constant matrix independent of the position x. In a D-dimensional flat spacetime with metric

$$g_{\mu\nu} = \begin{cases} \delta_{\mu\nu} = & \mathrm{diag}(1, 1, \dots, 1) \text{ if Euclidean,} \\ \eta_{\mu\nu} = & \mathrm{diag}(-1, 1, \dots, 1) \text{ if Minkowskian,} \end{cases}$$

the **conformal condition** can be explicitly written as:

$$\partial_\mu x'^\rho \partial_\nu x'^\sigma g_{\rho\sigma} = \Omega^2(x) g_{\mu\nu}. \tag{1.3.1}$$

Thus if we have already Wick rotated to Euclidean signature, we will be working with the Euclidean metric $\delta_{\mu\nu}$, if not, we will be working with the Minkowski metric $\eta_{\mu\nu}$. In any case, the conformal condition can be expressed as shown above, where $g_{\mu\nu}$ is the appropriate metric for the signature we are working with.

1.3.2 | Conformal Killing Equation

Since we are interested in finding the Lie algebra of this transformations group, thus it is convenient to consider infinitesimal transformations. For an infinitesimal transformation we can write

$$x'^\mu = x^\mu + \epsilon^\mu(x), \quad \Omega^2(x) = 1 + f(x) + \dots$$

and, since $\Omega^2(x)$ is a positive function, we chose to write it as the exponential of another function $f(x)$ (this is just a convenient parametrization, not a restriction). Thus we can write

$$\Omega^2(x) = e^{f(x)} \simeq 1 + f(x) + \dots$$

Taking the conformal condition (1.3.1) and inserting the infinitesimal transformation (recognizing that the r.h.s. is the metric in the new coordinates, i.e. $g'_{\mu\nu}$), we get

$$g'_{\mu\nu} = g_{\mu\nu} - (\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu),$$

and now expanding $g'_{\mu\nu} = e^{f(x)} g_{\mu\nu}$ to first order in ϵ, we get

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = f(x) g_{\mu\nu}.$$

We now take the trace of both members, multiplying by $g^{\mu\nu}$ (saturting the indices), getting to

$$f(x) = \frac{2}{D} \partial_\sigma \epsilon^\sigma(x),$$

which implies that the function $f(x)$ is determined by the divergence of $\epsilon^\mu(x)$. Inserting this back into the previous equation we get

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = \frac{2}{D} (\partial \cdot \epsilon) g_{\mu\nu}. \tag{1.3.2}$$

This is the conformal Killing equation, which is the fundamental equation defining the conformal Killing vectors $\epsilon^\mu(x)$ that generate the conformal transformations. The solutions to this equation will give us the explicit form of the infinitesimal conformal transformations in flat spacetime.

Solution of Conformal Killing Equation

To explicitly solve the conformal Killing equation, one can apply a standard tensorial manipulation. By taking a further derivative of both sides, one obtains an equation with three spacetime indices. By cyclically permuting these indices to generate three distinct equations (adding the first two and subtracting the third) one can isolate the second derivatives of the displacement vector $\epsilon(x)$.

After performing these algebraic steps, one finds that for spacetime dimensions $D > 2$, the second derivatives of the scalar function $f(x)$ must strictly vanish ($\partial_\mu \partial_\nu f = 0$). This implies that $f(x)$ can be at most a linear function of the coordinates. Since $f(x)$ is defined through the first derivative of $\epsilon(x)$, it naturally follows that the third derivatives of $\epsilon(x)$ must also vanish. Consequently, the components of $\epsilon^\mu(x)$ are restricted to be at most *quadratic* polynomials in the coordinates.

Indeed, for $D > 2$ the general solution for $\epsilon(x)$ is precisely quadratic in x^μ, and it takes the exact form:

$$\epsilon^\mu(x) = a^\mu + \omega^\mu{}_\nu x^\nu + \lambda x^\mu + (b \cdot x) x^\mu - \frac{1}{2} x^2 b^\mu.$$

If we now restrict the form of the solution, we can get some informations about the structure of the conformal group. In particular, we can identify four different types of transformations, based on the parameters appearing in the solution:

- **Translations** $x'^\mu = x^\mu + a^\mu$: these are the simplest transformations, corresponding to a uniform shift of the coordinates by a constant vector a^μ.

- **Rotations** $x'^\mu = x^\mu + \omega^\mu{}_\nu x^\nu$: this comes from the fact that we can always decompose the transformationmatrix into a symmetric and an antisymmetric part, and the latter corresponds to rotations.

- **Dilatations** $x'^\mu = (1 + \lambda) x^\mu$: this corresponds to an infinitesimal scaling of the coordinates (inflation or deflation), and it is indeed proportional to the coordinates themselves. This is the symmetric part we cited before, which is not a rotation but a scaling. It is an important transformation, since it is related to the scale invariance of the theory, and it is peculiar among the conformal transformations.

- **Special conformal transformations** $x'^\mu = x^\mu + 2(b \cdot x)x^\mu - b^\mu x^2$: these are the most non-trivial transformations, featuring a quadratic dependence on the coordinates. It is a D-parameter family of transformations (which reduces to 4 parameters in $D = 4$ dimensions), parameterized by the constant vector b^μ. Geometrically, this complex non-linear transformation can always be gracefully decomposed into a sequence of three fundamental operations: an inversion ($x^\mu \to x^\mu/x^2$), followed by a constant translation by the vector b^μ, and finally a second inversion.

The first two types of transformations (translations and rotations) are the usual Poincaré transformations, which are symmetries of any relativistic field theory. The last two types (dilatations and special conformal transformations) are additional symmetries that arise in scale-invariant theories, and they are responsible for the enhancement from scale invariance to full conformal invariance.

We already know the form of the generators of translations and rotations, since they are the usual ones, and they give us meaningful information about the structure of the theory of a Poincaré invariant QFT. We are here considering systems at criticality, which are scale invariant, thus it seems natural to expect that the generators of dilatations will also give us important information about the structure of the theory at criticality. Finally, the generators of special conformal transformations are the most non-trivial ones, and they are responsible for the full conformal symmetry of the theory, even if now we do not have a clear physical intuition about their meaning, but they are crucial for the structure of the conformal group and for the constraints that conformal invariance imposes on the theory.

Considering the application of these transformation on generic functions of the coordinates, we can obtain the form of the **generators**

- **Translations**: $P_\mu = -i\partial_\mu$, which is the usual generator of translations in field theory, corresponding to the momentum operator.
- **Rotations**: $L_{\mu\nu} = i\epsilon_{\mu\nu\rho\sigma}x^\rho\partial^\sigma$, which is the usual generator of rotations (angular momentum) in field theory, corresponding to the angular momentum operator.
- **Dilatations**: $D = -ix_\mu\partial^\mu$, which is the generator of dilatations (scale transformations) in field theory, and it is related to the scaling dimension of fields and operators in the theory.
- **Special conformal transformations**: $K_\mu = -i(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu)$, which is the generator of special conformal transformations in field theory; we do not have a simple physical interpretation for this generator, but its importance will become clear when we discuss the structure of the conformal group.

1.3.3 | Conformal Algebra

The generators of the first two types satisfy the usual roto-translation (Poincaré) algebra:

$$\begin{cases} [P_\mu, P_\nu] = 0, \\ [P_\rho, L_{\mu\nu}] = \delta_{\rho\mu}P_\nu - \delta_{\rho\nu}P_\mu, \\ [L_{\mu\nu}, L_{\rho\sigma}] = \delta_{\nu\rho}L_{\mu\sigma} + \delta_{\mu\sigma}L_{\nu\rho} - \delta_{\mu\rho}L_{\nu\sigma} - \delta_{\nu\sigma}L_{\mu\rho}, \end{cases} \tag{1.3.3}$$

but now also the commutations with the dilatation operator D have to be added:

$$\begin{cases} [D, P_\mu] = P_\mu, \\ [D, L_{\mu\nu}] = 0, \end{cases} \tag{1.3.4}$$

and finally those involving the special conformal transformations K_μ:

$$\begin{cases} [K_\mu, K_\nu] = 0, \\ [K_\rho, L_{\mu\nu}] = \delta_{\rho\mu} K_\nu - \delta_{\rho\nu} K_\mu, \\ [K_\mu, P_\nu] = 2(\delta_{\mu\nu} D - L_{\mu\nu}), \\ [D, K_\mu] = -K_\mu. \end{cases} \tag{1.3.5}$$

Note, in particular, that the rotation (spin) operator and the dilatation one commute, which means that we can find a basis of local fields that diagonalizes both operators simultaneously (this should not come as a surprise, since this two transformations are built from the symmetryc and antisymmetric parts of the general infinitesimal transformations).

This algebra is isomorphic to $\mathfrak{so}(D+1,1)$ and the corresponding conformal group is $SO(D+1,1)$ in euclidean spacetime (it would become $SO(D,2)$ if a Minkowski metric is adopted instead). In any case, it is a group with $(D+1)(D+2)/2$ parameters. Physically, every parameter of the symmetry group translates into a mathematical constraint that the Lagrangian (or the correlation functions) of the theory must satisfy. As a general rule, the larger the symmetry group, the tighter the constraints imposed on the theory.

For example, in $D = 4$ dimensions, requiring standard Poincaré invariance imposes 10 constraints on the theory (corresponding to the 10 generators of spacetime translations and Lorentz rotations). However, if we upgrade our requirement to full conformal invariance, the parameters of the conformal group become 15. Therefore, the Lagrangian is subjected to 15 strict constraints: the original 10 from the Poincaré subgroup, plus 5 additional ones arising directly from the extended conformal symmetry (1 from dilatations and 4 from special conformal transformations). This highly constrained framework is precisely what restricts the possible forms of the theory and makes CFTs so mathematically powerful.

Little Group

Generators can be realized in various (irreducible) representations of the conformal group. The **little group** is the subgroup of the conformal group that leaves invariant a specific point, conventionally chosen as the origin of coordinates $x = 0$ (any point can be used as such, due to translational invariance).

We can classify the behavior of generic fields by studying how they transform exactly at this origin. For any other point in spacetime, the transformation rules can be deduced simply by applying a spatial translation to shift the field from 0 to x.

Let us illustrate this mechanism by considering rotations. When an infinitesimal Lorentz transformation $L_{\mu\nu}$ acts on a field evaluated at $x = 0$, it operates strictly on its internal indices:

$$L_{\mu\nu}\phi(0) \coloneqq S_{\mu\nu}\phi(0).$$

The operator $S_{\mu\nu}$ represents the **intrinsic spin** matrix of the field ϕ. Physically, a general coordinate transformation changes not only the spacetime coordinates themselves but also the internal structure of the object evaluated at those coordinates (the components in such basis). If the field is a simple scalar (a single function), this extra intrinsic term vanishes ($S_{\mu\nu} = 0$). However, if the field is a vector or a tensor (comprising multiple component functions), $S_{\mu\nu}$ actively mixes these components, representing the total internal angular momentum.

Analogously, we can define the action of the dilatation operator at the origin:

$$D\phi(0) \coloneqq \tilde{D}\phi(0).$$

To find out what these operators do to a field at an arbitrary point $x \neq 0$, we must conjugate them using the finite translation operator, $e^{ix^\rho P_\rho}$, which is obtained by exponentiating the infinitesimal translation generator P_ρ. Applying this transformation yields:

$$e^{ix^\rho P_\rho} L_{\mu\nu} e^{-ix^\rho P_\rho} = S_{\mu\nu} + i\epsilon_{\mu\nu\rho\sigma} x^\rho \partial^\sigma,$$

and

$$e^{ix^\rho P_\rho} D e^{-ix^\rho P_\rho} = \tilde{D} - ix_\mu \partial^\mu.$$

To explicitly compute these conjugations, we make use of the Baker-Campbell-Hausdorff formula:

$$e^{-A} B e^{A} = B + [B, A] + \frac{1}{2!}[[B, A], A] + \frac{1}{3!}[[[B, A], A], A] + \dots$$

These relations mathematically demonstrate the split of the total angular momentum into an orbital part $(-x_\mu P_\nu + x_\nu P_\mu)$ and an intrinsic spin part $(S_{\mu\nu})$, as well as the split of dilatations into a spacetime coordinate scaling $(x_\mu P^\mu)$ and an intrinsic scaling dimension $(\tilde{D})$:

$$\begin{aligned} L_{\mu\nu} &= S_{\mu\nu} + i\epsilon_{\mu\nu\rho\sigma} x^\rho \partial^\sigma, \\ D &= \tilde{D} - ix_\mu \partial^\mu. \end{aligned}$$

Of course, being strictly internal operators evaluated at the origin, $S_{\mu\nu}$ and $\tilde{D}$ also commute.

The same can also be done for the special conformal transformations, resulting in

$$K_\mu = -i(2x_\mu x^\nu \partial_\nu - x^2 \partial_\mu) + \tilde{K}_\mu,$$

where one could try to derive the explicit form of the internal operator $\tilde{K}_\mu$ by applying the same procedure, but it is not necessary for our purposes.

1.4 | Stress-Energy Tensor and Conformal Invariance

In this section we discuss the classical relation between scale invariance and conformal invariance, following the standard Noether approach.

1.4.1 | Noether Current for Scale Transformations

What are the Noether's currents when we apply the conformal group as the symmetry group of a classical field theory? To answer this, we apply these transformations to the Lagrangian and define an action.

Consider a local field theory in D Euclidean dimensions with action

$$S = \int d^D x \mathcal{L}(\phi_a, \partial_\mu \phi_a).$$

where the index a labels the various fields. Any continuous transformation involving coordinates and fields that leaves the action invariant ($\delta S = 0$) is a symmetry of the theory. Such transformations can include purely internal symmetries (which do not affect spacetime coordinates) or spacetime transformations like rotations, which can also induce an internal transformation of the field itself (its intrinsic spin).

Note that the Lagrangian density $\mathcal{L}$, under such a transformation, needs not to be strictly invariant; it can change by a total derivative

$$\mathcal{L} \to \mathcal{L} + \partial_\mu K^\mu,$$

of a quantity K_μ and still ensure that $\delta S = 0$. Under these circumstances, Noether's theorem dictates that for any continuous symmetry (parameterized by elements of a continuous group lying on a manifold whose dimension is independent of spacetime) there exists a corresponding current

$$j_\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta\phi_a - K_\mu.$$

This current turns out to be a conserved current, satisfying the continuity equation

$$\partial_\mu j^\mu = 0,$$

which in turn implies the conservation of the corresponding charge

$$Q = \int j^0 d^{D-1} x.$$

In the case of space-time translation invariance, we define the canonical **stress-energy tensor**. It carries two indices: one (μ) acting as the standard current index, and a second (ν) acting as a sort of internal index labeling the specific translation parameter. The canonical tensor is defined as:

$$T_{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial_\nu \phi_a - \delta_{\mu\nu} \mathcal{L}.$$

If the theory is invariant under space-time translations, $T_{\mu\nu}$ is conserved:

$$\partial^\mu T_{\mu\nu} = 0.$$

We now ask: what is the variation of the field when we apply a dilatation? Under an infinitesimal scale transformation

$$x^\mu \to x'^\mu = (1+\epsilon) x^\mu,$$

a field $\phi_a(x)$ of classical scaling dimension Δ_a transforms as

$$\delta\phi_a(x) = -\epsilon(x^\mu\partial_\mu + \Delta_a)\phi_a(x).$$

The associated Noether current is the **dilatation current** (j_μ of the dilatation), given by:

$$D_\mu = x^\nu T_{\mu\nu} - V_\mu,$$

where V_μ is the so-called **virial current**. Its divergence is

$$\partial^\mu D_\mu = T^\mu_\mu - \partial^\mu V_\mu.$$

Therefore, requiring scale invariance ($\partial^\mu D_\mu = 0$) implies:

$$T^\mu_\mu = \partial^\mu V_\mu. \tag{1.4.1}$$

This is the general Noether condition for scale invariance: the trace of the energy-momentum tensor does not need to vanish identically, but only up to a total divergence.

While this analysis can be done in a huge variety of theories in any dimension, it is important to note that the most general variation of the action under an arbitrary conformal transformation will ultimately depend on this trace property.

1.4.2 | From Scale Invariance to Conformal Invariance

We now ask under which conditions scale invariance implies full conformal invariance. For an infinitesimal conformal transformation we may write

$$x^\mu \to x^\mu + \epsilon^\mu(x).$$

The variation of the action is controlled by the energy-momentum tensor:

$$\delta S = -\int d^D x\ T^{\mu\nu}\partial_\mu\epsilon_\nu = -\frac{1}{2}\int d^D x\ T^{\mu\nu}(\partial_\mu\epsilon_\nu + \partial_\nu\epsilon_\mu)$$

where we assume that the stress-energy tensor is symmetric or it has been symmetrized à la Belinfante (as shown in section 1.4.3). If $\epsilon^\mu(x)$ is a conformal Killing vector, then

$$\partial_\mu\epsilon_\nu + \partial_\nu\epsilon_\mu = \frac{2}{D}(\partial\cdot\epsilon)\delta_{\mu\nu},$$

and therefore

$$\delta S = -\frac{1}{D}\int d^D x(\partial\cdot\epsilon)T^\mu_\mu.$$

Hence, if the trace vanishes

$${T_\mu}^\mu = 0,$$

then the action is invariant under all conformal transformations, since its variation vanishes for any conformal Killing vector $\epsilon^\mu(x)$. We conclude that <u>a sufficient condition for conformal invariance is the existence of a symmetric, conserved and traceless energy-momentum tensor</u>.

1.4.3 | Belinfante Energy Momentum Tensor

Equation (1.4.1) shows that in a scale invariant theory the trace is a divergence. The next question is whether one can redefine the energy-momentum tensor so as to make the trace vanish identically. Suppose that the virial current is itself a divergence,

$$V_\mu = \partial^\nu \sigma_{\nu\mu},$$

or, more specifically, that it can be written in a form allowing an improvement term. Then one can define an improved (Belinfante) energy-momentum tensor

$$T^{Bel}_{\mu\nu} = T_{\mu\nu} + \partial^\rho B_{\rho\mu\nu}, \tag{1.4.2}$$

where $B_{\rho\mu\nu}$ is antisymmetric in the first two indices,

$$B_{\rho\mu\nu} = -B_{\mu\rho\nu}.$$

so that conservation is preserved:

$$\partial^\mu \Theta_{\mu\nu} = 0.$$

With a suitable choice of the improvement term (CCJ = Callan-Coleman-Jackiw), one may achieve

$$\Theta^\mu{}_\mu = 0.$$

In that case the theory is not only scale invariant, but conformally invariant. The CCJ choice cannot always be implementable, leading to the fact that for $D > 2$ there can be some (very special) cases of theories that are scale invariant but their energy-momentum tensor cannot be made traceless. In $D = 2$ however, this is always possible.

1.4.4 | Polyakov theorem

We can now state the classical result in the form relevant for our purposes.

Theorem 1.1 (Classical Polyakov theorem)**.** *Consider a local D-dimensional field theory admitting a symmetric conserved energy-momentum tensor. If the theory is scale invariant and if the trace of the energy-momentum tensor can be improved, so that there exists an improved tensor $\Theta_{\mu\nu}$ satisfying*

$$\partial^\mu \Theta_{\mu\nu} = 0, \quad \Theta_{\mu\nu} = \Theta_{\nu\mu}, \quad \Theta^\mu{}_\mu = 0,$$

then the theory is conformally invariant.

Proof. For an infinitesimal conformal transformation generated by a conformal Killing vector $\epsilon^\mu(x)$, the variation of the action is

$$\delta S = -\frac{1}{2}\int d^D x \Theta^{\mu\nu}(\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu).$$

Since

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = \frac{2}{D}(\partial \cdot \epsilon) g_{\mu\nu}$$

one gets

$$\delta S = -\frac{1}{D}\int d^D x (\partial \cdot \epsilon)\Theta^\mu_\mu$$

But $\Theta^\mu_\mu = 0$, hence

$$\delta S = 0.$$

Therefore the action is invariant under all infinitesimal conformal transformations. This proves classical conformal invariance of any QFT having a traceless improved energy-momentum tensor.

□

Comments

The theorem should be understood as follows:

- scale invariance implies that the trace is a total divergence;
- if this divergence can be removed by improving the energy-momentum tensor, then the improved tensor is traceless;
- tracelessness plus conservation is exactly what is needed to generate the full local conformal symmetry.

Thus the passage from scale invariance to conformal invariance is controlled by the trace of the energy-momentum tensor.

> **Remark (Quantum comment).** *At the quantum level the situation is subtler. Even when the classical theory is conformally invariant, quantization may spoil the tracelessness of the energy-momentum tensor through an anomaly. In two dimensions this anomaly is encoded in the central extension of the symmetry algebra and, equivalently, in the non-vanishing central charge appearing in the* TT *operator product expansion. We postpone the quantum discussion to later chapters, where the Virasoro algebra and the central charge will emerge naturally.*

1.5 | Algebra of local fields

Local fields $A(x) \in \mathcal{A}$ are operators in a QFT depending on only one space-time point. This set includes fundamental fields, but also their powers and derivatives. However, they are not freely generated: the algebra of local fields is subject to constraints (for example, equations of motion like $\partial^2\phi = \phi^5$). The set $\mathcal{A}$ is called the algebra of local fields. Indeed:

- Local fields add up and can be multiplied by scalars, so $\mathcal{A}$ has the structure of a vector space. We can select a preferred basis $\{\varphi_n(x)\}$ for this space, such that all fields can be expressed as a linear combination of this basis.
- Usually, we choose a basis of local fields that diagonalizes two commuting operators (since they commute, they can be simultaneously diagonalized):
  - **Dilatation**: $\tilde{D}\varphi_n(0) = \lambda^{d_n}\varphi_n(0)$. The eigenvalue d_n (or Δ_n) is the scale dimension. This is the crucial number we are interested in, as it dictates how the field rescales and ultimately determines the critical exponents that characterize specific universality classes.
  - **Spin**: $S\varphi_n(0) = e^{is_n\theta}\varphi_n(0)$. Local fields can have various spins ($s_n = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$).
- $\varphi_n(x)$ is therefore uniquely characterized by this pair of numbers $\{d_n, s_n\}$.
- All other fields are linear combinations

$$A(x) = \sum_{n=0}^{\infty} a_n\varphi_n(x), \quad \varphi_0 \equiv \mathbb{I},$$

  where the equality here has to be understood in the *weak sense*. This means that two operators are considered equal if they yield the exact same result when inserted into *any* correlation function of the theory. You can have two formally different expressions, but if they do the same job on the full algebra of fields, they represent the same physical field. Formally, for any string of local fields $X = A_1(x_1)\ldots A_n(x_n) \in \mathcal{A}^{\otimes n}$, we have

$$A(x) = B(x) \Leftrightarrow \langle A(x)X(x_1,\ldots,x_n)\rangle = \langle B(x)X(x_1,\ldots,x_n)\rangle.$$

2-point functions of local fields, at criticality, are fixed by roto-translation and scale invariance:

$$\mathcal{D}_{ij}(x_1 - x_2) = \langle\varphi_i(x_1)\varphi_j(x_2)\rangle = \frac{A_{ij}}{|x_1 - x_2|^{d_i+d_j}},$$

where $A_{ij} = A\delta_{ij}$. So they are all zero if $\phi_j \neq \phi_i$. The constant A can be fixed by normalizing the field ϕ_i:

$$\mathcal{D}_{ij}(x_1 - x_2) = \langle\varphi_i(x_1)\varphi_j(x_2)\rangle = \frac{\delta_{ij}}{|x_1 - x_2|^{2d_i}}.$$

Near criticality we have a mass scale $m \sim \xi^{-1}$, i.e. the inverse of the correlation length. The expectation values of local fields (think e.g. to the order parameters) rescale as:

$$\langle\varphi_i(x)\rangle = \mu_i m^{d_i}.$$

At criticality, i.e. in presence of scale invariance and $\xi \to \infty$:

$$\langle\phi_0\rangle = \langle\mathbb{I}\rangle = 1, \quad \langle\phi_i(x)\rangle = 0 \quad (i = 1, 2, \ldots).$$

1.5.1 | Operator Product Expansion

The vector space of local fields is raised to an algebra if a suitable notion of product between fields is established: the **Operator Product Expansion (OPE)**. Any product of fields that are separated by some distance can be expanded as a sum of individual fields. Consider a composite (non-local) field $A(x_1)B(x_2)$. In the limit $x_1 \to x_2$, we expect that it reproduces the properties of a local field $C(x_2)$, which can then be expanded in the basis of local fields. This is conceptually vital to understand objects like ϕ^2, which is rigorously defined as $\phi(x_1)\phi(x_2)$ evaluated as $x_1 \to x_2$. Note that x_1 and x_2 do not need to be infinitesimally close; the OPE holds as long as there are no other field singularities in the middle (i.e., within a certain radius of convergence).

Wilson assumed that for any pair of local fields $A(x_1)$ and $B(x_2)$ the OPE holds:

$$A(x_1)B(x_2) = \sum_{k=0}^{\infty} c_k(x_1, x_2)\varphi_k(x_2).$$

In particular, for our chosen basis:

$$\varphi_p(x_1)\varphi_q(x_2) = \sum_r c^r_{pq}(x_1, x_2)\varphi_r(x_2).$$

The functions $c^r_{pq}(x_1, x_2)$ are called structure functions of the OPE algebra. Because the theory is *translationally invariant*, these functions cannot depend on the absolute coordinates x_1 and x_2, but only on their difference $x_1 - x_2$. Furthermore, *rotational invariance* (Poincaré invariance in the Wick rotated Euclidean description) implies they depend only on the distance between the two points (the modulus, not the full vector):

$$c^r_{pq}(|x_1 - x_2|) = c^r_{pq}(|x_{12}|).$$

Scale invariance at criticality further restricts the function:

$$c^r_{pq}(x_1, x_2) = \frac{C^r_{pq}}{|x_{12}|^{d_p + d_q - d_r}}.$$

The numbers C^r_{pq} are called structure constants.

Here and in the following we denote, for short,

$$x_{ij} = x_i - x_j.$$

The set of all scaling dimensions d_n (the spectrum of the theory) together with the structure constants C^r_{pq} form the so-called **CFT data**. These quantities completely define the dynamical content of the theory. If they are known exactly, the theory is considered exactly solved. This is because, by recursively applying the Operator Product Expansion, the product of two fields inside any correlation function can be systematically replaced by a sum over single fields. Through this contraction procedure, any N-point correlation function can be reduced to a combination of $(N-1)$-point functions, and eventually completely decomposed down to known 2-point or 1-point functions.

However, in a generic conformal field theory, these structure constants and scaling dimensions are not known *a priori*. One must rely on sophisticated non-perturbative methods (such as the conformal bootstrap mentioned in the previous chapter) to constrain and compute these numbers.

Furthermore, the physical importance of the OPE is most evident in the short-distance limit, when two fields are brought arbitrarily close to each other ($x_1 \to x_2$). In this limit, the expansion typically exhibits a divergence dictated by the power-law denominator $|x_{12}|^{d_p+d_q-d_r}$. While the OPE naturally contains both regular and singular terms, it is precisely this *singular part* that encodes the most vital algebraic information about the correlation functions and the symmetries of the theory. To fully appreciate how these short-distance singularities dictate the entire structure of the theory, it is highly advantageous to restrict our attention to the incredibly rich case of two dimensions.

2 Conformal Group in Two Dimensions

If the dimension of spacetime is larger than two ($D > 2$), the conformal group is a finite-dimensional Lie group. While it is larger than the standard Poincaré group therefore providing more constraints on the theory, the number of these constraints remains strictly finite. However, in exactly two dimensions, the situation changes drastically and the conformal symmetry becomes infinite-dimensional. The two-dimensional case is thus special because the condition for conformal invariance is much more powerful.

This dramatic enhancement makes CFT_2 exactly solvable in many cases. All these results were established in a seminal breakthrough paper by Belavin, Polyakov and Zamolodchikov in 1984 [3] and later reviewed in Lecture notes by Ginsparg [4] and by Cardy [5] and in books by Di Francesco, Mathieu and Sénéchal [6] and by Mussardo [7] and in many other contributions.

TODO: add references.

To see this, let us work in 2D Euclidean space (if we start from a Minkowski metric, we can always perform a Wick rotation). In a 2D Minkowski spacetime, the natural variables would be the *lightcone coordinates*; in Euclidean space, the analogous operation is to map the 2D Cartesian plane into the complex plane.

Complex Coordinates

We take a new basis by changing coordinates from the real Cartesian coordinates x^1 and x^2 to the complex coordinates z and $\bar{z}$:

$$\begin{cases} z = x^1 + ix^2, \\ \bar{z} = x^1 - ix^2, \end{cases} \implies \begin{cases} x^1 = \dfrac{z+\bar{z}}{2}, \\ x^2 = \dfrac{z-\bar{z}}{2i}. \end{cases}$$

While physically one coordinate is the complex conjugate of the other, it is mathematically very convenient to treat z and $\bar{z}$ as two formally independent variables characterizing a point in $\mathbb{C}^2$.

In these new coordinates, the flat Euclidean line element becomes:

$$\mathrm{d}s^2 = (\mathrm{d}x^1)^2 + (\mathrm{d}x^2)^2 = \mathrm{d}z\mathrm{d}\bar{z},$$

where $\mathrm{d}z\mathrm{d}\bar{z}$ also acts as the area element ($\mathrm{d}^2\mathbf{x}$) up to a constant factor. The corresponding derivatives with respect to the complex coordinates are given by:

$$\begin{cases} \partial_z = \dfrac{1}{2}(\partial_1 - i\partial_2), \\ \partial_{\bar{z}} = \dfrac{1}{2}(\partial_1 + i\partial_2), \end{cases} \implies \begin{cases} \partial_z z = 1, \quad \partial_z \bar{z} = 0, \\ \partial_{\bar{z}} z = 0, \quad \partial_{\bar{z}} \bar{z} = 1. \end{cases}$$

From the line element $\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu$, we can read off the metric tensor in the $(z, \bar{z})$ basis, which takes the off-diagonal form:

$$g = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}.$$

Consequently, any field in our theory can be redefined as a function of these two independent variables in the complex plane:

$$\varphi(x) \to \varphi(z,\, \bar{z}),$$

letting us treat the two-dimensional space as a complex manifold. This choice of coordinates is not just a mathematical convenience; it is crucial for revealing the rich structure of conformal transformations in two dimensions, which are intimately connected to the properties of analytic functions.

With these tools at our disposal, we can now analyze the conformal transformations in two dimensions and their profound implications for the structure of the theory.

2.1 | Conformal Transformations in Two Dimensions

Let us now consider conformal transformations. A coordinate transformation is conformal if it simply rescales the line element by a strictly positive local factor Ω:

$$(\mathrm{d}s')^2 = \Omega(z,\, \bar{z})\mathrm{d}s^2,$$

which means *the transformation is preserving the metric up to a local scaling.* This is a much weaker condition than the isometry condition, which requires the metric to be invariant under the transformation. The conformal condition allows for a much larger class of transformations, as it only requires angles to be preserved locally, while lengths can be rescaled by an arbitrary function.

In complex coordinates, we can achieve this by applying independent transformations to z and $\bar{z}$. Then the idea is to solve the conformal condition locally introducing the following analytic maps:

$$\begin{cases} z \mapsto w = f(z), \\ \bar{z} \mapsto \bar{w} = \bar{f}(\bar{z}), \end{cases} \implies \begin{cases} \mathrm{d}w = f'(z)\mathrm{d}z, \\ \mathrm{d}\bar{w} = \bar{f}'(\bar{z})\mathrm{d}\bar{z}. \end{cases}$$

Here, $f(z)$ is an analytic (holomorphic) function, meaning its derivative $f'(z)$ exists, and $\bar{f}(\bar{z})$ is an anti-holomorphic function. The new line element becomes:

$$(\mathrm{d}s')^2 = \mathrm{d}w\mathrm{d}\bar{w} = f'(z)\bar{f}'(\bar{z})\mathrm{d}z\mathrm{d}\bar{z},$$

which perfectly matches the conformal condition with the scale factor completely factorized into a holomorphic and an anti-holomorphic part:

$$\Omega(z,\, \bar{z}) = f'(z)\bar{f}'(\bar{z}).$$

This reveals a profound result: *any* analytic transformation of the complex plane is a conformal transformation. Geometrically, analytic functions preserve local angles, maintaining the shape of infinitesimally small figures, even though globally they can stretch and severely distort areas.

Since there are infinitely many analytic functions, the conformal group in two dimensions is **infinite-dimensional**. This implies we have infinitely many symmetries and, consequently, infinitely many constraints on the algebra of local fields. When a theory is subjected to infinitely many constraints, it often becomes exactly solvable, allowing us to compute quantities far beyond standard perturbation theory.

2.1.1 | Infinitesimal Conformal Transformations

To rigorously link this back to the conformal Killing equation, consider an infinitesimal conformal map:

$$z \mapsto z + \epsilon(z),$$
$$\bar{z} \mapsto \bar{z} + \bar{\epsilon}(\bar{z}),$$

where $\epsilon(z)$ and $\bar{\epsilon}(\bar{z})$ are infinitesimal holomorphic and anti-holomorphic functions, respectively. We define the displacement vector components as:

$$\epsilon(z) = \epsilon_1 + i\epsilon_2, \quad \bar{\epsilon}(\bar{z}) = \epsilon_1 - i\epsilon_2.$$

By definition, an infinitesimal transformation is conformal if it satisfies the conformal Killing equation. In $D = 2$ dimensions, this equation reads:

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = \frac{2}{2}(\partial \cdot \epsilon)\delta_{\mu\nu} = (\partial_1 \epsilon_1 + \partial_2 \epsilon_2)\delta_{\mu\nu}. \tag{2.1.1}$$

Evaluating this tensor equation component by component ($(\mu,\nu) = (1,1)$, $(2,2)$ and $(1,2)$), we obtain exactly the celebrated **Cauchy-Riemann equations**:

$$\begin{cases} \partial_1\epsilon_1 = \partial_2\epsilon_2, \\ \partial_1\epsilon_2 = -\partial_2\epsilon_1. \end{cases} \tag{2.1.2}$$

These equations are the necessary and sufficient condition for $\epsilon(x^1, x^2) = \epsilon_1 + i\epsilon_2$ to be a complex analytic function $\epsilon(z)$. This confirms that in two dimensions, the set of all conformal transformations coincides precisely with the set of all holomorphic and anti-holomorphic mappings.

Furthermore, if we were to write the Cauchy-Riemann equations in complex coordinates, we would find that they reduce to the simple conditions:

$$\begin{aligned} \partial_{\bar z}\epsilon(z,\bar z) = 0 \quad &\Longrightarrow \epsilon(z,\bar z) = \epsilon(z), \\ \partial_z\bar\epsilon(z,\bar z) = 0 \quad &\Longrightarrow \bar\epsilon(z,\bar z) = \bar\epsilon(\bar z), \end{aligned}$$

which explicitly shows that the holomorphic and anti-holomorphic sectors are completely decoupled, and the conformal transformations can be treated independently in each sector. This factorization is a fundamental structural property of two-dimensional conformal field theory.

Since $\epsilon(z)$ is analytic, it can be expanded in a Laurent series around any point in its domain of analyticity (for simplicity, we can choose the origin $z = 0$) as:[1]

$$\epsilon(z) = \sum_{n=-\infty}^{\infty} \epsilon_n z^{n+1}.$$

This series represents the most general form of such an analytic function and is convergent within its specific annulus of convergence. By isolating the terms of this expansion, we can extract a convenient basis of infinitesimal holomorphic transformations:

$$\epsilon_n(z) = -z^{n+1}, \quad n \in \mathbb{Z},$$

then the generators of the transformations on scalar fields $\varphi(x)$ as

$$\ell_n = -z^{n+1}\partial_z, \quad \bar\ell_n = -\bar z^{n+1}\partial_{\bar z},$$

where the generators for the other sector were derived by the same reasoning, but in the $\bar z$ direction. The generators ℓ_n and $\bar\ell_n$ form a convenient basis for the infinite-dimensional algebra of conformal transformations in two dimensions.

We now compute the commutator $[\ell_n, \ell_m]$ to determine the structure constants of the algebra. By applying the operators to a test function $f(z)$, we obtain:

$$[\ell_n, \ell_m] = (n-m)z^{n+m+1}\partial_z = (n-m)\ell_{n+m} \tag{2.1.3}$$

This relation defines the **Witt algebra**, which is the algebra of infinitesimal conformal transformations in two dimensions. The exact same commutation relation holds for the anti-holomorphic generators $\bar\ell_n$. Because the indices n and m can take any integer value, this forms an infinite-dimensional Lie algebra, which along with the other sector's algebra, constitutes the full symmetry algebra of two-dimensional conformal field theories:

$$[\ell_n, \ell_m] = (n-m)\ell_{n+m}, \quad [\bar\ell_n, \bar\ell_m] = (n-m)\bar\ell_{n+m}, \quad [\ell_n, \bar\ell_m] = 0. \tag{2.1.4}$$

[1] More rigorously, any function holomorphic in an annulus $r < |z| < R$ admits a unique Laurent expansion that converges uniformly on compact subsets of the annulus. The existence of this series is a direct mathematical consequence of holomorphy in that domain, without requiring additional global assumptions about the function's boundedness or divergence.

The Witt algebra serves as the classical precursor to the Virasoro algebra, which represents the quantum version of the 2D conformal algebra. As an interesting historical note, Miguel Virasoro introduced this algebraic structure in 1970 in the context of early string theory (dual resonance models), as a PhD student in Turin. Furthermore, while the global conformal transformations (the Möbius transformations, which we will introduce shortly) had already been known for over a century in classical mathematics, Virasoro's profound contribution was uncovering the physical relevance of this infinite-dimensional algebra.

2.1.2 | Global Conformal Transformations

Not all holomorphic maps are globally well-defined on the Riemann sphere. The globally defined conformal transformations, i.e. those mapping a plane into a plane, are the Möbius transformations:

$$z \mapsto w = \frac{az+b}{cz+d}, \quad ad - bc = 1, \tag{2.1.5}$$

which form a six-parameter family of transformations. These transformations are generated by the three generators ℓ_{-1}, ℓ_0 and ℓ_1 (and their anti-holomorphic counterparts), which correspond to the following specific transformations:

1. $\ell_0 = -z\partial_z$ is the generator of **dilatations**,
2. $\ell_{-1} = -\partial_z$ is the generator of **translations**,
3. $\ell_1 = -z^2\partial_z$ is the generator of **special conformal transformations**.

Since the same applies to the anti-holomorphic sector, we have a total of six generators: the global conformal algebra is generated by

$$\{\ell_{-1}, \ell_0, \ell_1\} \oplus \{\bar{\ell}_{-1}, \bar{\ell}_0, \bar{\ell}_1\}, \tag{2.1.6}$$

which is isomorphic to the algebra of the 2×2 special linear group $\mathfrak{sl}(2, \mathbb{C})$, which is the double cover of the Lorentz group $SO(3,1)$: the conformal group in two dimensions is isomorphic to $\mathrm{SU}(2) \oplus \mathrm{SU}(2)$. The foundamental difference of the two-dimensional case with respect to higher dimensions is that there exist infinitely many local conformal transformations (not mapping plane to plane but changing the geometry of the two-dimensional spacetime), but only a finite number of global conformal transformations.

Conformal Generators and the Poincaré Group

We can also identify the generators of the Poincaré group in two dimensions: the translation generators of the conformal algebra ℓ_{-1} and $\bar{\ell}_{-1}$ give us the energy and the momentum:

$$\begin{cases} H = \ell_{-1} + \bar{\ell}_{-1}, \\ P = i(\ell_{-1} - \bar{\ell}_{-1}), \end{cases}$$

where the Hamiltonian H generates **time translations** and the momentum P generates **spatial translations**. The eigenvalues of these operators correspond to the energy and momentum of states in the theory, respectively.

From the dilatation generator ℓ_0 and its anti-holomorphic counterpart $\bar{\ell}_0$, we can construct the **generator of rotations** (or spin) S and the **generator of scale transformations** D:

$$\begin{cases} S = \ell_0 - \bar{\ell}_0, \\ D = \ell_0 + \bar{\ell}_0. \end{cases}$$

The eigenvalues of S correspond to the spin of states, while the eigenvalues of D correspond to their scaling dimensions.

We have also generators for another type of transformations, the special conformal transformations, which are generated by ℓ_1 and $\bar{\ell}_1$. These generators do not have a direct analogue in the Poincaré group, but they are essential for the full conformal symmetry.

2.1.3 | Plane to Cylinder Map

As we have seen, the local conformal algebra in two dimensions is infinite-dimensional, since any analytic function generates a valid local conformal transformation. However, one might initially wonder how this reconciles with the fact that the *global* conformal group (the Möbius group $\mathrm{SL}(2, \mathbb{C})$) is strictly 6-dimensional. The resolution lies in the fact that the global conformal transformations are the only ones that are globally well-defined and invertible on the entire Riemann sphere (the compactified plane), without introducing singularities.

While the formulation of Conformal Field Theory on the complex plane is geometrically elegant and natural, the physical interpretation of quantum states and their time evolution can be somewhat obscure. In standard Quantum Field Theory, we are accustomed to defining states on spatial slices and evolving them forward in time. To recover this intuitive Hamiltonian picture, we can map the complex plane to a cylindrical geometry.

`img/plane_cylinder_map.png`

Figure 2.1: The conformal mapping from the complex plane to the cylinder. Concentric circles on the plane correspond to constant-time slices on the cylinder, while radial lines correspond to spatial slices.

We introduce a conformal mapping from the plane coordinate z to a cylinder coordinate w:

$$z = e^{\frac{2\pi}{L} w} \quad \text{or equivalently} \quad w = \frac{L}{2\pi} \log z$$

The logarithm function maps the complex plane onto an infinite strip. By imposing periodic boundary conditions, this strip rolls up seamlessly into a cylinder of circumference L. If we decompose the cylinder coordinate into its real and imaginary parts, $w = \tau + i\sigma$, and express the plane coordinate z in polar form, $z = |z| e^{i \arg z}$, we find the relations:

$$\tau = \frac{L}{2\pi} \log |z| \, , \quad \sigma = \frac{L}{2\pi} \arg z$$

Here, $\tau \in [-\infty, \infty]$ represents the Euclidean time evolving along the infinite length of the cylinder, while $\sigma \in [0, L]$ is the compactified spatial coordinate wrapping around the cylinder.

This geometric mapping has profound physical implications for our quantization scheme:

- Concentric circles on the z-plane (curves of constant $|z|$) are mapped to constant-time slices (constant τ) on the cylinder.
- Radial dilatations on the plane ($|z| \to \lambda|z|$) correspond to time translations on the cylinder ($\tau \to \tau + \frac{L}{2\pi} \log \lambda$).

Consequently, the six generators of the global conformal group on the plane map directly to the six global generators on the cylinder. This transformation provides a concrete framework to quantize CFT_2 using the standard QFT approach, allowing us to interpret states living on a circle and evolving radially. It also provides a powerful mechanism to control and calculate finite-size effects in critical statistical models.

2.2 | Classification of Fields and Stress-Energy Tensor

We now classify local fields according to their transformation properties under conformal mappings.

A field $\Phi(z, \bar{z})$ is defined as a **conformal field** if it transforms homogeneously under independent global dilatations of the coordinates, namely $z \to w = \lambda z$ and $\bar{z} \to \bar{w} = \bar{\lambda}\bar{z}$. Under this scaling, the conformal field transforms as:

$$\Phi'(w, \bar{w}) = \lambda^{-h}\bar{\lambda}^{-\bar{h}}\Phi(z, \bar{z})$$

The numbers h and $\bar{h}$ are called the holomorphic and anti-holomorphic *conformal weights* (sometimes also referred to as right and left conformal dimensions). Because the dilatation operator and the spin operator commute, any conformal field possesses both a definite **scale dimension** and a definite **spin**:

$$\Delta = h + \bar{h}, \quad s = h - \bar{h}.$$

We can directly relate these quantum numbers to the generators of the global conformal algebra. The dilatation operator D and the rotation operator S are given by the sum and difference of the zero-modes of the Witt algebra:

$$D = \ell_0 + \bar{\ell}_0 \ , \quad S = \ell_0 - \bar{\ell}_0$$

In terms of differential operators on the complex plane, these generators take the form $D = z\partial_z + \bar{z}\partial_{\bar{z}}$ and $S = z\partial_z - \bar{z}\partial_{\bar{z}}$. It is therefore straightforward to verify that applying these operators to a field yields precisely its scaling dimension and spin: $D \to h + \bar{h} = \Delta$ and $S \to h - \bar{h} = s$. Consequently, under a pure global scaling $z \to \lambda z$ and $\bar{z} \to \lambda\bar{z}$, a field with well-defined weights transforms as:

$$\Phi(z, \bar{z}) \to \lambda^{-h}\lambda^{-\bar{h}}\Phi(z, \bar{z}) = \lambda^{-\Delta}\Phi(z, \bar{z})$$

This explicitly shows that Δ strictly governs the scaling behavior of the field.

A conformal field represents an element of the conformal basis within the algebra of local fields.

Primary Fields

Building upon this, a field $\Phi(z, \bar{z})$ is classified as a **primary field** of conformal weights $(h, \bar{h})$ if it transforms covariantly under *any* finite local conformal transformation $z \mapsto w = f(z)$ and $\bar{z} \mapsto \bar{w} = \bar{f}(\bar{z})$:

$$\Phi'(w, \bar{w}) = \left(\frac{\mathrm{d}w}{\mathrm{d}z}\right)^{-h} \left(\frac{\mathrm{d}\bar{w}}{\mathrm{d}\bar{z}}\right)^{-\bar{h}} \Phi(z, \bar{z}). \tag{2.2.1}$$

This is the fundamental transformation law in two-dimensional CFT. We can easily derive its infinitesimal form. Consider a small local conformal transformation parameterized by

$$w = z + \epsilon(z), \quad \bar{w} = \bar{z} + \bar{\epsilon}(\bar{z}).$$

The Jacobian factors expand as:

$$\frac{\mathrm{d}w}{\mathrm{d}z} = 1 + \partial_z\epsilon(z) \ , \quad \frac{\mathrm{d}\bar{w}}{\mathrm{d}\bar{z}} = 1 + \partial_{\bar{z}}\bar{\epsilon}(\bar{z}).$$

Keeping only the first-order terms in ϵ, the conformal weight factors become:

$$\left(\frac{\mathrm{d}w}{\mathrm{d}z}\right)^{-h} \approx 1 - h\partial_z\epsilon \ , \quad \left(\frac{\mathrm{d}\bar{w}}{\mathrm{d}\bar{z}}\right)^{-\bar{h}} \approx 1 - \bar{h}\partial_{\bar{z}}\bar{\epsilon},$$

and, at the same time, we can Taylor expand the field itself:

$$\Phi(z,\bar{z}) = \Phi(w-\epsilon, \bar{w}-\bar{\epsilon}) \approx \Phi(w,\bar{w}) - \epsilon\partial_w\Phi - \bar{\epsilon}\partial_{\bar{w}}\Phi.$$

Finally, by substituting these expansions into the finite transformation law and isolating the variation $\delta\Phi = \Phi'(z,\bar{z}) - \Phi(z,\bar{z})$, we obtain the infinitesimal transformation rule for a primary field:

$$\delta\Phi = -\epsilon(z)\partial_z\Phi - \bar{\epsilon}(\bar{z})\partial_{\bar{z}}\Phi - h(\partial_z\epsilon(z))\Phi - \bar{h}(\partial_{\bar{z}}\bar{\epsilon}(\bar{z}))\Phi. \tag{2.2.2}$$

This transformation law shows that the field Φ transforms as a tensor under all the local conformal transformations, with weights h and $\bar{h}$. The primary fields are the subset of conformal fields that transform in this simple way, and they play a central role in the structure of the theory, as they are the building blocks from which all other fields can be generated by applying the generators of the conformal algebra.

Secondary and Quasi-Primary Fields

Many fields in a CFT, however, do not obey this strict local transformation rule. Fields that are not primary are collectively called **secondary fields**.

Among these, a special and very important subcategory exists: if a conformal field transforms covariantly *only* under the restricted subset of global Möbius transformations (the $SL(2,\mathbb{C})$ group defined in (2.1.5)), it is called a **quasi-primary field**. Under such a transformation, a quasi-primary field behaves as:

$$\Phi'(w,\bar{w}) = (cz+d)^{2h}(\bar{c}\bar{z}+\bar{d})^{2\bar{h}}\Phi(z,\bar{z}). \tag{2.2.3}$$

Every primary field is trivially quasi-primary, but the converse is generally false. A prominent example of a field that is quasi-primary but not primary is the energy-momentum tensor itself, which falls into the larger set of secondary fields because its transformation law acquires an anomalous shift under generic local mappings.

When we classify the operator content of a Conformal Field Theory, we will see that all fields naturally organize themselves into representations of the conformal algebra. The primary fields act as the fundamental building blocks of the theory. In representation theory, they correspond to the **highest-weight states**, as we will see.

By acting on a primary field with the generators of local conformal transformations, we can generate an infinite family of quasi-primary and secondary fields known as the **descendants** of the primary field. In the algebraic language we will introduce later, a primary field is annihilated by the "lowering" generators (those with $n > 0$), while its descendants are constructed by repeatedly applying the "raising" operators (the generators with $n < 0$). Together, a primary field and all its generated descendants form what is called a complete conformal family or conformal module, but all of this will be discussed in detail in the next chapters.

It is crucial to understand that the labels $(h,\bar{h})$ do not exclusively belong to primary fields. Even secondary or quasi-primary fields possess well-defined scaling properties locally. Under a general conformal map, the transformation law of any field always begins with the tensorial component $(\mathrm{d}w/\mathrm{d}z)^{-h}(\mathrm{d}\bar{w}/\mathrm{d}\bar{z})^{-\bar{h}}\Phi$, followed by anomalous terms. The primary fields are simply the special building blocks for which these anomalous terms vanish completely.

Furthermore, the simple derivative of a primary field is generally not a primary field. If we were to take the derivative of the primary transformation law, the chain rule would generate second-order derivative terms (like $\partial_z^2\epsilon$), which spoil the strict definition of a primary field. However, taking derivatives shifts the conformal weights in a predictable way: the holomorphic derivative $\partial_z\Phi$ of a primary field has weights $(h+1,\bar{h})$, while the anti-holomorphic derivative $\partial_{\bar{z}}\Phi$ has weights $(h,\bar{h}+1)$.

Example (Transformation of the derivative of a field). Consider the holomorphic derivative $\partial_z\Phi(z,\bar{z})$ of a primary field of weight $(h,\bar{h})$. Under a conformal map $z\mapsto w(z)$, the derivative transforms according to the chain rule:

$$\partial_z\Phi(z,\bar{z})\mapsto\partial_z\left[\left(\frac{dw}{dz}\right)^{-h}\Phi(w,\bar{w})\right]$$

Applying the product rule, we obtain:

$$=\left(\frac{dw}{dz}\right)^{-h}\frac{dw}{dz}\partial_w\Phi(w,\bar{w})-h\left(\frac{dw}{dz}\right)^{-h-1}\frac{d^2w}{dz^2}\Phi(w,\bar{w})$$

The first term evaluates to $(dw/dz)^{-(h+1)}\partial_w\Phi$, which is exactly the transformation of a primary field with a shifted weight $(h+1,\bar{h})$. However, the second term introduces an anomaly proportional to the second derivative of the transformation, $\partial_z^2 w$. Because of this extra term, the derivative of a primary field is not a primary field itself, but rather a secondary field.

TODO: check computation.

To conclude this geometric classification, it is useful to recall a few specific classes of conformal fields based on their weights:

- **Scalar fields:** A scalar field has zero spin ($s=0$), which implies that its holomorphic and anti-holomorphic weights must be equal:

$$h=\bar{h}=\frac{\Delta}{2}$$

- **Holomorphic (chiral) fields:** A field $\Phi(z)$ that depends exclusively on z has $\bar{h}=0$. Its scale dimension and spin are completely determined by its holomorphic weight: $\Delta=h$ and $s=h$. Analytically, it satisfies the equation $\partial_{\bar{z}}\Phi(z)=0$.

- **Anti-holomorphic (anti-chiral) fields:** Conversely, a field $\bar{\Phi}(\bar{z})$ depending only on $\bar{z}$ has $h=0$, yielding $\Delta=\bar{h}$ and $s=-\bar{h}$. It satisfies the condition $\partial_z\bar{\Phi}(\bar{z})=0$.

Physically, the differential equations satisfied by chiral and anti-chiral fields can be interpreted as continuity equations for a conserved current splitting into two helicity components, deeply tying these fields to underlying conservation laws (à la Noether) and fundamental symmetries.

2.2.1 | Stress-Energy Tensor

We know that the stress-energy tensor $T_{\mu\nu}$ is a local field, symmetric (or at least it can be made symmetric), and it is a conserved current, so it must satisfy the following conditions:

$$T_{\mu\nu}(x)=T_{\nu\mu}(x),\quad\partial^\mu T_{\mu\nu}(x)=0,\quad T^\mu_{\ \mu}(x)=0,$$

where the continuity equation $\partial^\mu T_{\mu\nu}=0$ expresses the *conservation of energy and momentum*, and the traceless condition $T^\mu_{\ \mu}=0$ is a consequence of *scale invariance* at criticality (thus for Polyakov, since we have both rototranslation and scale invariance we know it to be conformal invariant). Conformal invariant theories can be improved with a stress-energy tensor which is traceless.

We can again change to complex coordinates

$$z = x^1 + ix^2, \quad \bar{z} = x^1 - ix^2,$$

and by trasnforming the components of the stress-energy tensor accordingly

$$T_{\mu\nu} = \frac{\partial x^\alpha}{\partial x^\mu}\frac{\partial x^\beta}{\partial x^\nu}T_{\alpha\beta},$$

where we have clearly defined $x^\alpha = (z, \bar{z})$; we can thus write the components of the stress-energy tensor in complex coordinates as

$$\begin{cases} T(z,\bar{z}) = T_{zz}(z,\bar{z}) = 2\pi(T_{11} - T_{22} + 2iT_{12}), \\ \bar{T}(z,\bar{z}) = T_{\bar{z}\bar{z}}(z,\bar{z}) = 2\pi(T_{11} - T_{22} - 2iT_{12}), \\ \Theta(z,\bar{z}) = T_{z\bar{z}}(z,\bar{z}) = 2\pi(T_{11} + T_{22}) = 0. \end{cases}$$

where the 2π factor comes from the original work of Polyakov, and it is just a convention. The last component Θ is zero due to the traceless condition, and we get a further condition on the components:

$$T_{11} = -T_{22}.$$

We can write the conservation equation $\partial^\mu T_{\mu\nu} = 0$ in complex coordinates, and we find that it decouples into two independent equations:

$$\begin{cases} \partial_z T_{zz} + \partial_z T_{z\bar{z}} = 0, \\ \partial_{\bar{z}} T_{\bar{z}\bar{z}} + \partial_{\bar{z}} T_{z\bar{z}} = 0, \end{cases}$$

but since in a conformal theory $\Theta = T_{z\bar{z}} = 0$ is zero, we can develop and

TODO: check with professor, my result is the opposite.

$$\partial_z T(z,\bar{z}) = 0, \quad \partial_{\bar{z}}\bar{T}(z,\bar{z}) = 0,$$

and find out that T is independent of $\bar{z}$ and $\bar{T}$ is independent of z:

$$T = T(z), \quad \bar{T} = \bar{T}(\bar{z}), \quad \begin{cases} \partial_{\bar{z}} T(z) = 0, \\ \partial_z \bar{T}(\bar{z}) = 0. \end{cases}$$

Thus, the first two components T and $\bar{T}$ are the holomorphic and anti-holomorphic components of the stress-energy tensor, and they are the most important fields in the theory, since they generate the conformal transformations.

Classification and Scaling Dimension of the Stress-Energy Tensor

Next let us take into account the action as the integral of the lagrangian density, which is a local field, and thus we can write

$$\mathcal{S} = \int \mathrm{d}^2x\, \mathcal{L}(x) = \int \mathrm{d}z\mathrm{d}\bar{z}\, \mathcal{L}(z,\bar{z}),$$

and since the action is supposed to have zero mass dimension, the lagrangian density must have scaling dimension $\Delta = 2$. Thus, the lagrangian density is a local field with scaling dimension 2, and since we derive the stress-energy tensor from the lagrangian through

$$T_{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\partial_\nu \phi_a - \delta_{\mu\nu}\mathcal{L},$$

then we know the stress-energy tensor must have the same scaling dimension, and thus it is a local field with scaling dimension 2. We would have expected it to have three components, but the traceless condition reduces it to two independent components, which are the holomorphic and anti-holomorphic components $T(z)$ and $\bar{T}(\bar{z})$.

We know that T is $(2,0)$ while $\bar{T}$ is $(0,2)$, so they are not primary fields, since they do not transform tensorially under the conformal transformations, hence they transform covariantly only under the global conformal transformations, being secondary fields.

We know that a two point function at the critical point should rescale by a power law, and thus we can write

$$\langle T(z)T(w)\rangle \sim \frac{1}{|z-w|^4},$$

thus if we do a mobius transformation when approaching the forth order pole, it is fine, but a slightly more general transformation will give us some extra terms, which are not the transformation of a primary field, so T is not a primary field, but it is a secondary field. The same holds for $\bar{T}$. Thus the stress-energy tensor is not a primary field.

2.2.2 | Relation between OPE and commutators

Let $a(z)$ and $b(z)$ be holomorphic fields (we do not consider here the dependence on $\bar{z}$). Consider two operators defined by contour integrals:

$$A = \oint_0 \mathrm{d}z\, a(z), \quad B = \oint_0 \mathrm{d}w\, b(w).$$

To relate their commutator to the OPE, we first consider the integral of their radially ordered product $\mathcal{R}\{a(z)b(w)\}$. In radial quantization (as we will see in section 2.6), radial ordering acts exactly like time-ordering: for $|z| > |w|$ it yields $a(z)b(w)$, while for $|z| < |w|$ it yields $\eta b(w)a(z)$, where $\eta = \pm 1$ dictates the particle statistics ($+1$ for bosons, -1 for fermions).

By defining $C_\pm$ as contours around the origin with radius $|w| \pm \epsilon$, we can write the integral along a small loop enclosing w as the limit of the difference between these two orderings:

$$\oint_w \mathrm{d}z\, \mathcal{R}\{a(z)b(w)\} = \lim_{\epsilon\to 0}\left\{\oint_{C_+} \mathrm{d}z\, a(z)b(w) - \eta \oint_{C_-} \mathrm{d}z\, b(w)a(z)\right\} = [A, b(w)]_\eta.$$

The contours $C_\pm$ can be continuously deformed à la Cauchy to obtain the single circular integral around w, as depicted in Figure 2.2. Since on these fixed contours the operators no longer cross each other radially, the right-hand side directly becomes the action of the (anti-)commutator, denoted by $[\cdot,\cdot]_\eta$.

img/OPE_commutators.jpg

Figure 2.2: Contours C_+ and C_- can be deformed à la Cauchy to obtain the circle integral around w.

Integrating both sides over w, the (anti-)commutator of the two global fields A and B can be computed as:

$$[A, B]_\eta = \oint_0 \mathrm{d}w \oint_w \mathrm{d}z\, \mathcal{R}\{a(z)b(w)\}.$$

This crucial result implies that the commutator of two operators is entirely determined by the singular terms of their OPE. Thanks to Cauchy's residue theorem, integrating z around w guarantees that all regular terms vanish, isolating only the pole contributions.

2.2.3 | Conserved Charges, OPE and Classification of Fields

According to Noether's theorem, every continuous symmetry of the action yields a conserved current. In our cylindrical geometry, the conserved charge associated with the continuity equation $\partial_\mu j^\mu = 0$ is obtained by integrating the temporal component of the current over the spatial slice:

$$\mathcal{Q} = \int_0^L d\sigma j_\tau(\sigma, \tau).$$

This charge acts as the generator of the corresponding symmetry. The infinitesimal variation of any local field under this symmetry is determined by its commutator (or anti-commutator, depending on the statistics) with the charge:

$$\delta\Phi(\sigma, \tau) = [\mathcal{Q}, \Phi(\sigma, \tau)]_\eta.$$

To make contact with the powerful methods of complex analysis, we translate this formulation from the cylinder to the complex plane. In terms of the complex coordinates z and $\bar{z}$, the components of the current vector j_μ (where $\mu = \tau, \sigma$) can be decoupled into purely holomorphic and anti-holomorphic parts, $j(z)$ and $\bar{j}(\bar{z})$:

$$j_\tau(\tau, \sigma) = \frac{j(z) + \bar{j}(\bar{z})}{2}\ , \quad j_\sigma(\tau, \sigma) = \frac{j(z) - \bar{j}(\bar{z})}{2i}.$$

Under the exponential map, the spatial integral over σ from 0 to L on the cylinder seamlessly transforms into a closed contour integral in the complex plane, circulating around the origin. The total charge $\mathcal{Q}$ naturally splits into holomorphic and anti-holomorphic sectors:

$$\mathcal{Q} = \frac{Q + \bar{Q}}{2} \quad \text{with} \quad Q = \oint_0 \frac{dz}{2\pi i} j(z)\ , \quad \bar{Q} = \oint_0 \frac{d\bar{z}}{2\pi i} \bar{j}(\bar{z}).$$

We can now apply this general formalism to the specific case of local conformal symmetry. For an infinitesimal conformal transformation parameterized by independent holomorphic and anti-holomorphic functions:

$$z \to z + \epsilon(z)\ , \quad \bar{z} \to \bar{z} + \bar{\epsilon}(\bar{z}),$$

the corresponding conserved Noether currents are directly proportional to the components of the energy-momentum tensor. Thus, the conserved charges that generate these transformations are exactly given by:

$$Q_\epsilon = \frac{1}{2\pi i} \oint dz\, \epsilon(z) T(z),$$

and similarly in the anti-holomorphic sector:

$$\bar{Q}_{\bar{\epsilon}} = -\frac{1}{2\pi i} \oint d\bar{z}\, \bar{\epsilon}(\bar{z}) \bar{T}(\bar{z}).$$

A crucial feature of these contour integrals is their topological invariance: because the stress-energy tensor $T(z)$ is purely holomorphic away from singularities, the contour can be continuously deformed without altering the value of the integral, provided no operator insertions are crossed.

By taking the commutator of these charges with a generic field, we obtain the infinitesimal conformal variation of that field. Representing the commutator via a contour integral tightly wrapping the insertion point w, we find:

$$\delta_\epsilon \Phi(w,\bar{w}) = [Q_\epsilon, \Phi(w,\bar{w})] = \oint_w \frac{dz}{2\pi i}\epsilon(z)T(z)\Phi(w,\bar{w}),$$
$$\delta_{\bar{\epsilon}} \Phi(w,\bar{w}) = [\bar{Q}_{\bar{\epsilon}}, \Phi(w,\bar{w})] = -\oint_{\bar{w}} \frac{d\bar{z}}{2\pi i}\bar{\epsilon}(\bar{z})\bar{T}(\bar{z})\Phi(w,\bar{w}).$$

This profound relation explicitly demonstrates that the variation of any field under local conformal transformations is completely dictated by its Operator Product Expansion (OPE) with the energy-momentum tensor.

Definition of Primary Field from the OPE with the Stress Tensor

Given the relation between variations and the energy-momentum tensor, we can now provide a purely algebraic, quantum-mechanical definition of a primary field. A local field $\Phi(w,\bar{w})$ is formally defined as a **primary field** of conformal weights $(h,\bar{h})$ if its short-distance Operator Product Expansion with the stress tensor takes the precise form:

$$T(z)\Phi(w,\bar{w}) \sim \frac{h\Phi(w,\bar{w})}{(z-w)^2} + \frac{\partial_w \Phi(w,\bar{w})}{z-w} + \text{regular terms}$$

and analogously for the anti-holomorphic sector:

$$\bar{T}(\bar{z})\Phi(w,\bar{w}) \sim \frac{\bar{h}\Phi(w,\bar{w})}{(\bar{z}-\bar{w})^2} + \frac{\partial_{\bar{w}} \Phi(w,\bar{w})}{\bar{z}-\bar{w}} + \text{regular terms}$$

We must now verify that this algebraic definition is perfectly equivalent to the geometric transformation law we introduced earlier. To prove this, we insert the postulated OPE directly into the contour integral expression for the infinitesimal variation $\delta_\epsilon \Phi(w,\bar{w})$:

$$\delta_\epsilon \Phi(w,\bar{w}) = \frac{1}{2\pi i}\oint_w dz\, \epsilon(z)T(z)\Phi(w,\bar{w})$$

Substituting the singular terms of the OPE, the regular terms smoothly vanish upon contour integration, leaving:

$$\delta_\epsilon \Phi(w,\bar{w}) = \frac{1}{2\pi i}\oint_w dz\, \epsilon(z)\left[\frac{h\Phi(w,\bar{w})}{(z-w)^2} + \frac{\partial_w \Phi(w,\bar{w})}{z-w}\right]$$

We can evaluate these terms individually using the Cauchy integral formula and its derivatives:

$$\frac{1}{2\pi i}\oint_w dz \frac{\epsilon(z)}{z-w} = \epsilon(w)\ , \quad \frac{1}{2\pi i}\oint_w dz \frac{\epsilon(z)}{(z-w)^2} = \partial_w \epsilon(w)$$

Inserting these results back into the variation, we immediately find:

$$\delta_\epsilon \Phi(w,\bar{w}) = \epsilon(w)\partial_w \Phi(w,\bar{w}) + h\partial_w \epsilon(w)\Phi(w,\bar{w})$$

By performing the identical procedure for the anti-holomorphic variation $\delta_{\bar{\epsilon}}\Phi$ we obtain

$$\delta_{\bar{\epsilon}} \Phi(w,\bar{w}) = \bar{\epsilon}(\bar{w})\partial_{\bar{w}} \Phi(w,\bar{w}) + \bar{h}\partial_{\bar{w}} \bar{\epsilon}(\bar{w})\Phi(w,\bar{w}),$$

and combining the two parts, we exactly recover the infinitesimal conformal transformation law of a primary field. Thus, the analytic structure of the OPE perfectly encapsulates the geometric symmetries of the theory.

2.3 | Free Theories in Two Dimensions

Before continuing with the analysis of CFT in 2D, we pause to consider some simple examples, crucial in learning features of CFT and at the same time of great utility. The massless free boson is essentially the Gaussian theory considered and illustrated in the RG framework where it represents one of the most important sets of universality classes of critical phenomena. The free massless fermion is also a very interesting CFT in itself, but its importance will appear later, when it will be related to the Ising universality class.

2.3.1 | Free Massless Boson in $D = 2$

We consider the theory of a free massless scalar boson in two dimensions. The Euclidean action is given by:

$$\mathcal{S} = \frac{1}{8\pi} \int \mathrm{d}^2x \, \partial_\mu \varphi(x) \partial^\mu \varphi(x)$$

where the overall normalization factor $1/8\pi$ is a standard convention (often chosen following Ginsparg's notes); while arbitrary, it simplifies the coefficients in subsequent calculations. Notice the absence of a mass term $m^2\varphi^2$, which ensures the classical scale invariance of the action: if there were a mass term, it would introduce a dimensionful parameter that explicitly breaks scale invariance, and thus conformal invariance.

Using standard Quantum Field Theory methods, we can evaluate the propagator (the two-point correlation function) of the fundamental field φ:

$$\langle \varphi(x)\varphi(y) \rangle = -\log |x-y|^2 + \text{const.}$$

where x and y are two-dimensional Euclidean vectors, and the arbitrary constant can be fixed by choosing an infrared cutoff or a specific normalization. By switching to complex coordinates z and $\bar{z}$, we can rewrite the distance squared as $(z-w)(\bar{z}-\bar{w})$, splitting the propagator into:

$$\langle \varphi(z,\bar{z})\varphi(w,\bar{w}) \rangle = -\log(z-w) - \log(\bar{z}-\bar{w}) + \text{const.}$$

This effectively decouples the holomorphic and anti-holomorphic sectors.

> **Remark (Coleman's Theorem).** *The presence of the logarithm in the two-point function is a direct consequence of the massless nature of the boson in two dimensions, which leads to severe infrared divergences. This perfectly aligns with the Mermin-Wagner-Coleman theorem, which precludes the spontaneous breaking of continuous symmetries in two dimensions due to these overwhelming large-scale quantum fluctuations.*

Because its two-point function is not a strict power law, $\varphi(z,\bar{z})$ lacks a definite scaling dimension and is *not* a conformal field (if we were to treat it as such we would encounter inconsistencies). However, it serves as a fundamental building block to construct composite fields (such as its derivatives or vertex operators) that *do* behave as perfectly well-defined conformal primaries.

Furthermore, the classical equations of motion (the d'Alembert equation $\Box\varphi = 0$) imply that the most general solution separates completely into a chiral and an anti-chiral part:

$$\varphi(z,\bar{z}) = \phi(z) + \bar{\phi}(\bar{z}).$$

Since the two components are totally decoupled, we can focus on the holomorphic sector z; identical results will hold symmetrically for the $\bar{z}$ sector. We define the derivative field $\partial\phi(z) \equiv \partial_z\phi(z)$, which instead is a well-behaved conformal field: indeed, we can compute its two-point function by differentiating the fundamental propagator in order to prove that it has a well-defined scaling dimension:

$$\langle\partial\phi(z)\partial\phi(w)\rangle = \partial_z\partial_w\langle\phi(z)\phi(w)\rangle = -\partial_z\partial_w\log(z-w) = -\partial_w\frac{1}{z-w} = -\frac{1}{(z-w)^2}.$$

Now, to absorb the minus sign and work with a positive correlator, we define the current:[2]

$$J(z) = i\partial\phi(z),$$

For this new field, the two-point function is strictly conformal:

$$\langle J(z)J(w)\rangle = \frac{1}{(z-w)^2}.$$

This demonstrates that $J(z)$ is a highly well-behaved conformal field with scaling dimension $\Delta = 1$. Because it depends solely on z, it has conformal weights $(h,\bar{h}) = (1,0)$, meaning it has spin

$$s = h - \bar{h} = 1.$$

Symmetrically, $\bar{J}(\bar{z}) = i\bar{\partial}\bar{\phi}(\bar{z})$ has weights $(0,1)$ and spin $s=-1$.

Crucially, the equations of motion $\Box\varphi \implies \partial J + \bar{\partial}\bar{J} = 0$, identify the two currents as **conserved chiral currents**:

$$\begin{cases}\partial J(z) = \bar{\partial}J(z) = 0,\\ \partial\bar{J}(\bar{z}) = \bar{\partial}\bar{J}(\bar{z}) = 0.\end{cases}$$

Although the deriving conserved charges may lack a direct physical interpretation, their existence rigorously implies an underlying continuous symmetry within the theory.

Chiral Current OPE and the Energy-Momentum Tensor

Let us now study the Operator Product Expansion (OPE) of $J(z)$ with itself. Generalizing *Wick's theorem* to this conformal context, the product of two local fields can be expanded into its singular contractions plus a normal-ordered regular part:

$$J(z)J(w) = \overbrace{J(z)J(w)} + :J(z)J(w):$$

The singular contraction perfectly corresponds to the two-point function we just calculated. Thus, we write the OPE as:

$$J(z)J(w) = \langle J(z)J(w)\rangle + \text{reg.} = \frac{1}{(z-w)^2} + \text{reg.}$$

where the regular terms can be expanded in a Taylor series in $(z-w)$. When we compute the vacuum expectation value $\langle J(z)J(w)\rangle$, we isolate exactly the singular contraction. This happens because, by strict definition in quantum field theory, the vacuum expectation value of any normal-ordered product is identically zero ($\langle :J(z)J(w):\rangle = 0$). The regular part simply vanishes under the VEV, leaving only the singularity.

Finally, we construct the energy-momentum tensor from the Lagrangian density. Classically, T is symmetric and traceless, and its holomorphic component is $-\frac{1}{2}(\partial\phi)^2$. At the quantum level,

[2] It is just a convention, one can also decide to "be russian about it" and write the action with a minus sign in front, but it is not the standard convention.

however, the product of two local operators evaluated at the exact same point is highly divergent. To define a finite, meaningful composite field $T(z)$, we must employ a point-splitting normal ordering procedure to rigorously subtract the infinity (the contraction):

$$T(z) \equiv -\frac{1}{2} : (\partial\phi(z))^2 := \frac{1}{2} : J(z)^2 :$$

Analytically, this normal ordering is defined via the limit:

$$T(z) = -\frac{1}{2} \lim_{w \to z} \left[\partial\phi(z)\partial\phi(w) - \langle \partial\phi(z)\partial\phi(w) \rangle \right],$$

since we have seen that the VEV of the product $\partial\phi(z)\partial\phi(w)$ captures the singular part of the OPE, and thus we are removing the divergence. This subtraction isolates the regular, finite quantum operator that consistently generates conformal transformations for the free boson theory.

Current and Energy-Momentum Tensor OPE

We now compute the Operator Product Expansion of the stress-energy tensor $T(z)$ with the conserved current $J(w)$. Using the normal-ordered definition $T(z) = \frac{1}{2} : J(z)^2 :$ we can write

$$T(z)J(w) = \frac{1}{2} : J(z)J(z) : J(w).$$

To evaluate this, we apply Wick's theorem. We must contract the fields inside the normal ordering with the field outside. Since the fields inside the normal ordering $: J(z)J(z) :$ are already regularized by definition, they cannot be contracted with each other. We can only contract one of the $J(z)$ fields with the $J(w)$ field. Because there are two identical $J(z)$ fields inside the normal ordering, there are exactly 2 equivalent ways to perform this single contraction:

$$\begin{aligned} T(z)J(w) &= \frac{1}{2} : J(z)J(z) : J(w) + \frac{1}{2} : J(z)J(z) : J(w) + \text{reg.} \\ &= 2 \times \frac{1}{2} \langle J(z)J(w) \rangle J(z) + \text{reg.} = \frac{J(z)}{(z-w)^2} + \text{reg.} \end{aligned}$$

This is not yet the final formula for the OPE, because the right-hand side still depends on the local field evaluated at z, while an OPE must express everything in terms of local operators evaluated strictly at w times the powers of $(z-w)$. Since $J(z)$ is analytic everywhere except at the insertions, we can Taylor-expand it around w:

$$J(z) = J(w) + (z-w)\partial_w J(w) + \mathcal{O}((z-w)^2).$$

Inserting this expansion back into our result, the $(z-w)$ factor cancels one power of the denominator, enforcing the final expression for the OPE:

$$T(z)J(w) = \frac{J(w) + (z-w)\partial_w J(w)}{(z-w)^2} + \text{reg.} = \frac{J(w)}{(z-w)^2} + \frac{\partial_w J(w)}{z-w} + \text{reg.}$$

This is a very important result: as we will see in the next section, it explicitly proves that the chiral current $J(w) = i\partial\phi(w)$ is indeed a primary field of conformal dimensions $(h, \bar{h}) = (1, 0)$.

The same calculation can be repeated for the anti-holomorphic sector for $\bar{T}$ and $\bar{J}$, showing that $\bar{J}(\bar{w}) = i\bar{\partial}\bar{\phi}(\bar{w})$ is a primary field of conformal dimensions $(0, 1)$.

Energy-Momentum Tensor OPE

For the next fundamental OPE, we compute $T(z)T(w)$, the product of the stress-energy tensor with itself. Substituting the definitions, we have:

$$T(z)T(w) = \frac{1}{4} : J(z)J(z) :: J(w)J(w) :$$

Applying Wick's theorem, we must sum over all possible cross-contractions between the two pairs of normal-ordered fields. We can group the results into three distinct contributions based on the number of contractions:

$$\begin{aligned} T(z)T(w) &= \frac{1}{4} : J(z)J(z)J(w)J(w) : && (0) \\ &+ : J(z)J(z) : J(w)J(w) : && (1) \\ &+ \frac{1}{2} : J(z)J(z) :: J(w)J(w) : && (2) \end{aligned}$$

1. **Zero contractions:** This term is simply the normal-ordered product of all four fields, $: T(z)T(w) :$ and by definition this is entirely non-singular as $z \to w$, so it only contributes to the regular terms of the OPE:

$$: T(z)T(w) :=: T^2 : (w) + (z-w) : T\partial T : (w) + \cdots = \text{reg.}$$

2. **One contraction:** We contract one $J(z)$ with one $J(w)$. Since there are two choices for the first field and two choices for the second, there are $2 \times 2 = 4$ equivalent ways to perform this single contraction. This leaves one $J(z)$ and one $J(w)$ uncontracted inside a normal ordering:

$$4 \times \frac{1}{4} \langle J(z)J(w) \rangle : J(z)J(w) := \frac{: J(z)J(w) :}{(z-w)^2}.$$

To express this entirely at w, we Taylor-expand the uncontracted $J(z)$ around w: $: J(z)J(w) :=: J(w)^2 : +(z-w) : \partial_w J(w)J(w) : + \ldots$ Substituting this expansion yields:

$$\frac{: J(w)^2 :}{(z-w)^2} + \frac{: \partial_w J(w)J(w) :}{z-w} + \text{reg.}$$

We immediately recognize the first numerator as $2T(w)$. For the second term, using the algebraic identity $\partial_w T(w) = \frac{1}{2}\partial_w : J(w)^2 :=: J(w)\partial_w J(w) :$, we recognize the exact derivative of the stress tensor. Thus, the 1-contraction contribution is:

$$\frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} + \text{reg.}$$

If the expansion stopped here, $T(w)$ would be a perfect primary field of dimension $\Delta = 2$. However, we have one final combinatorial term.

3. **Two contractions:** We must fully contract both $J(z)$ fields with both $J(w)$ fields. For the first $J(z)$, there are 2 possible $J(w)$ fields to contract with. The second $J(z)$ is then forced to contract with the remaining 1 field. Thus, there are $2 \times 1 = 2$ equivalent ways to perform this double contraction:

$$2 \times \frac{1}{4} \langle J(z)J(w) \rangle \langle J(z)J(w) \rangle = \frac{1}{2} \frac{1}{(z-w)^2} \frac{1}{(z-w)^2} = \frac{1/2}{(z-w)^4}$$

Summing all three contributions together, we obtain the exact fundamental OPE for the stress-energy tensor:

$$T(z)T(w) = \frac{1/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} + \text{reg.}$$

The presence of the highly singular fourth-order pole explicitly demonstrates that $T(z)$ is not a true primary field, but rather a quasi-primary field carrying an anomalous transformation rule.

By comparing this result with the universal definition of the TT OPE, the constant numerator of the fourth-order pole strictly identifies the **central charge** $c/2$. For the free massless scalar boson, we read off that the central charge is exactly $c = 1$. In a broader context, c acts as a vital parameter characterizing the specific universality class of the underlying Conformal Field Theory.

This derivation highlights why the central charge is a fundamental defining parameter of a CFT. Because it emerges directly from the OPE of the stress-energy tensor (a universal operator intimately connected to the underlying symmetries and the action of the theory) c inherently characterizes the specific universality class we are studying.

A primary goal of two-dimensional Conformal Field Theory is the exact classification and solution of theories purely through algebraic data (like the central charge c and the conformal weights $(h, \bar{h})$ of the primary fields), bypassing the traditional Lagrangian approach entirely. While a complete classification for all c is immensely difficult, the subset of theories with $0 < c < 1$, known as **minimal models**, has been completely classified and exactly solved. In these models (among which the critical Ising model is the most famous example) the finite number of primary fields and the tight algebraic constraints allow for the exact analytic computation of all physical correlation functions.

2.3.2 | Free Majorana Fermion

Another fundamental model in two-dimensional conformal field theory is the free massless Majorana fermion. Its Euclidean action is defined using a two-component spinor Ψ and the standard Dirac matrices:

$$\mathcal{S} = \frac{1}{4\pi}\int d^2x \Psi^\dagger \gamma^0 \gamma^\mu \partial_\mu \Psi,$$

where we can choose the representation $\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\gamma^1 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$.

Writing the two-component spinor explicitly as $\Psi = \begin{pmatrix} \psi \\ \bar{\psi} \end{pmatrix}$, the action decouples:

$$\mathcal{S} = \frac{1}{4\pi}\int d^2x (\bar{\psi}\partial_z \bar{\psi} + \psi \partial_{\bar{z}} \psi),$$

which are two independent free fermionic actions for the holomorphic field $\psi(z)$ and the anti-holomorphic field $\bar{\psi}(\bar{z})$ (by direct computation). Varying this action yields the classical equations of motion:

$$\begin{cases} \partial_z \bar{\psi} = 0 \implies \bar{\psi} = \bar{\psi}(\bar{z}), \\ \partial_{\bar{z}} \psi = 0 \implies \psi = \psi(z), \end{cases}$$

where the first equation for each sector is a conservation law, indicating that ψ and $\bar{\psi}$ are **conserved chiral currents**. This cleanly demonstrates the holomorphic factorization: $\psi(z)$ is purely chiral, and $\bar{\psi}(\bar{z})$ is purely anti-chiral.

Fermionic Propagator and OPE

We wish to find the fundamental two-point function $\langle\psi(z)\psi(w)\rangle$. In the path integral formalism, the propagator is the Green's function of the kinetic operator. For the holomorphic component ψ, the relevant operator from the action is $\partial_{\bar{z}}$. We must solve $\partial_{\bar{z}}G(z,w) \propto \delta^{(2)}(z-w)$. Using the standard complex analysis identity $\partial_{\bar{z}}\left(\frac{1}{z}\right) = \pi\delta^{(2)}(z)$, and fixing the normalization from the $1/4\pi$ factor in the action, we obtain the exact propagator:

TODO: check procedure

$$\langle\psi(z)\psi(w)\rangle = \frac{1}{z-w} = \psi(z)\psi(w).$$

From the propagator, we immediately read off the fundamental Operator Product Expansion (OPE) for the free fermion, representing the Wick contraction:

$$\psi(z)\psi(w) = \frac{1}{z-w}.$$

Taking a derivative with respect to z allows us to find the contraction between a derivative and a field:

$$\partial\psi(z)\psi(w) = -\frac{1}{(z-w)^2}.$$

From these, we can infer the **conformal dimensions** of the fermionic field. The leading singularity in the OPE $\psi(z)\psi(w) \sim \frac{1}{z-w}$ indicates that ψ has a scaling dimension of $\Delta = 1/2$. This is because the two-point function of a primary field with dimension Δ behaves as $\langle\mathcal{O}(z)\mathcal{O}(w)\rangle \sim \frac{1}{(z-w)^{2\Delta}}$. Thus, for the free fermion, we have:

$$\begin{cases} s = \frac{1}{2}, & \Delta = \frac{1}{2}, \\ s = -\frac{1}{2}, & \Delta = \frac{1}{2}, \end{cases} \implies \begin{cases} (h,\bar{h}) = (\frac{1}{2}, 0) \text{ for } \psi(z), \\ (h,\bar{h}) = (0, \frac{1}{2}) \text{ for } \bar{\psi}(\bar{z}). \end{cases}$$

Energy-Momentum Tensor and OPE with Fermionic Field

We now construct the holomorphic stress-energy tensor $T(z)$ by applying the standard procedure to the Lagrangian. To ensure finite quantum operators, we define it using normal ordering:

$$T(z) = \frac{\partial\mathcal{L}}{\partial(\partial_z\Psi)}\partial_z\Psi = -\frac{1}{2} : \psi(z)\partial\psi(z) :$$

To understand the conformal properties of $\psi(z)$, we must compute its OPE with $T(z)$. Applying Wick's theorem to $T(z)\psi(w) = -\frac{1}{2} : \psi(z)\partial\psi(z) : \psi(w)$, we must be careful with the anti-commuting nature of the fermionic fields. Evaluating the two possible single contractions yields:

$$\begin{aligned} T(z)\psi(w) &= -\frac{1}{2} : \psi(z)\partial\psi(z) : \psi(w) - \frac{1}{2} : \psi(z)\partial\psi(z) : \psi(w) \\ &= \frac{\frac{1}{2}\psi(z)}{(z-w)^2} + \frac{\frac{1}{2}\partial\psi(w)}{z-w} + \text{regular terms}, \end{aligned}$$

where the second term is positive since we must anti-commute the uncontracted $\partial\psi(z)$ past the contracted $\psi(w)$ to bring it next to $\psi(z)$.

Because we need the right-hand side expressed entirely at the target point w, we Taylor expand the remaining field $\psi(z)$ around w:

$$\psi(z) = \psi(w) + (z-w)\partial\psi(w) + \dots$$

Substituting this expansion into the first term of the contraction gives:

$$\frac{\frac{1}{2}[\psi(w)+(z-w)\partial\psi(w)]}{(z-w)^2}=\frac{\frac{1}{2}\psi(w)}{(z-w)^2}+\frac{\frac{1}{2}\partial\psi(w)}{z-w}$$

Combining this result with the second term of the original contraction, the two halves sum to a whole derivative:

$$T(z)\psi(w)=\frac{\frac{1}{2}\psi(w)}{(z-w)^2}+\frac{\partial\psi(w)}{z-w}+\text{regular terms.}$$

This is precisely the defining OPE of a primary field.

By comparing it to the general definition, we deduce that the holomorphic fermion ψ is a primary operator of conformal dimension $(h,\bar{h})=(\frac{1}{2},0)$. Symmetrically, the anti-holomorphic fermion $\bar{\psi}$ is a primary operator of dimension $(0,\frac{1}{2})$. Both fields have the same spin module $s=h-\bar{h}=\pm\frac{1}{2}$ and the same scaling dimension $\Delta=h+\bar{h}=\frac{1}{2}$, as we had already inferred from the two-point function.

Energy-Momentum Tensor OPE

To evaluate the full Operator Product Expansion of the stress-energy tensor with itself for the free Majorana fermion, we must compute:

$$T(z)T(w)=\frac{1}{4}:\psi(z)\partial\psi(z)::\psi(w)\partial\psi(w):$$

Applying Wick's theorem, we group the expansion by the number of contractions:

1. **Zero contractions:** This leaves all four fields inside a single normal ordering, producing only regular terms that do not contribute to the singular part of the OPE.

2. **One contraction:** This yields four distinct pairings (contracting ψ with ψ, $\partial\psi$ with $\partial\psi$, and the two mixed terms). Evaluating these requires extreme care with the minus signs arising from the anticommutation of the fermionic fields during operator exchanges. After evaluating the contractions, one must Taylor-expand the remaining normal-ordered fields around w to correctly extract the singular contributions proportional to $T(w)$ and $\partial_w T(w)$.

3. **Two contractions:** This fully contracts the operators, generating the leading c-number singularity. There are two non-vanishing pairings: contracting the fields together and the derivatives together ($\langle\psi\psi\rangle\langle\partial\psi\partial\psi\rangle$), and the cross-contraction ($\langle\psi\partial\psi\rangle\langle\partial\psi\psi\rangle$). To compute the first pairing, we evaluate the correlator of the derivatives by extracting the operators from the fundamental propagator:

$$\begin{aligned}\langle\partial\psi(z)\partial\psi(w)\rangle&=\partial_z\partial_w\langle\psi(z)\psi(w)\rangle=\partial_z\partial_w\left(\frac{1}{z-w}\right)\\&=\partial_z\left(\frac{1}{(z-w)^2}\right)=-\frac{2}{(z-w)^3}.\end{aligned}$$

During the evaluation, we will need to identify the derivative of the stress-energy tensor by applying the Leibniz rule, noting that the $:\partial\psi(w)\partial\psi(w):$ term vanishes identically due to fermionic statistics:

$$\partial_w T(w)=\partial_w\left(-\frac{1}{2}:\psi(w)\partial_w\psi(w):\right)=-\frac{1}{2}:\psi(w)\partial_w^2\psi(w):$$

By rigorously summing all the evaluated contributions, the OPE beautifully assembles into the final result:

$$T(z)T(w) = \frac{1/4}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} + \text{reg.}$$

Comparing this isolated leading singularity to the universal defining OPE $T(z)T(w) \sim \frac{c/2}{(z-w)^4}$, we formally conclude that the free Majorana fermion possesses a central charge of $c = 1/2$. This fundamentally contrasts with the free real scalar boson, which we previously found to have $c = 1$. Consequently, a Dirac fermion (which is mathematically constructed as a complex linear combination of two independent real Majorana fermions) possesses an additive central charge of $1/2 + 1/2 = 1$.

2.4 | Stress-Energy OPE, Central Charge and Ward Identities

As we have established, the holomorphic component of the stress tensor, $T(z)$, generates infinitesimal conformal transformations and controls the local Ward identities. The operator product expansion of $T(z)$ with itself, $T(w)$, plays a distinguished role in two-dimensional conformal field theory. It determines:

- the algebra of infinitesimal conformal transformations,
- the quantum modification of the classical Witt algebra,
- the appearance of the central charge.

This is precisely the point where the quantum nature of the theory manifests itself most clearly.

2.4.1 | General Form of the $T(z)T(w)$ OPE

The stress-energy tensor $T_{\mu\nu}$ is a conserved current. Under rescaling, its scale dimension and spin are fixed to be:

$$\Delta_T = 2, \quad s_T = 2,$$
$$\Delta_{\bar{T}} = 2, \quad s_{\bar{T}} = -2,$$

which immediately imply the conformal weights:

$$(h_T, \bar{h}_T) = (2, 0)\ , \quad (h_{\bar{T}}, \bar{h}_{\bar{T}}) = (0, 2).$$

As a quasi-primary field, it rescales globally as $T(\lambda z) = \lambda^2 T(z)$. This fixes its two-point correlation function to be strictly of the form:

$$\langle T(z)T(w)\rangle = \frac{\text{const.}}{(z-w)^4}.$$

By convention, this constant is denoted as $c/2$, where c is called the **central charge**.

The full OPE of $T(z)T(w)$ for $z \to w$ is tightly constrained. Both scale invariance and the requirement that correlation functions of local fields must be meromorphic[3] dictate the general structure of the singular terms:

$$T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{J(w)}{(z-w)^3} + \frac{Q(w)}{(z-w)^2} + \frac{R(w)}{z-w} + \text{regular terms},$$

where the generic operators must have specific holomorphic dimensions: $h_J = 1$, $h_Q = 2$, and $h_R = 3$, since the total conformal weight of the singular terms must match the sum of the weights of the two T operators, which is 4.

Because $T(z)$ is a bosonic field, the OPE must be symmetric under the exchange of coordinates and operators, meaning $T(z)T(w) = T(w)T(z)$. We can write the reversed OPE for $w \to z$:

$$T(w)T(z) = \frac{c/2}{(w-z)^4} + \frac{J(z)}{(w-z)^3} + \frac{Q(z)}{(w-z)^2} + \frac{R(z)}{w-z} + \text{regular terms}.$$

Assuming analyticity of the fields for z near w, we can express the right-hand side entirely at the coordinate w by Taylor expanding the fields $J(z)$ and $Q(z)$ around w. For instance:

$$J(z) = J(w) + (z-w)\partial_w J(w) + \frac{1}{2}(z-w)^2\partial_w^2 J(w) + \dots$$

[3] Meromorphic means that the function is analytic except for isolated poles.

Substituting these expansions into the reversed OPE, we get:

$$T(w)T(z) = \frac{c/2}{(z-w)^4} - \frac{J(w)}{(z-w)^3} + \frac{Q(w) + \partial_w J(w)}{(z-w)^2} - \frac{\partial_w Q(w) + \frac{1}{2}\partial_w^2 J(w) - R(w)}{z-w} + \text{regular terms.}$$

By matching the singularities term by term with the original $T(z)T(w)$ expansion, we find strict algebraic constraints on the unknown fields:

$$J(w) \equiv 0\ , \quad R(w) = \frac{1}{2}\partial_w Q(w),$$

which means that $J(w)$ along with its derivatives must vanish, and $R(w)$ is not an independent operator but is determined by $Q(w)$.

If we now make the highly reasonable physical hypothesis that <u>$T(z)$ is the *only* conformal field of spin and dimension 2 in the algebra of local fields</u>, then $Q(w)$ must be proportional to $T(w)$ itself. Indeed, we know that $[Q(w)] = 2$ and it must be holomorphic thus chiral:

$$\partial_{\bar{w}} Q(w) = 0, \quad \partial_w \bar{Q}(\bar{w}) = 0,$$

which looks exactly like the structure of $T(w)$. We can thus write $Q(w) = \kappa T(w)$. The OPE then simplifies to:

$$T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{\kappa T(w)}{(z-w)^2} + \frac{\frac{\kappa}{2}\partial_w T(w)}{z-w} + \text{regular terms.}$$

To fix the proportionality constant κ, we must look at the physical meaning of the second-order pole in an OPE with the stress-energy tensor.

- For any general field, the coefficient of the $(z-w)^{-2}$ pole strictly corresponds to its conformal weight h, which dictates its transformation under the global dilatation generator L_0.
- Since we analytically know that the stress-energy tensor $T(w)$ is defined as a field of conformal dimension $h = 2$, requiring dimensional consistency under a global dilatation $z, w \to \lambda z, \lambda w$ immediately fixes $\kappa = 2$.
- Furthermore, we know the OPE of the stress-energy tensor with a generic primary field $\Phi(w)$ of weight h must contain a second-order pole with coefficient $h\Phi(w)$:

  $$T(z)\Phi(w) = \frac{h\Phi(w)}{(z-w)^2} + \frac{\partial_w \Phi(w)}{z-w} + \text{regular terms,}$$

  which is the defining OPE for a primary field. Now, $T(w)$ is a quasi-primary field of weight $h = 2$, so intuitively, the second-order pole in the $T(z)T(w)$ OPE must also have coefficient $2T(w)$ to be consistent with the transformation properties of T itself.

This consequently sets the coefficient of the first-order pole to 1, yielding the fundamental TT operator product expansion valid for any two-dimensional Conformal Field Theory:

$$T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} + \text{regular terms.} \tag{2.4.1}$$

This is the general form of the $T(z)T(w)$ OPE, which holds for all two-dimensional CFTs. It encodes the algebraic structure of the conformal symmetry and the presence of the central charge c, which characterizes the quantum anomaly in the conformal symmetry.

Central Charge

The constant c is a fundamental characteristic of each specific theory, rightfully named the **central charge**. It is fixed once the stress-energy tensor is defined, which in turn is known when the underlying Lagrangian is specified.

As we have seen in our practical calculations for free models, different theories possess different central charges: for a free massless boson $c = 1$, while for a free massless Majorana fermion $c = 1/2$.

For physical, unitary models defined by Hermitian Hamiltonians, the Hermiticity of the stress-energy tensor T requires that its two-point correlator be strictly positive, leading to the bound:

$$c > 0 \quad \text{for Hermitian (unitary) models.}$$

An entirely analogous $\bar{T}\bar{T}$ OPE defines an anti-holomorphic central charge $\bar{c}$. If the theory is parity invariant (meaning it is completely symmetric under the spatial reflection $z \leftrightarrow \bar{z}$), the two central charges must coincide:

$$c = \bar{c}.$$

Variation of T under Conformal Transformations

To understand how the energy-momentum tensor transforms, we first recall the Cauchy integral formula and its generalization for derivatives:

$$f(z) = \oint_z \frac{\mathrm{d}w}{2\pi i} \frac{f(w)}{w - z}\,, \qquad \frac{d^n f(z)}{dz^n} = n! \oint_z \frac{\mathrm{d}w}{2\pi i} \frac{f(w)}{(w - z)^{n+1}},$$

where the contour integral is taken over a path in the complex w-plane encircling the point z.

We also recall that the infinitesimal variation of any local conformal field $A(z, \bar{z})$ under a generic holomorphic conformal transformation $\epsilon(z)$ is generated by its commutator with the conserved charge (from section 2.2.3), which can be expressed as a contour integral of the OPE:

$$\delta_\epsilon A(z, \bar{z}) = \oint_z \frac{\mathrm{d}w}{2\pi i} \epsilon(w) T(w) A(z, \bar{z}).$$

We now apply this general formula to the specific case where the field itself is the stress-energy tensor, $A(z, \bar{z}) = T(z)$. By inserting the fundamental TT operator product expansion (expanded around z) into the integral:

$$T(w)T(z) \sim \frac{c/2}{(w - z)^4} + \frac{2T(z)}{(w - z)^2} + \frac{\partial_z T(z)}{w - z},$$

we obtain:

$$\delta_\epsilon T(z) = \oint_z \frac{\mathrm{d}w}{2\pi i} \epsilon(w) \left[\frac{c/2}{(w - z)^4} + \frac{2T(z)}{(w - z)^2} + \frac{\partial_z T(z)}{w - z} \right].$$

Using the Cauchy derivative formulas on the test function $\epsilon(w)$, we can evaluate each term of the singular expansion exactly. The simple pole yields $\epsilon(z)\partial_z T(z)$, the double pole yields $2(\partial_z \epsilon(z))T(z)$, and the fourth-order pole produces $\frac{1}{3!}\partial_z^3 \epsilon(z)$ multiplied by $c/2$. Putting it all together, we find:

$$\delta_\epsilon T(z) = \epsilon(z)\partial_z T(z) + 2(\partial_z \epsilon(z))T(z) + \frac{c}{12}\partial_z^3 \epsilon(z).$$

This represents the infinitesimal conformal transformation of $T(z)$. The first two terms perfectly match the expected transformation law for a primary field of conformal weight $h = 2$. However, the third term is an inhomogeneous anomaly proportional to the central charge c, explicitly demonstrating that the stress-energy tensor is only a quasi-primary field, not a primary one.

By an exponentiation procedure, one can integrate this infinitesimal variation to obtain the exact finite transformation law under a macroscopic conformal map $z \mapsto w(z)$:

$$T(z) = \left(\frac{\partial w}{\partial z}\right)^2 T(w) + \frac{c}{12}\{w, z\}_S,$$

where the symbol $\{w, z\}_S$ denotes the **Schwarzian derivative**, defined as:

$$\{w, z\}_S = \frac{\partial_z^3 w}{\partial_z w} - \frac{3}{2}\left(\frac{\partial_z^2 w}{\partial_z w}\right)^2.$$

TODO: check everything.

Example (Infinitesimal Limit of the Schwarzian Transformation). We want to expand the finite transformation law for a small coordinate change $w = z + \epsilon(z)$ to reproduce the infinitesimal variation $\delta_\epsilon T(z)$.

First, we compute the derivatives of w with respect to z: $\partial_z w = 1 + \partial_z \epsilon$, $\partial_z^2 w = \partial_z^2 \epsilon$, and $\partial_z^3 w = \partial_z^3 \epsilon$.

For the Jacobian prefactor, we expand to first order in ϵ:

$$\left(\frac{\partial w}{\partial z}\right)^2 = (1 + \partial_z \epsilon)^2 \approx 1 + 2\partial_z \epsilon$$

For the Schwarzian derivative, the denominator is $1 + \partial_z \epsilon \approx 1$, so to first order:

$$\{w, z\}_S \approx \frac{\partial_z^3 \epsilon}{1} - \frac{3}{2}(\partial_z^2 \epsilon)^2 \approx \partial_z^3 \epsilon$$

(The squared term $(\partial_z^2 \epsilon)^2$ is second-order in ϵ and thus vanishes).

Now, substituting these into the finite transformation law $T(z) = (\partial_z w)^2 T(w) + \frac{c}{12}\{w, z\}_S$:

$$T(z) \approx (1 + 2\partial_z \epsilon) T(z + \epsilon) + \frac{c}{12}\partial_z^3 \epsilon$$

Taylor expanding $T(z + \epsilon) \approx T(z) + \epsilon \partial_z T(z)$ and keeping only first-order terms:

$$T(z) \approx (1 + 2\partial_z \epsilon)(T(z) + \epsilon \partial_z T(z)) + \frac{c}{12}\partial_z^3 \epsilon$$

$$T(z) \approx T(z) + \epsilon \partial_z T(z) + 2(\partial_z \epsilon) T(z) + \frac{c}{12}\partial_z^3 \epsilon$$

Subtracting $T(z)$ from both sides isolates the variation $\delta_\epsilon T(z)$:

$$\delta_\epsilon T(z) = \epsilon(z)\partial_z T(z) + 2(\partial_z \epsilon(z)) T(z) + \frac{c}{12}\partial_z^3 \epsilon(z)$$

This exactly matches the result derived from the contour integral of the TT OPE.

2.4.2 | Physical Meaning of the Central Charge

To fully grasp the physical significance of the central charge c, we investigate how the vacuum of the theory behaves when we move from the infinite flat plane to a space with a finite geometry. We do this by mapping the complex z-plane to a cylinder of circumference L using the standard conformal transformation:

$$z = e^{\frac{2\pi}{L} w} \quad \text{or equivalently} \quad w = \frac{L}{2\pi} \log z$$

where $w = \tau + i\sigma$ is the coordinate on the cylinder.

To see how the stress-energy tensor transforms under this map, we first compute the necessary derivatives. The first derivative is simply $\partial_z w = \frac{L}{2\pi z}$. The Schwarzian derivative $\{w, z\}_S$ measures the "non-Möbius" nature of the transformation. By plugging the derivatives of the logarithm into the definition of the Schwarzian, the scale factor $L/2\pi$ neatly cancels out, leaving:

$$\{w, z\}_S = \frac{1}{2z^2}$$

Substituting these ingredients into the finite transformation law for the stress-energy tensor, we obtain the exact form of the tensor on the cylinder:

$$T_{\text{cyl}}(w) = \left(\frac{2\pi}{L}\right)^2 \left\{T_{\text{pl}}(z)z^2 - \frac{c}{24}\right\}$$

We can now evaluate the vacuum expectation value of this operator. On the infinite plane, global conformal invariance dictates that the one-point function of any quasi-primary field with non-zero dimension must vanish. Therefore, $\langle T_{\text{pl}}(z)\rangle = 0$.

However, inserting this vanishing value into our transformation yields a strictly non-zero expectation value on the cylinder:

$$\langle T_{\text{cyl}}(w)\rangle = -\frac{\pi^2 c}{6L^2}$$

This is a profound physical result. The expectation value of the diagonal component $T + \bar{T} = T_{11}$ is directly proportional to the vacuum energy density. While the vacuum energy is exactly zero on the infinite plane, the periodic boundary conditions of the cylindrical geometry induce a macroscopic shift in the zero-point energy.

This non-zero vacuum energy density is the conformal field theory realization of the **Casimir energy**, caused entirely by finite-size boundary effects. Including both the holomorphic and anti-holomorphic sectors, the total Casimir energy density is given by:

$$\langle T_{11}\rangle = \frac{1}{2\pi}\{\langle T\rangle + \langle \bar{T}\rangle\} = -\frac{\pi c}{6L^2}$$

This explicitly demonstrates that the central charge c is not just an abstract algebraic parameter governing the Virasoro anomaly, but a measurable physical quantity that dictates the finite-size scaling behavior of the theory's ground state.

2.4.3 | Ward Identities and Correlation Functions

We now derive the conformal Ward identities. Consider a correlation function of local fields:

$$\langle \Phi_1(z_1, \bar{z}_1) \cdots \Phi_n(z_n, \bar{z}_n)\rangle$$

Inserting the stress tensor $T(z)$ into the correlator, and using its role as the generator of conformal transformations, one finds the local Ward identity:

$$\langle T(z)\Phi_1(z_1, \bar{z}_1) \cdots \Phi_n(z_n, \bar{z}_n)\rangle = \sum_{i=1}^{n} \left[\frac{h_i}{(z - z_i)^2} + \frac{1}{z - z_i}\partial_{z_i}\right] \langle \Phi_1(z_1, \bar{z}_1) \cdots \Phi_n(z_n, \bar{z}_n)\rangle$$

provided that all the fields Φ_i are primary fields. Similarly, in the anti-holomorphic sector:

$$\langle \bar{T}(\bar{z})\Phi_1(z_1, \bar{z}_1) \cdots \Phi_n(z_n, \bar{z}_n)\rangle = \sum_{i=1}^{n} \left[\frac{\bar{h}_i}{(\bar{z} - \bar{z}_i)^2} + \frac{1}{\bar{z} - \bar{z}_i}\partial_{\bar{z}_i}\right] \langle \Phi_1(z_1, \bar{z}_1) \cdots \Phi_n(z_n, \bar{z}_n)\rangle$$

These fundamental relations are known as the local conformal Ward identities.

Proof. The derivation follows straightforwardly from the Operator Product Expansion (OPE). Since the stress-energy tensor $T(z)$ is purely holomorphic everywhere away from operator insertions, the correlator $\langle T(z)\Phi_1 \cdots \Phi_n \rangle$ behaves as a meromorphic function of the coordinate z. Its singularities arise solely when z approaches one of the insertion points z_i. By applying the defining OPE between the stress tensor and a primary field for each Φ_i, and subsequently summing all the singular contributions, one exactly recovers the expression for the local Ward identity. □

Global Ward Identities

If we restrict our attention to the three globally defined holomorphic conformal transformations (the Möbius transformations), which correspond to the choices $\epsilon(z) = 1$, $\epsilon(z) = z$, and $\epsilon(z) = z^2$, the local Ward identities reduce to a set of global constraints:

$$\sum_{i=1}^{n} \partial_{z_i} \langle \Phi_1(z_1, \bar{z}_1) \cdots \Phi_n(z_n, \bar{z}_n) \rangle = 0$$

$$\sum_{i=1}^{n} (z_i \partial_{z_i} + h_i) \langle \Phi_1(z_1, \bar{z}_1) \cdots \Phi_n(z_n, \bar{z}_n) \rangle = 0$$

$$\sum_{i=1}^{n} (z_i^2 \partial_{z_i} + 2h_i z_i) \langle \Phi_1(z_1, \bar{z}_1) \cdots \Phi_n(z_n, \bar{z}_n) \rangle = 0$$

There are perfectly analogous equations governing the anti-holomorphic sector.

These relations constitute the Ward identities for the global conformal group in two dimensions. They impose stringent differential constraints on the correlation functions of primary fields, strictly reflecting how these fields must transform under global conformal mappings. Physically, each identity enforces the invariance of the theory under a specific subgroup: the first equation guarantees translation invariance, the second enforces invariance under dilations and rotations, and the third corresponds to special conformal transformations. Solving this set of differential equations is a crucial step in understanding the structure of two-dimensional Conformal Field Theories, as it allows us to completely determine the exact analytical form of two- and three-point correlation functions up to overall multiplicative constants.

Correlation Functions from Ward Identities

We can start from the two-point function of primary fields, which is constrained by conformal symmetry to take the form:

$$G(z_1, \bar{z}_1; z_2, \bar{z}_2) = \langle \phi_1(z_1, \bar{z}_1) \phi_2(z_2, \bar{z}_2) \rangle = \frac{C_{12}}{(z_1 - z_2)^{2h} (\bar{z}_1 - \bar{z}_2)^{2\bar{h}}},$$
$$G(|z_{12}|, |\bar{z}_{12}|) = \frac{C}{|z_{12}|^{h_1 + h_2} |\bar{z}_{12}|^{\bar{h}_1 + \bar{h}_2}}, \quad z_{ij} = z_i - z_j,$$

where from the first line we see that the two-point function is non-zero only if the conformal weights of the two fields are equal, i.e., $h_1 = h_2$ and $\bar{h}_1 = \bar{h}_2$. This is a consequence of the Ward identities, which require that the correlation function be invariant under conformal transformations. If the weights were different, the correlation function would not be invariant under dilations or special conformal transformations, leading to a contradiction.

So the only two points correlation functions which can be different from zero are those in which the fields enter with the same weight; it may even be the same kind of operator, or a multiplet of operators with the same conformal weights. In this case, the two-point function is fixed up to a constant C. Indeed now, we can rewrite the constant as

$$C = \delta_{h_1,h_2}\delta_{\bar{h}_1,\bar{h}_2},$$

which leads us to the final form of the two-point function:

$$G(|z_{12}|, |\bar{z}_{12}|) = \frac{\delta_{h_1,h_2}\delta_{\bar{h}_1,\bar{h}_2}}{|z_{12}|^{h_1+h_2}|\bar{z}_{12}|^{\bar{h}_1+\bar{h}_2}} \simeq \frac{1}{(x_1-x_2)^{2\Delta}},$$

where in the last step we have used the fact that the two-point function depends only on the distance between the two points, and we have defined $\Delta = h + \bar{h}$ as the scaling dimension of the field. This form of the two-point function is a direct consequence of conformal invariance and is a fundamental result in two-dimensional conformal field theory. It shows that the correlation functions of primary fields are highly constrained by symmetry, and it provides a starting point for understanding more complex correlation functions in these theories.

If we now insert the stress energy tensor $T(z)$ into the correlation function, we can derive the operator product expansion (OPE) between $T(z)$ and a primary field $\phi(w, \bar{w})$. The OPE takes the form:

$$\langle T(z)\phi(z_1, \bar{z}_1)\cdots\phi(z_n, \bar{z}_n)\rangle = \dots$$

where we used the fact that the OPE of $T(z)$ with a primary field $\phi(w, \bar{w})$ is given by:

$$T(z)\phi(w, \bar{w}) \sim \frac{h}{(z-w)^2}\phi(w, \bar{w}) + \frac{1}{z-w}\partial_w\phi(w, \bar{w}) + \text{regular terms},$$

where h is the conformal weight of the primary field ϕ. This OPE encodes how the stress energy tensor acts on the primary field, and it is a key ingredient in understanding the structure of conformal field theories. The first term in the OPE represents the scaling dimension of the primary field, while the second term represents the action of the Virasoro algebra on the primary field. Then we have a method to compute the correlation functions of primary fields by using the OPE with the stress energy tensor, which allows us to derive the Ward identities and ultimately determine the form of the correlation functions in two-dimensional conformal field theories.

We need correlation function to be invariant under the global conformal transformations, which are generated by the modes of the stress energy tensor. This leads to the Ward identities, which impose constraints on the correlation functions of primary fields.

For the three-point function, the Ward identities imply that it must take the form:

$$\langle\phi_1(z_1, \bar{z}_1)\phi_2(z_2, \bar{z}_2)\phi_3(z_3, \bar{z}_3)\rangle = \frac{C_{123}}{(z_{12})^{h_1+h_2-h_3}(z_{23})^{h_2+h_3-h_1}(\bar{z}_{31})^{\bar{h}_1+\bar{h}_3-\bar{h}_2}(\bar{z}\setminus\backslash)},$$

where now the three-point function is fixed up to a constant C_{123}, which is known as the structure constant of the theory. The form of the three-point function is determined by conformal symmetry, and holds for all quasi-primary fields. The structure constant C_{123} encodes the interactions between the three fields and is a fundamental quantity in conformal field theory: we need a method to find it.

Since solving a theory means solving or finding a way to compute all the correlation functions, we need to find a method to compute the structure constants. This is where the operator product expansion (OPE) comes into play. The OPE allows us to express the product of two fields as a sum over all possible fields in the theory, with coefficients that depend on the positions of the fields. By using the OPE, we can relate the three-point function to the two-point function and extract the structure constant C_{123}. It is not guaranteed that the OPE will still work in the same way as in higher dimensions, but in two-dimensional conformal field theories, the OPE is a powerful tool that allows us to compute correlation functions and determine the structure constants of the theory. This is a key step in solving two-dimensional conformal field theories and understanding their properties.

2.5 | Virasoro Algebra

2.5.1 | Mode Expansion of the Stress Energy Tensor

Starting from an infinitesimal conformal transformation in the complex coordinate parameterized by a holomorphic function

$$w = z + \epsilon(z),$$

we recall that the conserved Noether charge associated with this local symmetry can be expressed as a contour integral of the stress-energy tensor $T(z)$ around a closed path C:

$$Q_\epsilon = \frac{1}{2\pi i} \oint_C \mathrm{d}z\, \epsilon(z) T(z).$$

To systematically study these charges and the algebraic structure they form, it is extremely convenient to expand our local fields into discrete modes. On the cylinder, where the spatial coordinate is naturally compact and periodic, any generic local field $\phi(\sigma, \tau) = \phi(w, \bar{w})$ admits a standard Fourier expansion:

$$\phi(w, \bar{w}) = \sum_{n,\bar{n}=-\infty}^{+\infty} \phi_{n,\bar{n}} e^{-\frac{2\pi i}{L}(nw + \bar{n}\bar{w})}.$$

When we map this theory back to the complex plane, we must take into account the transformation law for a primary field of conformal weights $(h, \bar{h})$. Since the scaling dimensions dictate how the field rescales, this Fourier series translates directly into a Laurent expansion around the origin:

$$\phi(z, \bar{z}) = \sum_{n,\bar{n}=-\infty}^{+\infty} \phi_{n,\bar{n}} z^{-n-\Delta} \bar{z}^{-\bar{n}-\bar{\Delta}}.$$

By using the Cauchy integral theorem, we can isolate and extract the individual mode operators $\phi_{n,\bar{n}}$ via the inversion formula:

$$\phi_{n,\bar{n}} = \oint_0 \frac{\mathrm{d}z}{2\pi i} \oint_0 \frac{\mathrm{d}\bar{z}}{2\pi i} z^{n+\Delta-1} \bar{z}^{\bar{n}+\bar{\Delta}-1} \phi(z, \bar{z}).$$

We now apply this general formalism specifically to the stress-energy tensor. As we established earlier, the tensor splits into independent chiral ($T(z)$) and anti-chiral ($\bar{T}(\bar{z})$) components, which possess conformal weights $(2, 0)$ and $(0, 2)$ respectively. Therefore, setting $\Delta = 2$, their Laurent expansions take the special form:

$$T(z) = \sum_{n=-\infty}^{+\infty} L_n z^{-n-2}\ , \quad \bar{T}(\bar{z}) = \sum_{n=-\infty}^{+\infty} \bar{L}_n \bar{z}^{-n-2}.$$

The expansion coefficients L_n and $\bar{L}_n$ are the celebrated Virasoro modes. Inverting the series through contour integration, these modes are given explicitly by:

$$L_n = \oint_0 \frac{\mathrm{d}z}{2\pi i} z^{n+1} T(z)\ , \quad \bar{L}_n = \oint_0 \frac{\mathrm{d}\bar{z}}{2\pi i} \bar{z}^{n+1} \bar{T}(\bar{z}).$$

Notice the profound connection between these modes and the conserved charges we introduced at the beginning. Because the transformations $\epsilon(z)$ are holomorphic, they can be expanded in a basis of polynomials $\epsilon_n(z) = -z^{n+1}$. Inserting this expansion into the integral for Q_ϵ, the general conserved conformal charge becomes a direct linear combination of the Virasoro modes:

$$Q_\epsilon = \sum_{n=-\infty}^{+\infty} \epsilon_n L_n. \tag{2.5.1}$$

Consequently, the holomorphic conformal generators $\{L_n\}$ form the exact basis of a linear Lie algebra, just as the anti-holomorphic modes $\{\bar{L}_n\}$ form an independent copy of this algebra for the anti-chiral sector.

2.5.2 | Virasoro Algebra from Stress Energy Tensor OPE

We can now compute the exact commutation relations of the Virasoro modes L_n. Starting from their definition via contour integrals:

$$L_n = \oint_0 \frac{\mathrm{d}w}{2\pi i} w^{n+1} T(w) \ , \quad L_m = \oint_0 \frac{\mathrm{d}z}{2\pi i} z^{m+1} T(z),$$

we can express their commutator $[L_m, L_n]$ using the radial ordering formalism we developed earlier. The commutator translates into the difference of two contour integrals, which can be deformed into a single integral of the radially ordered product, where the z contour tightly encloses the point w:

$$[L_m, L_n] = \frac{1}{(2\pi i)^2} \oint_0 \mathrm{d}w\, w^{n+1} \oint_w \mathrm{d}z\, z^{m+1} T(z)T(w).$$

To evaluate this, we insert the singular part of the $T(z)T(w)$ Operator Product Expansion into the integral:

$$[L_m, L_n] = \frac{1}{(2\pi i)^2} \oint_0 \mathrm{d}w\, w^{n+1} \oint_w \mathrm{d}z\, z^{m+1} \left[\frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} \right].$$

We can evaluate the inner integral over z using Cauchy's integral formula and its derivatives:

$$f(w) = \oint_w \frac{\mathrm{d}z}{2\pi i} \frac{f(z)}{z-w} \ , \quad \frac{\mathrm{d}^k f(w)}{\mathrm{d}w^k} = k! \oint_w \frac{\mathrm{d}z}{2\pi i} \frac{f(z)}{(z-w)^{k+1}}.$$

We divide the calculation into three parts, corresponding to the three poles in the OPE.

Fourth-Order Pole

The fourth-order pole gives a purely numerical contribution proportional to the central charge c:

$$\begin{aligned} I_1 &= \frac{1}{2\pi i} \oint_0 \mathrm{d}w\, w^{n+1} \oint_w \frac{\mathrm{d}z}{2\pi i} z^{m+1} \frac{c/2}{(z-w)^4} \\ &= \frac{c/2}{3!} \frac{1}{2\pi i} \oint_0 \mathrm{d}w\, w^{n+1} \partial_w^3 (w^{m+1}) \\ &= \frac{c}{12}(m+1)m(m-1) \frac{1}{2\pi i} \oint_0 \mathrm{d}w\, w^{m+n-1}. \end{aligned}$$

Since the contour integral of w^k around the origin vanishes unless $k = -1$ (where it equals $2\pi i$), this term is non-zero only when $m + n = 0$:

$$I_1 = \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

Second-Order Pole

The second-order pole contributes:

$$\begin{aligned}
I_2 &= \frac{1}{2\pi i}\oint_0 \mathrm{d}w\, w^{n+1} \oint_w \frac{\mathrm{d}z}{2\pi i} z^{m+1} \frac{2T(w)}{(z-w)^2} \\
&= \frac{2}{2\pi i}\oint_0 \mathrm{d}w\, w^{n+1} \partial_w (w^{m+1}) T(w) \\
&= 2(m+1)\frac{1}{2\pi i}\oint_0 \mathrm{d}w\, w^{m+n+1} T(w).
\end{aligned}$$

Recognizing the integral as the exact definition of the mode L_{m+n}, we get:

$$I_2 = 2(m+1)L_{m+n}.$$

First-Order Pole

Finally, the first-order pole yields:

$$\begin{aligned}
I_3 &= \frac{1}{2\pi i}\oint_0 \mathrm{d}w\, w^{n+1} \oint_w \frac{\mathrm{d}z}{2\pi i} z^{m+1} \frac{\partial_w T(w)}{z-w} \\
&= \frac{1}{2\pi i}\oint_0 \mathrm{d}w\, w^{m+n+2} \partial_w T(w).
\end{aligned}$$

Using integration by parts on the w contour, this becomes:

$$I_3 = -\frac{1}{2\pi i}\oint_0 \mathrm{d}w\, \partial_w (w^{m+n+2}) T(w) = -(m+n+2)L_{m+n}.$$

Virasoro Commutation Relations

Summing the three contributions ($I_1 + I_2 + I_3$), we obtain the fundamental commutation relations of the modes:

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}. \tag{2.5.2}$$

This equation defines the **Virasoro algebra**. It is the infinite-dimensional algebra generating the holomorphic conformal transformations in two-dimensional CFT.

By repeating the exact same procedure for the anti-holomorphic stress tensor $\bar{T}(\bar{z})$, we find that its modes $\bar{L}_n$ satisfy an identical algebra:

$$[\bar{L}_m, \bar{L}_n] = (m-n)\bar{L}_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}. \tag{2.5.3}$$

Furthermore, because the holomorphic and anti-holomorphic sectors are completely independent, their respective generators commute:

$$[L_m, \bar{L}_n] = 0.$$

Thus, the full local conformal symmetry in two dimensions is dynamically governed by two independent, commuting copies of the Virasoro algebra. This rich and rigidly constrained algebraic structure is the cornerstone that makes two-dimensional conformal field theories exactly solvable in many cases.

The Full Conformal Algebra and the Classical Limit

In other words, the complete local conformal algebra in two dimensions is given by the direct product of the holomorphic and anti-holomorphic sectors:

$$\Gamma = \mathrm{Vir}_c \otimes \overline{\mathrm{Vir}}_c$$

where Vir_c denotes the Virasoro algebra with central charge c, explicitly written as:

$$\begin{aligned}
[L_n, L_m] &= (n-m)L_{n+m} + \frac{c}{12}n(n^2-1)\delta_{n,-m}, \\
[L_n, \bar{L}_m] &= 0, \\
[\bar{L}_n, \bar{L}_m] &= (n-m)\bar{L}_{n+m} + \frac{c}{12}n(n^2-1)\delta_{n,-m}.
\end{aligned} \tag{2.5.4}$$

Consequently, the physical conformal fields of the theory must lie in irreducible representations of this full algebra Γ.

It is particularly illuminating to consider the limit where the central charge vanishes. If we set $c = 0$, the Virasoro algebra rigorously reduces to:

$$[L_m, L_n] = (m-n)L_{m+n},$$

which is precisely the **Witt algebra**, namely the classical Lie algebra of holomorphic vector fields defined on the circle. Therefore, mathematically, the Virasoro algebra represents the unique non-trivial central extension of the Witt algebra. Physically, the central term proportional to c is a purely quantum effect, arising directly from the short-distance singularities in the operator product expansion.

This profound observation perfectly clarifies the relation between the classical discussion of conformal symmetry and its quantum counterpart:

- **Classically**, the infinitesimal conformal transformations strictly form the Witt algebra without any anomalous extension;
- **Quantum mechanically**, the presence of the conformal anomaly forces the physical generators to satisfy the Virasoro algebra.

Historically, the profound understanding of these transformations originated in the context of string theory, where the fields defined on the two-dimensional worldsheet of the string transform precisely under the action of the Virasoro algebra. Mathematically, this constitutes a quintessential example of an infinite-dimensional Lie algebra. Its internal product is given by the mode commutator $[L_m, L_n]$, which strictly satisfies the Jacobi identity. The emergence of this algebraic structure not only revolutionized theoretical physics but also sparked an intense, dedicated mathematical effort to study and classify infinite-dimensional Lie algebras.

The overarching strategy to completely solve a two-dimensional Conformal Field Theory relies entirely on this algebraic framework. Despite the immense size of the Hilbert space and the local operator algebra, every physical state and field must fall into specific irreducible representations of the Virasoro algebra. Consequently, the full Hilbert space rigidly decomposes into a direct sum of these infinite-dimensional representations. By exploiting this representation theory, we can systematically classify the operator content of the theory and analytically compute correlation functions, which is the fundamental mechanism making two-dimensional conformal models exactly solvable.

The Global Conformal Subalgebra

An important finite-dimensional subalgebra of the full Virasoro algebra is generated by the following three modes:

$$L_{-1}, \quad L_0, \quad L_1.$$

It is crucial to notice that the central extension term, $\frac{c}{12}m(m^2-1)\delta_{m+n,0}$, strictly vanishes for $m, n \in \{-1, 0, 1\}$. Because $m^3 - m = 0$ on this specific subspace, the central anomaly disappears entirely. Consequently, if we restrict our algebra to these generators, the commutation relations close without any central charge contribution:

$$[L_0, L_{-1}] = L_{-1} \ , \quad [L_0, L_1] = -L_1 \ , \quad [L_1, L_{-1}] = 2L_0. \tag{2.5.5}$$

These three specific modes are precisely the generators of the **global conformal group** (the Möbius transformations). Algebraically, they form the Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ in the holomorphic sector, which is exactly what we expect from the global conformal transformations defined on the Riemann sphere.

This elegantly confirms a fundamental physical property: <u>the central extension affects only the genuinely local, infinite-dimensional part of the conformal symmetry</u>. While the local conformal symmetry inevitably acquires a quantum anomaly, *the global conformal symmetry remains perfectly unbroken in the quantum theory.*

2.5.3 | Interpretation of the Central Charge

The central charge c is arguably the most important macroscopic parameter of a two-dimensional conformal field theory. As we established, while the classical algebra of infinitesimal conformal transformations is the Witt algebra, the quantum mechanical framework modifies this algebra by introducing a central extension. This quantum modification mathematically originates from the singular fourth-order pole in the TT Operator Product Expansion. The coefficient c precisely measures the size of this quantum anomaly.

It is extremely useful to emphasize the following fundamental points regarding c:

- c is a dynamical parameter and is not uniquely fixed by conformal symmetry alone.
- Different physical theories generally correspond to distinct values of c.
- c actively counts, in a broad sense, the *number of effective degrees of freedom of the theory.* We can plainly see this additive property from our earlier concrete examples:
  - a single free real scalar boson has $c = 1$,
  - a single free Majorana fermion has $c = 1/2$.

This intuitive statement (that the central charge counts the effective degrees of freedom) can be made mathematically rigorous with advanced methods that go beyond the immediate scope of these lectures (most notably through Zamolodchikov's famous c-theorem, which proves that a generalized function c strictly decreases along Renormalization Group flows, formalizing the physical loss of high-energy degrees of freedom as a system cools down).

Ultimately, the physical significance of the central charge is multifold and pervasive across the whole theory:

- It formally measures the **conformal anomaly** (the quantum breaking of classical scale invariance when placing the theory on a curved background).
- It actively appears in the finite-size geometric dependence of the vacuum energy on the cylinder (generating the **Casimir effect**).
- It universally controls the asymptotic growth of the density of states at extremely high energies through **Cardy's formula**, effectively determining the high-energy thermodynamic entropy of the system.
- It plays a central, rigid role in the exact classification of the so-called **minimal models** (rational CFTs with a finite number of primary fields).

Thus, the precise numerical value of c is not merely an algebraic curiosity, but a quantity that encapsulates the deepest dynamical and thermodynamic information about the physical theory.

2.5.4 | Action of Virasoro Generators on Primary Fields

We now want to determine the explicit action of the Virasoro generators on a local primary field. To do this, we must compute the commutator $[L_n, \Phi(w, \bar{w})]$. In the framework of radial quantization, we must be incredibly careful with the ordering of operators: the commutator translates geometrically into the difference of two contour integrals over z, one where $|z| > |w|$ and one where $|z| < |w|$. Applying the exact same contour deformation procedure we used for the TT commutator, this difference collapses into a single contour integral over z tightly wrapping the insertion point w:

$$[L_n, \Phi(w, \bar{w})] = \oint_w \frac{\mathrm{d}z}{2\pi i} z^{n+1} T(z)\Phi(w, \bar{w}).$$

We can now evaluate this integral by starting from the defining Operator Product Expansion of the stress-energy tensor with a primary field:

$$T(z)\Phi(w, \bar{w}) \sim \frac{h\Phi(w, \bar{w})}{(z-w)^2} + \frac{\partial_w \Phi(w, \bar{w})}{z-w}.$$

Inserting the singular part of this OPE directly into our contour integral, we obtain:

$$[L_n, \Phi(w, \bar{w})] = \oint_w \frac{\mathrm{d}z}{2\pi i} z^{n+1} \left[\frac{h\Phi(w, \bar{w})}{(z-w)^2} + \frac{\partial_w \Phi(w, \bar{w})}{z-w} \right].$$

By applying Cauchy's integral formula and its first derivative formula to the analytic function $f(z) = z^{n+1}$ evaluated at the pole $z = w$, we can easily extract the residues for both terms:

$$[L_n, \Phi(w, \bar{w})] = (w^{n+1}\partial_w + h(n+1)w^n)\Phi(w, \bar{w}). \tag{2.5.6}$$

This formula is extremely important and represents a key result in two-dimensional conformal field theory. It shows explicitly how the infinite set of local Virasoro generators act dynamically on primary fields as differential operators.

Furthermore, we can observe the consistency of this result with the global subgroup. If we restrict this action to the modes $n = -1, 0, 1$, the differential operators accurately reproduce the classical action of the globally defined transformations. Specifically, L_{-1} generates spatial translations (∂_w), L_0 generates dilatations and rotations ($w\partial_w + h$), and L_1 generates special conformal transformations ($w^2\partial_w + 2hw$).

2.6 | Radial Quantization and State-Operator Correspondence

Up to this point, we have mostly described conformal field theory in terms of local fields, their transformation properties, and their correlation functions. We now move to a more intrinsic operator formulation.

The key idea is that in two-dimensional Euclidean field theory it is natural to quantize the theory not with respect to one of the Cartesian coordinates, but with respect to the radial distance from the origin. In this framework:

- circles centered at the origin play the role of equal-time slices;
- local operator insertions define states on such circles;
- the generators of conformal transformations act as operators on the Hilbert space.

This geometric setup naturally leads to the operator-state correspondence, one of the most remarkable and powerful features of two-dimensional conformal field theory.

2.6.1 | Transformation of the Stress Tensor under the Cylinder Map

Unlike a primary field, the stress tensor does not transform homogeneously under general conformal maps because of the central charge anomaly. Under a finite conformal transformation

$$z \mapsto w = f(z),$$

the transformation law of the stress tensor acquires an inhomogeneous shift:

$$T(z) = \left(\frac{dw}{dz}\right)^2 T(w) + \frac{c}{12}\{w, z\},$$

where the shift $\{w, z\}$ is the Schwarzian derivative, defined as:

$$\{w, z\} = \frac{w'''(z)}{w'(z)} - \frac{3}{2}\left(\frac{w''(z)}{w'(z)}\right)^2.$$

To study finite-size effects, we can map the complex plane to a cylindrical geometry. We define the transformation from the plane coordinate z to the cylinder coordinate w as $w = \frac{L}{2\pi}\log z$. Because the complex logarithm is a multi-valued function, we must choose a branch cut (for example, along the negative real axis, giving a phase shift of $\pm i\pi$). By identifying the values on the two sides of the cut via periodic boundary conditions, the mapped strip wraps around itself seamlessly, defining a cylinder (as we have seen in section 2.1.3).

If we decompose the new coordinate into its real and imaginary parts, $w = \tau + i\sigma$, we can identify $\tau \in (-\infty, \infty)$ as the infinite Euclidean time, and $\sigma \in [0, L]$ as the compactified spatial coordinate. We refer to this geometry as being infinite in time and finite in space. The finite size of the space is entirely dictated by the periodicity in σ, which is controlled by the scale factor $\frac{L}{2\pi}$.

For the sake of mathematical simplicity, let us explicitly compute the transformation of the stress-energy tensor using the normalized map $w = \log z$ (which corresponds to setting the circumference to $L = 2\pi$). The derivatives of this map are simply:

$$w'(z) = \frac{1}{z}\ , \quad w''(z) = -\frac{1}{z^2}\ , \quad w'''(z) = \frac{2}{z^3}.$$

Inserting these derivatives into the definition of the Schwarzian derivative, we find:

$$\{w, z\} = \frac{2/z^3}{1/z} - \frac{3}{2}\left(\frac{-1/z^2}{1/z}\right)^2 = \frac{2}{z^2} - \frac{3}{2z^2} = \frac{1}{2z^2}.$$

Substituting this result, along with the first derivative $\mathrm{d}w/\mathrm{d}z$, back into the general transformation law, we obtain:

$$T_{\text{plane}}(z) = \frac{1}{z^2}T_{\text{cyl}}(w) + \frac{c}{24z^2}.$$

Equivalently, we can invert this relation to isolate the stress tensor on the cylinder:

$$T_{\text{cyl}}(w) = z^2 T_{\text{plane}}(z) - \frac{c}{24}.$$

This final equation carries profound physical implications when we evaluate it on the vacuum state. On the infinite, flat complex plane, global conformal invariance dictates that the one-point function of any field with a non-zero conformal weight must strictly vanish, $\langle\varphi\rangle = 0$. The only field allowed to have a non-zero expectation value is the identity operator $\mathbb{I}$, which is perfectly invariant under all transformations and has a conformal weight of zero. Therefore, evaluating the stress tensor on the plane yields $\langle T_{\text{plane}}(z)\rangle = 0$.

However, if we take the vacuum expectation value of our inverted equation, we find a non-zero result on the cylinder. How can we have a vacuum expectation value different from zero? This occurs because the vacuum state on the cylinder is physically distinct from the vacuum state on the plane due to the non-trivial topology of the mapping. The cylindrical geometry is finite in space, which introduces boundary conditions that restrict the quantum fluctuations.

This constant shift implies that the vacuum energy on the cylinder is not zero. Summing the holomorphic and anti-holomorphic contributions yields a macroscopic shift in the ground-state energy:

$$E_0 = -\frac{c}{12},$$

(up to normalization conventions for the cylinder circumference). This is the conformal field theory realization of the **Casimir effect**: *a generalized vacuum energy driven purely by the finite geometry and directly proportional to the central charge c of the theory.*

Understanding transformations to finite geometries and the role of the central charge in these mappings is a cornerstone of modern theoretical physics. It has critical practical implications for the study of critical phenomena (especially when predicting finite-size scaling in numerical simulations constrained by finite memory or lattice sizes) as well as in string theory, where the fundamental worldsheet is frequently modeled precisely as a cylinder or a torus.

2.6.2 | Hamiltonian on the Cylinder

The ultimate goal of introducing this geometric machinery is to properly quantize the conformal field theory on the cylinder. By transforming the generator of time translations from the plane to the cylinder, we can identify the Hamiltonian of the theory in this geometry. Because τ represents the Euclidean time direction, the operator generating translations along τ is precisely the Hamiltonian H_{cyl}. Consequently, equal-time commutators in this quantization scheme correspond to commutators evaluated on the same cylindrical slice at a constant τ. Under the plane-to-cylinder map, this Hamiltonian is identified as:

$$H_{\text{cyl}} = L_0 + \bar{L}_0 - \frac{c}{12},$$

up to normalization conventions depending on the circumference of the cylinder.

Similarly, we can consider the momentum operator P_{cyl}, which generates translations along the compact spatial direction σ (i.e., rigid translations along the circumference of the cylinder). Interestingly, for the momentum operator, the anomalous displacement due to the central charge perfectly cancels out between the holomorphic and anti-holomorphic sectors. Thus, the momentum on the cylinder is simply given by:

$$P_{\text{cyl}} = L_0 - \bar{L}_0.$$

Mapping this dynamics back to the complex plane reveals a profound geometric correspondence. Because $\tau \propto \log |z|$, increasing the time τ on the cylinder corresponds exactly to increasing the modulus $|z|$ on the plane. Therefore, time evolution on the cylinder is precisely equivalent to radial evolution on the plane, and constant-time slices on the cylinder seamlessly map to concentric circles on the plane. In terms of **continuous symmetries**, time translations on the cylinder correspond to dilatations on the plane, while spatial translations along the cylinder's circumference correspond to purely angular rotations on the plane. The explicit identification of the Hamiltonian and momentum operators on the cylinder is thus crucial for quantizing the theory and understanding how its dynamics dynamically manifest across these equivalent geometries.

For a physical state characterized by conformal weights $(h, \bar{h})$, the energy on the cylinder is given by the eigenvalue of H_{cyl}:

$$E = h + \bar{h} - \frac{c}{12}$$

This relation elegantly shows that the eigenvalues of energy and momentum on the cylinder correspond directly to the scaling dimension $\Delta = h + \bar{h}$ and the spin $s = h - \bar{h}$ of the associated fields on the plane. The central charge c plays a fundamental role in determining the entire energy spectrum on the cylinder, leading directly to the presence of a non-zero vacuum energy (the Casimir effect) that we discussed earlier. Finally, this energy formula plays an extremely important role in finite-size scaling techniques, providing a direct analytical method to extract the central charge of physical statistical systems at criticality simply by studying their finite-volume spectrum.

2.6.3 | Time Ordering and Radial Ordering

In the standard formulation of Quantum Field Theory, the products of quantum operators inside correlation functions are physically meaningful only if they are properly time-ordered. The time-ordering operator $\mathcal{T}$ ensures that *operators acting at earlier times are applied to the state first* (i.e., placed to the right). On our newly defined cylinder, where τ acts as the time coordinate, the time-ordered product of two generic fields is defined as:

$$\mathcal{T}\{A_1(\sigma_1,\tau_1)A_2(\sigma_2,\tau_2)\} = \begin{cases} A_1(\sigma_1,\tau_1)A_2(\sigma_2,\tau_2) & \text{if } \tau_1 > \tau_2 \\ \eta A_2(\sigma_2,\tau_2)A_1(\sigma_1,\tau_1) & \text{if } \tau_1 < \tau_2 \end{cases}$$

where the statistical factor η accounts for the quantum statistics of the fields involved:

$$\eta = \begin{cases} +1 & \text{if at least one of the two fields is bosonic,} \\ -1 & \text{if both fields are fermionic.} \end{cases}$$

Because we have established that time τ on the cylinder corresponds exactly to the radial distance $|z|$ on the plane ($\tau \propto \log |z|$), the dynamical concept of time ordering is naturally translated into the geometric concept of **radial ordering** on the complex plane.

We define the radial ordering operator $\mathcal{R}$ such that operators located closer to the origin act first, evolving radially outwards. For two fields inserted at complex coordinates z_1 and z_2, this reads:

$$\mathcal{R}\{A_1(z_1,\bar{z}_1)A_2(z_2,\bar{z}_2)\} = \begin{cases} A_1(z_1,\bar{z}_1)A_2(z_2,\bar{z}_2) & \text{if } |z_1| > |z_2| \\ \eta A_2(z_2,\bar{z}_2)A_1(z_1,\bar{z}_1) & \text{if } |z_1| < |z_2| \end{cases}$$

This radial quantization is particularly natural in Conformal Field Theory because the radial ordering of operator insertions is strictly preserved by conformal transformations. Physically, a quantum state is associated with a circle of fixed radius r, and is defined by performing the path integral over the interior of the disk. Inserting local operators inside this disk systematically modifies the state defined on the boundary circle. Furthermore, beyond preserving causality, this geometric setup provides an incredibly powerful analytical tool: radial ordering is precisely the natural framework required to perform contour manipulations and evaluate Operator Product Expansions (OPEs) using complex analysis.

When we compute correlation functions or evaluate the Operator Product Expansion (OPE) of two fields as they approach each other, the radial ordering is strictly and implicitly assumed. Furthermore, it plays a vital role when translating commutators of conserved charges into contour integrals, where the difference between integrating over a larger circle versus a smaller circle precisely encodes the commutator (or anti-commutator) of the enclosed operators. We will see this in more detail in the following sections.

2.6.4 | The State-Operator Correspondence

To make the construction of radial quantization concrete, consider a circle of radius $|z| = R$ on the complex plane. A state on that boundary circle is defined by performing the Euclidean path integral over the interior disk $|z| < R$, keeping the field configurations on the boundary fixed.

If no operator is inserted inside the disk, the path integral naturally defines the vacuum state of the theory:

$$|0\rangle = \int_{|z|<R} \mathcal{D}\varphi \, e^{-S[\varphi]}.$$

However, if a local operator $\Phi(z,\bar{z})$ is inserted inside the disk, the path integral defines a new, modified state on the boundary. In particular, inserting the operator exactly at the origin defines a corresponding "ket" state:

$$|\Phi\rangle = \Phi(0,0)\,|0\rangle = \lim_{z,\bar{z}\to 0} \Phi(z,\bar{z})\,|0\rangle\,.$$

This geometric picture perfectly matches the cylinder mapping. The origin of the plane ($z = 0$) corresponds to the infinite past on the cylinder ($\tau \to -\infty$). Therefore, inserting an operator at the origin is equivalent to preparing an asymptotic “in” state in the standard scattering formalism. Because the conformal map from the cylinder lands on the Riemann sphere (the compactified complex plane), the operator creating the state remains strictly local. This requirement of locality ensures that the correlation functions are mathematically well-defined and physically consistent with the axioms of quantum field theory.

This powerful relation is known as the **state-operator correspondence**. While a mapping between states and local operators can sometimes appear in non-conformal theories, it is absolutely guaranteed to hold in two-dimensional Conformal Field Theory. It implies that *every state in the Hilbert space can be represented by some local operator inserted at the origin.* Crucially, this allows us to fully classify both the fields and the states of the theory using the irreducible representations of the Virasoro algebra.

Bra States and the Inversion Map

To formulate a complete quantum mechanics framework, we also need to define "bra" states, which correspond to the asymptotic "out" states at infinite future ($\tau \to +\infty$ on the cylinder). On the complex plane, this corresponds to inserting an operator at spatial infinity ($z \to \infty$). To define this rigorously without dealing with divergences, we use the inversion map:

$$z \mapsto w = \frac{1}{z}.$$

This inversion is a global Möbius transformation that beautifully maps the point at infinity back to the origin on the Riemann sphere. By transforming the field from z to this new coordinate w and taking the limit as $w \to 0$, we define a local field acting on the vacuum to create the "out state". When we transform a primary field of conformal weights $(h, \bar{h})$ through this inversion, the Jacobian factors introduce powers of z and $\bar{z}$. Thus, the conjugate bra state is defined as:

$$\langle\Phi| = \lim_{z,\bar{z}\to\infty} \langle 0| \, z^{2h}\bar{z}^{2\bar{h}}\Phi(z,\bar{z}).$$

These compensating scale factors are strictly necessary to produce a finite, non-zero limit and ensure consistency with the normalization of correlation functions.

Inner Products and Hilbert Space Sectors

The state-operator correspondence implies a direct and profound relation between the algebraic inner product of states in the Hilbert space and the analytic two-point correlation functions evaluated on the plane. Let Φ and Ψ be two primary fields. Using our definition for the bra state, the inner product is written as:

$$\langle\Psi|\Phi\rangle = \lim_{z,\bar{z}\to\infty} z^{2h_\Psi}\bar{z}^{2\bar{h}_\Psi} \langle 0| \Psi(z,\bar{z})\Phi(0,0) |0\rangle .$$

The expectation value on the right is exactly the standard two-point function of the fields. As we have derived previously from the global Ward identities, the two-point function of primary fields is highly constrained:

$$\langle\Psi(z,\bar{z})\Phi(0,0)\rangle = \frac{C_{\Psi\Phi}\delta_{h_\Psi,h_\Phi}\delta_{\bar{h}_\Psi,\bar{h}_\Phi}}{z^{2h_\Psi}\bar{z}^{2\bar{h}_\Psi}}.$$

Inserting this known form into the inner product, a beautiful cancellation occurs: the z^{2h_Ψ} and $\bar{z}^{2\bar{h}_\Psi}$ factors from the bra state definition exactly cancel the identical coordinate dependence in the denominator of the two-point function. The spatial dependence completely vanishes, leaving only the overall constant:

$$\langle\Psi|\Phi\rangle = C_{\Psi\Phi}\delta_{h_\Psi,h_\Phi}\delta_{\bar{h}_\Psi,\bar{h}_\Phi}.$$

Thus, the two-point function seamlessly defines the natural scalar product on the space of states.

This result carries a vital physical meaning: states created by operators with different conformal weights are strictly orthogonal. This orthogonality principle is extremely powerful, as it provides a natural mechanism to divide the entire Hilbert space of the theory into distinct, non-overlapping superselection sectors, each built upon a different primary state.

2.6.5 | Action of Virasoro Generators on States

We now examine how the Virasoro generators act on the states defined by local operators. Recall that a state associated with a primary field is defined by inserting the field at the origin in the path integral definition of the vacuum, $|\Phi\rangle = \Phi(0,0)\,|0\rangle$. To find the explicit action of the Virasoro modes L_n on this state, we use their commutation relation with the primary field:

$$[L_n, \Phi(w,\bar{w})] = (w^{n+1}\partial_w + h(n+1)w^n)\Phi(w,\bar{w}).$$

We evaluate this action by taking the limit as $w, \bar{w} \to 0$. First, we must rely on the fundamental properties of the vacuum state $|0\rangle$. The physical requirement that the stress-energy tensor $T(z)$ must be perfectly regular (finite) both at the origin and at spatial infinity imposes that the vacuum state must be annihilated by all modes L_n with $n > 0$. Furthermore, the conditions $L_{-1}\,|0\rangle = L_0\,|0\rangle = L_1\,|0\rangle = 0$ directly express the *invariance of the vacuum under the global conformal subgroup* (translations, dilatations, and special conformal transformations). Thus we know that the vacuum state gets annihilated by the application of any generator L_n for $n \geq -1$.

Since the vacuum is annihilated by these modes, applying the commutator to the vacuum state eliminates the term where the mode acts on the right:

$$\lim_{w,\bar{w}\to 0}(L_n\Phi(w,\bar{w}) - \Phi(w,\bar{w})L_n)\,|0\rangle = \lim_{w,\bar{w}\to 0} L_n\Phi(w,\bar{w})\,|0\rangle = L_n\,|\Phi\rangle\,.$$

For any $n \geq 1$, evaluating the right-hand side of the commutator at $w \to 0$ yields strictly zero, as all terms carry positive powers of w:

$$\begin{aligned} L_n\,|\Phi\rangle &= \lim_{w,\bar{w}\to 0}\left(w^{n+1}\partial_w\Phi(w,\bar{w}) + h(n+1)w^n\Phi(w,\bar{w})\right)|0\rangle \\ &= \lim_{w,\bar{w}\to 0}(w^{n+1}\,|\partial\Phi\rangle + h(n+1)w^n\,|\Phi\rangle) = 0 \end{aligned}$$

where now it is not important to specify the precise form of the state created by the derivative of the field, since the positive powers of w ensure that all contributions vanish in the limit. Thus, all positive modes L_n with $n \geq 1$ annihilate the primary state.

However, for the specific case of $n = 0$, the derivative term vanishes (due to the w^1 factor), but the zero-th power of w in the second term leaves a non-zero contribution. The commutator evaluation reduces precisely to $h\Phi(0,0)$, yielding:

$$L_0\,|\Phi\rangle = h\,|\Phi\rangle\,.$$

An identical derivation holds for the anti-holomorphic sector, yielding $\bar{L}_n\,|\Phi\rangle = 0$ for $n \geq 1$ and $\bar{L}_0\,|\Phi\rangle = \bar{h}\,|\Phi\rangle$. Because the state is annihilated by all positive modes and is an eigenstate of the zero-modes, we conclude that the state associated with a primary field acts as a **highest-weight state** for the Virasoro algebra.

Dilatations as the Hamiltonian of Radial Quantization

A crucial consequence of radial quantization is that *radial evolution is dynamically generated by dilatations.* Under a global scale transformation $z \to \lambda z$ and $\bar{z} \to \lambda\bar{z}$, the radial coordinate transforms as $r \to \lambda r$. In terms of the Euclidean cylinder time $\tau = \log r$, this purely geometric rescaling corresponds to a standard time translation $\tau \to \tau + \log\lambda$.

Therefore, the generator of dilatations intrinsically plays the role of the Hamiltonian in radial quantization. Combining the holomorphic and anti-holomorphic sectors, this generator is given by $L_0 + \bar{L}_0$. Similarly, the angular momentum operator, which generates rigid rotations, is given by the difference $L_0 - \bar{L}_0$.

We can now readily compute the action of these macroscopic physical operators on our primary states: using the eigenvalue equations we just derived from the Virasoro algebra, for a state associated with a primary field of conformal weights $(h, \bar{h})$, we find:

$$(L_0 + \bar{L}_0) |\Phi\rangle = (h + \bar{h}) |\Phi\rangle = \Delta |\Phi\rangle \,,$$
$$(L_0 - \bar{L}_0) |\Phi\rangle = (h - \bar{h}) |\Phi\rangle = s |\Phi\rangle \,.$$

Thus, the scaling dimension Δ becomes the exact energy of the state in radial quantization, while the conformal spin s corresponds directly to its physical angular momentum.

Highest-Weight States and Primary Fields

Motivated by our previous discussion on the action of the Virasoro modes, we can formally define a **highest-weight state** $\left|h, \bar{h}\right\rangle$ as a state satisfying the eigenvalue equations for the zero-modes:

$$L_0 \left|h, \bar{h}\right\rangle = h \left|h, \bar{h}\right\rangle , \quad \bar{L}_0 \left|h, \bar{h}\right\rangle = \bar{h} \left|h, \bar{h}\right\rangle ,$$

while being strictly annihilated by all positive modes:

$$L_n \left|h, \bar{h}\right\rangle = 0, \quad \bar{L}_n \left|h, \bar{h}\right\rangle = 0 \quad (n > 0).$$

These conditions perfectly match the properties we just derived for the states created by inserting primary fields at the origin. Therefore, we establish a fundamental structural identification of conformal field theory: <u>primary fields on the complex plane strictly correspond to highest-weight states in radial quantization</u>.

Descendant States

In standard Lie algebras like SU(2), applying lowering operators to a highest-weight state generates a finite multiplet of states with progressively lower eigenvalues. In the infinite-dimensional Virasoro algebra, however, the situation is different. Starting from a highest-weight state (the primary state), one obtains new states by acting with the negative Virasoro modes L_{-n} (with $n > 0$).

To understand the physical nature of these states, we evaluate their energy (the eigenvalue of L_0). We start from the fundamental commutation relation:

$$[L_0, L_{-n}] = nL_{-n}.$$

By applying this commutator to a primary state $|\Phi\rangle = \left|h, \bar{h}\right\rangle$ of weight h, we find:

$$L_0(L_{-n} |\Phi\rangle) = (L_{-n}L_0 + nL_{-n}) |\Phi\rangle = (h + n)(L_{-n} |\Phi\rangle).$$

This reveals a crucial physical property: while the operators L_{-n} act algebraically as lowering operators, they physically *increase* the conformal weight by n units. Because the conformal weight dictates the energy on the cylinder, the state $L_{-n} |\Phi\rangle$ has n units of energy more than the primary state. Therefore, the "highest-weight" state actually represents the *stable, lowest-energy ground state of its representation sector.*

We can apply these negative modes multiple times to the primary state, generating an infinite "tower" of descendant states. A generic descendant state takes the form:

$$L_{-n_1} L_{-n_2} \cdots L_{-n_k} \left|h, \bar{h}\right\rangle \quad \text{with } n_i > 0$$

This generic state is characterized by a **level** $N = n_1 + \cdots + n_k$, and its total holomorphic weight is $h + N$. The primary state and its infinite tower of descendants together form a complete representation of the Virasoro algebra, known as a **Verma module**. This structure is fundamental for understanding the spectrum of states in two-dimensional conformal field theories and for organizing the resolution of correlation functions (through the OPE and the Ward identities) in terms of contributions from primary fields and their descendants.

Example (Examples of Descendant States). At the lowest levels, the first independent descendant states in the holomorphic sector are:

$$L_{-1} \left|h, \bar{h}\right\rangle \ , \quad L_{-2} \left|h, \bar{h}\right\rangle \ , \quad L_{-1}^2 \left|h, \bar{h}\right\rangle \ , \quad \ldots$$

Under the state-operator correspondence, these descendant states correspond to local operators obtained by acting on the primary field with derivatives (and more general conformal descendants). For instance, consider the state $L_{-1} \left|\Phi\right\rangle$. We know from the action of Virasoro generators on fields that:

$$[L_{-1}, \Phi(w, \bar{w})] = \partial_w \Phi(w, \bar{w}).$$

Evaluating this at the origin directly maps the state to the derivative field:

$$L_{-1} \left|\Phi\right\rangle = (\partial\Phi)(0,0) \left|0\right\rangle .$$

Similarly, in the anti-holomorphic sector, $\bar{L}_{-1} \left|\Phi\right\rangle = (\bar{\partial}\Phi)(0,0) \left|0\right\rangle$. This confirms our earlier geometric observation: the derivative of a primary field is not primary itself, but rather the first descendant in its conformal family.

2.6.6 | Conformal Families and Verma Modules

As we have established, applying negative Virasoro modes to a highest-weight state (primary state) $\left|h\right\rangle$ generates an infinite tower of descendant states with increasing energy. Let us examine the lowest energy levels of this construction:

- Applying L_{-1} to $\left|h\right\rangle$ generates a state $L_{-1} \left|h\right\rangle$ with weight $h + 1$.
- Applying operators to reach weight $h+2$, we can use either L_{-2} or apply L_{-1} twice, generating two states: $L_{-2} \left|h\right\rangle$ and $L_{-1}^2 \left|h\right\rangle$.
- To reach weight $h+3$, we could seemingly construct multiple combinations: $L_{-3} \left|h\right\rangle$, $L_{-1}^3 \left|h\right\rangle$, $L_{-2}L_{-1} \left|h\right\rangle$, and $L_{-1}L_{-2} \left|h\right\rangle$.

However, the Virasoro algebra imposes strict linear dependencies among these combinations. For example, using the commutation relation $[L_{-1}, L_{-2}] = -L_{-3}$, we can rewrite $L_{-1}L_{-2} \left|h\right\rangle = L_{-2}L_{-1} \left|h\right\rangle - L_{-3} \left|h\right\rangle$. This proves that $L_{-1}L_{-2} \left|h\right\rangle$ is not a linearly independent state.

To systematically avoid overcounting and correctly identify the basis of independent states, we must choose and maintain a strict ordering convention. The standard choice is to order the modes with decreasing indices:

$$L_{-n_1}L_{-n_2}\cdots L_{-n_k}\left|h\right\rangle, \quad \text{with } n_1 \geq n_2 \geq \cdots \geq n_k \geq 1.$$

We can summarize the basis states and the dimensionality of the vector space at each low-lying level N in the following table:

Level N	**Basis States**	**Dimension** $p(N)$
0	$\left\|h\right\rangle$	1
1	$L_{-1}\left\|h\right\rangle$	1
2	$L_{-2}\left\|h\right\rangle,\ L_{-1}^2\left\|h\right\rangle$	2
3	$L_{-3}\left\|h\right\rangle,\ L_{-2}L_{-1}\left\|h\right\rangle,\ L_{-1}^3\left\|h\right\rangle$	3
4	$L_{-4}\left\|h\right\rangle,\ L_{-3}L_{-1}\left\|h\right\rangle,\ L_{-2}^2\left\|h\right\rangle,\ L_{-2}L_{-1}^2\left\|h\right\rangle,\ L_{-1}^4\left\|h\right\rangle$	5

While the dimension of the space of descendants matches the level N up to $N = 3$, this pattern breaks down at $N = 4$ (where the dimension is 5). The exact dimensionality of the subspace at level N is given mathematically by $p(N)$, the number of integer partitions of N. Euler provided an elegant solution to compute these partitions using a generating function:

$$\Phi(q) = \sum_{N=0}^{\infty} p(N)q^N = \prod_{n=1}^{\infty} \frac{1}{1-q^n}$$

By expanding the infinite product on the right-hand side, the non-negative integer coefficients of the Taylor expansion exactly predict the number of linearly independent descendant states at each level (miracolously the coefficients are all positive integers).

Of course, the Virasoro algebra allows us to navigate this tower in both directions. While negative modes L_{-n} act as creation operators moving us up to higher energy levels, applying generators with positive n to these descendants acts as lowering operators (in the sense of the energy spectrum), pulling us back down towards the primary state or to other lower-level descendants.

We started from primary states (generated by the application of a primary field to the vacuum) and we generated an infinite tower of descendant states. Under the state-operator correspondence, these descendants are the states generated by secondary fields (which are precisely the fields obtained by applying the modes L_{-n} to the primary field). The set of all descendants generated from a given primary state forms a representation of the Virasoro algebra. This representation is called a **conformal family** or **conformal tower**. Thus every primary field gives rise to:

- A primary state (the highest-weight state) at the top, with conformal weights $(h, \bar{h})$.
- An infinite tower of descendant states generated by the action of negative Virasoro modes, each with increasing conformal weight.

Schematically, we can represent a conformal family as:

$$\left|h,\bar{h}\right\rangle \quad \to \quad L_{-1}\left|h,\bar{h}\right\rangle, \bar{L}_{-1}\left|h,\bar{h}\right\rangle, L_{-2}\left|h,\bar{h}\right\rangle, \bar{L}_{-2}\left|h,\bar{h}\right\rangle, L_{-1}^2\left|h,\bar{h}\right\rangle, \ldots$$

This is the representation-theoretic organization underlying the operator content of the theory.

Verma Modules

The most general representation obtained by freely acting with all negative modes on a highest-weight state is called a **Verma module**. Each conformal family forms a representation of the Virasoro algebra, providing the complete algebraic framework needed to describe every state and field in the theory.

In the holomorphic sector, the Verma module built on a highest-weight state $|h\rangle$ is the linear span of states of the form

$$L_{-n_1}L_{-n_2}\cdots L_{-n_k}, \quad n_1 \geq n_2 \geq \cdots \geq n_k \geq 1,$$

exactly as represented in the table above (Table 2.6.6). The anti-holomorphic sector is constructed in an identical way using the $\bar{L}_{-n}$ modes. The full Verma module is then the tensor product of the holomorphic and anti-holomorphic sectors. The ordering convention avoids overcounting, since different products of negative modes are generally not independent unless ordered, as seen.

A Verma module (in the holomorphic sector), denoted as $\mathcal{V}_{c,h}$, is formally defined as the infinite direct sum of the finite-dimensional subspaces of descendant states at each level N:

$$\mathcal{V}_{c,h} = \bigoplus_{N=0}^{\infty} \mathcal{V}_{c,h}^{(N)}$$

where $\mathcal{V}_{c,h}^{(N)}$ is the subspace at level N, having dimension $p(N)$. The entire module is uniquely labeled by the central charge c of the theory and the conformal weight h of the original primary state. The Verma modules are the fundamental building blocks of the representation theory of the Virasoro algebra and are the essential tool for classifying the operator content in two-dimensional conformal field theories.

Null States and Reducibility

In general, not all descendant states generated within a Verma module are physically independent or physically allowed. It may happen that some specific linear combination of descendants possesses a strictly zero norm. Such states are called **null states** or **singular vectors**.

A singular vector $|\chi\rangle$ at level N exhibits a deeply peculiar algebraic property: while it is built from negative modes acting on $|h\rangle$, it is simultaneously annihilated by all positive Virasoro modes ($L_n |\chi\rangle = 0$ for $n > 0$). This means a singular vector behaves exactly like a new, independent primary state embedded within the descendant tower. Because its inner product with any other state is zero, *it completely decouples from all physical correlation functions.*

When null states are present, the original Verma module is no longer irreducible. Because the singular vector acts as a highest-weight state itself, it generates its own entire sub-Verma module of descendants. To preserve the probabilistic interpretation of quantum mechanics (unitarity) and obtain a physically sound irreducible representation, one must mathematically quotient out the entire submodule generated by these null states.

This reducibility issue is of paramount importance: setting the singular vectors to zero imposes incredibly powerful differential equations on the correlation functions. This mathematical truncation process will become central later, as it is the exact mechanism that governs the finite operator content and the exact solvability of the so-called **minimal models**. For the moment, we simply note that the representation theory of the infinite-dimensional Virasoro algebra is profoundly richer and more intricate than that of an ordinary finite-dimensional Lie algebra.

2.7 | Boh

[...]

$$\mathcal{H}_c = \bigoplus_{h,\bar{h}} N_{h,\bar{h}} \mathrm{Vir}_c(h) \otimes \bar{\mathrm{Vir}}_c(\bar{h}),$$

[...]

Let us consider the unitarity of the Virasoro generators. We have

$$L_n^\dagger = L_{-n},$$

it should be an hermiticity condition....

We can assume the normalization of primary tates as

$$\langle h|h\rangle = 1, \quad \langle h|h'\rangle = \delta_{h,h'}.$$

If we have this primary state, its scalar product is given by the two-point function as we have seen. what about the product of the descendant states? We can write for the secondary state

$$\langle h| L_{-1}^\dagger L_{-1} |h\rangle = \langle h| L_1 L_{-1} |h\rangle\,,$$

and since the commutator is given by

$$[L_1,\, L_{-1}] = \cdots = 2L_0,$$

we can write the previous expression as

$$\begin{aligned}\langle h| L_{-1}^\dagger L_{-1} |h\rangle &= \langle h| (2L_0 + L_{-1}L_1) |h\rangle \\ &= 2h \langle h|h\rangle + \langle h| L_{-1}L_1 |h\rangle = 2h.\end{aligned}$$

This is a positive number, so we can say that the state $L_{-1}|h\rangle$ is a physical state (its norm is positive). Thus we have found a first condition for the unitarity of the theory, which is that the conformal dimension $h \geq 0$ must be positive.

There are many interesting situations of non unitary CFTs or minimal models, in which we will not have this constarint of positivity for h. But we will not discuss this topic here.

We have also that the central charge c must be positive since ...

2.7.1 | Gram Matrix

The Gram matrix is defined as the matrix of scalar products of the descendant states:

$$G_{ij}^{(N)} = \left\langle v_i^{(N)} \middle| v_j^{(N)} \right\rangle,$$

where $\left| v_i^{(N)} \right\rangle$ are the descendant states at level N.

As we have already seen, the first level is given by the state $L_{-1}|h\rangle$, and its norm is $2h$:

$$G^{(1)} = \langle h| L_1 L_{-1} |h\rangle = 2h \geq 0,$$

while already at the second level we have two descendant states, $|1\rangle = L_{-2}|h\rangle$ and $|2\rangle = L_{-1}^2|h\rangle$, and we can compute

$$\begin{aligned}
\langle 1|1\rangle &= \langle h| L_2 L_{-2} |h\rangle = \langle h| \left(L_{-2}L_2 + 4L_0 + \frac{c}{2}\right)|h\rangle = 4h + \frac{c}{2},\\
\langle 1|2\rangle &= \langle h| L_2 L_{-1}^2 |h\rangle = \cdots = 6h,\\
\langle 2|2\rangle &= \langle h| L_{-1}^2 L_1^2 |h\rangle = \cdots = 4h(2h+1).
\end{aligned}$$

TODO: Finish the calculation of the Gram matrix.

Thus if we were to compute the Gram matrix at level 2, we would find

$$G^{(2)} = \begin{pmatrix} 4h + \frac{c}{2} & 6h \\ 6h & 4h(2h+1) \end{pmatrix},$$

TODO: do it

and if we were to stop at level three, we would find a 3×3 matrix, and so on.

From this matrix we can obtain some useful information about the theory. For example, we can compute its determinant, which will give us insight about the linear independence of the descendant states. If the determinant is zero, it means that there is a linear dependence among the descendant states, which can lead to the presence of null states in the theory. Null states are states that have zero norm and are orthogonal to all other states in the theory, including themselves. They play an important role in the structure of CFTs and can lead to constraints on the allowed values of h and c.

If $h = 0$ we have the identity operator, which corresponds to the vacuum state. And the first descendant state is given by $L_{-1}|0\rangle$, which is the application of the translation generator on the vacuum. Since the vacuum is invariant under translations, we have $L_{-1}|0\rangle = 0$, which means that the first descendant state is a null state. This is consistent with the fact that the determinant of the Gram matrix at level 1 is zero when $h = 0$.

In the conformal family of the identity operator (equivalently in the verma module of the vacuum) the first seconday state is absent (null state), while the second descendant state is given by $L_{-2}|0\rangle$, which is an operator of dimension 2, and it is the energy-momentum tensor $T(z)$. This is consistent with the fact that the determinant of the Gram matrix at level 2 is zero when $h = 0$ and c is arbitrary.

Let us indeed compute the determinant of the Gram matrix at level 2:

$$\det G^{(2)} = (4h + \frac{c}{2})(4h(2h+1)) - (6h)^2 = 16h^3 + 8h^2 - 2hc = 2h(16h^2(2c-10)h + c).$$

Usually, when solving for the unitarity constraints, we look for the values of h and c such that the determinant of the Gram matrix is positive, which ensures that all descendant states have positive norm. If we find a particular set of h and c which annihilates the determinant, it means that there is a null state at that level, i.e. a particular linear combination of descendant states at that level that has zero norm. It is orthogonal to all other states in the theory, including itself. If we want to build an irreducible representation of the Virasoro algebra, we need to quotient out by the null states, which means that we need to set the null states to zero in the theory. But a null vector gives birth to an entire submodule of the Verma module, which is generated by the action of the Virasoro generators on the null state, and is made up of null states. Thus, if we want to build an irreducible representation, we need to quotient out by the entire submodule generated by the null state.

One can also find another null state at an higher level, which generates another submodule, which can intersect with the previous one, and so on. This is the structure of the Verma module, which can be reducible or irreducible depending on the presence of null states. It is a difficult problem to find and quotient out all the null states in a given Verma module, but it is an important problem, since it allows us to classify the representations of the Virasoro algebra and to understand the structure of CFTs.

If we know the general form of the Gram Matrix determinant at level N, we can find the values of h and c for which the determinant is zero, which gives us the unitarity constraints on the theory. This general form is given by the Kaz determinant formula, which is a formula that gives the determinant of the Gram matrix at level N in terms of the conformal dimension h and the central charge c. The Kaz determinant formula is given by

$$\det G^{(N)} = \prod_{r,s\geq 1}^{rs\leq N} (h - h_{r,s}(c))^{P(N-rs)}, \quad h_{r,s}(c) = \frac{[(m+1)r - ms]^2 - 1}{4m(m+1)},$$

where $P(N)$ is the number of partitions of the integer N, and m is a parameter related to the central charge by the formula

$$c = 1 - \frac{6}{m(m+1)}.$$

[*graph h vs c*]

at each fixed value of c we have a discrete set of values of h for the same theory, and we want to find the values of h and c for which the determinant is positive, which gives us the unitarity constraints on the theory. We can already exclude the region of the parameter space where $h < 0$ or $c < 0$, since we have already seen that these values lead to negative norm states. Thus, we can focus on the region of the parameter space where $h \geq 0$ and $c \geq 0$.

In this region, we can find the values of h and c for which the determinant is zero, which gives us the unitarity constraints on the theory. The Kaz determinant has a polynomial structure, and the zeros of the determinant correspond to the values of h and c for which there are null states in the theory. The idea is to find out the roots of the determinant, and there are different ways to do this. The most common way is to parametrize the central charge c in terms of a parameter m as we have seen, and then to find the values of $h_{r,s}(c)$ for which the determinant is zero at each fixed value of c. This gives us a discrete set of values of h for each fixed value of c, which correspond to the unitarity constraints on the theory.

We get some useful constraints: the two parameters r and s are positive integers, and the product rs must be less than or equal to N, which means that we can only have a finite number of null states at each level. Then they have to satisfy

$$1 \leq r \leq m-1, \quad 1 \leq s \leq m,$$

thus we can write

$$(r,\, s) = (m-r,\, m+1-s),$$

which leads us to a table of values for r and s for each fixed value of m, which gives us the unitarity constraints on the theory. This creates different parabulas for each value of m on the left side of the constant $c = 1$ which intersect the line $c = 1$ in their minimum. Where the parabulas intersects among each other, we have unitary theories with the same central charge but different conformal dimensions.

We know from other results that for $c > 1$ al theories are unitary, while for $c < 1$ we have a discrete set of unitary theories, which are the minimal models. The minimal models are characterized by the values of m and the corresponding values of $h_{r,s}(c)$ for which the determinant is zero, which gives us the unitarity constraints on the theory. The minimal models are important because they are exactly solvable CFTs, and they have a rich structure of primary fields and correlation functions. They also have applications in statistical mechanics and condensed matter physics, where they describe critical phenomena in two dimensions.

Systems can have other symmetries apart from the conformal symmetry, and these symmetries can lead to additional constraints on the theory, which can further restrict the allowed values of h and c and expand the structure of the algbera (thus the theory). Indeed for $c > 1$ there is an huge amount of theories, which becomes tractable only when we have additional symmetries, which can be used to classify the theories and to find their correlation functions. But for $c < 1$ we have a discrete set of theories, which are the minimal models, and they are already classified by the values of m and the corresponding values of $h_{r,s}(c)$ for which the determinant is zero, which gives us the unitarity constraints on the theory (with additional symmetries). Thus, for $c < 1$ we have a complete classification of unitary CFTs, while for $c > 1$ we have an infinite number of theories, which can be classified only with additional symmetries.

With $m = 2$ we have only one unitary theory, which has only the vacuum state and is trivial. From $m = 3$ we have the first non-trivial unitary theory, and the central charges start from $c = 1/2$ and increase as m increases, approaching $c = 1$ as $m \to \infty$.

For example we have the Ising model with a distribution of vacancies (in some sites on the lattice we have a vacancy, which means that there is no spin in that site), which has an interesting phase diagram, and it has a tri-critical point.

Let us consider the case $m = 3$, we can compute a table of values for r and s at $c = \frac{1}{2}$

$$[\dots]$$

and we get only 3 values of h, which correspond to the three primary fields of the Ising model, which are the identity operator, the spin operator, and the energy operator. The identity operator has $h = 0$, the spin operator has $h = 1/16$, and the energy operator has $h = 1/2$. These three primary fields generate the entire spectrum of the theory, and all other fields can be obtained as descendants of these primary fields. The correlation functions of these fields can be computed using the conformal symmetry and the structure of the Verma modules, which allows us to find exact results for the correlation functions in this theory.

We should imagine to have this Kaz table (left table) multiplied by another identical Kaz table for the anti-holomorphic sector (right table), which gives us the full spectrum of the theory, which is given by the tensor product of the holomorphic and anti-holomorphic sectors. Thus, for each value of c we have a discrete set of values of h and $\bar{h}$ for which the determinant is zero, which gives us the unitarity constraints on the theory. And once we get h and $\bar{h}$ we can compute the conformal dimension $\Delta = h + \bar{h}$ and the spin $s = h - \bar{h}$ of the primary fields, which gives us the full spectrum of the theory. Looking at the universality classes given by these tables, we strongly suspect that this is the Ising model (as well as the free-majorana fermion theory, which is equivalent to the Ising model).

we can have a bosonic description of the Ising model, which is given by a ϕ^4 lagrangian, as well as the fermionic description, which is given by a free Majorana fermion lagrangian. These two descriptions are equivalent and dual.

Appendices

A | Notation and Conventions

B | Computations and further developments